



**GREEN
CLIMATE
FUND**

Meeting of the Board

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Port Moresby, Papua New Guinea

Provisional agenda item 11

GCF/B.42/02/Add.17

9 June 2025

Consideration of funding proposals – Addendum XVII

Funding proposal package for FP275

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase 1) Programme” by Caribbean Development Bank;
- b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- f) Gender documentation of the funding proposal.

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Funding Proposal

Project/Programme title:	Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase 1) Programme
Country(ies):	Barbados, Belize, Jamaica
Accredited Entity:	Caribbean Development Bank
Date of first submission:	2025/04/28
Date of current submission	2025/04/28
Version number	V.1.3



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

ACRONYMS / ABBREVIATIONS

AC	Air Conditioner
APRs	Annual Performance Reports
AQI	Air Quality Index
AE	Accredited Entity
BAU	Business as Usual
BCCI	Barbados Chamber of Commerce and Industry
BEL	Belize Electricity Limited
BLPC	Barbados Light & Power Company
BMCs	Borrowing Member Countries (collectively Barbados, Belize & Jamaica)
BNEP	Barbados National Energy Policy
BREA	Barbados Renewable Energy Association
CCKP	Climate Change Knowledge Portal
CDB	Caribbean Development Bank
CHP	Combined heat and power
CN	Concept Note
COP	Community of Practice
DERs	Distributed Energy Resources
DG	Distributed Generation
DPM	Division of Portfolio Management
DREAM	Disaster Risk and Energy Access Management
E&S	Environmental and Social
EE	Energy Efficiency
ERCAP	Energy Resilience for Climate Adaptation
ESCOs	Energy Service Companies
ESD	Energy for Sustainable Development in the Caribbean
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESMS	Environmental and Social Management System
ESS	Environmental and Social Safeguards
ESU	Environmental Sustainability Unit
FIT	Feed-in Tariff
FP	Funding Proposal
FX	foreign exchange
GAP	Gender Action Plan
GCF	Green Climate Fund
GESI	Gender Equality and Social Inclusion
GHG	greenhouse gas
GRM	grievance redress mechanism
ICA	Integrity, Compliance and Accountability
IEU	Independent Evaluation Unit
IPPs	Independent Power Producers
IUS	Integrated Utility Services
JPS	Jamaica Public Services Company Limited
KYC	Know Your Customer
LCDS	Low-Carbon Development Strategy
LEDs	Low Emissions Development Strategy
M&R	Monitoring and Reporting
MFFSD	Ministry of Forestry, Fisheries and Sustainable Development
ML/TF	money laundering/terrorist financing
MSET	Ministry of Science, Energy, Telecommunications and Transport
NCCPSAP	National Climate Change Policy, Strategy and Action Plan
NDA	National Designated Authorities

NDC	Nationally Determined Contributions
NPV	Net Present Value
OCR	Ordinary Capital Resources
OIE	Office of Independent Evaluation
OPPM	Operational Policies and Procedures Manual
PMC	Projects Complaints Mechanism
PPF	Project Preparation Facility
PPU	Procurement Policy Unit
PSSEP	Public Sector Smart Energy Programme
PV	Photovoltaic
PWDs	persons with disabilities
RE	renewable energy
RE/FIT	Renewable Energy Feed-In Tariff
REDD	Reduced Emissions from Deforestation and Forest Degradation
RER	Renewable Energy Rider
RFP	Request for Proposals
SDF	Special Development Fund
SDGs	Sustainable Development Goals
SEAH	Sexual Exploitation, Abuse and Harassment
SEFB	The Sustainable Energy Framework for Barbados
SIDS	Small Island Developing States
TA	Technical Assistance
TSPs	Technology Service Providers
UNFCCC	United Nations Framework Convention on Climate Change

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Programme	A.2. Public or private sector	Private	
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. Not applicable			
A.4. Result area(s)b	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF contribution	Co-financers' contribution¹	
	Mitigation total	Enter number %	Enter number %	
	☒ Energy generation and access	50 %	50 %	
	☐ Low-emission transport	Enter number %	Enter number %	
	☒ Buildings, cities, industries and appliances	25 %	25 %	
	☐ Forestry and land use	Enter number %	Enter number %	
	Adaptation total	Enter number %	Enter number %	
	☐ Most vulnerable people and communities	Enter number %	Enter number %	
	☐ Health and well-being, and food and water security	Enter number %	Enter number %	
	☒ Infrastructure and built environment	25 %	25 %	
☐ Ecosystems and ecosystem services	Enter number %	Enter number %		
A.5. Expected mitigation outcome <i>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</i>	601,550 tCO₂e (over a 25-year period, which captures the productive lifespan of all technologies/assets to be supported through the Programme).	A.6. Expected adaptation outcome <i>(Core indicator 2: direct and indirect beneficiaries reached)</i>	Approx 40,655 direct and indirect beneficiaries.	
			Approx 10,655 direct beneficiaries	Approx 30,000 indirect beneficiaries
			About 0.3% of the total population	About 1% of the total population

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

A.7. Total financing (GCF + co-finance²)	68,746,295 USD	A.9. Project size	Medium (Upto USD 250 million)
A.8. Total GCF funding requested	26,736,295 USD <i>For multi-country proposals, please fill out annex 17.</i>		
A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i> ☒ Grant <u>Enter number</u> ☐ Equity <u>Enter number</u> ☒ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	6 years	A.12. Total lifespan	25 years
A.13. Expected date of AE internal approval	Within 3 months after GCF Board approval	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☐ No ☒		
A.20. Executing Entity information	- Caribbean Development Bank (via the CDB TA Execution Team) - Barbados Light and Power Limited is a regulated electric utility in Barbados. It is a limited liability company incorporated under the Laws of Barbados, and is a wholly (100%) owned subsidiary of Emera Inc., a Canadian utilities conglomerate. - Belize Electricity Limited is a regulated electricity utility in Belize. It is a limited liability company incorporated under the laws of Belize, and is majority owned by the Belize public sector. The Government of Belize owns 32.58%, while the Social Security Board owns 31.27%. The remaining shares are owned by Fortis Cayman Inc. (33.3%) and 1,500 small shareholders (2.85%). - Jamaica Public Services Company Limited is a regulated electricity utility in Jamaica. It is a limited liability company incorporated under the laws of Jamaica, and is majority owned by a consortium comprising the Marubeni Corporation from Japan and the East-West Power Corporation from South Korea (combined 80%). The Government of Jamaica owns 19% of the shares while a small group of minority shareholders control the remaining 1%.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

² Refer to the Policy of Co-financing of the GCF.

The Caribbean region is currently facing the adaptation & mitigation challenges of climate change, evidenced by intensified storms and rising temperatures, and an energy sector heavily reliant on imported fossil fuels, contributing significantly to greenhouse gas (GHG) emissions.

Climate: The Borrowing Member Countries (collectively “Barbados, Belize and Jamaica” or the “BMCs”) are facing significant climate change impacts, including intensified storms, floods, droughts, sea-level rise, and rising temperatures. These changes are affecting not only the availability and supply of energy resources but also the transmission, distribution, operations, and energy demand. The historical and projected impacts of climate change indicate a trend of rising temperatures, escalating the risk of heat stress and severe weather events, which are expected to continue and intensify. For instance, the International Energy Agency has noted that extreme heat boosts the demand for cooling, with sustained average daily temperatures of 30°C typically resulting in a 16% spike in weekly sales of electricity.³ Moreover, the BMCs are committed to ambitious goals to reduce greenhouse gas emissions by increasing the share of renewables in their energy mix and improving energy efficiency across all sectors.

Energy: The BMCs heavily rely on imported fossil fuels for electricity, which constitutes a significant portion of their annual greenhouse gas emissions. The energy sectors in these countries are critical to their GHG profiles, with other sectors such as road transport, industry, agriculture, and land-use change also being notable contributors. The Integrated Utility Services (IUS) Programme intends to drive investment in renewable energy and energy efficiency measures to reduce this dependency, which is increasingly important as the demand for electricity is expected to rise with economic growth. The integration of Distributed Energy Resources (DERs) is also a focus for enhancing the resilience and diversification of the electricity grid.

The introduction of an IUS model in BMCs is seen as a strategic step toward addressing various challenges in their energy sectors. These include reducing GHG emissions, decentralizing energy systems, enhancing resilience to climate change impacts, and decreasing dependence on imported fossil fuels. Additionally, the model supports adaptation to heat through providing cooling solutions to ensure comfortable indoor temperatures, using energy-efficient air conditioners and decentralized renewable energy supply to reduce peak electricity demand during times of high temperatures, thereby supporting the reliability of the grid. The Programme aims to democratize energy policy, increase access to clean energy, reform utility business models, and enhance capabilities for financing DERs. Furthermore, it seeks to foster regional learning and knowledge sharing while developing a sustainable market ecosystem for behind-the-meter solutions in the Caribbean.

Programme Approach

The Governments of Barbados, Belize, and Jamaica aim to accelerate decarbonization efforts in their electricity sector and increase the resilience of the communities to climate change impacts. They aim to meet international climate change commitments by focusing on renewable energy (RE) and energy efficiency (EE) technologies. The proposed IUS Programme will be administered by electric utilities to support the deployment of these technologies. The Programme will involve education for customers, financing for installations, and repayment through monthly bills, overcoming the barrier of high upfront costs. This Programme constitutes the first of two phases for the overall IUS Programme in the region. While Phase I would provide support to the three countries to implement the Programme, Phase II will result in a broader implementation and scale-up to other member-countries of the Caribbean Development Bank (CDB).

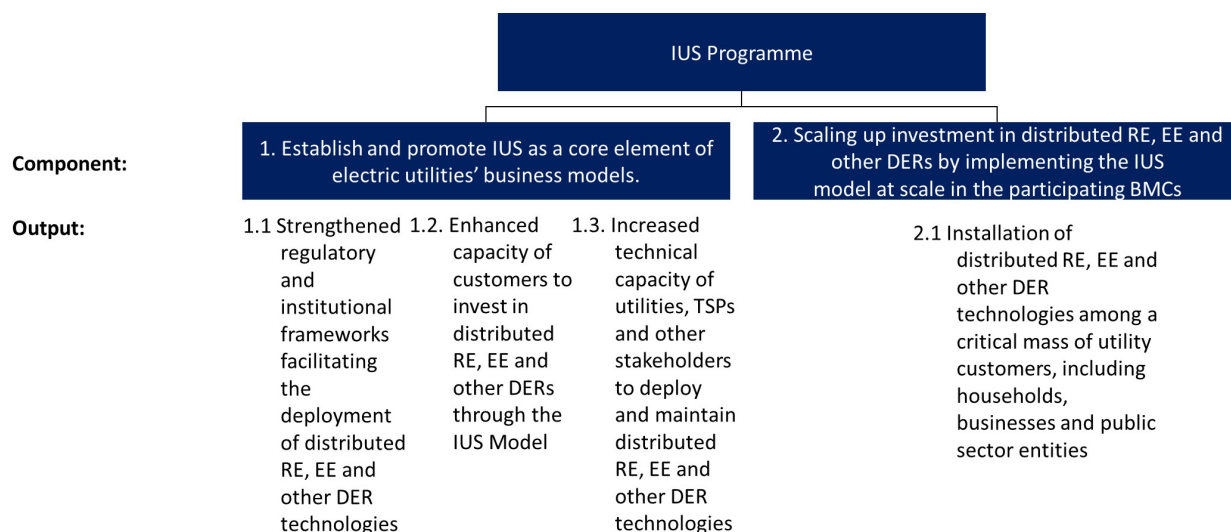
This IUS model is a mechanism to directly implement demand-side RE and EE projects through utilities in an open market system.

The IUS Programme will include a “national project” in each country to be executed by the national electric utility and overseen by the Ministry and regulator responsible for oversight of the electricity sector. Each “national project” will include a series of sub-projects at the level of the utility’s customers, notably residential customers (i.e. households) and commercial customers (i.e. local businesses). Through each of these small-scale sub-projects, eligible RE, EE and/or other DER technologies will be installed at the premises of the interested utility customers. Utility customers will first receive education and awareness about the options available to them and then they will choose to implement the technologies through registered qualified service providers that work closely with the utility. The utility will provide the upfront financing for the installation of the sub-projects and support the service providers, while the customers will then repay the amount owed to the utility for the installation through their monthly bill. A major barrier for the uptake of these technologies has historically been the high upfront cost to install them, so this Programme is expected to alleviate that pressure.

Programme Components:

The IUS Programme is structured in a way that it not only addresses all the key risks and barriers assessed during the programme preparation phase but also supports the much-needed ecosystem development for a sustainable, scalable, and replicable growth of behind the meter energy solutions in the BMCs. The below figure outlines the key components of the IUS Programme.

Figure 1: Structure of the IUS Programme



Source: CDB/CPCS

The Programme, through Component 1, establishes and promotes IUS as a core element of each Executing Entity's business model. It underpins timely and effective delivery of country-level support and enables the BMCs to guide and support one another during and after the Programme. Component 2 aims to build on this foundation to scale up investments in distributed RE, EE and other DERs through widespread use of the IUS model by the Executing Entities in the BMCs.

Each programme component is broken down into sub-components, which themselves comprise several activities that support a robust programme design and delivery.

Programme Benefits:

The IUS Programme anticipates significant climate-related benefits:

- **Reduction in GHG Emissions:** The Programme is projected to make a significant reduction in the carbon footprint of the participating countries by enabling the avoidance of an estimated 601,550 tons of CO₂ equivalent (tCO₂e) over the lifetime of the Programme. This results in an estimated cost of **USD 70 per tCO₂e avoided**,⁴ and for the GCF's contribution, the cost is approximately **USD 43 per tCO₂e avoided**.⁵
- **Increasing Resilience Against Heat Stress:** The Programme is projected to provide energy efficient air conditioners (AC) to about **9,444 residential** and **1,211 commercial** beneficiaries.
- **Decreased Burden on Utilities and Increased Renewable Energy Generation:** Energy efficiency interventions are expected to reduce energy consumption across various sectors, including residential, commercial, and public. The Programme anticipates considerable energy savings which translate into lower utility bills for consumers and decreased overall demand on energy infrastructure. A shift towards renewables is anticipated, with a targeted increase in the contribution of clean energy to the national grids. This includes

³ IEA (2023). [Keeping cool in a hotter world is using more energy, making efficiency more important than ever.](#)

⁴ Estimated cost per tCO₂e avoided refers to the total cost of the GCF loan, GCF grant and CDB loan (USD41.95 mil), excluding parallel investments by the utilities, divided by the total GHG emissions avoided.

⁵ Estimated GCF cost per tCO₂e avoided refers to the total GCF cost (USD25.76 mil) divided by the total GHG emissions avoided.

the deployment of technologies like solar photovoltaic systems, solar water heating, and possibly wind and geothermal energy, depending on the country-specific strategies and natural resource availability.

- **Economic Development:** The transition to a greener energy sector is intended to stimulate local economies through job creation in the renewables sector, foster innovation, and provide opportunities for local businesses. Energy cost savings and the potential for new energy-related businesses are additional economic benefits.
- **Capacity Building and Knowledge Sharing:** A core outcome of the Programme is the enhancement of institutional capacities and regulatory frameworks in each participating country. This involves training programs, development of standards, and the establishment of best practices for EE and RE deployment.
- **Scaling Up of Investments:** By de-risking investments and establishing favourable conditions for private and international financing, the Programme aims to catalyse additional resources for the energy sector transformation. This will enable a scaling up of clean energy projects and attract further investments into the region.
- **Gender and Social Inclusion:** The Programme places an emphasis on ensuring that the benefits of the energy transition are equitably distributed, with specific measures to include women and vulnerable communities in the opportunities created by the shift to sustainable energy sources. The Programme aims to provide a more inclusive way to support sustainable energy investments, particularly for women who often face more economic barriers than men due to traditional divisions of labour and social norms. Through the Gender Action Plan (GAP), the Programme will address the following gender-specific needs: governance and participation (through targeted actions focused on policies, utilities, and other key actors), and access, education, and assets (through targeted actions focused on the marginalised beneficiaries).

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

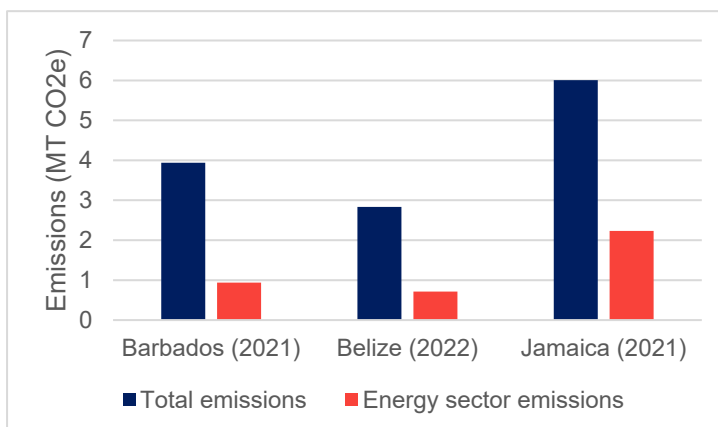
i. Climate change problem

Mitigation needs

B.1.1. Barbados, Belize, and Jamaica heavily rely on imported fossil fuels for electricity, contributing significantly to their GHG emissions:

The energy sectors in the BMCs that are participating in Phase I are characterized by a continued dependence (to varying degrees) on imported fossil fuels. Despite recent efforts to reduce this dependency, fossil fuels remain an important element of each country's electricity generation mix. As a result, electricity generation is responsible for a significant proportion of total annual GHG emissions in Barbados, Belize, and Jamaica. Other sectors that contribute significantly to total annual GHG emissions in these BMCs include road transport, industry, agriculture and land-use change & forestry.

Figure 2: Total emissions vis-a-vis energy sector emission (excl. industrial processes, etc.) in the three BMCs



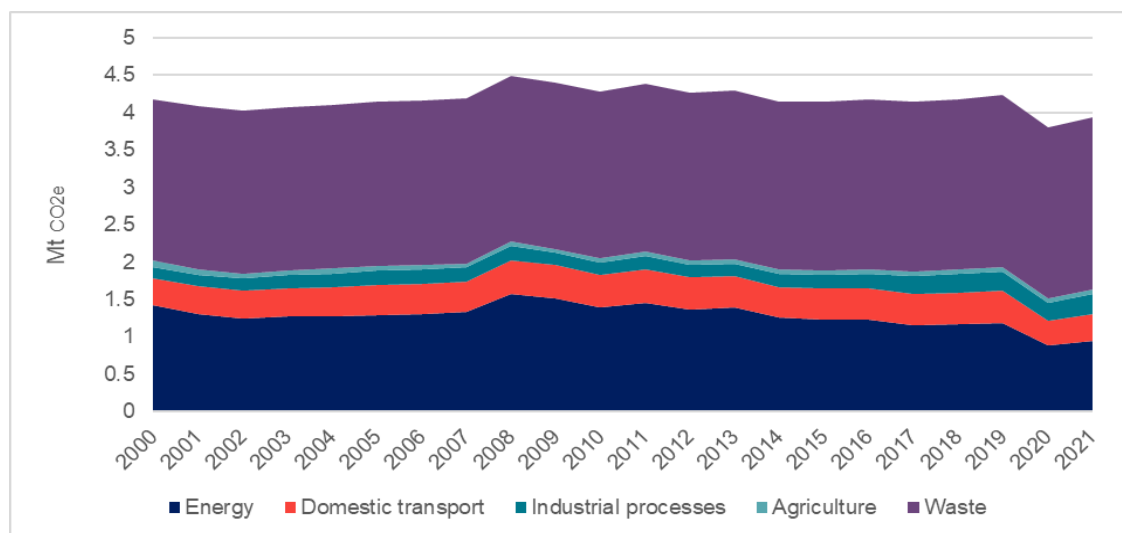
B.1.2. The Governments of Barbados, Belize, and Jamaica have each outlined clear strategies for addressing climate change and reducing emissions, initiatives that will be supported by the IUS programme. While Belize acts as a net carbon sink and the emissions from Barbados and Jamaica are comparatively lower than those of many high-emitting countries worldwide, all three nations have committed to ambitious goals to contribute to global carbon emission reduction targets. The following section provides a summary of the mitigation needs of these BMCs as derived from the Baseline Assessment. For detailed information, please refer to Annex 2b.

Mitigation rationale

Barbados:

B.1.3. Emissions profile: Barbados' GHG emissions are mostly attributed to four key sectors: energy, waste, industrial processes, and agriculture.

Figure 3: Barbados's Energy Greenhouse Gas Emissions by Source Category (2000 – 2021)



Source: Climatewatch Data

B.1.4. Emission reduction targets: Barbados has established climate targets as well as international commitments and policies to reduce its GHG emissions across all sectors, including the energy sector. The table below summarizes these commitments:

Table 1: Barbados' relevant international and climate commitments, policies, and climate targets

Document	Institution (date)	Key points
NDC	Ministry of Environment and National Beautification (2021)	By 2025: ↓ 20%* emissions ↓ 35%** emissions. Total reduction of 705 CO ₂ e*** By 2030: ↓ 35%* emissions ↓ 70%** emissions. Total reduction of 1,459 CO ₂ e***
Barbados National Energy Policy (BNEP) 2019-2030	Ministry of Energy and Water Resources (2019)	100% renewable energy by 2030: Solar centralised (205 MW); Solar distributed (105 MW); Wind onshore (150 MW); Wind offshore (150 MW); Biomass and Waste-To-Energy (15 MW); and Centralised and distributed energy storage (200 MW)
The Sustainable Energy Framework for Barbados (SEFB)	Government of Barbados (2010)	The framework seeks to achieve a reduction of Barbados's dependence on fossil fuel through the diversification of the country's energy matrix and its reduction to oil price volatility through the formulation of policy and regulation, promote measure to mitigate GHGs, and support institutional strengthening.

*unconditional, ** conditional, upon international support, *** against the 2008 base year

B.1.5. Associated current and projected emissions from the sector without the program: The Second National Communication from Barbados submitted to the UNFCCC in 2018 reported total emissions of 1.4 Mt CO₂e in 2010 for the energy sector. While the NDC included an economy-wide Business as Usual (BAU) emission projection, it did not disaggregate by sector. The economy-wide BAU emission projections for 2030 is 1.9 Mt CO₂e, while the emissions for the NDC base year (2008) is 2.1Mt CO₂e. Hence, the projected economy-wide emission reductions by 2030 is 165 kt CO₂e⁶, which is an average annual decrease in emissions by 7.5 kt CO₂e since 2009.⁷ This is equivalent to a 0.35% annual decrease in emissions from the 2008 base year emission values, based on the assumption that the emissions reduction is uniform across the years. In order to project energy emissions, the 0.35% annual emission reduction was then applied to the 2010 emissions for the energy sector (1.4 Mt CO₂e). Based on this, the projected emissions from Barbados's energy sector in 2030 is estimated at 1.3 Mt CO₂e.

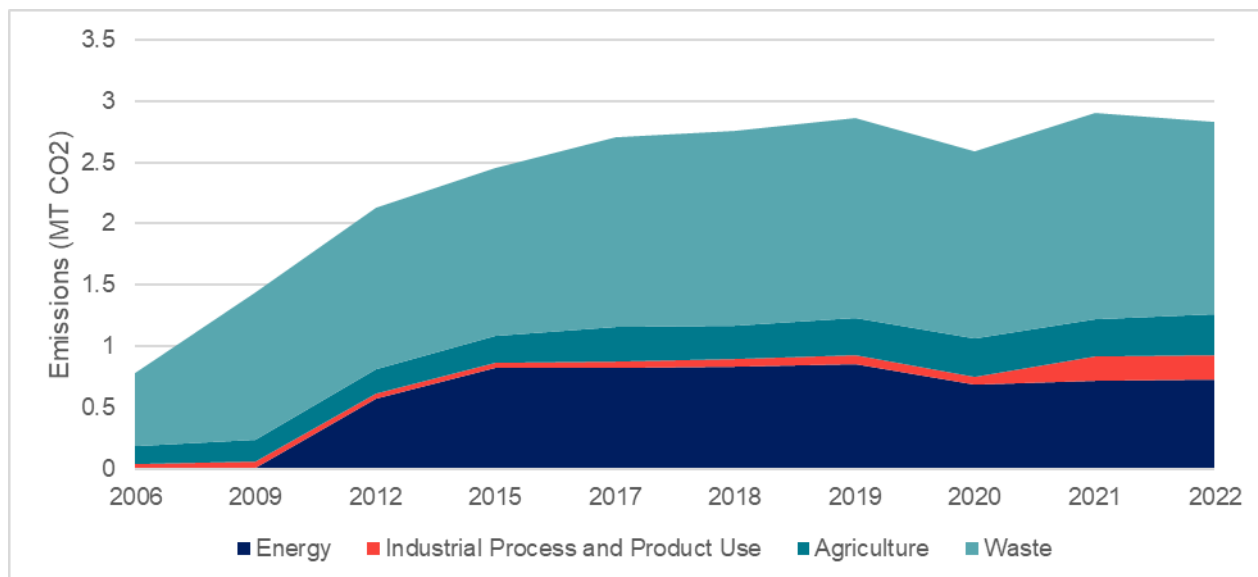
Belize

B.1.6. Emissions profile: Emissions from Belize are attributed to two key sectors, energy and transport. The following figure highlights GHG emissions across key sectors over the past years.

⁶ Obtained by subtracting the 2008 emissions from the 2030 projected economy-wide BAU emissions

⁷ Obtained by dividing

Figure 4: Belize emissions across key sectors



Source: CPCS analysis – data from UNFCCC⁸

B.1.7. Emission reduction targets: Belize has established climate targets as well as international commitments and policies to reduce its GHG emissions across all sectors, including the energy sector. The table below summarizes these commitments:

Table 2: Belize's relevant international and climate commitments, policies, and climate targets

Document	Institution (date)	Key points
NDCs	Government of Belize (2021)	Between 2021 and 2030: Total avoidance of 5,647 KtCO ₂ e By 2030: Avoidance of 44 KtCO ₂ e in the national electricity supply sector and 75% share of renewables in generation.
Belize Low Emissions Development Strategy (LED) and Action Plan	Belize National Climate Change Office (2019)	By 2050: Elimination of the majority of its gross carbon emissions. Proposes 12 actions for reducing emissions in the energy sector by 2050.
Horizon 2030	Ministry of Economic Development (2010)	Reference to the promotion of green energy and proposes 5 actions.
The National Energy Policy Framework	Government of Belize (2011)	The policy has the goal to: Promote sustainable energy for national development; Foster energy-efficient culture; Minimize energy costs and disaster impacts; Mitigate supply shocks and price volatility.
The National Climate Resilience Investment Plan 2013	Ministry of Finance and Economic Development (2013)	Proposes actions to render the energy sector of Belize more resilient to the impacts of climate change.
The National Climate Change Policy, Strategy and Action Plan (NCCPSAP), 2015-2020	Ministry of Forestry, Fisheries and Sustainable Development (MFFSD) (2014)	The policy seeks to achieve the mainstreaming and integration of Climate Change adaptation and mitigation at all levels of development planning. The strategy for the energy sector entails the elaboration of appropriate policy tools and instruments to advance energy efficiency, speed up the adoption of renewable energy and transit to clean production.

⁸ Belize. 2024 National Inventory Document; Emissions data not available for 2007, 2008, 20010, 2011, 2013, 2014 and 2016.

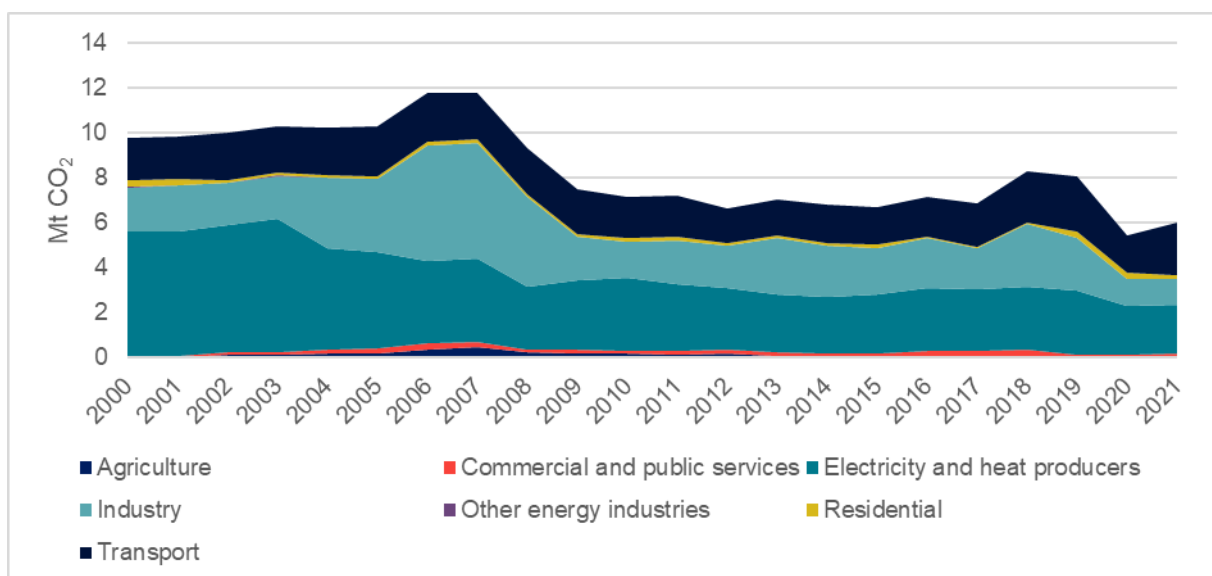
B.1.8. Associated current and projected emissions from the sector without the program

The emissions attributable to Belize's electricity sector in 2020 were estimated at 170 kt CO₂e, excluding emissions associated with imported electricity whose value for the same year was estimated at 106 kt CO₂e⁹. Hence, the 2020 total emissions from the electricity sector including imported and domestic production were 276 kt CO₂e. The Low Emission Development Strategy (LEDS) of Belize assumes an annual BAU energy consumption growth rate of 3%. Based on this, the BAU emissions for the electricity sector (domestic plus imports) in 2030 is projected to be 371 kt CO₂e.

Jamaica

B.1.9. Emissions profile: Jamaica's energy sector has been the largest source of CO₂ emissions in the last two decades.

Figure 5: Jamaica emissions across key sectors



Source: International Energy Agency (2023)

B.1.10. Emission reduction target: Jamaica's international commitments, policies and climate targets related to the energy sector are presented below.

Table 3: Jamaica's relevant international and climate commitments, policies, and climate targets

Document	Institution (date)	Key points
NDCs	Government of Jamaica (2020)	GHGs reductions from the electricity generation and energy use sectors and land use change and forestry, of between 25.4%* and 28.5%**.
National Policy on Environmental Management Systems	Ministry of Economic Growth and Job creation (2017)	The policy has as its overall goal the promotion, certification, implementation, monitoring and evaluation of environmental management systems in order to curb environmental impacts and enhance efficiencies.
National Energy Policy 2009-2030	Ministry of Energy and Mining (2010)	Revision of the Energy Policy Green Paper 2006-2020. 20% of renewable energy in the energy mix by 2030 by exploring the further development of renewable resources such as biofuels, wind, solar and hydro.

*unconditional, ** conditional, upon international support

⁹ Belize Low Emission Development Strategy (2021)

B.1.11. Associated current and projected emissions from the sector without the program

The emissions from the energy sector of Jamaica in 2012 was reported to be 7.4 Mt CO₂e, out of which 2.8 Mt CO₂e was attributed to heat and electricity production¹⁰. As per Jamaica's revised NDCs, BAU emissions in the energy sector are projected to reach 8.2 Mt CO₂e by 2030.

B.1.12. The IUS Programme will support the BMCs to achieve this vision through the provision of concessional financing which will drive investment in RE and EE measures. The attributable emissions from the energy sectors¹¹ for each of the three countries are shown in Figure 2.

B.1.13. To deliver on their emission reduction targets, it is of paramount importance that BMCs scale up the use of RE and EE measures. The need for such investments is reinforced by the fact that electricity demand in these countries is expected to increase over the coming decades as their economies grow. Consequently, the countries are working to reduce the dependence of their energy systems and energy-intensive sectors (e.g. tourism, agriculture, industry) on fossil fuels. Opportunities for doing so are increasing and seen as critical to the attainment of climate change and sustainable development goals but may simultaneously further increase overall demand for electricity.

Adaptation needs

B.1.14. The Caribbean region, including Barbados, Belize, and Jamaica, is already facing, and will continue to face, more intense storms, floods, droughts, rising sea levels, and higher temperatures.¹² The rise in temperature in the Caribbean is expected to gradually increase across all seasons consistent with global warming, through to the end of the century.¹³ All BMCs broke temperature records in 2023, experiencing the hottest days in the last 20 years. More recently, in April 2024, Belize recorded the highest temperature observed in any month for the second consecutive year. These extreme temperatures can lead to heat stress, negatively impacting human health and well-being. Active cooling solutions, such as energy-efficient air conditioning, can help alleviate these issues. Additionally, pairing PV generation with AC demand can flatten the consumption curve during peak hours. The IUS Programme will support beneficiaries in accessing these adaptation measures, increasing resilience against heat stress.

Table 4: Historical and projected climate change impacts for the BMCs

Relevant changes in climatic conditions
Barbados Historical: Since the 1960s, Barbados has experienced a notable increase in air temperature by approximately 0.14°C per decade, largely influenced by the El Niño Southern Oscillation, which significantly impacts the island's climate by dictating weather patterns and intensity. Additionally, located within the hurricane belt, Barbados is prone to natural hazards like hurricanes, floods, droughts, and landslides, especially during the hurricane season from June to November, with a peak in activity from September to November, as evidenced by severe damage from hurricanes such as Elsa in 2021 and Tomas in 2010. Projected: Modelling conducted under the CMIP6 framework, featured in the World Bank Climate Change Knowledge Portal (CCKP), projects a significant rise in the mean annual temperatures of Barbados under a high emissions scenario (SSP3-7.0). By the end of the century, temperatures are expected to reach 29.07°C, up from the 1995 baseline of 26.11°C (see Figure 7). Additionally, maximum temperatures are projected to increase by 3.5% to 13% by the 2100s across scenarios, relative to the same baseline. This data suggests a general rise in maximum temperatures throughout the year, peaking slightly from August to October. Minimum temperatures are expected to see a greater increase than maximum temperatures,

¹⁰ Jamaica's Third National Communication (2018)

¹¹ The emissions data is as per the officially published document by the three participating BMCs. These documents include the NDCs, the national communications. None of the participating BMCs have officially published their latest emissions data.

¹² UNFCC (2017). [Caribbean Marine Climate Change Report Card](#).

¹³ The Caribbean Development Bank (2020). [The state of the Caribbean Climate](#).

with projections showing a 4% to 18% rise across different scenarios compared to the baseline temperature of 25.43°C in 1995. These projections are consistent with data from more detailed, downscaled modelling efforts.¹⁴ Despite these projections, the current risk associated with extreme heat is considered low, indicating a 5% to 25% chance of experiencing at least one instance of prolonged extreme heat exposure, potentially leading to heat stress, in the next five years.¹⁵

Barbados is projected to start to face heat stress from the 2040s onwards, as well as an increase in humidity during the summer season is anticipated. Although there is a lack of detailed data on the adoption of cooling technologies in Barbados, the tourism sector, a significant part of the country's economy, has highlighted in its National Cooling Strategy the rising demand for air conditioning.

Figure 6: Projected (CMIP6) changes in average annual temperatures in Barbados

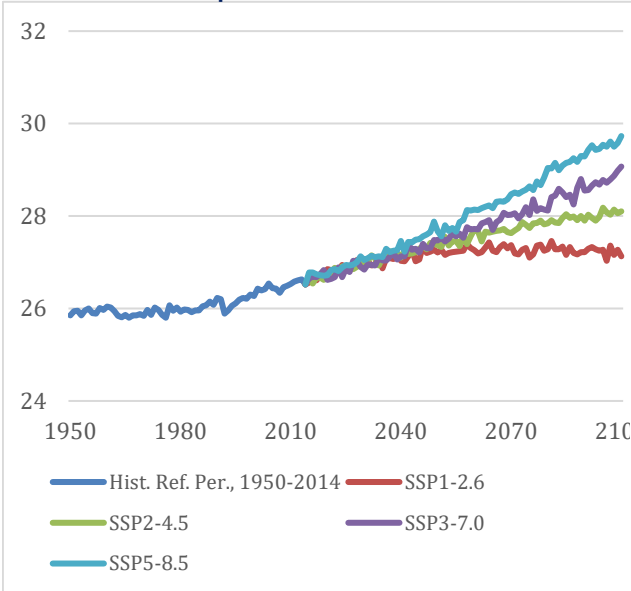
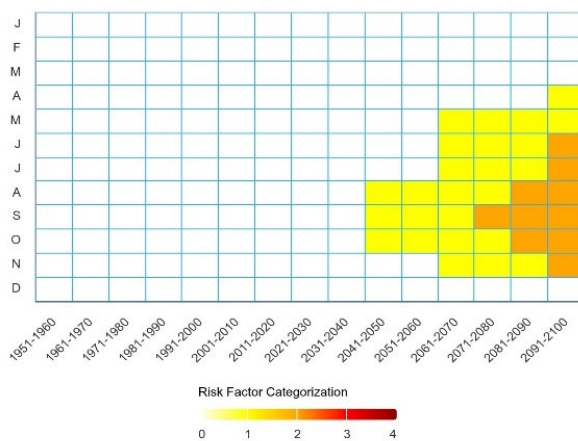


Figure 7: Heatplot for Heat Index Heat Risk Categorization (SSP2-4.5) – Barbados



Source: World Bank. Climate Change Knowledge Portal. Information retrieved on February 2024.

Belize

Historical: According to the World Bank CCKP, Belize has experienced a significant warming trend, with mean annual temperatures rising by 0.45°C since 1960, at an average rate of 0.10°C per decade. From 1960 to 2022, surface temperatures increased by 0.81°C. This period also saw a marked change in temperature patterns, with hot days and nights increasing by 67 days per year from 1960 to 2003, while cold days and nights decreased by 21 days per year.

Projected: Modelling conducted under the CMIP6 framework, featured in the World Bank CCKP, projects a significant rise in the mean annual temperatures of Belize under a high emissions scenario (SSP3-7.0). Temperatures are anticipated to increase by 1.05°C (0.88°C-1.33°C) by 2030 and are projected to reach 29.45°C (28.07°C-30.68°C) by the end of the century, from a baseline of 25.4°C (24.95°C-25.73°C) in 1995 (see Figure 9). Days with temperatures above 30°C are projected to increase by 74-175% across scenarios by the 2100s from 1995 baseline of 122 days (90 days-149 days). Days with temperatures above 35°C are projected to increase an average of 6 days per decade by 2100s (SSP3-7.0), with higher-than-average increase in days from April to September. Nights with temperatures above 26°C are projected to increase by 307-430% across scenarios from 1995 baseline of 7 days (3 days-13 days) by the end of the century. Beginning in the 2030s, Belize is anticipated to face a significant escalation in heat stress risks, even under the more optimistic SSP2-4.5 scenario (see Figure 10).

¹⁴ Caribbean Development Bank (2020). [The state of the Caribbean Climate.](#)

¹⁵ GFDRR (2020). [ThinkHazard. Barbados.](#)

Heat stress is likely to constitute a major challenge for Belize and other Caribbean countries as the climate changes over the coming decades. These impacts are expected to be particularly pronounced in the summer months. Indeed, although humidity levels in Belize tend to be consistently high throughout the year (often between 75-85%), they are typically highest during the rainy/wet season that occurs during the summer. These are the same months during which temperatures and temperature extremes (hot days, hot nights) are projected to increase the most over the coming decades, this highlights the need for Belize to increase uptake of/access to cooling solutions. In 2022, penetration of cooling technologies in Belize was 67%.

Figure 8: Projected (CMIP6) changes in average annual temperatures in Belize

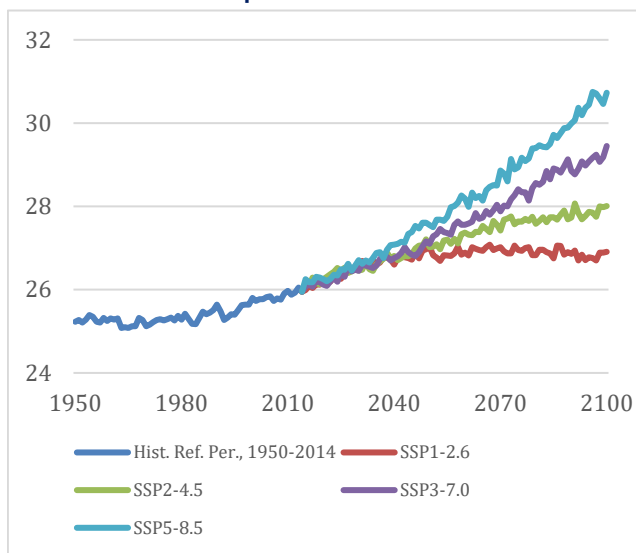
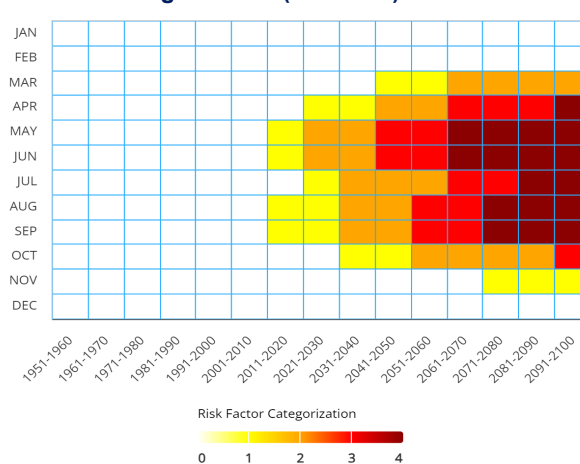


Figure 9: Heatplot for Heat Index Heat Risk Categorization (SSP2-4.5) – Belize



Source: World Bank. Climate Change Knowledge Portal. Information retrieved on February 2024.

Jamaica

Historical: As outlined in the World Bank CCKP maximum, average, and minimum temperatures are all exhibiting an increasing linear trend. Since 1960, average surface temperatures have increase by an average of 0.11°C per decade, maximum surface temperatures have increased by 0.08°C and minimum temperatures have increased by 1.04°C.

Projected: Modelling undertaken in the context of CMIP6, as featured in the World Bank CCKP, projects that the mean annual temperatures in Jamaica will increase moderately from 25.58°C (25.35°C - 25.8°C) in 1995 under the most optimistic scenario (SSP1-2.6) reaching 26.52°C (26.12°C - 26.89°C) by the end of the century. However, this scenario is highly unlikely given the current global GHG emissions trajectories. Comparatively, under an intermediate scenario (SSP3-7.0), temperature is anticipated to increase by 0.88°C (0.6°C-1.11°C) by 2030 and is projected to reach 28.77°C (28.01°C-29.64°C) by the end of the century. Additionally, the risk of extreme heat is classified as medium, indicating that there is more than a 25% chance of experiencing at least one period of prolonged exposure to extreme heat, leading to heat stress, within the next five years.¹⁶

The effects of heat stress have become increasingly evident in Jamaica. Schools, for instance, have reported that extreme heat has negatively impacted students' learning and well-being due to inadequate ventilation and cooling systems. In response, the Jamaican government has initiated pilot programs to install air-conditioning units in 30 schools. Additionally, the Jamaica Public Service Company has noted a significant increase in electricity demand, attributed to the rising use of cooling technologies. Furthermore, the Government of Jamaica has issued warnings to the population about the dangers of heat exposure, including heat-related illnesses such as heat exhaustion, heat rash, heat cramps, and most severely, heat stroke, which can be fatal.

¹⁶ GFDRR (2020). [ThinkHazard. Jamaica.](#)

Figure 10: Projected (CMIP6) changes in average annual temperatures in Jamaica

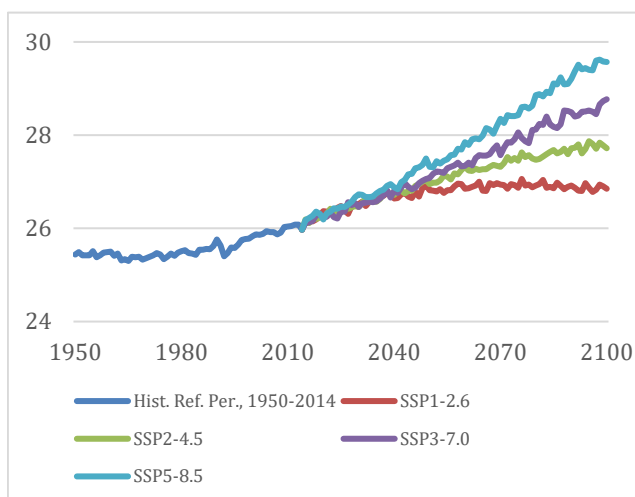
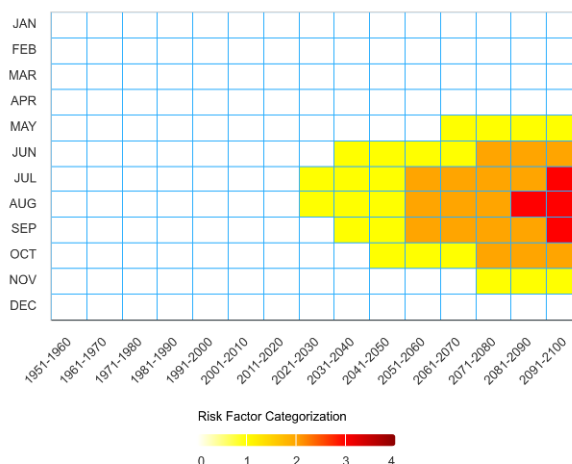


Figure 11: Heatplot for Heat Index Heat Risk Categorization (SSP2-4.5) – Jamaica



Source: World Bank. Climate Change Knowledge Portal. Information retrieved on February 2024.

The programme is expected to increase the adaptation potential of all three countries by building resilience for customers and the grid. **The region is expected to experience more frequent and severe storms, including flooding, storm surges, and other severe weather events that have the high potential to damage power sector infrastructure. Customers who are part of the program to install distributed energy resources such as solar rooftops with energy storage will be far more resilient to power disruptions from severe weather events. In addition, DER will support grid resilience by alleviating pressure on the grid during increased demand for active cooling solutions like air conditioning due to high temperatures through localized energy generation and storage.**

Table 5: Risk of climate hazard

Climate impact/ Hazard	Risk of climate hazard ^{17,18}			Examples of adaptation co-benefits ^{19,20}	
	Ba	B	J	Key risk	Energy services
Temperature rise/ Extreme heat	M	H	H	Heat-related illnesses, cardiovascular and respiratory problems, increased risk of infectious diseases, and strain on infrastructure and services.	Air conditioning technologies can alleviate the risks associated with extreme heat by providing a cool and comfortable indoor environment, offering relief from heat-related stressors, and enhancing overall well-being, while also providing a refuge during heatwaves.
Natural Disaster/Extreme events: Cyclone, Floods, Droughts, Heat waves	H	H	H	Loss of power due to infrastructure disruptions on the power supply caused by infrastructure damage,	Relocatable Distributed off-grid renewable energy technologies (RETs), mitigate disruptions in power supply by decentralizing generation, providing islanded

¹⁷ GFDRR (2020). [ThinkHazard](#)

¹⁸ World Bank, Climate Knowledge Portal. [Heat Risk](#).

¹⁹ IRENA (2021). [Bracing for Climate Impact: Renewable as a climate change adaptation strategy](#).

²⁰ Assessment of the country as outlined in the previous sections.

				overload, or overdemand, and load shedding. .	operation during grid outages, reducing the overall demand through load reduction and peak shaving, and serving as a redundant source of power. provide immediate energy access. Installation of flexible systems, such as solar PV and battery storage systems, that can be quickly deployed or removed in the event of natural disasters.
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Labels: H = High, M = Medium, L = Low. Countries: Ba = Barbados, B = Belize, J = Jamaica

B.1.15. The IUS Programme aims to facilitate the adoption of renewable energy and energy-efficient technologies, supporting countries to address the above-described needs/challenges by: (i) reducing greenhouse gas emissions and accelerating the transition away from reliance on fossil fuels; and (ii) increasing resilience against the mounting risk of heat stress in the Caribbean.

Additionally, the integration of Distributed Energy Resources (DERs) into the grid will enhance its resilience by diversifying the sources of electricity generation and peak shaving the demand. This diversification means that the grid will be powered by electricity generated from various energy resources and generation facilities.

ii. Baseline and context

B.1.16. Barbados, Belize, and Jamaica, as Small Island Developing States (SIDS) in the Caribbean, face common challenges inherent to their geographical and socio-economic contexts. These include vulnerability to natural disasters, reliance on tourism and agriculture, and the impacts of climate change, such as increasing temperatures, sea-level rise and extreme weather events. Despite varying geographic characteristics—with Belize on the mainland and Barbados and Jamaica as islands—all three countries strive for sustainable development, environmental conservation, and climate resilience. They also share developmental challenges such as resource management, food security, and economic diversification, underscored by a collective commitment to overcoming these obstacles through regional cooperation and policy initiatives.

B.1.17. A summary of relevant baseline conditions and characteristics in the three participating BMCs is included below. More information is available in the Baseline Assessment Report – see Annex 2b.

Table 6: Context/baseline conditions in the three participating BMCs

Context of targeted BMCs	
Barbados	
	Barbados, located in the eastern Caribbean, is a 34 km long and 23 km wide ²¹ island nation characterized by its limestone composition, which contributes to its renowned white sand beaches and clear waters. With a diverse population of approximately 290,000, ²² mainly of African descent, ²³ the nation boasts a high-income economy, heavily reliant on tourism, finance, and international business. ²⁴ Despite its economic success, Barbados encounters environmental challenges like climate change, yet maintains a commitment to sustainable practices. The geography, featuring a relatively flat terrain rising gently to the central highland region, allows for a concentrated population along the coasts. ²⁵

²¹ Government of Barbados (2010). [Barbados National Assessment Report](#).

²² Countrymeters (2023). Barbados Population.

²³ PopulationU (2023). [Barbados Population](#).

²⁴ Nations Encyclopedia (2023). [Barbados – Services](#).

²⁵ Government of Barbados (2010). [National Environmental Summary](#).



Electricity sector: The Barbados Light & Power Company (BLPC) holds a monopoly on electricity generation, transmission, and distribution. Since 2018, Independent Power Producers (IPPs) have been allowed, enhancing the solar energy contribution.²⁶ As of January 2023, distributed generation systems had a capacity of approximately 80 MW, with about 500 MW of renewable projects under development.²⁷ As of 2020, BLPC served 131,522 customers, with a peak demand of 141 MW and an installed capacity of about 276 MW, predominantly oil-fired.²⁸ Renewable energy contributed 7.20% of total generation in 2021. Efforts to expand renewable generation include the addition of solar PV farms and energy storage devices to improve grid stability. The energy sector aims for 100% renewable sources by 2030, with distributed generation expected to play a key role. The Renewable Energy Rider (RER) and Feed-in Tariff (FIT) programs have been instrumental in increasing solar PV and wind contributions.²⁹ Energy efficiency measures and the diversification of the energy mix are ongoing to reduce emissions and enhance sustainability.

Belize



Belize, located on the Caribbean coast of Central America, is distinguished by its diverse ecosystems, including the Belize Barrier Reef, the second largest in the world. With a land area of 22,960 km² and a population of approximately 421,754,³⁰ it boasts a multicultural society with a mix of Indigenous, Mestizo, Creole, and Afro-Amerindian heritage.³¹ Belize's economy is classified as upper-middle-income by the World Bank,³² despite facing challenges such as poverty and unemployment. Its economy is supported by agriculture, tourism, and services, with significant contributions from the barrier reef through tourism and fishing.³³ The country is characterized by its tropical climate, experiencing significant variability in rainfall and temperatures, which, along with its location, makes it susceptible to climate change and natural disasters like hurricanes. Belize's commitment to protecting its natural heritage is evident in its extensive network of protected areas, encompassing marine reserves, national parks, and nature reserves. Despite its beauty and biodiversity, Belize confronts issues related to food security and safety, with high rates of violent crime in certain areas.³⁴



Electricity sector: Belize Electricity Limited (BEL), regulated by the Public Utilities Commission, is Belize's sole electricity provider, selling approximately 561 GWh in 2021 with a peak demand of 104 MW. While Belize has made progress relating to use of alternative energy sources such as biomass and hydropower, petroleum-based fuels still represent the main sources of energy. Electricity generation witnessed slight increases in the use of petroleum fuels combined with the continuous use of biomass, hydropower and electricity importation from Mexico. Despite low solar PV contribution, efforts are being made to increase distributed generation, with regulatory frameworks for Distributed Generation (DG) under development. BEL, connecting over 170,000 customers across all major municipalities, is expanding its service range to include DG, allowing customers to sell excess energy back to the grid. Electricity consumption patterns show commercial and residential sectors as primary consumers, with street lighting demand expected to decrease due to an ongoing LED replacement program. Belize aims for a 75% renewable energy target by 2030, encouraging solar PV adoption despite the lack of a formal interconnection policy.³⁵ Projections suggest an increase in electricity demand, with BEL planning to enhance grid electrification, particularly in rural areas, and reduce emissions through increased renewable energy use and energy efficiency measures. Electricity rates in Belize are relatively low, with tiered pricing to support various customer classes.³⁶

Jamaica



Jamaica, located in the West Indies, is the third largest island in the Caribbean, positioned south of Cuba and west of Haiti. With a multi-racial population of over 2.9 million,³⁷ Jamaica's demographic fabric includes African, European, East Indian, and Chinese ancestries.³⁸ The island's population density stands at 272 persons per square kilometre, reflecting a diverse yet densely settled nation. The economy of Jamaica, as of the third quarter of 2022, showcased a 5.6 percent year-on-year

²⁶ From BLPC during virtual stakeholder consultation dated January 25, 2023.

²⁷ From BLPC during virtual stakeholder consultation dated January 25, 2023.

²⁸ EMERA 2021 Annual Report.

²⁹ Barbados Light and Power Company Ltd - Application Pursuant to Section 16 Of the Utilities Regulation Act Cap. 282 Of the Laws of Barbados for A Review of Electricity Rates Volume 3. September 2021

³⁰ Countrymeters, 2022. [Belize Population.](#)

³¹ UNFPA (2023). [About Belize.](#)

³² World Bank (2023). [GDP per Capita \(Current US\\$\)-Belize.](#)

³³ Britannica (2023). [Economy of Belize.](#)

³⁴ GOV.UK (2023). [Foreign Travel Advice Belize.](#)

³⁵ Government of Belize. [Belize's Nationally Determined Contributions \(NDCs\)](#), United Nations Framework Convention on Climate Change.

³⁶ 2021 Annual Report. Belize Electricity Ltd.

³⁷ Countrymeters (2022). [Jamaica's population.](#)

³⁸ Minority Rights (2023). [Minorities and Indigenous Peoples in Jamaica.](#)

expansion,³⁹ evidencing recovery from the COVID-19 pandemic's impacts with contributions from agriculture, forestry, fishing, and manufacturing.⁴⁰ Despite the economy's gradual recovery, Jamaica grapples with challenges such as income inequality and a high vulnerability to climate change, with significant risks including sea level rise and changing rainfall patterns impacting its agriculture-dependent economy. Jamaica's natural beauty and cultural heritage, alongside its challenges, portray a complex tapestry of development, resilience, and ongoing efforts to mitigate environmental and economic vulnerabilities.



Electricity sector: The Jamaican electricity sector includes the Jamaica Public Services Company Limited (JPS) as the primary utility and various privately owned Independent Power Producers (IPPs), servicing 98% of the population. JPS, owned by Marubeni and East West Power, alongside the Government of Jamaica, holds a monopoly on transmission and distribution with a license until 2027.

The sector has seen a renewable energy increase to almost 18% by 2022, with the national target set at 30% by 2030, potentially increasing to 50% as suggested by the Prime Minister.⁴¹ Despite the shift towards renewables, fossil fuels dominate the energy mix, contributing to high CO₂ emissions. Electricity demand is expected to grow, driven by industrial expansion and increased residential and commercial usage. Distributed generation initiatives like the Net Billing Programme aim to enhance renewable energy contributions.⁴² However, high installation costs and financing challenges limit broader access to renewable technologies. The government's efforts to integrate more renewables and improve energy efficiency are anticipated to shape future energy demand and supply, aiming for significant energy savings and reduced emissions by 2035. Electricity costs, among the highest globally due to reliance on expensive fuel sources, are subject to government interventions to alleviate impacts on residential customers.

iii. BMCs readiness for programme deployment

B.1.18. All three countries have expressed a strong interest in implementing IUS to enhance their investment in distributed energy, EE, and related sectors. This commitment is evidenced by their proactive efforts towards institutionalising and launching IUS frameworks. By adopting IUS, these nations aim to scale up their energy initiatives, fostering sustainable development and reducing their carbon footprints. In addition, the BMCs are making positive institutional and regulatory improvements, motivated to support a strong and effective IUS rollout.

B.1.19. Many participating BMCs have sought regulatory approval from their respective electricity regulators for the approval of an IUS Pilot Programme, supported by GIZ and Caricom, to advance the BMCs' readiness for full-scale programme deployment. The programme design of the IUS Pilot Programme is similar to the programme design as elaborated under Section B3 of this funding proposal i.e., the utilities funding energy efficiency retrofits and renewable energy installations through an on-bill repayment mechanism. However, the regulatory approval in the BMCs is delayed primarily due to capacity constraints and the regulatory clarity required for a disruptive and innovative business model for the promotion of DERs. The following presents the current readiness status of each BMC studied in terms of their readiness to support and implement the IUS programme, specifically in terms of the legal, regulatory and institutional frameworks. Detailed information about the readiness of BMCs can be found in Annex 2, Feasibility Study. The TA planned under Component 1 of the IUS Programme is designed to accelerate country readiness by providing adequate capacity support to the key stakeholders in the first year of Programme Implementation ("Programme Year 1" or "PY1").

B.1.20. Barbados: Our assessment of the level of IUS readiness in Barbados in terms of the legal, regulatory and institutional frameworks is outlined below.

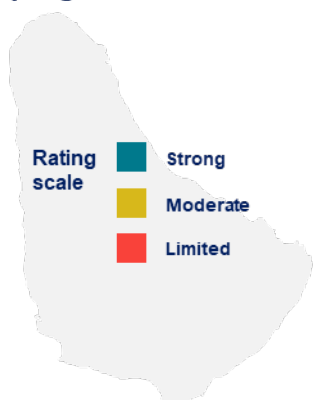
³⁹ Trading Economics (2023). [Jamaica GDP Annual Growth Rate](#).

⁴⁰ Britannica (2023). [Economy of Jamaica](#).

⁴¹ [Government of Jamaica \(2018\)](#).

⁴² Ministry of Science Energy and Technology (2016).

Assessment of readiness for IUS programme – Barbados



Legal readiness

Clarity required from the Ministry for the IUS programme roll-out with the BLPC acting as a financier.



Regulatory readiness

Regulatory approval required for the programme

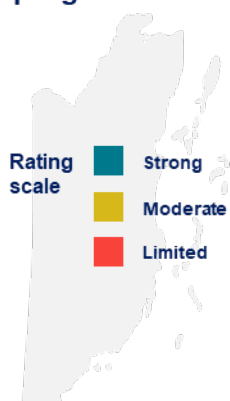


Institutional readiness

All the key institutions are aligned with the common goal to decarbonize Barbados' grid and support behind the meter solutions to lower energy consumption and customer's bills.

B.1.21. Belize: Our assessment of the level of IUS readiness in Belize in terms of the legal, regulatory and institutional frameworks is outlined below.

Assessment of readiness for IUS programme – Belize



Legal readiness

Clarity required from the Ministry for the IUS programme roll-out with BEL acting as a financier.



Regulatory readiness

Pending regulatory approval for the programme. The PUC also deferred its decision to publish DG tariffs.

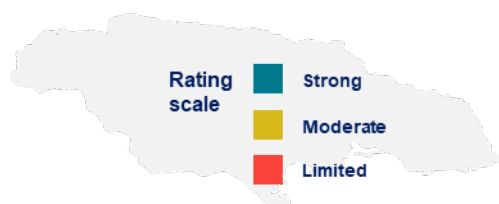


Institutional readiness

Need for additional inter-stakeholder discussions to ensure alignment with the underlying goals of energy security and grid decarbonization.

B.1.22. Jamaica: Our assessment of the level of IUS readiness in Jamaica in terms of the legal, regulatory and institutional frameworks is outlined below.

Assessment of readiness for IUS programme – Jamaica



Legal readiness

Clarity from the MSET on the applicability of IUS programme under existing legal frameworks.



Regulatory readiness

Regulatory approval from the OUR on programme applicability



Institutional readiness

Most of the institutions have prior experience in delivering clean energy and energy efficiency programmes of similar nature.

Related projects and interventions

B.1.23. National stakeholders and international development partners have been implementing (and are continuing to implement) various initiatives to support the sustainable energy transition. Key initiatives are summarized below, along with the lessons learnt that have been considered (and informed the design of) the proposed IUS Programme as well as a brief description of how it is complementary to these other efforts.

Table 7: Key national initiatives to promote RE and EE in the participating BMCs.

Country/Programme	Description	Lessons learnt	How the IUS programme complements these initiatives
Barbados Sustainable Energy Investment Programme (Energy Smart Fund 1)	The Energy Smart Fund, implemented by EGFL in coordination with the Ministry, offered technical and financial support for renewable energy and energy efficiency projects, including pre-investment studies, subsidized loans, and rebates for energy-efficient lighting and Acs.	From February 2011 to June 2017, the \$10 million program offered technical assistance, energy-efficient LED vouchers, and AC rebates, benefiting businesses with 28 grants for energy studies, distributing over 57,000 LEDs. Similar design elements have been considered when preparing the IUS Programme.	The Energy Smart Fund 1 concentrated only on promoting energy efficient technologies through the Ministry. The IUS Programme on the other hand covers wider range of both renewable energy and energy efficiency technologies and services, establishes a long-term sustainable market ecosystem through necessary reforms, and deploys a disruptive, facilitative, and inclusive business model.
Barbados Sustainable Energy Investment Programme (Energy Smart Fund 2)	The program expands on its predecessor to enhance renewable energy and energy efficiency among SMEs and the public sector, with a focus on project implementation, an AC rebate program, e-mobility, and capacity	The Energy Smart Fund 2, valued at USD 45 million, expands on its predecessor's work by enhancing renewable energy and energy efficiency in SMEs and the public sector, introducing e-mobility solutions, and emphasizing capacity	The Energy Smart Fund 2 is supporting SMEs to access funding through financial institutions – whereas the IUS programme aims to establish a business model and service offering that will make these technologies accessible to other

	building for the Project Execution Unit.	building. Approved in 2019, its design features are incorporated into the IUS Programme, particularly the need to support pre-investment audits.	segments of the market (i.e. those that are not as well served by existing financial institutions)
Barbados Public Sector Smart Energy Programme (PSSEP)	The project provides a package of economic instruments targeted towards SMEs to improve renewable energy and energy efficiency adoption. This includes retrofits of government buildings, a pilot project for converting government fleets to EVs, and capacity building.	Need for capacity building, institutional strengthening and public awareness particularly around professional and technical skills, etc. The IUS Programme design holistically considers these elements as part of the TA component.	The PSSEP is focussed on improving energy consumption in government buildings with _____. The IUS Programme on the other hand focuses on multiple end beneficiaries, multiple technologies and services, a TA component that supports market development and regulatory policy reforms, among other benefits.
Barbados Disaster Risk and Energy Access Management (DREAM) project	The program aimed to reduce greenhouse gas emissions and enhance disaster risk reduction by improving the renewable energy policy framework, developing clean energy capacity, and demonstrating solar PV feasibility through a model tendering process and installation.	The program highlighted the necessity for technical training in solar PV installation, public awareness, and capacity building. It developed procurement documents and a PV monitoring app to enhance system performance analysis. These insights and tools are integrated into the IUS program design for Barbados' implementation. Many learnings from this project like the need for training, etc. is included in the IUS Programme design. Furthermore, documentation and tools already developed under this project may be leveraged by Barbados to implement its IUS Programme.	The DREAM project compliments the IUS Programme in many ways. The IUS Programme however offers more technologies and services, provides a team of international and local experts to support key energy sector stakeholders through the TA component, provides access to low-cost finance, ensures democratization of clean energy deployment.
Belize Energy for Sustainable Development in the Caribbean (ESD)	Being implemented in Antigua & Barbuda, Belize, Grenada, St. Lucia, and St. Vincent and the Grenadines, this project aims to improve energy efficiency and renewable energy technologies through policy and project demonstrations. It	The IUS Programme both complements and supplements the multi country ESD programme in several ways. For instance, the IUS Programme design supports TSPs to enhance their service offering (moving toward being well capacitated	Belize's ESF project compliments the IUS programme in many ways. The IUS Programme however offers more technologies and services with the possibility to add newer technologies as per local demand, supports stakeholder's capacity

	involves retrofitting public buildings, implementation of new energy efficiency codes, and establishing blended-grant mechanisms via development banks to promote ESCOs.	ESCOs), empowers end users to benefit from distributed RE and EE solutions, provides access to low-cost finance, improves the legal and regulatory environment.	strengthening through the international and local experts under TA component, addresses cross-cutting elements such as gender and E&S, provides access to low-cost finance, ensures democratization of clean energy deployment.
Belize Energy Resilience for Climate Adaptation (ERCAP)	The project aims to demonstrate solutions to enhance resilience of the energy system to climate change and extreme weather events. It includes long-term planning, demonstration measures, and project implementation support.	BEL has already set up grievance redress mechanism under this project which may be leveraged / tailored to be used for the IUS programme.	The IUS Programme is a market ecosystem development mechanism that also includes grievance redressal mechanism. The IUS Programme focusses on behind the meter distributed renewable energy and energy efficiency solutions primarily for residential and commercial customers.
Jamaica Energy efficiency and conservation programme	The programme is being implemented by the Ministry of Science, Energy, Telecommunications and Transport (MSET) at an approved funding of US\$ 20 million. The general objective of this program is to support the reduction of Jamaica's dependency on oil consumption through the design and implementation of cost savings, EE and energy conservation (EC) measures in the public sector and reduce GHG emissions. The Program also aims at materializing the potential for EE and EC in public buildings and thus subsequently reduces energy consumption.	Programme's objectives broadly align with the IUS Programme objectives i.e. to lower reliance on fossil fuel and facilitate EE in the public sector. MSET can ensure programme coordination with the IUS Programme and leverage any frameworks and processes developed under this programme for expedited implementation of the IUS Programme.	The energy efficiency and conservation programme focus on only on one sector and beneficiary. On the other hand, the IUS Programme covers both distributed renewable energy as well as energy efficiency. The IUS Programme focusses on a range of end beneficiaries. More technologies, services, and end beneficiaries may be added as per local requirements.
Jamaica Jamaica's energy management and efficiency programme	The programme is being implemented by the MSET and ended in November 2023. The programme seeks to promote energy efficiency in government facilities and fuel conservation in road	Jamaica is experienced in reducing energy consumption in government buildings through this programme. Frameworks, processes, and procedures developed under the programme could be tailored and	Jamaica's energy management and efficiency programme focussed on reducing energy consumption in government buildings. On the other hand, the IUS Programme is broader as it covers both distributed renewable

	transportation by reducing fuel imports.	used for expedited roll-out of the IUS Programme.	energy efficiency. The IUS Programme also focusses on more end beneficiaries. More technologies, services, and end beneficiaries may be added as per local requirements.
Jamaica Smart LED streetlight programme	JPS is aiming to install 110,000 LED streetlights, as well as intelligent controls and metering capabilities for revenue generation. LED streetlights deliver enhanced reliability and significantly longer lifespans than traditional streetlights ⁴³ . As of 2022, JPS has reported to have installed 105,000 streetlights	JPS can leverage its experience to deliver other distributed renewable energy and energy efficiency solutions as proposed under the IUS Programme.	Smart LED streetlight programme focussed exclusively on installing LED streetlights to lower energy consumption. The IUS Programme, in addition to financing LEDs, also provides residential and commercial customers the option to avail other clean and energy efficient technologies and services at their premises.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Climate problem

B.2a.1 The Caribbean region is currently grappling with two significant challenges: climate change, characterized rising temperatures, and an energy sector heavily dependent on imported fossil fuels, which substantially contribute to greenhouse gas emissions. Reflecting regional and global trends, the BMCs are experiencing severe climate change impacts, including intensified storms, floods, droughts, sea-level rise, and increasing temperatures. These changes are affecting not only the availability and supply of energy resources but also the transmission, distribution, operations, and energy demand.

B.2a.2 Historical and projected climate change impacts show a trend of rising temperatures, increasing the risk of heat stress and severe weather events, which are expected to continue and intensify. For example, the International Energy Agency has noted that extreme heat increases the demand for cooling, with sustained average daily temperatures of 30°C typically leading to a 16% spike in weekly sales. Additionally, the BMCs are committed to ambitious goals to reduce greenhouse gas emissions by increasing the share of renewables in their energy mix and improving energy efficiency across all sectors.

Goal

B.2b.3 The IUS Programme aims to address the climate problem through a paradigm shift impact in the BMCs energy sector. This paradigm shift can be achieved **IF the project establishes and operationalizes an innovative Integrated Utility Services (IUS) model by facilitating access to concessional financing, capacity building and regulatory reforms, THEN a paradigm shift will occur, leading to an accelerated transition to decentralized clean energy, increased energy security, and climate resilience across Barbados, Belize and Jamaica, BECAUSE a conducive and inclusive ecosystem for DER utilization will have been established that will lower GHG emissions, improve energy security and enhance climate resilience.**

B.2a.4 The proposed IUS Programme aims to reduce GHG emissions from the energy sector and enhance resilience to heat stress (cooling systems coupled with PV generation) in the three participating countries by scaling up investment in (and adoption/use of) distributed RE, EE and other DER technologies/measures. The Programme will accomplish this by addressing key barriers currently inhibiting businesses, homeowners, and other stakeholders from making such investments, including limited

⁴³ USTDA. [Success in Jamaica: LED Street Lights](#)

awareness, knowledge, and capacity; limited access to financing; limited risk appetite; limited availability of qualified ESCOs; and policy and regulatory frameworks that do not sufficiently incentivize investment in DERs.

B.2a.5. To overcome these barriers to investment, the Programme will support the electric utilities in Barbados, Belize, and Jamaica to establish and operate the IUS Model⁴⁴ as part of their regular business operations. To accomplish this, the Programme will also address key barriers currently preventing utilities from implementing this model at scale, including internal capacity constraints; limited access to the concessional financing needed for the 'launch' phase of the IUS scheme; the need for complementary technical assistance (TA) to successfully begin operating such a scheme at scale; and other policy and regulatory barriers. In so doing, the programme will directly assist the three national electric utilities to support behind-the-meter investment in distributed RE, EE and other DERs among a critical mass of customers (mostly households and businesses, as well as some public sector entities), while also equipping the utilities and other stakeholders to continue scaling up investment using the IUS Model after Programme closure.

Paradigm shift

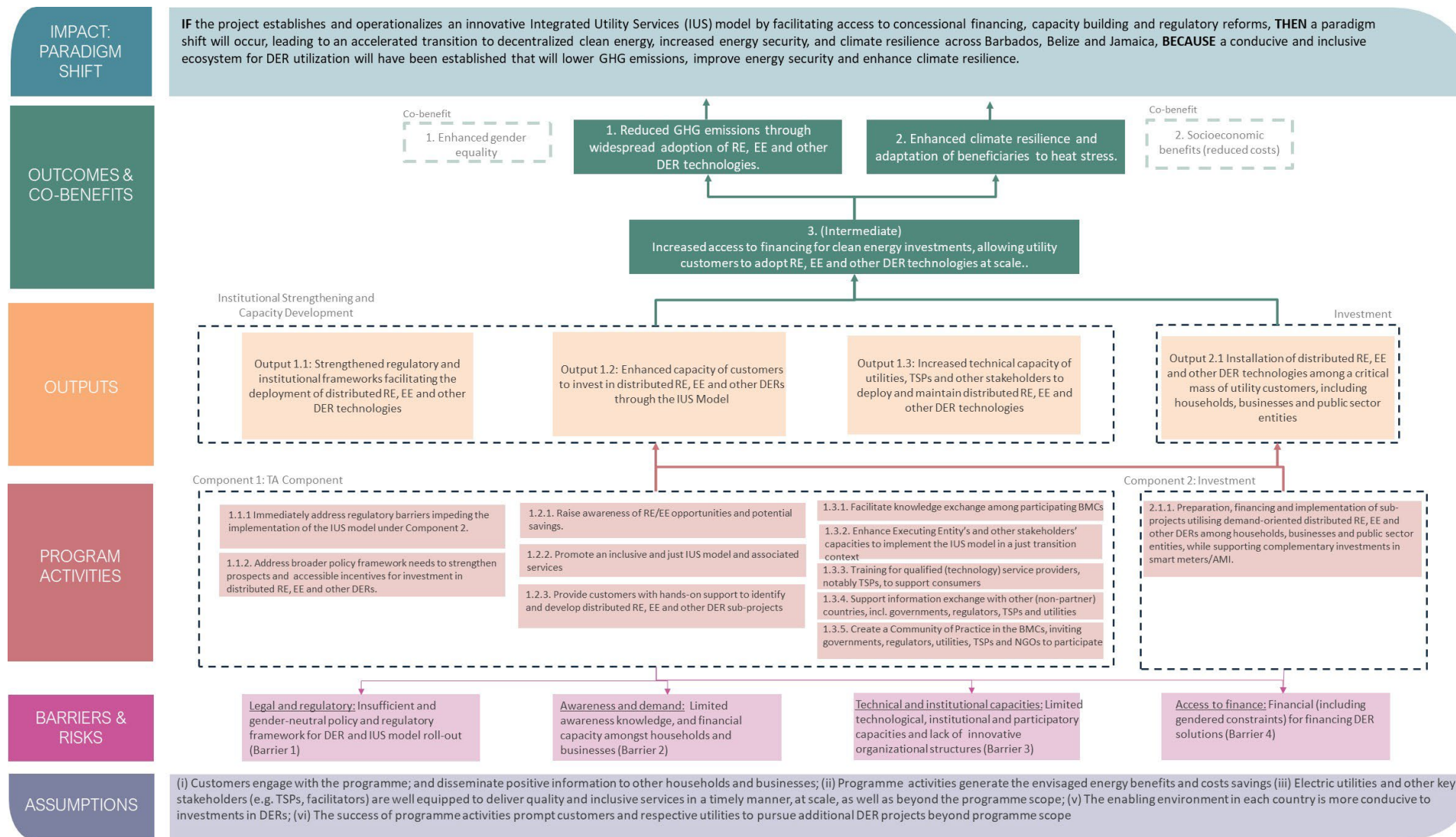
B.2a.6. The Programme aims to be a long-term sustainable, scalable, and replicable solution to address the BMCs, and, by extension the region, climate challenges. The Programme is envisaged to achieve the shift impact, as elaborated in the programme's Goal Statement, by:

- (a) ensuring a robust market ecosystem to accelerate deployment of renewable energy technologies as well as energy efficiency solutions;
- (b) strengthening stakeholders' institutional capacity to advance clean energy in the participating countries and help achieve climate targets;
- (c) strengthening the technical capacity of Technology Service Providers (TSPs) to cater to the needs of the market;
- (d) providing equal opportunity for women and other social groups (such as low-income households, women led households, etc.) to adopt clean energy solutions; and
- (e) providing simplified access to more affordable finance; etc.

B.2a.7. The Theory of Change figure below elaborates the broader goals of the IUS programme.

⁴⁴ Detailed programme design is available B.3.

Figure 12: IUS Programme Theory of Change



Outcomes

B.2a.8. To achieve the over-arching goal the Programme aims to deliver the following two main Outcomes and a corresponding intermediate Outcome:

Outcome 1: Reduced GHG emissions through widespread adoption of RE, EE and other DER technologies. The Programme is expected to increase reliance on local energy resources to meet its energy needs. Since the local energy resources are primarily clean sources of energy, in the long-term the Programme is expected to continue to mitigate emissions from the energy sector.

Outcome 2: Enhanced climate resilience and adaptation of beneficiaries to heat stress. The Programme will support energy efficiency solutions such as efficient ACs. The programme therefore provides direct long term heat stress adaptation benefits to the end beneficiaries.

Outcome 3 (intermediate): Increased access to financing for clean energy investments, allowing utility customers to adopt RE, EE and other DER technologies at scale.

Outputs

B.2a.10: To achieve the two main Outcomes and one intermediate Outcome described above, the Programme will aim to deliver four Outputs.

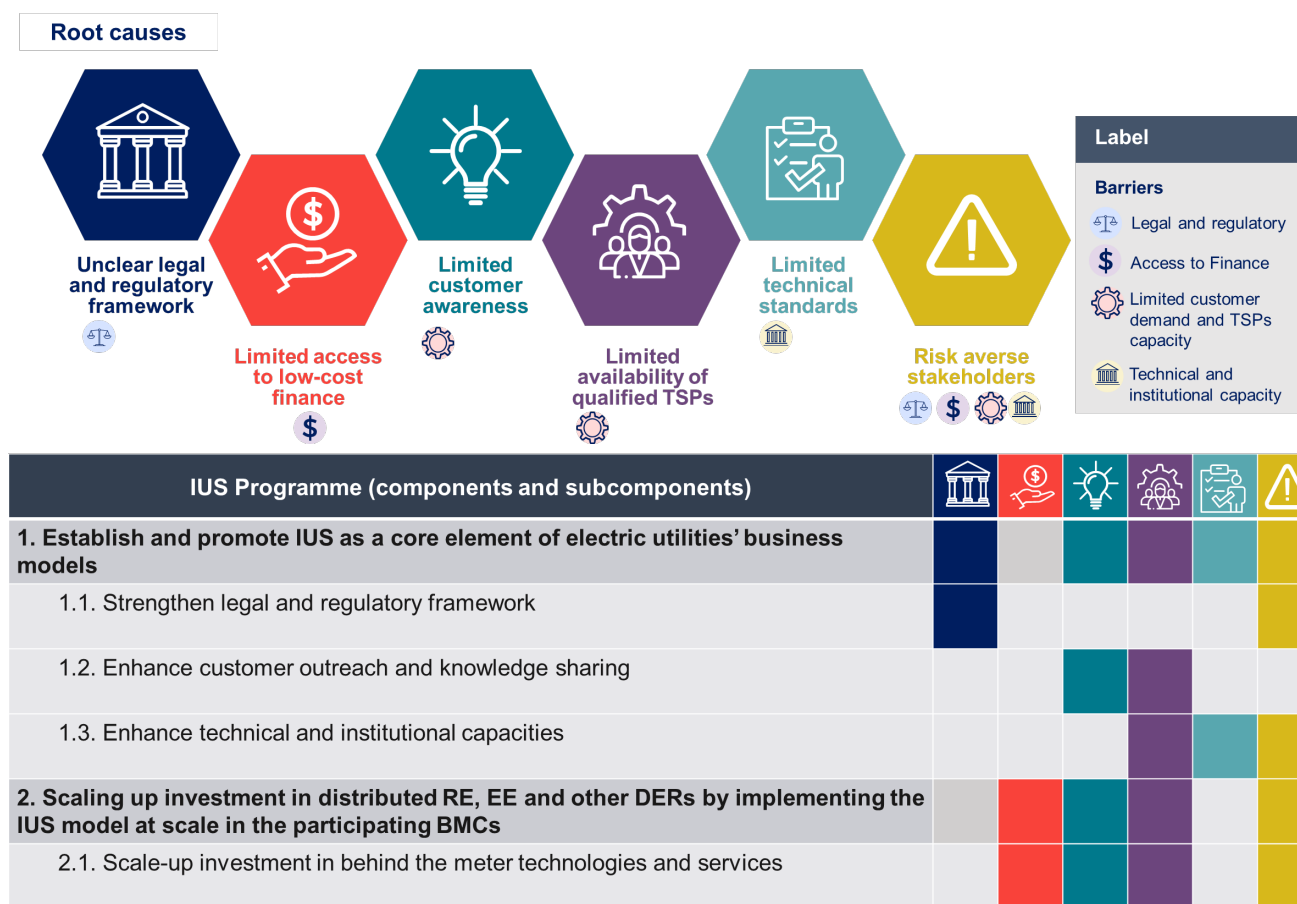
1. Output 1.1: Strengthened regulatory and institutional frameworks facilitating the deployment of distributed RE, EE and other DER technologies. The Programme will support the partner countries and the key stakeholders (i.e., energy ministry and the electricity regulator) to develop the necessary capacity and strengthen the regulatory policy environment in the country for smooth implementation and roll-out of the IUS programme. The programme will implement two Activities to support Output 1.1: – *Activity 1.1.1: Immediately address regulatory barriers impeding the implementation of the IUS Model under Component 2, and Activity 1.1.2: Address broader policy framework to strengthen prospects and incentives for investment in distributed RE, EE, and DERs* – which are described in more detail in Section B.3 (below).
2. Output 1.2: Enhanced capacity of customers to invest in distributed RE, EE and other DERs through the IUS Model. The Programme will also sensitize the customers in the participating countries to support the longer-term societal shift toward activities and investments that contribute to countries' climate change goals and resilience. The Programme will implement three Activities to support Output 1.2: – *Activity 1.2.1: Raise awareness of RE/EE opportunities and potential savings, Activity 1.2.2: Promote an inclusive and just IUS Model and associated services, and Activity 1.2.3: Provide customers with hands-on support to identify and develop distributed RE, EE, and DER sub-projects* – which are described in more detail in Section B.3 (below).
3. Output 1.3: Increased technical capacity of utilities, TSPs and other stakeholders to deploy and maintain distributed RE, EE and other DER technologies. The Programme will enhance regional capacity to promote and implement the IUS model. The Programme aims to achieve this through collaborative learning and knowledge exchange among participating countries, developing programme guiding documents like tailored implementation manuals, developing additional capacity for business development and cross-cutting areas such as Environmental and Social (E&S) management, Monitoring and Reporting (M&R), Gender Equality and Social Inclusion (GESI), etc. The Programme will implement five Activities to support Output 1.3: *Activity 1.3.1: Facilitate knowledge exchange among the targeted BMCs, Activity 1.3.2. Enhance Executing Entities' and other stakeholder capacities to implement the IUS Model in a just transition context, Activity 1.3.3: Training for qualified (technology) service providers, notably TSPs, to support customers, Activity 1.3.4: Support information exchange with other (non-partner) countries including government, regulators, TSPs, and utilities, and Activity 1.3.5: Create a Community of Practice* – which are described in more detail in Section B.3 (below).

4. Output 2.1: Installation of distributed RE, EE and other DER technologies among a critical mass of utility customers, including households, businesses and public sector entities. The Programme will support the execution of a critical mass of distributed renewable energy and energy efficiency sub-projects at customer level that successfully generate or save energy and create cost savings for customers (described in more detail in Section B.3 below). There are many financial and non-financial barriers inhibiting the flow of financing to local private sector stakeholders for such investments, which the Programme aims to address through an accessible and concessional climate financing (Component 2) and complementary technical assistance and capacity-building support (Component 1). It is therefore important for the Programme to contribute to measurable progress in relation to the provision of sub-loans and adoption rates of eligible technologies and practices.

Root causes and barriers

B.2a.11: Based on a detailed review of available literature, as well as stakeholder consultations that were organised during the programme preparation period (between Q3 2022 and Q3 2023), numerous key factors were identified that are affecting and limiting the uptake of distributed renewable energy and energy efficiency solutions in the residential and small to medium commercial groups in the participating countries. Detailed barrier assessment and analysis is elaborated in Section B.3 Project/programme description and Annex 2: Feasibility Study. These factors are grouped into a more concise set of barriers, which are described below and reflected in the Theory of Change diagram above:

Figure 13: Key barriers hindering the growth of behind the meter services



Source: CPCS

B.2a.12. Figure 16 summarises root causes hindering the growth of behind the meter services like distributed RE and EE in the participating BMCs, which include:

- **Unclear legal and regulatory framework:** BMCs need clearer legal or regulatory guidance on issues like DG tariffs/FIT and whether utilities can offer on-bill financing. There is also a lack of a standardized application/interconnection process.
- **Limited customer awareness and capacity:** Customers often lack awareness of the benefits of the proposed technologies, affecting implementation smoothness.
- **Limited availability of TSPs providing end-to-end solutions:** While there are TSPs with the necessary credentials, few can deliver comprehensive solutions.
- **Limited access to low-cost finance:** Difficulty in securing financing due to being considered high-risk, cumbersome processes for loan applications, and a lack of equitable financial products are significant barriers.
- **Risk adverse stakeholders:** Both small electricity customers and financial institutions are hesitant to invest in behind-the-meter solutions, partly due to unclear risk assessments and high interest rates.
- **Limited technical standards:** There is a need for globally accepted technical standards to ensure the standardization and quality of services.

B.2a.13. In addition to the typical behind the meter barriers as detailed above, there are other areas specific to the BMCs that may impact IUS Programme roll-out.

- **Capacity constraints and model understanding:** Regulatory authorities face capacity issues and apprehensions about approving novel business models, requiring additional stakeholder training.
- **Limited locally-available financing options:** High local financing costs and limited access to low-cost finance make it challenging for utilities to support the financial aspects of distributed RE, EE, and other DERs.
- **Limited distributed RE and EE ecosystem:** A supportive ecosystem development is critical, requiring technical assistance for regulatory frameworks, TSP certification, customer engagement, and overcoming other ecosystem development barriers.

B.2a.14 The activities of the IUS Programme, described in detail in Section B.3, have been designed to address the above barriers by targeting the root causes that have hindered the adoption of RE and EE technologies in the targeted BMCs. The following table illustrates how the IUS Programme activities will help overcome the identified barriers throughout the 6-year programme implementation period.

Table 8: IUS Programme responses to barriers and root causes

Barriers	Root causes	How the IUS Programme addresses these barriers and root causes
Legal and regulatory	• Unclear legal and regulatory framework • Risk adverse stakeholders	The programme will address this under sub-component 1.1, building on the work and assessments done through the ongoing Pilot Programme supported by GIZ and the CARICOM Secretariat, and guided by the baseline assessment that was conducted during the programme design and feasibility work. The TA that responds to these barriers and root causes include: • Prepare legal and regulatory note on existing laws/framework. • Assist stakeholders to prepare and secure approval of legal and regulatory amendments in the existing framework. • Support to develop any regulatory framework amendments necessary for smooth IUS roll-out (e.g., RE/FIT regulations, etc.) • Update/draft administrative or interconnection processes. • Streamline energy accounting and settlement processes. • Capacity building support to the regulatory authorities to review IUS related regulatory matters. • Support national stakeholders (Ministry, Regulator) to strengthen the policy and incentive framework to better support IUS investment.

		- TA execution support including contract management supervision, reporting, and invoicing, etc. - Capacity building on matters related to GESI, E&S, M&R.
Access to Finance	- Limited access to low-cost finance - Risk averse stakeholders	The IUS model eliminates the need for households and businesses to secure a loan from (often risk averse) financial institutions to invest in DERs. The model also minimizes the investment risk to be borne by the customer. The IUS model shifts investment risks from customers as the customers can now avail clean energy technologies or services with little to no capital investment. Interested customers can avail low-interest loans through utilities and install pre-assessed technologies from reliable technology providers, making DERs more affordable and secure. This reduces the financial burden and exposure to potential losses for customers.
Limited customer demand and TSPs capacity	- Limited customer awareness - Limited availability of qualified TSPs - Risk averse stakeholders	Under output 1.2, the Programme will engage with a large pool of customers to increase their awareness of suitable DERs, as well as how the Programme can support them to adopt such technologies/measures. The Programme will also engage National Energy Consultants and strengthen the utility help desk to assist customers prepare sub-project applications. The programme also aims to connect customer with TSPs through in-person customer awareness sessions. The customers may connect with a TSP of their choice who may also assist in sub-project application preparation and submission. The TA that will address these barriers and root causes include: - Develop inclusive outreach strategies – direct to customer, technology specific. - Develop marketing and communication material. - Organize customer awareness sessions. - Set-up/strengthen utility customer help desk to address IUS related queries. - Develop basic energy savings calculator for customers and provide capacity building support to the TSPs Furthermore, IUS model also overcomes the barrier posed by the limited availability of qualified ESCOs by capacitating the electric utilities to take on similar functions (i.e. providing innovative financing solutions to deployment of DERs) – a role that they are already well suited to fulfil due to their technical capacity/expertise, as well as their existing networks and contractual relationships with households and businesses in the participating BMCs.
Technical and institutional capacity	- Limited technical standards - Risk averse stakeholders	- The programme will strengthen the capacity of each participating electric utility to fully and effectively administer their IUS scheme through output 1.3. In doing so, the Programme will reinforce the post-programme sustainability of these schemes and equip the utilities to expand these services to reach additional customers over the medium/long term. - The programme will address this barrier through output 1.1, which aims to support the broader range of policy and regulatory reforms needed to unlock investment in distributed RE, EE and other DERs. In doing so, the programme will strengthen incentives for additional customers to engage with their electric utilities to develop/deliver IUS sub-projects in the BMCs after programme closure, thereby strengthening the sustainability and transformational potential of this scheme in each country.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services

						water security		
Outcome 1	☒	☐	☒	☐	☐	☐	☐	☐
Outcome 2	☐	☐	☐	☐	☐	☐	☒	☐
Outcome 3	☒	☐	☒	☐	☐	☐	☒	☐

Co-benefit

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Average annual cost reductions (nominal) per beneficiary due to energy savings.	☐	☒	☒	☐	☐	☐
Co-benefit 2: Number of female beneficiaries of the IUS Programme	☐	☒	☒	☒	☐	☐

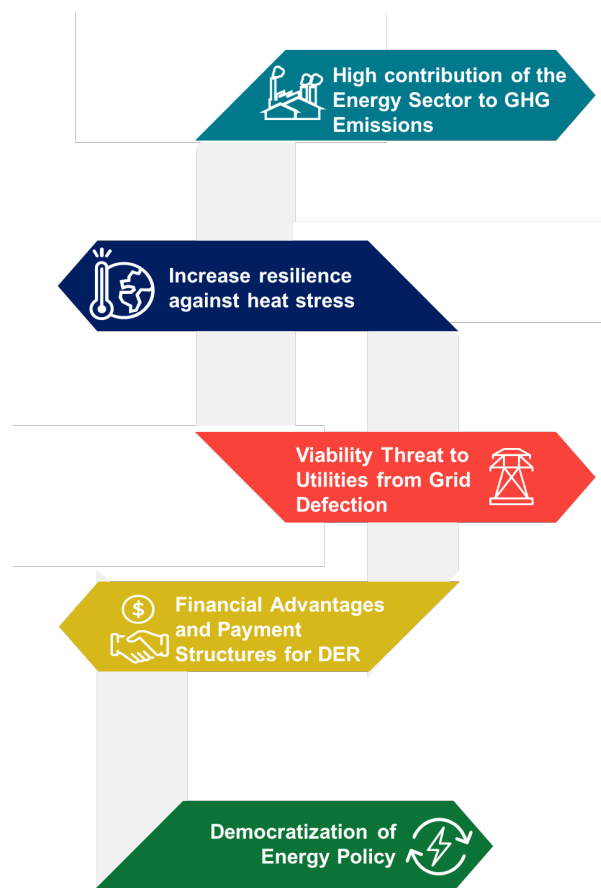
B.3. Project/programme description (max. 2500 words, approximately 5 pages)

i. Programme objectives

B.3.1 The IUS Model is a mechanism to directly implement demand-side renewable energy and energy efficiency through utilities in an open market system. It is a model to support renewable energy and energy efficient technology deployment through a financing program that is administered by electric utilities. This programme would have the utilities launch and operate the IUS at a country-wide scale by supporting energy service companies, educating customers, and financing the installation of renewable energy and energy efficiency interventions. This programme funding proposal constitutes the first of two phases for the overall IUS Programme in the region; the current **Phase I proposal will support three countries (Barbados, Belize, Jamaica) to implement IUS, while Phase II will result in a broader implementation and scale-up to other borrowing member countries in the Caribbean Region.** Throughout the remainder of this funding proposal, “IUS Programme” is used to refer specifically to Phase I.

B.3.2 The Programme aims to address the need for democratizing energy policy, increasing customers access to clean energy, reforming the business models of traditionally vertically integrated utilities, and enhancing utility capabilities for commercial financing of DERs. The IUS Programme is not only expected to foster an ecosystem conducive to sectoral sustainability but also facilitate regional learning and knowledge sharing, crucial for a unified approach to energy challenges in the Caribbean. The IUS Programme helps develop the much-needed market ecosystem for a sustained behind the meter solutions market in the Caribbean. The graphic below summarizes how the IUS Programme can help overcome current challenges in the transition towards more sustainable sources of energy.

Figure 14: Summary of IUS programme rationale

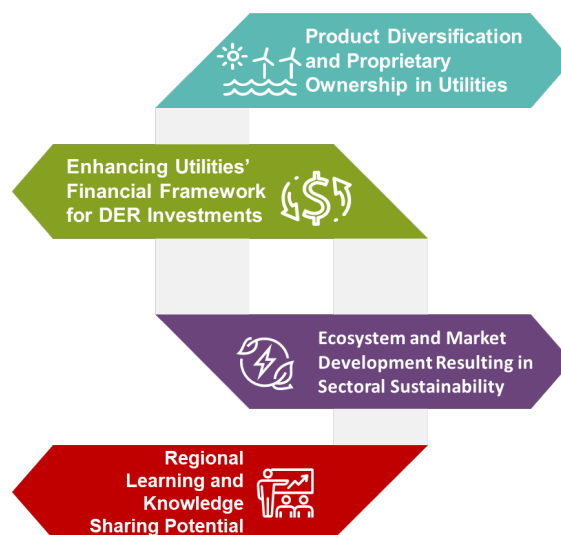


Source: CPCS

- High contribution of the Energy Sector to GHG Emissions**
 While BMCs are transitioning to RE and EE solutions to mitigate their emissions, they still rely heavily on traditional energy. The IUS model can play a crucial role in this transition by making RE and EE technologies more accessible and will also contribute to developing the market by maintaining a reliable and permanent group of experts to provide these services
- Increase resilience against heat stress**
 The IUS programme will strengthen BMCs' resilience to heat stress and enhance power generation capacity to meet peak AC demand. By promoting energy-efficient AC units and renewable energy technologies, the programme reduces grid strain during high demand, lowers energy consumption, and integrates sustainable energy sources. This approach ensures a stable, reliable power supply, enabling BMCs to better manage rising temperatures and improve energy security and sustainability.
- Viability Threat to Utilities from Grid Defection**
 Consumers are increasingly opting for cleaner and more cost-effective energy sources, such as rooftop solar, reducing their dependence on the traditional grid. The IUS model supports utilities in transitioning to RE and EE, ensuring customer retention and financial stability. Furthermore, the IUS model enhances the accessibility of these technologies for utility customers
- Financial Advantages and Payment Structures for DER**
 The IUS model, through on-bill financing, makes the adoption of DERs financially accessible to a wider customer base. By spreading the cost over time with scheduled payments, the model enhances the affordability of these energy solutions, encouraging broader adoption.
- Democratization of Energy Policy**
 The IUS model contributes to the democratization of energy policy by empowering consumers with choices in their energy consumption and production. It allows customers to select from various energy services and technologies, promoting a more consumer-centric approach and facilitating broader participation in energy decision-making processes.

- Product Diversification and Proprietary Ownership in Utilities**
 The IUS model will enable utilities in the BMCs to diversify their revenue streams, engage in new business models, and adapt to a changing energy landscape while maintaining their central role in the energy supply chain.
- Enhancing Utilities' Financial Framework for DER Investments**
 The IUS model's financing mechanism can provide a solution, enabling utilities to invest in DER projects and recover costs through on-bill financing, thus expanding their financial capabilities
- Ecosystem and Market Development Resulting in Sectoral Sustainability**
 The Caribbean's energy transition has been slow, primarily due to existing energy systems that are not equipped to incorporate renewables. The adoption of the IUS model can stimulate the development of a sustainable energy ecosystem and market. By integrating renewable energy and energy efficiency services, the model promotes sectoral sustainability. It encourages innovation, fosters the growth of related industries, and contributes to the creation of a resilient energy market.
- Regional Learning and Knowledge Sharing Potential**
 Implementing the IUS model in the BMCs offers potential for regional learning and knowledge sharing through Caribbean governments, the CCREEE, and other regional organizations. As each country adapts the model to their specific contexts, they can share insights and best practices, accelerating the transition to sustainable energy solutions across the region and contributing to global efforts in combating climate change.

Source: CPCS

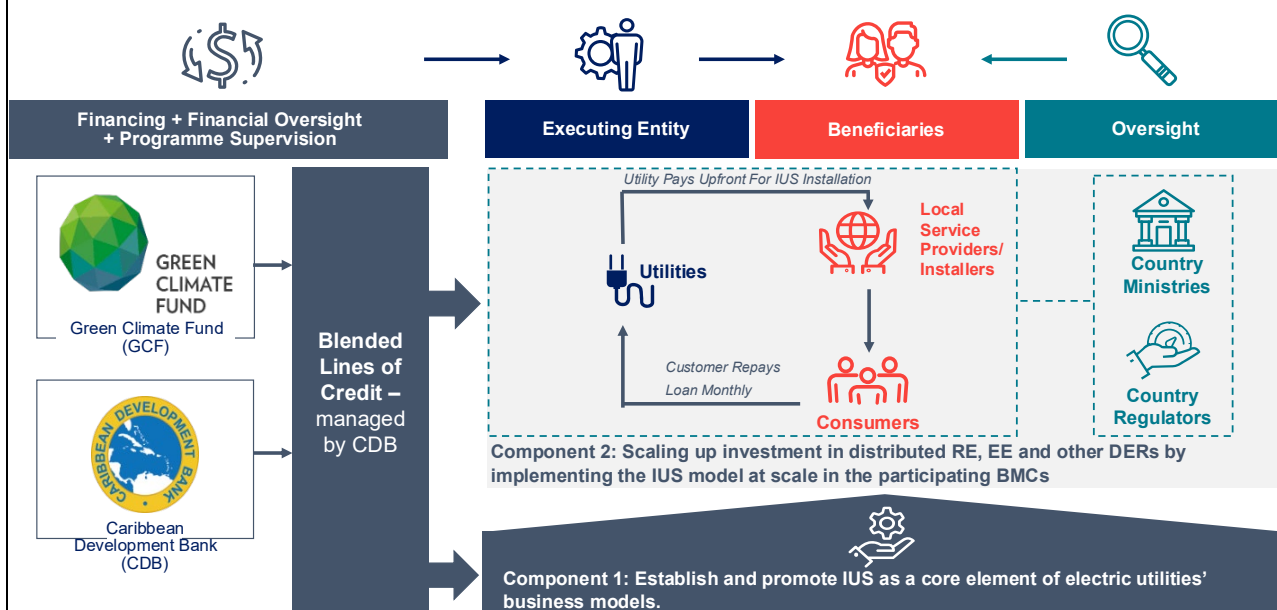


ii. Programme approach

B.3.3 The IUS Programme will include a “national project” in each of the three countries to be executed by the national electric utility and overseen by the Ministry and regulator responsible for oversight of the electricity sector. Each project will include a series of sub-projects that supports various renewable energy, energy efficiency, and other DER technologies. Utility customers will first receive education and awareness about the options available to them and then they will choose to implement the technologies through registered qualified service providers that work closely with their electric utility. The utility will provide the upfront financing and contract qualified TSPs to procure and install priority technologies at the premises of interested customers/consumers, who will then repay the amount owed to the utility for the installation through their monthly electricity bill. A major barrier for the uptake of these technologies has historically been the high upfront cost to install them, so this Programme is expected to alleviate that pressure.

B.3.4 The IUS Programme is an innovative solution that fosters collaboration. A typical behind-the-meter transaction involving business models like the IUS, involves government ministries, departments and agencies, utilities, system operators, independent power producers, technology service providers, customer and business associations, other clean energy facilitators and outreach providers, and lenders, among others. The following graphic provides an overview of the operational elements of the IUS Programme, including the key actors and components.

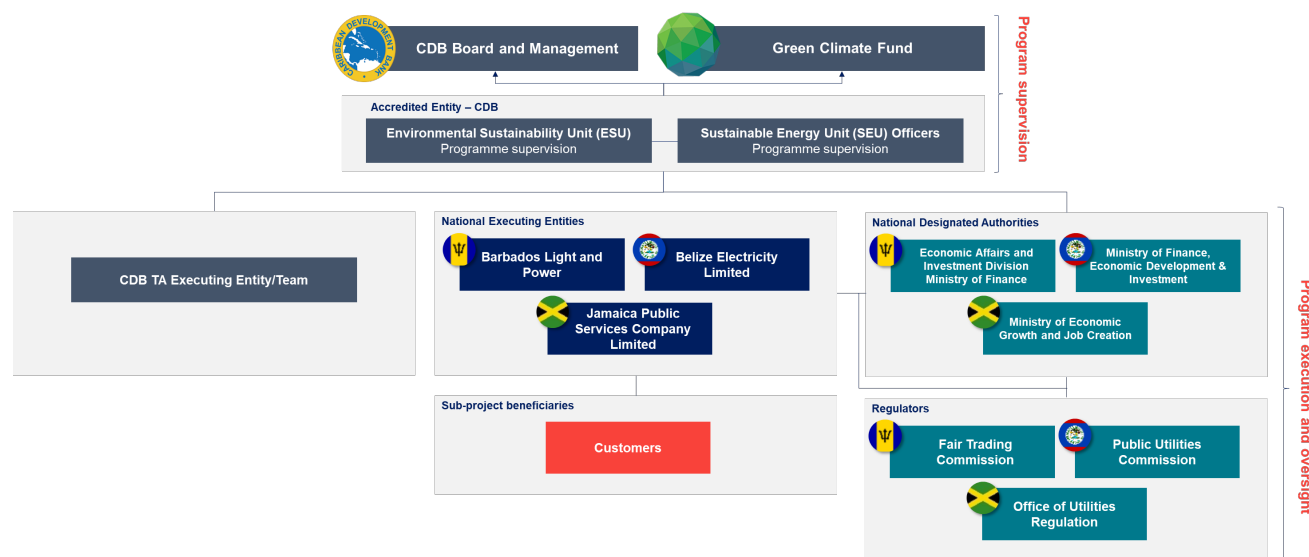
Figure 15: IUS Programme Schematic



Source: CPCS

As indicated in the above graphic, the Caribbean IUS transaction, like other behind the meter transactions, involves several actors. These actors are public entities like the government ministries, electricity regulators, and private entities like service providers, customers, etc. The CDB-led IUS programme however is unique given the regional nature of the Programme and the involvement of GCF. This uniqueness results in the involvement of multiple critical actors, as listed below, to the transaction unlike a typical behind the meter transaction.

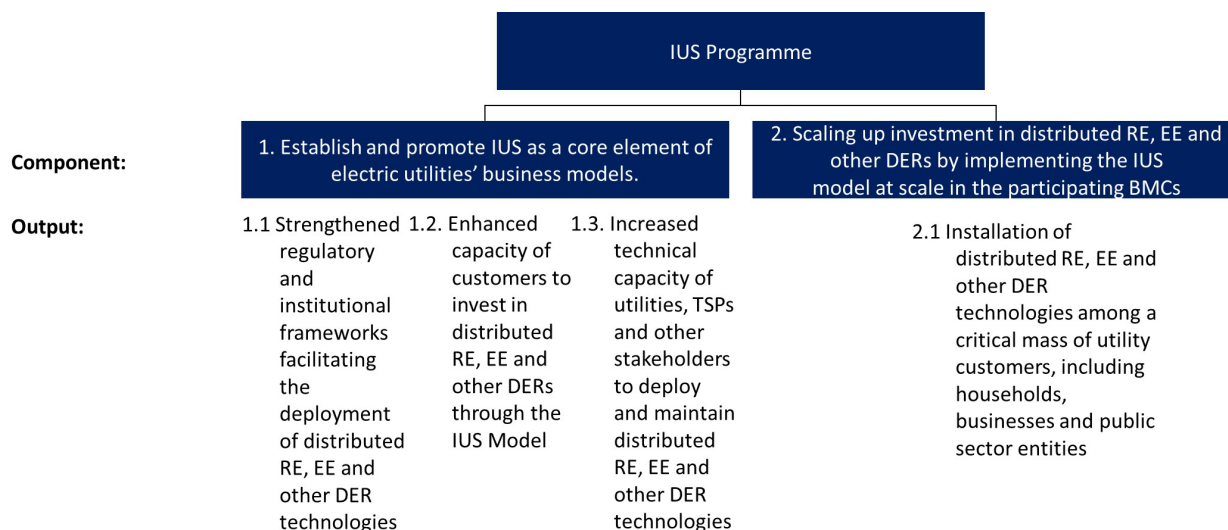
Figure 16: Programme governance arrangements



iii. Programme description

B.3.5 The IUS Programme is designed in a way that it not only addresses all the key barriers that are otherwise inhibiting investment in the transition to a more sustainable and distributed energy system (as outlined in Section B.2 and further elaborated in Annex 2: Feasibility Study), but also supports the much-needed ecosystem development for sustainable, scalable, and replicable growth of behind the meter energy solutions in the BMCs. Figure 20 below outlines the key components of the IUS programme.

Figure 17: Components of the IUS Programme



Component 1: Establish and promote IUS as a core element of electric utilities' business models

1. The Programme, through Component 1, aims to establish and promote IUS as a core element of electric utilities' business models in the BMCs to underpin timely and effective delivery of country-level support and enable the BMCs to guide and support one another during and after the Programme. The support provided under this Component will build directly on the considerable progress that has already been made in the three participating BMCs (as outlined above in Section B.1 and further elaborated in Annex 2: Feasibility Study). As a result, Component 1 will address the pending regulatory and institutional barriers that need to be overcome to begin financing IUS sub-projects at scale. Component 2 will then build on this foundation to scale up investments in distributed RE, EE and other DERs

through widespread use of (and financing of sub-projects through) the IUS model by electric utilities in the three participating BMCs.

2. This component lays down the groundwork for the electric utilities in the three participating BMCs to implement IUS at scale. This will ensure that all the key stakeholders have the necessary tools and frameworks to facilitate the proposed investments under Component 2 of the IUS programme.

This component is sub-divided into three Outputs:

Output 1.1: Strengthened regulatory and institutional frameworks facilitating the deployment of distributed RE, EE and other DER technologies.

3. The Programme will support regulatory authorities in the participating countries to enhance and strengthen their regulatory framework through legal and regulatory amendments, update or develop new regulatory frameworks specifically for the distributed renewable energy and energy efficiency sector, update or develop a timebound sub-project approval and administrative process, increase access to global experience and learnings, provide regulatory authorities the opportunity to apply lessons learnt for sub-project appraisal, among others. The regulatory authority will also leverage expertise of an international regulatory expert as well as a national regulatory specialist to build and strengthen its capacity to cater to the distributed renewable energy sector for long-term sectoral sustainability. Furthermore, the Programme will also support the relevant energy ministry in the participating countries to strengthen its national policy of climate change and renewable energy and develop additional incentive mechanisms to increase clean energy adoption.
4. The CDB Technical Assistance (TA) Execution Team (a regional TA Executing Entity that will operate within CDB in a manner that is separate from the Accredited Entity team) will support key stakeholders from the BMCs to address any pending legal and regulatory and/or capacity issues at the institutional level that may inhibit the IUS roll-out. These pending matters will be resolved based on the IUS Pilot Programme experience and lessons learnt from similar utility-led behind the meter programmes delivered in comparable global jurisdictions. It is also informed by the detailed assessments of the current policy, regulatory and institutional context in each of the three participating BMCs, which is briefly summarized in Section B.1 (above) and further elaborated in Annex 2: Feasibility Study. This Output will include two activities.
5. **Activity 1.1.1.: Immediately address regulatory barriers impeding the implementation of the IUS model under Component 2:** This TA activity aims to assist national stakeholders in each BMC to finalize updates to their legal and regulatory frameworks such that they are fit-for-purpose to support effective roll-out of IUS. In particular, this activity will facilitate the process of preparing and approving legal and regulatory amendments that are needed to operationalize and institutionalize IUS in each country, while also strengthening the capacity of the regulators in each country to oversee and regulate IUS activities. This will be done by contracting external consultants/specialists who will directly assist national counterparts to address these pending barriers. Additionally, this activity will provide external expert advice to help the relevant stakeholders in each participating BMC to strengthen at least one aspect of their broader regulatory framework, such as Renewable Energy Feed-In Tariff (RE/FIT) regulations, grid codes, or other relevant areas.

Furthermore, this activity will provide support for updating or drafting key documents related to administrative procedures of the utility to receive, review, and approve sub-project application, energy accounting and settlement processes of the customers electricity bill availing benefits under the IUS Programme. These procedural documents are necessary to ensure a streamlined process for all the customers irrespective of their customer group, demography, etc. Standardised processes flow also helps increase efficiency and transparency. For instance, once the regulator or utility publishes the administrative procedures, a customer becomes fully aware about all application processing stages, the official responsible to review and approve the application at each stage, among others.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

6. **Activity 1.1.2 Address broader policy framework needs to strengthen prospects and incentives for investment in distributed RE, EE and other DERs:** The activity will assist national stakeholders (including the energy ministry, regulatory authority, and others as needed) in strengthening the national policy and incentive framework to support investment in priority technologies, including those to be supported/financed under the IUS programme. This will be done by contracting external consultants/specialists to directly assist the national counterparts with these reform processes. This support will align with NDCs and promote sector sustainability. Additionally, under this the activity, the CDB TA Execution Team will contract external consultants/specialists to develop TSP qualification guidelines to ensure quality installation of clean energy technologies, working in close collaboration with the utilities, energy ministries and regulatory authorities. This is needed to ensure that a minimum technical competence benchmark could be established in the BMCs to help reduce/eliminate sub-par sub-project installations.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

Output 1.2: Enhanced capacity of customers to invest in distributed RE, EE and other DERs through the IUS Model.

7. One key barrier inhibiting investment in sustainable energy and energy efficiency solutions is the lack of customer awareness, and limited outreach regarding the energy and cost saving potential associated with these technologies. Under this sub-component, each participating utility, National Designated Authority (NDA), the energy ministry, and other key actors will engage in customer outreach and knowledge sharing activities. Through the CDB TA Execution Team, these stakeholders will receive direct support from external service providers to assist in planning, preparing and delivering these outreach efforts. Customer outreach will be delivered through various mechanisms, for instance newspaper advertisements, webinars, public events, etc. The modalities of customer outreach and building awareness will be BMC- and stakeholder-specific. These awareness and outreach initiatives will serve multiple purposes, including: (a) informing customers about the benefits of behind the meter solutions; (b) explaining the modalities of the IUS Programme and how it is delivered; (c) the amount of public support available for sub-project enrolment, etc. As part of these outreach efforts, a particular emphasis will be placed on underserved customer groups like women, etc. The support provided under this Output will include developing clean energy technology and IUS Programme specific marketing material, developing inclusive marketing, communications and outreach strategies, conducting in-person outreach sessions for customer awareness, facilitating customers' access to available TSPs, helping customers understand the benefits of the priority technologies (i.e., lowering their electricity bill), helping customers understand the administrative/approval process, strengthening utility's capabilities to handle customer queries, facilitating free/low-cost energy audits for eligible customers, among others. This Output will include three activities:

8. **Activity 1.2.1 Raise awareness of RE/EE opportunities and potential savings:** Under this activity, the CDB TA Execution Team will contract external consultants/specialists to work with national counterparts (led by the energy ministry) in each of the three countries to develop and execute a marketing and outreach strategy to bolster demand for sustainable energy technologies. The consultants will first assist the national counterparts in each country to define and develop their marketing and outreach strategy. Once the strategies have been prepared, the consultants will also (guided by the priorities identified in the strategy) develop engaging, technology-specific content (including infographics, posters, pamphlets and other multimedia materials) that will showcase the benefits and functionalities of clean energy solutions. The national counterparts will then use these materials to deliver the outreach activities outlined in their respective strategies. This approach aims to reach a broad audience, including vulnerable groups, ensuring accessibility and inclusivity.

Under this activity, the CDB TA Execution Team will also (through a suitable service provider) work with the national counterparts to organize six outreach sessions (1 per country in PY1 and 1 per country in PY3) to educate end beneficiaries—both residential and commercial consumers—on the advantages/benefits of clean energy solutions. These sessions are also expected to cover information about the IUS Programme, including the sub-project preparation and application process. The outreach sessions will therefore contribute to Activities 1.2.1 and 1.2.2, but the costs have been clustered under Activity 1.2.1 to facilitate smooth implementation. These efforts will be championed by the national counterparts (with support from local NGOs and industry/business associations), and aim

to both catalyse demand for sub-projects under Component 2 *as well as* foster long-term clean energy market development and enhance Programme sustainability.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

9. **Activity 1.2.2 Promote an inclusive and just IUS model and associated services:** Through this activity, the programme will aim to increase demand for IUS services both during and beyond the Programme implementation period, thereby reinforcing sustainability. The CDB TA Execution Team will contract a suitable service provider to assist national counterparts (led by the electric utilities) to develop a Programme-/IUS-focused communications action plan. Similar to Activity 1.2.1, the service provider will then also develop specific outreach materials to be used by the electric utilities when delivering their respective communications action plans. Both the plans and the associated outreach materials will be inclusive and cater specifically to vulnerable groups to ensure wide-reaching benefits. By making the Programme's information accessible to everyone, including those who are blind, deaf, or from marginalized groups, the strategy seeks to raise awareness, encourage participation, and drive the Programme's long-term success.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

10. **Activity 1.2.3 Provide customers with hands-on support to identify and develop distributed RE, EE and other DER sub-projects:** Under this activity, each electric utility (as a national Executing Entity) will receive resources with which to contract a National Energy Consultant who will assist the utilities with business development and customer outreach/support, as well as other programme-related responsibilities (e.g. monitoring and reporting). In so doing, the National Energy Consultants are expected to contribute to both fostering a pipeline of suitable sub-project proposals in each country as well as helping improve the quality of the submitted proposals. As part of these efforts, the CDB TA Execution Team will also engage a consultant to assist each utility to establish or strengthen their helpdesk/customer support services to efficiently address queries from customers related to sub-projects applications. Among other responsibilities, the National Energy Consultants are expected to assist the utilities in responding to such help-desk requests in a timely manner. In addition, CDB-contracted consultants will also assist the utilities to develop simple, user-friendly energy savings calculators that customers can use for self-screening purposes. In so doing, this activity will enhance accessibility to energy-saving information and empower program participants to make informed decisions about their energy usage. Finally, under this activity, the Programme will also develop eligibility criteria to be used to guide the selection of specific beneficiaries (lower-income residential customers and vulnerable groups) who will receive free energy audits. Once the eligibility criteria are developed, the Programme will facilitate access to such energy audits. CDB estimates that a total of 300 low-income households will receive free energy audit services financed through the Programme, with 20 selected annually per country from PY1-5 based on the eligibility criteria.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

Output 1.3: Increased technical capacity of utilities, TSPs and other stakeholders to deploy and maintain distributed RE, EE and other DER technologies.

11. **Enhance technical and institutional capacity:** Insufficient technical and institutional capacity has been identified as a key barrier. This is attributable to several factors, notably that IUS constitutes a material shift from the conventional electric utility business model to more innovative approaches which are not common in the BMCs. Post IUS roll-out, the BMCs will actively promote behind the meter solutions resulting in a radical shift from the status quo. This sub-component aims to improve capacities and information exchange at the national and regional level to enable the partner utilities (and other stakeholders, both in and beyond the targeted countries) to make this transition, institutionalize the IUS model and learn from one another as they work to scale up IUS and transform their energy systems. In addition to the CDB, regional stakeholders like the CARICOM Secretariat, CARILEC, and

the CCREEE would play an active role to promote regional learning and ensure programme replicability and scalability. This Output will include five activities:

12. **Activity 1.3.1 Facilitate knowledge exchange among the participating BMCs:** Under this activity, the CDB TA Execution Team will work with an external service provider and national counterparts to document and disseminate good practices and lessons learnt related to programme implementation/execution among the stakeholders in the three countries. This will include developing knowledge exchange materials, such as presentations and briefing notes, and holding online/virtual knowledge-sharing sessions. The knowledge materials will be developed in consultation with (and using inputs from) the utilities, regulators and energy ministries, allowing them to learn from one another and support/strengthen each other's efforts to successfully implement IUS.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

13. **Activity 1.3.2 Enhance Executing Entities' and other stakeholders' capacities to implement the IUS model in a just transition context:** The roll-out of IUS by the three participating electric utilities will require them to adjust their business operations and establish new (or adjust existing) internal systems and procedures. The associated costs – particularly in the start-up phase of IUS implementation – can be significant and a risk to the utilities. This activity therefore focuses on enhancing the capacities of the electric utilities (as National Executing Entities) to implement the IUS model. As part of this support, the CDB TA Execution Team will contract a service provider to work with each electric utility to develop a tailored operations manual. These country-specific operations manuals will build on the overall Programme Operations Manual (see Annex 21), and will include clear guidance on: financial management and organizational matters; KYC, AML/CFT and other compliance-related risks; product/service offerings; reporting requirements; environmental & social risk management; and GESI -related actions. To further equip each utility to implement IUS, this activity will support the preparation of an IUS business development strategy for each national Executing Entity. Furthermore, the CDB TA Execution Team will contract an IT firm to assist each utility to develop a web-based digital application system to facilitate the receipt and processing of sub-project applications, leveraging existing utility-operated platforms to the extent possible. As utilities are processing sub-project applications, they will also have access to external consultant support to assist during the appraisal process, particularly for sub-projects that propose to use non-preapproved technologies (described in more detail under Component 2, below). Finally, under this activity, CDB-contracted specialists will support each utility to strengthen their internal policies, procedures and systems to better advance GESI goals and manage E&S risks – efforts that will be complemented by targeted trainings on E&S, GESI, and the mitigation/management of compliance-related risks (e.g. KYC, AML/CFT) for relevant utility staff (one for each utility in PY1, and a refresher training for each utility in PY2). This multifaceted approach aims to equip stakeholders with the necessary tools and knowledge for the successful execution of the IUS model.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

14. **Activity 1.3.3 Train qualified (technology) service providers, notably TSPs, to support consumers:** This activity aims to enhance TSPs' capabilities through specialized training modules, designed to support long-term market growth within the IUS programme. The training will cover a wide range of topics, including IUS programme requirements, technical and legal aspects, E&S risk mitigation/management, GESI, and M&R. Additional subjects such as contracts, customer outreach, quality standards, and familiarity with third-party certifications may also be included. Tailored to the specific needs of each BMC, the trainings will be delivered by CDB-contracted service providers, who are expected to deliver one training session in each country in PY1 and one additional training in each country (with a particular focus on E&S risk management) again in both PY2 and PY4. These trainings will collectively boost TSP expertise and their contribution to the sustainable energy transition. Indeed, better-capacitated TSPs are key to the long-term success of the energy transition, as the efficiency and effectiveness of their services are important determinants of whether/to what extent sustainable energy solutions are cost competitive.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

15. **Activity 1.3.4 Support information exchange with other (non-partner) countries, incl. governments, regulators, TSPs and utilities:** Under this activity, the CDB TA Execution Team (with support from external consultants) will develop and disseminate outreach materials to key stakeholders in BMCs across the region. These materials will share insights, challenges, and opportunities encountered in clean energy deployment through the IUS model. The goal is to provide BMCs across the Caribbean with a comprehensive understanding of the IUS model and its potential customization for their jurisdictions, thereby laying a strong foundation for scaling up and replication under Phase II of the IUS programme. To further support these efforts, under Activity 1.3.4 CDB will host four (4) virtual information exchange and knowledge-sharing sessions – two (2) sessions in PY2 and two (2) additional sessions in PY5. These sessions will offer an overview of the progress, challenges, and solutions identified in participating BMCs, enriched by firsthand experiences shared by key stakeholders. Each session will feature a Q&A segment to foster interaction and engagement. Finally, this activity will support three (3) representatives from each of the three (3) participating BMCs to attend an in-person knowledge sharing summit in PY3 to promote IUS and the Caribbean's experience in implementing this innovative model, covering travel and logistics costs.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

16. **Activity 1.3.5 Create a Community of Practice (COP) in the BMCs, inviting governments, regulators, utilities, TSPs and NGOs to participate:** This activity focuses on creating a Community of Practice (COP) to enhance peer-to-peer and south-south learning among stakeholders of the IUS programme. The COP aims to bring together TSPs, Executing Entity staff, and technical experts from different jurisdictions to share knowledge and experiences, thereby facilitating the advancement of the IUS Programme. The initiative will include virtual sessions with international experts on distributed energy and utility-led distributed energy models akin to IUS. Furthermore, an in-person field mission will be organized, hosted by one of the participating BMCs for stakeholders from the other participating BMCs. This mission will feature site visits to key IUS sub-projects and enable in-person, multi-stakeholder discussions, enhancing collaborative learning and the sharing of best practices within the IUS Programme framework.

Executing Entity: CDB (via the TA Execution Team)

Source of funds: GCF grant financing

Component 2: Scaling up investment in distributed RE, EE and other DERs by implementing the IUS model at scale in the participating BMCs.

17. This component builds on the work done under Component 1 to finance behind the meter technologies and services for participating customers. This component is also the core investment component of the IUS Programme. Participating utilities, in their capacity as national Executing Entities, are responsible for managing timely execution of the loan component of the IUS Programme in each BMC, including screening and approving sub-projects, disbursing to TSPs, collecting and processing repayments/reflows, monitoring and reporting, etc. In addition, the utilities/national Executing Entities will work in parallel to invest in the deployment of smart meters/advanced metering infrastructure (AMI) that is needed to improve the efficiency and effectiveness of customer-level investments in RE, EE and other DERs, and thereby support the broader transition toward a more sustainable, resilient and distributed energy system in the three countries. This component includes one Output.

Output 2.1: Installation of distributed RE, EE and other DER technologies, as well as complementary smart meters/AMI, among a critical mass of utility customers, including households, businesses and public sector entities.

Participating utilities, in their capacity as national Executing Entities, will receive, review, screen, and approve sub-project proposals, as well as finance the approved sub-projects and monitor (and report

on) their timely implementation. In parallel, they will invest in smart meters/AMI. This Output includes one activity:

- 18. Activity 2.1.1 Prepare, finance and implement sub-projects utilising distributed RE, EE and other DERs among households, businesses and public sector entities, while supporting complementary investments in smart meters/AMI:** This activity aims to deploy the loan component, enabling utility clients to obtain financing for sub-projects in BMCs. The national Executing Entities (electric utilities) will review sub-project proposals and only approve those that comply with the programme eligibility criteria, which are summarized in Paragraph B.3.11 (below) and further elaborated in Annex 21: Programme Operations Manual. If a sub-project proposal uses the pre-approved technologies – namely LED/efficient lighting, efficient AC/cooling, solar rooftop PV (with or without storage), and solar water heaters – the utility will not have to assess the technology eligibility as part of its review. Utilities will also be able to finance sub-projects that utilize other (non-preapproved) RE/EE/DER technologies, as per their local demand, but will have to undertake supplementary due diligence before approving such applications, as outlined in Annex 21. Note that CDB tentatively expects the funding under the lines of credit to be structured as follows: USD11,667,000 for BLPC in Barbados; USD9,667,000 for BEL in Belize; and USD15,666,000 for JPS in Jamaica. These figures include GCF financing and co-financing, and constitute initial estimates that are subject to change during the final CDB appraisal of each line of credit.

In parallel to their support for sub-projects financed through the IUS model (using their blended GCF-CDB lines of credit), the national Executing Entities (i.e. the three participating electric utilities) will make complementary investments in smart meters/AMI at the premises of their customers. This is expected to include customers that benefit from sub-projects under the IUS Programme as well as other customers that may not benefit from such sub-projects during the Programme implementation period. These investments in smart meters/AMI will be led by the electric utilities independently of the blended GCF-CDB lines of credit, and thus these contributions are reflected in the Programme as in-kind co-financing. The deployment of smart meters/AMI will improve the efficiency and effectiveness of investments in RE, EE and other DERs, and are critical to the long-term success of the broader transition toward a more sustainable, resilient and distributed energy system that the Programme is helping to catalyze. By increasing the use of smart meters/AMI broadly in their respective jurisdictions, the utilities will lay a foundation for the long-term sustainability of the IUS model and the priority/eligible technologies.

Executing Entities: Barbados Light and Power, Belize Electricity Limited, Jamaica Public Service Company

Source of funds:

Blended lines of credit for IUS sub-projects: GCF loan financing, CDB loan financing

Investments in smart meters/AMI: In-kind contributions from the national Executing Entities (i.e. the three participating electric utilities)

B.3.6 The IUS Programme design detailed above will directly address the root causes and barriers described in Section B.2a through a dedicate TA component that enables a market ecosystem development by strengthening the legal, regulatory, and procedural framework in the participating BMCs as well as increasing customer awareness and access to finance, among other benefits.

iv. Technologies alternatives and selection criteria

B.3.7 At the start of the programme design stage, programme stakeholders identified a set of potential technologies and services that could be financed under the Caribbean IUS Programme (subject to further review):

- LED lighting and controls
- Solar PV
- Solar water heating
- Surge protection at the meter
- E-mobility (partly as a demand response measure)
- HVAC (AC retrofits, sensors, low investment insulation)
- Building/energy management systems

- Meter socket base changeout programme
- Battery storage
- Motors and drives
- Combined heat and power
- Power-factor correction and other ancillary power quality services

B.3.8 During the detailed design and feasibility stage, the programme preparation team developed a set of criteria (see Table 5 below) with which to further assess the potential technologies and ultimately develop a list of preapproved⁴⁵ technologies that could be prioritized under the Programme and more easily/rapidly approved by the participating electric utilities (as national Executing Entities). The preapproved technologies are the most likely and most viable options based on prevalent market conditions, capacity of local TSPs, legal and regulatory framework, historical and envisaged demand, among others.

Table 9: Technology assessment

Technologies	(A) Provides direct mitigation and adaptation benefits	(B) Capital cost low Enough to be Financially Feasible	(C) Established/proven technology with likelihood of local capacity to implement	(D) Scalable and Replicable	(E) Relatable to the IUS Programme
LED/efficient lighting	✓	✓	✓	✓	✓
Efficient A/C, cooling/ HVAC	✓	✓	✓	✓	✓
Solar rooftop PV (with or without storage)	✓	✓	✓	✓	✓
Solar water heater	✓	✓	✓	✓	✓
Surge protection at the meter	●	✓	✓	✗	✓
Battery storage (grid scale)	●	✗	●	●	●
E-mobility (partly as a demand response measure)	●	✗	●	●	●
Building/energy management system	✓	●	✓	✓	●
Motors and drives	●	✗	✓	✗	✓
Combined heat and power (CHP)	✓	✗	✓	✗	●
Power factor correction and other ancillary	●	✓	✓	✗	✓

⁴⁵ Technologies classified as 'preapproved' are those that provide direct mitigation and adaptation benefits, are cost effective, scalable and relatable with the IUS Programme. These are the technologies/services that the utility may start financing without additional technology-focused due diligence once the Programme is officially launched.

power quality
services

Source: CPCS

Legend:

- ✓ High potential
- ✗ Not suitable at this stage
- Situation and market dependent

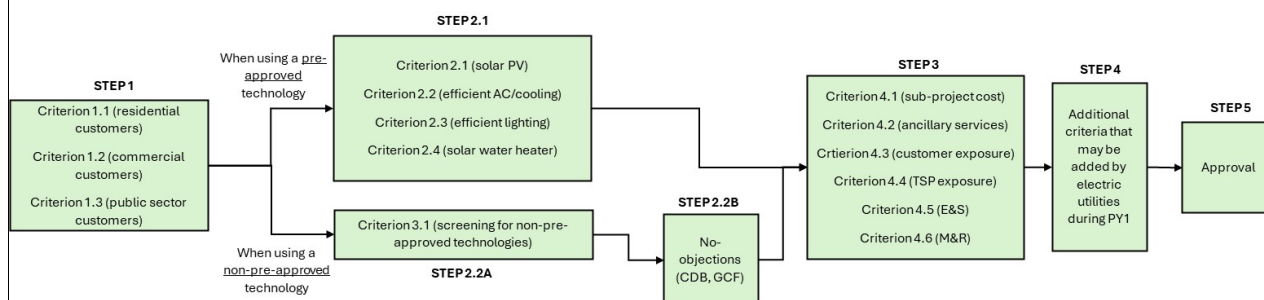
Preapproved technologies

B.3.9 Based on the assessment, four technologies have been preapproved for the IUS Programme, namely: **LED/efficient lighting, efficient air conditioning/cooling, solar rooftop PV (with or without storage), and solar water heaters**. Based on the current market conditions and the applicable legal and regulatory frameworks as highlighted in the earlier chapters, the overall ecosystem is relatively conducive for these technologies in the participating BMCs.

B.3.10 There are however associated services that may be bundled in a packaged service offering by the utilities with the four preapproved technologies. This includes surge protection, motors and drives, power factor correction and other ancillary power quality services, and building/energy management systems. As such, additional services/products could be unlocked for utilities by advancing the four preapproved technologies.

Eligibility criteria

B.3.11 The electric utilities will finance sub-projects that utilise one or more of the pre-approved technologies as long as the applications comply with the programme eligibility criteria, which are outlined below.



Category	Criterion Number	Criterion	Description	Metric(s)	How utilities should document this	Additional Notes or Conditions
Beneficiaries	1.1	Eligible residential customers	This criterion should be used to determine if a household is eligible to receive support, subject to fulfilling the other eligibility criteria as outlined below.	There are no limitations based on household size or income, or based on whether they rent vs. own the property.	N/A	In the event that the applicant/prospective beneficiary is a renter, the property owner must also consent to the transaction.
				The household must have been a utility customer (at their current or previous	Check the number of months for which billing	None.

				address) for at least the last 12 months.	records are available.	
				Any investments that the household previously made in the specific technology/ies for which they are applying for support through the IUS Programme cannot exceed the value/cost of their sub-project application.	Self-certification by the applicant, to be reviewed by the utility.	The accuracy of these self-certifications will be checked/verified for a sample of beneficiaries as part of the monitoring and reporting process for the Programme.
				The household must have fulfilled their payment obligations to the electric utility (i.e. paid on time, and in full) in at least 9 of the 12 months prior to the submission of the application.	Check the customer's payment history and track record.	If a customer has recently moved, the utility may either assess the customer's payment history from their previous address or check their credit score.
	1.2	Eligible commercial customers	This criterion should be used to determine if a local business is eligible to receive support, subject to fulfilling the other eligibility criteria as outlined below.	There are no limitations based on the size or annual revenue of the business, or based on whether they rent vs. own the property.	N/A	In the event that the applicant/prospective beneficiary is a renter, the property owner must also consent to the transaction.
				The business must have been a utility customer (at their current or previous address) for at least the last 12 months.	Check the number of months for which billing records are available.	None.
				Any investments that the business previously made in the specific technology/ies for which they are applying for support through the IUS Programme cannot exceed the value/cost of their sub-project application.	Self-certification by the applicant, to be reviewed by the utility.	The accuracy of these self-certifications will be checked/verified for a sample of beneficiaries as part of the monitoring and reporting process for the Programme.
				The business must have fulfilled their payment obligations to the electric utility (i.e. paid on time, and in full) in at least 9 of the 12 months prior to the submission of the application.	Check the customer's payment history and track record.	If a customer has recently changed address, the utility may either assess the customer's payment history from their previous address or check their credit score.
	1.3	Eligible public	This criterion should be used to determine if a	There are no limitations based on the size of the public	N/A	None.

		sector customers	public sector utility customer is eligible to receive support, subject to fulfilling the other eligibility criteria as outlined below.	sector utility customer.			
				The public sector entity must have been a utility customer for at least the last 12 months.	Check the number of months for which billing records are available.	None.	
				Any investments that the public sector utility customer previously made in the specific technology/ies for which they are applying for support through the IUS Programme cannot exceed the value/cost of their sub-project application.	Self-certification by the applicant, to be reviewed by the utility.	The accuracy of these self-certifications will be checked/verified for a sample of beneficiaries as part of the monitoring and reporting process for the Programme.	
				The public sector utility customer must have fulfilled their payment obligations to the electric utility (i.e. paid on time, and in full) in at least 9 of the 12 months prior to the submission of the application.	Check the customer's payment history and track record.	If the public sector entity is moving to a new location or opening a new office, their broader payment track can be reviewed.	
				Individual solar PV installations may not exceed: 10 kW for residential customers; and 100 kW for commercial and public sector customers.	Check installation size relative to customer type.	Utilities may request permission to exceed these limits. CDB no-objection will be required.	
				All solar PV installed must be new/additional (i.e. no replacement of existing solar PV installations).	Check and confirm (Y/N) compliance with this requirement.		
				For applications that include batteries, the battery size must not exceed: 20 kWh for residential customers; and 100 kWh for commercial and public sector customers.	Check battery size relative to customer type.	Utilities may request permission to exceed these limits. CDB no-objection will be required.	
Pre-approved technologies	2.1	Solar PV	This criterion should be used when assessing a sub-project that includes solar PV (with or without battery).	The utility must determine that the sub-distribution infrastructure that connects the customer to the grid is	Check and confirm (Y/N) compliance with this requirement.	Utilities will be given flexibility to make this determination, as they manage the sub-distribution infrastructure and will review/ appraise	

				able to accommodate the additional load.		the sub-project application.	
	2.2	Efficient AC/cooling	This criterion should be used when assessing a sub-project that includes efficient AC/cooling.	When an existing AC/cooling technology is to be replaced, the new technology must be expected to result in an overall reduction in the customer's total consumption of electricity from the grid.	Check and confirm (Y/N) compliance with this requirement.		
				When a new AC/cooling technology is to be installed (i.e. not replacing an existing AC/cooling technology), the new AC/cooling technology must not be expected to increase the customer's overall consumption of electricity from the grid by more than 20%.	Check and confirm (Y/N) compliance with this requirement.		
				Only AC/cooling technologies that utilize modern coolants are eligible (i.e. no use of R-22 refrigerant).	Check and confirm (Y/N) compliance with this requirement.		
	2.3	Efficient lighting	This criterion should be used when assessing a sub-project that includes efficient lighting.	When existing lighting is to be replaced, the new lighting must be expected to result in an overall reduction in the customer's total consumption of electricity from the grid.	Check and confirm (Y/N) compliance with this requirement.		
				When new lighting is to be installed (i.e. not replacing existing lighting), the new lighting must not be expected to increase the customer's overall consumption of electricity from the grid by more than 20%.	Check and confirm (Y/N) compliance with this requirement.		
				Only LED lighting is eligible.	Check and confirm (Y/N) compliance with this requirement.		

	2.4	Solar water heater	This criterion should be used when assessing a sub-project that includes a solar water heater.	When an existing hot water technology is to be replaced, the new technology must be expected to result in an overall reduction in the customer's total consumption of electricity from the grid or consumption of other primary energy (e.g. LPG).	Check and confirm (Y/N) compliance with this requirement.	
				When a new hot water technology is to be installed (i.e. not replacing an existing hot water technology), the new hot water technology must not be expected to increase the customer's overall consumption of electricity from the grid by more than 20%.	Check and confirm (Y/N) compliance with this requirement.	
Additional (non-pre-approved) technologies	3.1	Climate change relevance	This criterion should be used to determine whether a technology that is <u>not included in the list of pre-approved technologies</u> would be eligible for support through the Programme.	Additional technologies can only be financed if they deliver climate change mitigation benefits at a rate of USD85 per tCO ₂ e reduced/avoided or lower (i.e. 20% above the average for the Programme).	Record the USD cost per tCO ₂ e reduced or avoided.	This criterion is only applicable when a sub-project application includes a technology that is not included in the list of pre-approved technologies. No-objections will be required from both CDB and the GCF Secretariat before a utility can finance a sub-project that includes a non-pre-approved technology. Once this no-objection has been granted, the utility can continue financing other sub-projects that include this technology.
Other criteria	4.1	Sub-project cost	This criterion should be used when assessing the total cost of a sub-project application.	Maximum sub-project cost is USD400,000, including ancillary technologies.	Record the sub-project cost.	Utilities may finance larger sub-projects subject to receiving a no-objection from CDB. This limit is the overall limit that will be applied across the Programme. CDB may establish lower utility-specific limits

						if/as needed when finalizing the appraisal of each line of credit.
	4.2	Ancillary services	This criterion should be used when assessing the proportion of a sub-project application that is allocated for emissions-reducing technologies vs. ancillary services.	The cost of ancillary services should not exceed 30% of the total sub-project cost.	Record the cost of ancillary services.	'Ancillary services' includes any technologies or services that do not deliver direct climate change mitigation benefits.
	4.3	Customer exposure	This criterion should be used when assessing sub-project applications from customers that have already/previously secured sub-project financing through the Programme.	No individual utility customer shall receive more than USD1 million in financial support through the Programme.	Record the costs of the new sub-project application and those of any/all previously-approved sub-projects with the customer in question.	
	4.4	TSP exposure	This criterion should be used when determining which TSP will be contracted to procure and install the requested technology/ies at the premises of the applicant.	No individual TSP shall receive more than 20% of the total value of the contracts in each country.	Record the cost of the new sub-project application and those of any/all previously-approved sub-projects with the TSP in question.	
	4.5	E&S	This criterion should be used to determine whether potential E&S risks have been properly considered, and if suitable risk mitigation and management measures have been identified (as relevant).	All sub-projects must comply with the E&S due diligence requirements as outlined in Annex 6: ESMF.	Y/N	
	4.6	M&R	This criterion should be used to determine whether sufficient data is available to establish a sub-project baseline for M&R purposes.	All sub-projects must have a clearly defined baseline for M&R purposes.	Y/N	Sub-project baselines should be set in accordance with the requirements and methodologies described in Annex 22.

Indicative administrative process

B.3.19 Sub-project applications will be received by the National Executing Entity from eligible beneficiaries seeking to access programme funding. The sub-project application is then appraised by the relevant department/unit of the Executing Entity that is leading the IUS programme. The Executing Entity may seek support or advice from the TA Implementation Support Consultant while reviewing / appraising the sub-project application. Each Executing Entity, however, may not engage their National Energy Consultant in the sub-project application review / appraisal process. The three National Energy Consultants (1 per utility/national Executing Entity) will rather focus on supporting business development and sub-project origination, as well as monitoring and reporting. If these individuals were to also support the review/appraisal of submitted applications, this could lead to a potential conflict of interest as they may ultimately be supporting appraisal of applications that they helped originate and guide.

B.3.20 The Executing Entities will review if (a) the applicant meets the applicable Know Your Customer (KYC) criteria for each country; (b) the applicant satisfies the definition of final beneficiaries; and (c) the technology is preapproved. After the preliminary checks, the Executing Entities will undertake an in-depth review of the sub-project application, including but not limited to, (a) is the application made for one or more than one eligible technology/service, (b) historical energy consumption and bill payment frequency of at least three preceding years from the date of application, (c) in case of grid connected renewable energy sub-project the Executing Entity needs to check if there is enough available hosting capacity in the distribution feeder connecting the applicant with Executing Entity's grid, (d) has a TSP already been identified by the application, (e) assess creditworthiness of the applicant, (f) socio-economic background of the applicant (for inclusivity under the programme), (g) E&S and/or GESI considerations etc. A more detailed description of the eligibility criteria is included in Section 3.3.2 of this Operations Manual, refer to Annex 21.

B.3.21 After this review, the National Executing Entity will decide to either approve, reject, or undertake additional review of the sub-project application. The TA Implementation Support Consultant will assist the Executing Entity in sub-project application appraisal, as needed. Based on the results of the assessment at this stage, the Executing Entity will process the sub-project financing under the programme if all the conditions are fulfilled.

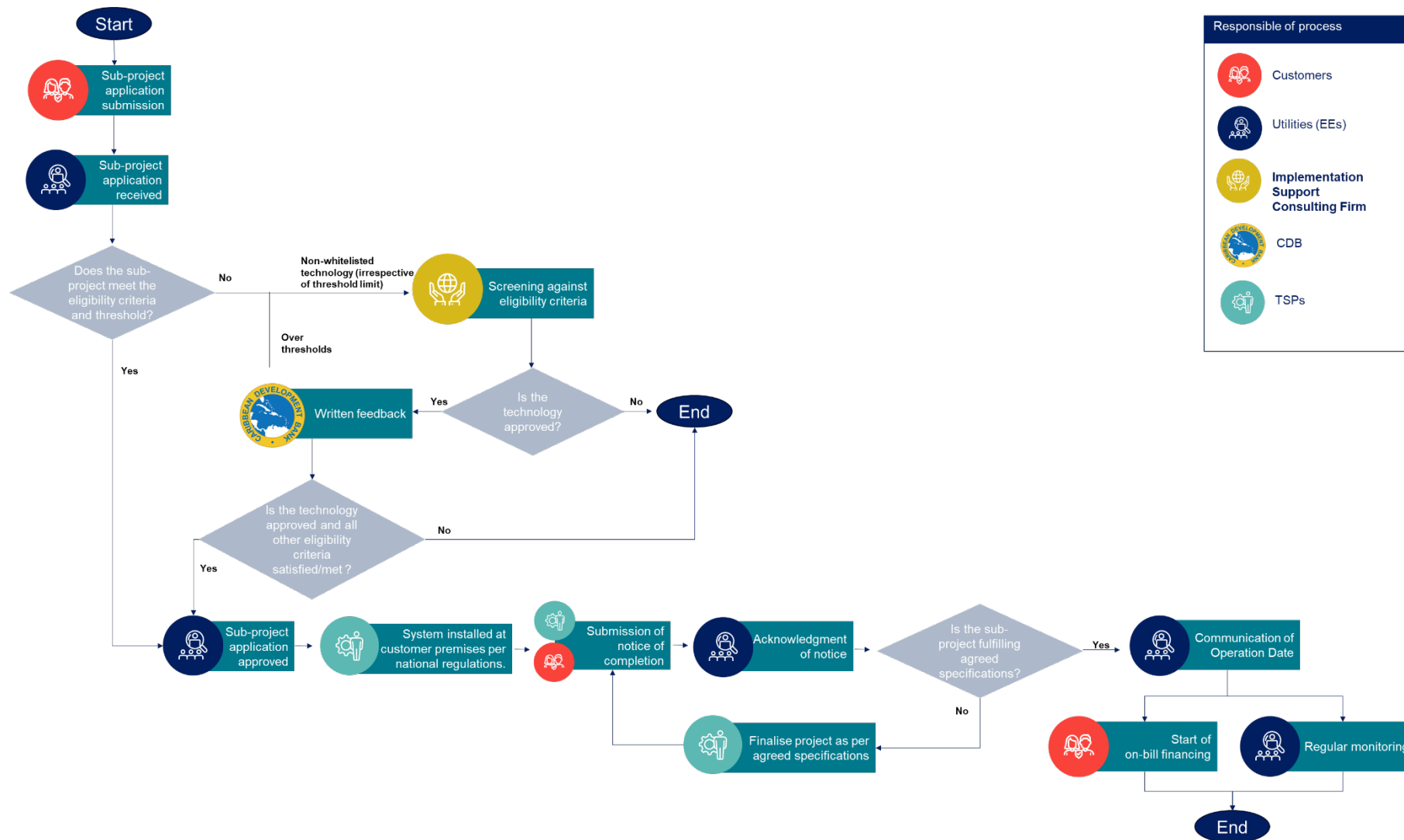
The generic flow to guide the Executing Entities has been presented in Figure 21 below. Detailed, time-bound, and tailored step-by-step procedures, along with the personnel responsible at each stage, will be developed and included in the Executing Entity-specific Operations Manuals that will be developed during the inception phase (PY1) of the programme implementation period.

Knowledge management

B.3.22 CDB will be responsible for knowledge management under the IUS Programme. In addition, the National Energy Consultants will be tasked with collecting and transmitting all the documents developed in their respective BMCs to the TA Execution Team on a quarterly basis. The CDB TA Execution Team will then share these documents and information with the CDB AE Team. The CDB AE Team may then also develop a knowledge sharing portal to disseminate knowledge with other stakeholders.

Figure 18: Application screening process flow

B



Source: CPCS

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

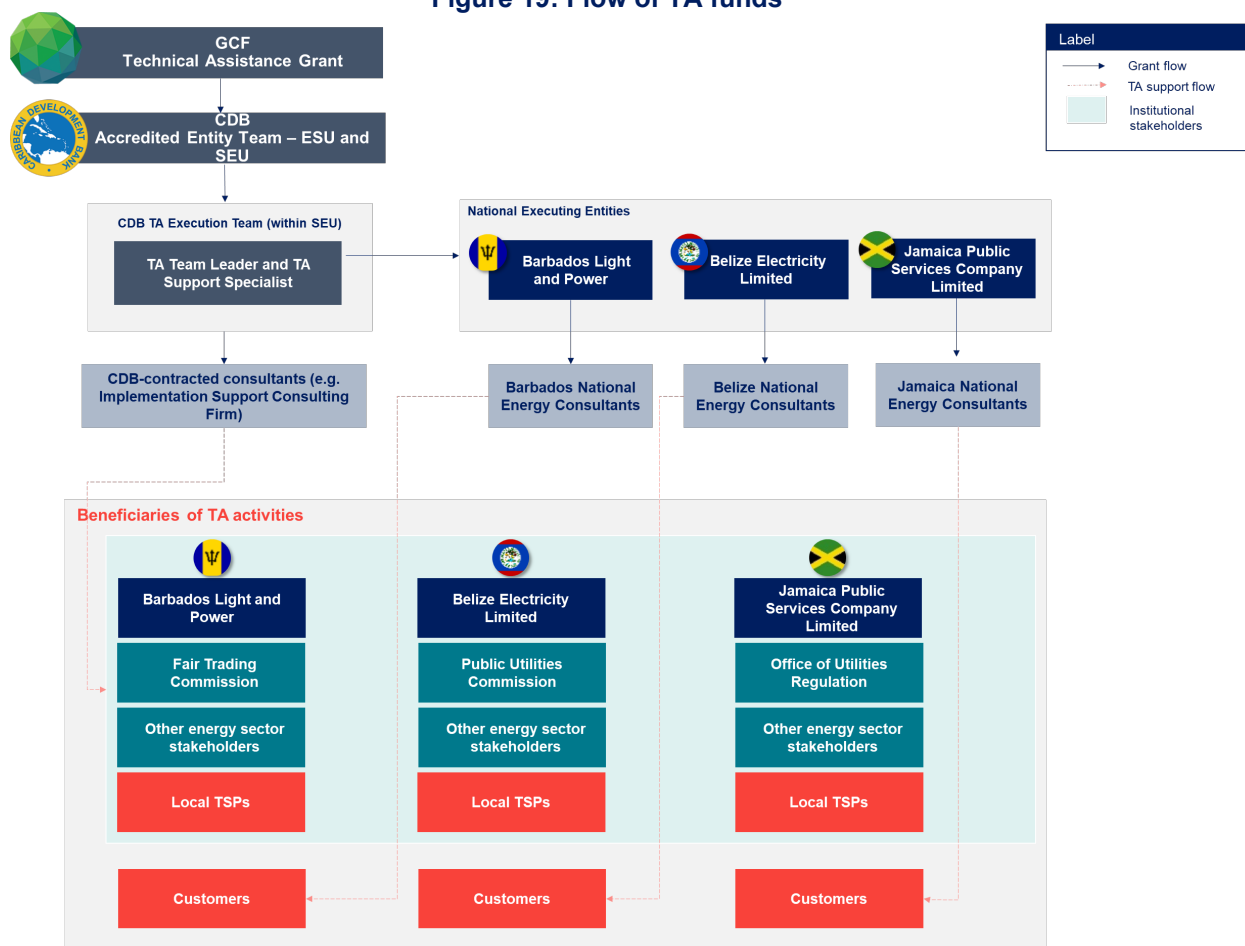
Implementation arrangements

B.4.1. The CDB is the Accredited Entity for this programme. CDB has been operating in the Caribbean Region for over 50 years and has a strong track record of providing financial and technical assistance to support the regional energy sector and sustainable energy transition. CDB has also worked with electric utilities throughout the region, including BLPC, BEL, and JPS. As the Accredited Entity, CDB will lead programme implementation, supervision, reporting, evaluation and closure. Different areas within CDB will contribute to fulfilling these functions, as outlined in Figure 22 (below).

- **TA component:** CDB, through its dedicated TA Execution Team, will directly execute most activities (and associated contracts and consultancies) under the TA component. The only exceptions are the National Energy Consultants, which will be procured and contracted by the national Executing Entities using GCF grant financing. This grant financing will be allocated and disbursed to the national Executing Entities by CDB and governed by subsidiary grant agreements between the Bank and these Executing Entities.
- Under this Component, the Programme will facilitate capacity building and execution capabilities of the regulatory body, the Executing Entities, and other key stakeholders through the TA grant financing.
- It will do so by addressing capacity, knowledge and awareness gaps that are inhibiting the flow of financing towards climate investment. Among others, the key outcome of the TA support would be an established distributed clean energy market and business ecosystem in the three BMCs which is self-sustainable beyond the GCF programme funding. The TA support will also aim to establish the IUS model as a viable option for accelerating utility led distributed clean energy deployment in other Caribbean nations. Furthermore, this Component will support the effective delivery of the financing and investments under Component 2 and help catalyse the broader process of transforming Caribbean finance – and the targeted productive sectors and energy systems – over the medium/long term.

B.4.2. CDB will support the partner BMCs, Executing Entities, and other stakeholders to better understand the links between their business operations and climate change, as well as apply this understanding when developing, appraising, financing, and implementing climate change-related sub-projects. The flow of activities is depicted in the figure below.

Figure 19: Flow of TA funds



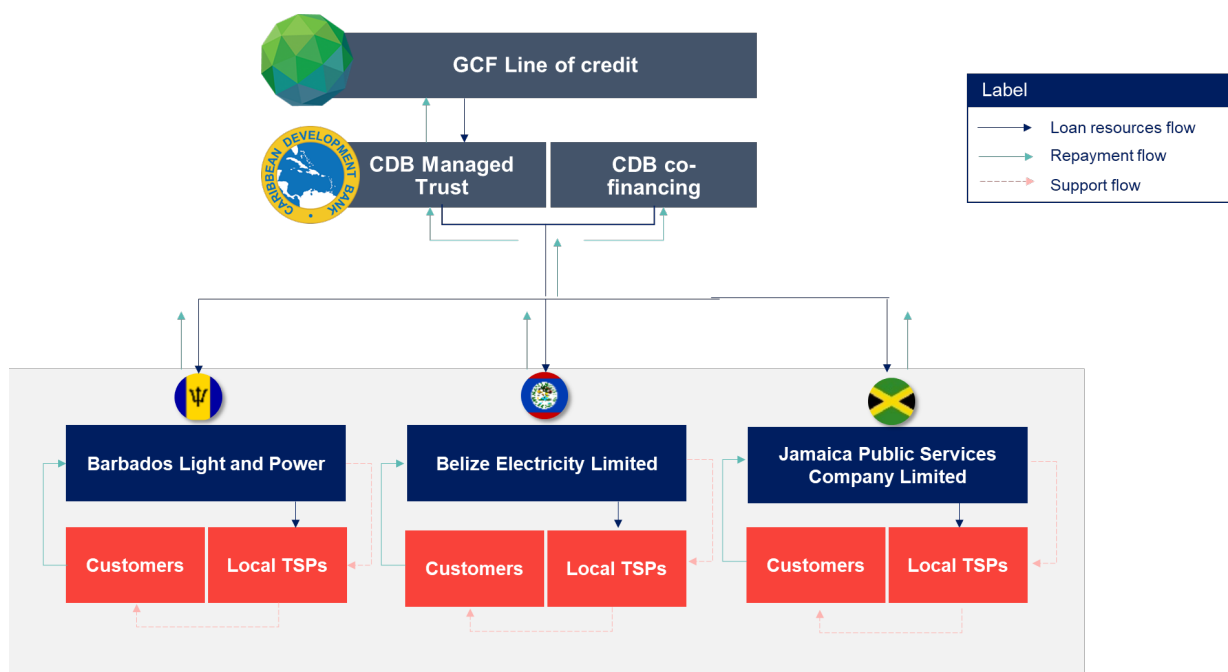
Source: CPCS

- Loan component:** CDB, through its Accredited Entity Team, will be responsible for managing relationships and credit lines with the National Executing Entities. Administration of the lines of credit will be done in accordance with relevant CDB policies and procedures. As is standard operating procedures of the CDB, the GCF loan resources will be deposited in a CDB-managed bank account (IUS Programme Account) and a unique identification code assigned to those resources. Aligned with the terms and conditions of the FAA and CDB's AMA, all transactions associated with GCF resources for the IUS Programme will be tagged with this unique identification code, allowing for ease for monitoring and tracking via an account ledger.
- CDB will extend a concessional line of credit to each national Executing Entity that includes both concessional GCF resources and CDB co-financing. CDB will enter into a subsidiary agreement with each national Executing Entity that governs the use of the loan resources, including the GCF resources and CDB resources which are expected to rank *pari-passu*.
- Disbursement requests from the national Executing Entities will be submitted to, and processed by, CDB in accordance with the Bank's OPPM. All disbursements from CDB and the IUS Programme Account will be made to the Executing Entities in *pari-passu* terms (funding will be disbursed simultaneously in the agreed proportion). The first disbursement of the loan will only be made once the Executing Entity has fulfilled all Conditions Precedent (CPs) as outlined in the relevant subsidiary agreement between CDB and the Executing Entity. Subsequent disbursements will be made once other programme-specific CPs have been satisfied. These may include, *inter alia*:
 - Compliance with Financial Covenants that CDB may require of Executing Entities;
 - Completion of programme-related reporting for the previous period;
 - Confirmation of the projected demand for financing for sub-projects in the forthcoming period.

National Executing Entities will be responsible for channeling resources provided by CDB to support/finance eligible sub-projects for eligible beneficiaries. To this effect, the National Executing Entities will enter into agreements with both the eligible beneficiaries and the TSPs to be contracted to provide the requisite services to the eligible beneficiaries.

B.4.3. The National Executing Entities will also be responsible for collecting repayments from the eligible beneficiaries, through an on-bill financing mechanism, where the amount of loan payment is collected by each Executing Entity through its regular billing cycle. The national Executing Entities will also be responsible for repaying their loans to the CDB (including CDB resources and GCF resources) in accordance with the subsidiary agreement.

Figure 20: Flow of line of credit



Source: CPCS

B.4.4. In addition to the above-described flow of funds and support for the IUS sub-projects to be financed and implemented under Activity 2.1.1, the electric utilities will also deploy and install smart meters/AMI at the premises of their customers. This may include the customers that will benefit from IUS sub-projects and/or other customers that do not benefit from IUS sub-projects during the Programme implementation period. These investments are not captured in the flow of funds diagrams because the investments are expected to be in-kind contributions by the participating electric utilities.

Contractual arrangements

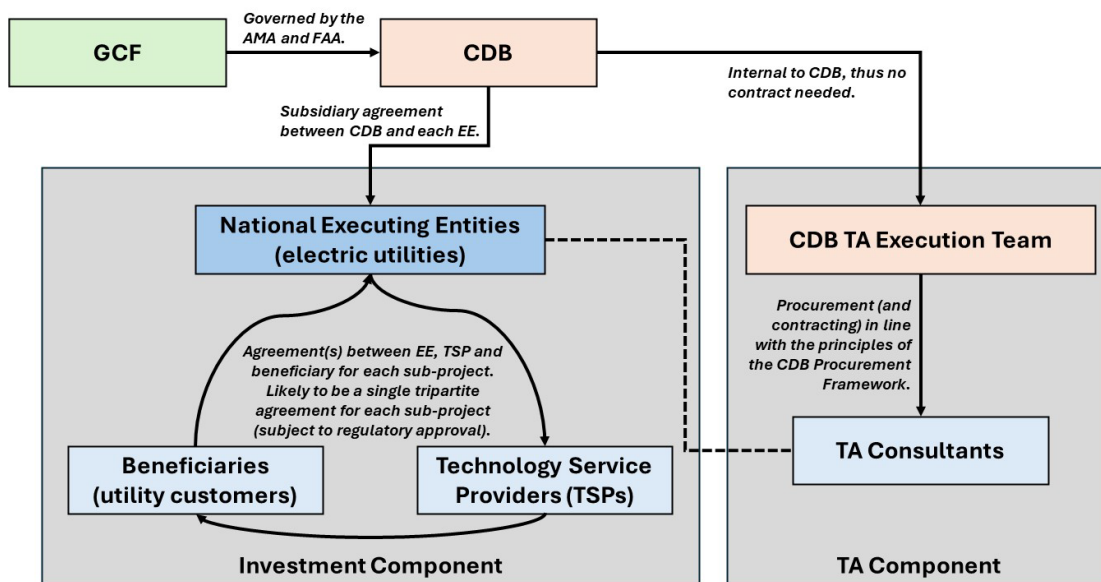
B.4.5. The relationship between CDB and the GCF will be guided by the Accreditation Master Agreement (AMA) between these two organizations, as well as the Funded Activity Agreement (FAA) that will be developed specifically for this programme. The FAA will reflect the fact that the GCF loan financing would be provided to the three national borrowers/Executing Entities through a CDB-managed trust. CDB will in turn enter into a subsidiary agreement⁴⁶ with each Executing Entity (i.e. electric utility) that will address, *inter alia*, their responsibilities as Executing Entities and their obligations as borrowers vis-à-vis their respective blended GCF-CDB lines of credit.

⁴⁶ This is tentatively expected to be two agreements per national Executing Entity: a subsidiary loan agreement for each line of credit; and a subsidiary grant agreement to provide the grant financing with which each national Executing Entity can procure and contract their National Energy Consultant.

B.4.6. For each sub-project that is financed by a national Executing Entity (i.e. utility), the Executing Entity will enter into an agreement the eligible beneficiary (i.e. utility customer) and the Technology Service Provider that is contracted to provide the requisite services (i.e. installation of eligible technologies) to the eligible beneficiary. These sub-project-level agreements will cover the obligations of all parties including, *inter alia*: installation of the eligible technologies and ensure they are in good working order (TSP); adherence to appropriate environmental and social standards when installing the eligible technologies (TSP); providing the data/information needed to verify the timely execution of the sub-project (TSP); providing the data/information needed to monitor benefits arising from sub-projects (customer); and repaying the investment cost to the utility (customer), among others. Based on consultations during the programme preparation phase, CDB generally expects that most utilities will opt to structure these as tripartite agreements between the utility, TSP and customer.

B.4.7. Specifically with regards to the TA component, the CDB TA Execution Team will directly procure and enter into contracts with almost all service providers, and will do so in line with CDB's Procurement Framework and using its standard forms of contract. The only exception will be the National Energy Consultants, one of which will be procured and contracted by each Executing Entity (i.e. utility). The nationally executed procurement processes will also be managed in line with CDB's Procurement Framework.

B.4.8. These contractual arrangements are summarized in the diagram below.



Key programme actors

B.4.9. Below we provide an overview of the role of each of the types of institutions which will be involved in the Caribbean IUS programme within each participating BMC.



The GCF

GCF is accountable to the Parties of the United Nations Framework Convention on Climate Change (UNFCCC) and is guided by the principles and provisions of the UNFCCC.



The CDB

As the Accredited Entity, the CDB will provide financing, financial oversight, overall programme supervision, and ensure the executing entities abide by the policies and standards of the GCF and CDB. CDB will also manage the blended lines of credit, ensure timely credit recovery, and ensure robust programme monitoring, reporting and evaluation.

Environmental Sustainability Unit (ESU): The ESU is the Focal Point for the Green Climate Fund in the CDB. It will bear overall responsibility for supervising all programme activities, including: the TA support to be executed by the national Executing Entities; the TA support to be executed by the TA Execution Team; and the execution of the Lines of Credit by the Executing Entities (for which the ESU will collaborate closely with other relevant CDB staff, notably from the SEU). The ESU will compile all relevant programme information to draft reports for the GCF and its stakeholders.

Board: The Board of Directors of the Caribbean Development Bank is the ultimate supervisory authority within the Bank and has the responsibility to ensure Management is administering the Bank's operations according to policy. The Board may, at its discretion, request/receive updates on the programme. Board decisions or resolutions will be executed by CDB management, who will then direct the ESU and other divisions/unit to follow through on Board decision or resolution.

Management: Includes the President, as well as Vice Presidents, Departmental Directors and Division/Unit Heads. Management acts upon Board decisions or resolutions, as well as on-demand of internal areas. Management will take action on programme-related activities if the Board or the ESU require this level of decision.

Other involved areas within CDB that will support the programme through oversight are the following:

Office of Independent Evaluation: The CDB Office of Independent Evaluation will be involved in supporting/quality-assuring the independent mid-term and final evaluations of this programme.

Procurement Policy Unit (PPU): The Procurement Policy Unit (PPU) will be involved in supporting CDB-managed procurement processes, including those managed by the TA Execution Team and Accredited Entity Team. The PPU also includes a Claims team that will support the processing of invoices and payments to consultants contracted by CDB (both in its Executing Entity and Accredited Entity roles).

Risk Management: The Risk Management Department monitors the Bank's exposure to operative, financial and portfolio risk. For the programme, the Risk Department will receive periodic reports on loan performance from the Accredited Entity Team and may request corrective actions to ensure financial good standing by the utilities.



Government

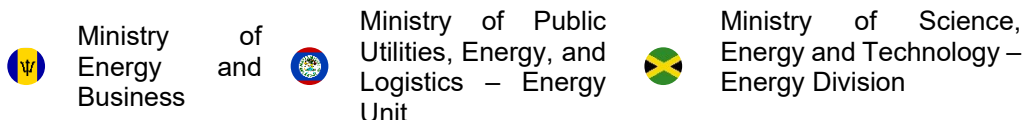
Government ministries, departments and agencies are pivotal in shaping energy policy and overseeing energy market participants. Decision-making authority is often delegated to respective ministries, oversight bodies like public utility commissions, etc. to ensure the underlying sectoral goals and objectives are met. For instance, in the case of DER, RE, and EE, the government establishes legal and regulatory frameworks for the sector, broader policy goals and objectives like the NDCs, etc., regulations for private sector involvement, sets renewable energy goals, defines codes and standards for energy asset operations, and determines tariff rules.

The success of a behind-the-meter RE/EE market is grounded in the government's policies that act as a facilitator for both market development and growth. Without

government intervention to eliminate obstacles, the underlying business models may be unviable.

The governments of the BMCs, through their energy ministries and departments, will be responsible to ensure facilitative legal and policy support as well as address policy, legal, and regulatory hurdles.

Relevant Government Ministries



NDAs

National Designated Authorities are government institutions that serve as the interface between each country and the GCF. NDAs provide broad strategic oversight of the GCF's activities in the country and communicate the country's priorities for financing low-emission and climate-resilient development.

The respective NDAs will ensure various programme requirements like (a) the programme's alignment with the local context, (b) stakeholder engagement, (c) liaison between key stakeholders, (d) support with GCF M&R, etc.

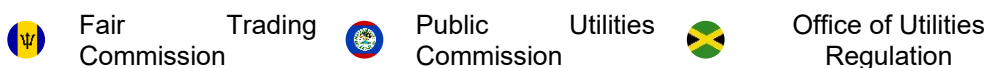
GCF NDAs



Regulator

Newer utility models such as the IUS create new lines of revenue for utilities. To integrate this revenue stream into the utility's established regulated business, regulatory interventions and clear guidelines are necessary. The regulators in all the participating BMCs will ensure that (a) the regulated business model does not violate or contradict any existing legal and regulatory framework; (b) the appropriate regulatory framework such as DER regulations like net metering/billing, etc. are well defined; (c) there is clarity regarding energy transactions, accounting, and compensation; (d) a well-defined customer focused and business-oriented interconnection/application process is in place; and (e) robust technical and grid standards are followed for equipment; etc.

Electricity Regulators



Utilities (Executing Entities)

There are three participating national Executing Entities under the IUS programme who are already familiar with DER and behind the meter RE and EE to a varying degree. For instance, the BLPC has been delivering the "buy-all, sell-all" initiative. JPS is among the leading Caribbean utilities delivering net-billing models. As a result, the utilities are already experienced with DER initiatives.

Under the IUS governance structure, the Executing Entities will ensure a fair and transparent programme delivery to their customers equitably. To support the programme roll-out the Executing Entities will facilitate (a) all the programme related regulatory

approvals, (b) changes in the billing system for on-bill financing, (c) adoption of a standardized and timebound application process, (d) programme outreach material, (e) customer's creditworthiness, among others.

Additionally, the Executing Entities will (a) review and approve sub-projects with the support of the TA implementation consultant(s), (b) administer their respective Line of Credit, finance eligible sub-projects to support eligible beneficiaries, collect and process repayments/reflows from beneficiary customers and (in turn) to CDB, (c) procure, contract, and manage, a full-time support staff – including a National Energy Consultant to be paid through the IUS programme budget – with expertise in distributed renewable energy and energy efficiency to support on day-to-day activities, customer outreach, business development, etc. under the IUS programme, (d) provide periodic updates and information to the CDB as per the monitoring and reporting framework, (e) set up/strengthen customer support centres, etc.

Each national Executing Entity will be required to nominate a programme focal point who will lead on communications with other programme stakeholders (notably CDB) and will share this individual's contact information with the CDB Accredited Entity Team and the TA implementation consultant during the start of the programme implementation period.

Participating Utilities



Barbados
and Power

Light



Belize
Limited

Electricity



Jamaica Public Services
Company Limited



Technology Service Providers

TSPs are private entities in the sector capable of providing various services like EE, solar rooftop, solar water heaters, battery storage for optimizing time-of-use, and operation and maintenance initiatives geared towards lowering customers' energy expenses, among others.

TSPs will play a wider role in the IUS programme. TSPs will not only provide both distributed generation installations and EE services but also support broader efforts to enhance customer outreach and assist customers to navigate the IUS application approval.

TSPs will enter into an agreement with customers and utilities to provide their services as per the technical, regulatory, and commercial specifications of the respective BMCs.

TSPs will also be receive additional training(s) on the technology, legal and regulatory, and commercial aspects of the programme through the TA support. By engaging with (and benefiting from) the IUS Programme, local TSPs will become capacitated to provide additional – and better – services, thereby transitioning toward being able to serve households and businesses in ways that are similar to the more comprehensive sets of energy services provided by ESCOs in other jurisdictions.



Other regional stakeholders

- **CCREEE:** It has been associated with the development of the IUS programme since it was conceived. During the programme implementation and roll-out, CCREEE will leverage its unique regional position to help the BMC institutions with expert inputs to develop IUS conducive regulatory policy frameworks and facilitate in programme outreach, among others.
- **CARILEC:** The Caribbean Electric Utility Services Corporation is an association of electric energy solutions providers and other stakeholders operating in the electricity industry in the Caribbean region, Central and South America and Globally. The CARILEC Secretariat plays a leading role in electric utility advocacy, growth, and sustainability in the Region. As a regional organisation CARILEC members include utilities, service providers, and training institutes. Consequently,

CARILEC can be a regional platform for knowledge sharing, capacity building, and training to strengthen the IUS programme's supply side ecosystem.



Business Associations

Association like Barbados Renewable Energy Association (BREA), Barbados Chamber of Commerce and Industry (BCCI), etc., are key institutions that represent industries, service providers, and business. Such organisations will be leveraged by the respective energy ministry, NDAs, and TA implementation consultant for customer outreach to promote the IUS programme.



Customers

The key beneficiary of the IUS programme in all BMCs reaping benefits like self-generation and energy independence, lower electricity costs, efficient use of energy, and contributing to BMCs' climate goals, among others. Customers will prepare and submit sub-project applications to their respective electric utilities.



TA Execution Team

The TA Execution Team will be established within the CDB and will execute most of the TA activities under the programme (and by extension, manage most procurement processes). The only exception is the National Energy Consultant, which will be contracted directly by the National Executing Entities (utilities). The TA Execution Team will include a full-time TA Lead who will provide overall technical leadership with regards to the execution of the TA component and will also include a TA Support Specialist to assist the TA Lead. The TA Execution Team will procure, contract, manage, monitor, and report on progress achieved by the other consultants that will be contracted under the TA component. The TA Execution Team will be established within the CDB Sustainable Energy Unit and will report directly to the Unit Coordinator. The TA Execution Team will also maintain a close dialogue with the CDB Accredited Entity Team to keep them updated on progress that is relevant to their overall programme supervision responsibilities, including supervision of the national Executing Entities' progress in executing their respective lines of credit.



TA Implementation Support Consultant

The TA Implementation Support Consultant will be procured, contracted, managed and monitored by the TA Execution Team. The TA Implementation Support Consultant will include a multi-disciplinary team comprised of international experts (tentatively expected to include a Technical/ Clean Energy Expert, Legal/ Regulatory Expert, Energy Policy Expert, Gender and Social Inclusion Expert, Environmental and Social Risk Management Expert, Monitoring and Reporting Expert, Capacity Building Expert, and Marketing and Outreach Expert) and national expertise (tentatively expected to include a National Regulatory Expert).

The TA Implementation Support Consultant will work in close consultation with three key national stakeholders in each country, namely the energy ministry, the regulatory authority, and the national Executing Entity. They will directly report their progress and be directly accountable to the TA Execution Team.



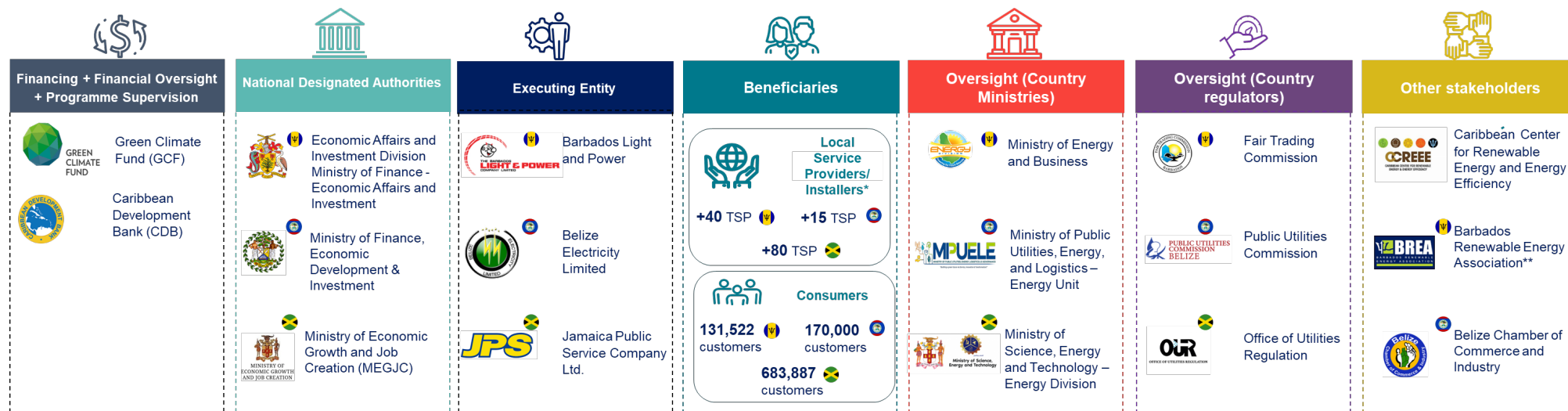
National Energy Consultants

One National Energy Consultant will be procured, contracted, and managed by each national Executing Entity, with financial support from the programme's TA budget. The National Energy Consultant will be stationed at the Executing Entity's office and will report to the appropriate department/ unit/ or office of the Executing Entity that leads the IUS service delivery. The National Energy Consultant will support day-to-day activities related to the IUS programme, customer support, business development, etc. The National

Energy Consultant, however, is not responsible to support the national Executing Entity with sub-project appraisal due to possibility of conflict of interest.

Figure 24 below provides a detailed stakeholder mapping for the IUS programme.

Figure 21: Detailed stakeholder mapping for the IUS programme



Source: CPCS

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

B.5.1. While the Caribbean region has embarked on significant efforts to enhance financing options for RE and EE projects, access to financing remains constrained, highlighting notable challenges across Barbados, Belize, and Jamaica. While some public and private financing exists in Barbados, Belize, and Jamaica, it is primarily focused on businesses and does not cover the diverse technologies included in the IUS programme. Additionally, there is a shortage of commercial banks offering loans for distributed RE, and those that do often charge high interest rates due to the perceived high risk of lending to local households and businesses (particularly the types of lower-income groups and smaller businesses that the IUS model and Programme are particularly well suited to support) to invest in these technologies. EE projects, in particular, are rarely considered for financing. One significant gap is the upfront cost barrier versus long-term savings, which is particularly challenging for lower-income households and businesses. Although RE and EE technologies promise substantial savings over time, the initial investment required can be prohibitively high. This financial challenge is a major deterrent for both public and private investors, who may not have the capital or willingness to invest heavily without immediate returns.

On the public side, capital is often constrained and must be allocated to competing priorities, leaving limited resources for RE and EE projects. Governments in these countries face numerous pressing needs, making it challenging to prioritize and fund energy transitions without substantial external support. Additionally, the lack of a clear regulatory framework and successful demonstrative cases further deters private investment.

Funding structure

B.5.2. To make the proposed programme (and its underlying projects and sub-projects) feasible, CDB is requesting concessional financing from the GCF. These concessional resources are needed in part because the programme will assist electric utilities to launch and institutionalize IUS – a new and innovative approach to supporting investment in distributed RE, EE and other DER technologies/measures among the utilities' customers – as part of their core business models. The process of launching an IUS scheme in each participating BMC and implementing it at scale involves risks and associated costs that the utilities and partner governments are not able to bear on their own. The following types of GCF financing are therefore critical to the success of this programme.

B.5.3. Line of credit/Loans: To launch and sustain their initial IUS schemes, partner utilities will need concessional loans to make RE/EE sub-projects appealing to customers, and to enable a more inclusive set of sub-project beneficiaries (including lower-income and vulnerable/marginalized groups). Despite the strong savings and short payback periods associated with distributed RE, EE, and other DERs, Caribbean households and businesses, wary of risk and often financially constrained, might hesitate to invest— a situation exacerbated by COVID-19's economic fallout. Recognizing this, CDB aims to encourage investment in DERs by ensuring initial modest on-bill rates to demonstrate value. Given the region's utilities' similar aversion to risk, introducing the IUS model with affordable rates and favourable terms is crucial. CDB will thus facilitate the transfer of GCF's concessional benefits to utilities, ensuring they, in turn, offer customers more accessible costs for RE, EE, and DER technologies than market rates. Note that several mechanisms are included in the programme design to ensure transfer of this concessionality to the end beneficiaries, including by establishing maximum spreads that can be charged by the electric utilities.

That being the case, CDB is proposing that the GCF Secretariat provide a concessional line of credit amounting to USD 20.8 million. CDB in turn expects to blend the GCF resources with USD 16.2 million in co-financing. As outlined in the Integrated Financial Model (see Annex 3), this would enable the Bank to extend a concessional line of credit to each electric utility/national Executing Entity. Even after the utilities have placed a spread on these resources to cover overhead costs and a risk premium – rates that will be determined jointly with national counterparts, notably the regulators during PY1 – the concessional GCF-CDB blended lines of credit will enable the utilities to provide financing to customers/end-beneficiaries at rates (tentatively 4.5-6.0%) that are competitive relative to the financing provided by traditional/commercial banks (whose products on average start at around 6.5%) while remaining strategically balanced – low enough to support greater social inclusion and incentivize investment, yet not so low as to distort the local markets. Additionally, this financing will allow access to funding for EE technologies, which are not as readily available as those for renewable energy.

B.5.4. Grant financing: The programme will facilitate capacity building and strengthen the execution capabilities of the regulatory authorities, the Executing Entities, and other key stakeholders through the TA grant financing. CDB will support the partner BMCs, Executing Entities, and other stakeholders to better understand the links between their business operations and climate change, as well as apply this understanding when developing, appraising, financing, and implementing climate change-related sub-projects. These non-revenue-generating activities are only viable at full concessionality. CDB is therefore requesting grant financing to support the implementation of these activities.

B.5.5. The GCF is uniquely suited to provide the above-mentioned types of financing for the IUS programme. CDB is not able to provide the necessary package of financing through its own resources. The Bank has modest concessional financing at its disposal through its Special Development Fund (SDF) window, and cannot use these resources to provide the volumes of financing that the three participating countries need to launch and implement IUS at scale. CDB also works with other partners to mobilize funds for climate action – including bilateral development partners – but they are typically unable/unwilling to provide such large amounts of funding for new/innovative approaches such as IUS. Indeed, many partners have a lower risk appetite than is required for the proposed programme. Given its risk appetite, emphasis on innovation and ability to provide large-scale financing (including grant financing), the GCF is therefore uniquely positioned to provide financial support to the proposed IUS programme alongside CDB co-financing. Without the GCF's support, the proposed programme could not be financed/approved – and without this programme, the participating countries would not be able to launch and implement IUS. Indeed, their fiscal space (while already historically constrained) has become even more limited since the onset of the COVID-19 pandemic, as well as the subsequent geopolitical instability and macroeconomic developments – notably the worldwide increase in interest rates that have significantly increased debt servicing costs in almost all developing countries. Discretionary resources with which to address the financial and non-financial barriers outlined in Section B.2 are simply not available.

B.5.6. Co-financing: The CDB expects to provide co-financing amounting to about USD 16.2 million as part of the blended line of credit facility that will be available to the BMCs for sub-project installations.

B.6. Exit strategy (max. 500 words, approximately 1 page)

B.6.1. The programme includes several elements that will help ensure services, benefit streams and outcomes are sustained after programme closure.

- (1) Project level:** Through this programme, key stakeholders in the participating countries will have established and operated the IUS model at scale over a multi-year time horizon, thereby providing them with valuable experience and a track record that can be leveraged to continue administering IUS after programme closure. Through the TA component, the programme will also strengthen the capacity of national stakeholders – notably the electric utilities and regulators – to continue implementing the IUS model without the need for external support. Furthermore, the programme will connect the participating electric utilities with commercial financiers and sensitize them about the potential and benefits of lending to the utilities to enable further financing of IUS sub-projects after the GCF and CDB financing have been fully utilized. In this sense, the proposed programme is seen as a 'bridge to bankability'. Having established a strong track record of successful projects that generate reflows and enable a profitable IUS business for the utilities, these key stakeholders will also have a strong incentive to continue implementing their respective IUS schemes – scaling up and out – after programme closure.
- (2) Sub-project level.** The programme promotes investments in financially viable renewable energy technologies and energy efficiency services that are economically beneficial for the end beneficiaries. Given the direct financial benefits that are expected to be accrued by the beneficiaries, these households and businesses are expected to continue utilizing these technologies and practices for their entire productive lifespan. This extent to which this expectation holds will be assessed by the national Executing Entities and broader programme implementation team, in line with the Theory of Change, logical framework and monitoring and reporting framework (see Annex 22). Given these direct financial incentives, the sub-project beneficiaries are also expected to properly operate and maintain their programme-supported technologies/assets and cover the associated O&M costs. The anticipated returns from these investments will further ensure that the end beneficiaries have the necessary resources to meet these O&M costs.
- (3) TSPs and other financial services providers.** The programme incorporates a comprehensive TA and capacity-building element designed to enhance the capabilities of TSPs and other financial service providers in all three participating countries. As a result, these entities will be able to better assess climate-

related risks and opportunities and engage in supporting/financing the sustainable energy transition – including by working with the electric utilities to continue administering IUS services after programme closure. By doing so, the programme aims to tackle significant knowledge and capacity barriers that currently hinder the financing flow for climate-related projects. This is especially crucial for certain customer segments, such as residential customers, who typically struggle to secure affordable financing for such investments.

- (4) **End beneficiaries.** The TA and capacity-building component of the programme places a significant focus on increasing awareness and capacity among potential sub-project beneficiaries (i.e. local households and businesses). This is expected address key barriers that are inhibiting demand for both the prioritized technologies and IUS-type services, thereby increasing the potential uptake of IUS-type services over the medium/long term. Notably, the programme intends to reach a wide range of eligible customers, both residential and commercial, through its outreach and awareness-raising activities. While the programme will only be able to finance sub-projects for a portion of these eligible customers, the broader group of beneficiaries is expected to be more aware of the benefits of the prioritized technologies, more inclined to borrow and invest in them, and have enhanced opportunities to do so. This heightened awareness is anticipated to stimulate demand for financing across a larger customer base during and after the programme. To reinforce this effect, the programme will also support eligible final beneficiaries in crafting high-quality sub-project proposals. This will enhance the overall quality of financed sub-projects, leading to better investment performance. Consequently, the programme aims to generate a strong demonstration effect in partner countries, especially considering their relatively small economies. Furthermore, the costs of many promoted technologies and energy efficiency solutions are expected to decrease over time, further driving demand. Coupled with the TA activities that aim to address supply-side barriers (i.e. by strengthening the capacity of electric utilities and regulators, as well as financial institutions and TSPs), the customer-focused support should help ensure that the benefit streams of the IUS model are sustained and scaled up after programme closure.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	26.74			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	20.8 million USD	20 years	05 years	%		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(iii) Equity	<u>Enter amount</u>			<u>Enter % equity return</u>		
(iv) Guarantees	<u>Enter amount</u>	<u>Enter years</u>				
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	5.94 million USD					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	42.0			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Caribbean Development Bank	<u>Senior Loans</u>	<u>11.2</u>	<u>million USD (\$)</u>	20 years 05 years	<u>Variable</u> ⁴⁷	<u>pari passu</u>
	<u>Senior Loans</u>	<u>5.0</u>	<u>million USD (\$)</u>	20 years 05 years	<u>Fixed</u>	<u>pari passu</u>
	<u>In-Kind</u>	<u>0.8</u>	<u>million USD (\$)</u>	<u>N/A</u>		
Electric utilities	<u>In-kind</u>	<u>25.0</u>	<u>million USD (\$)</u>	<u>N/A</u>		
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>68.74</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee). Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets. Please also include parallel financing associated with this project or programme (refer to the co-financing policy). N.A.					
C.2. Financing by component						

⁴⁷ CDB Ordinary Capital Resources (OCR), which are market-rate resources provided at a variable rate that is set by the CDB Board of Directors and adjusted on an as-needed basis.

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 1	Output 1.1	1,547,600	1,547,600	Grants	0	0	N.A.
	Output 1.2	2,015,000	2,015,000	Grants	0	0	N.A.
	Output 1.3	789,750	789,750	Grants	0	0	N.A.
Component 2	Output 2.1	62,000,000	20,800,000	Senior loans	16,200,000	Senior loans	CDB
					25,000,000	In-kind	Electric utilities (BLPC, BEL, JPS)
PMC and M&R	Audit costs	261,000	261,000	Grants	0	0	N.A.
	PMC	1,780,445	970,445	Grants	810,000	In-kind	CDB
	Supplementary data collection to support M&R	352,500	352,500	Grants	0	0	N.A.
Indicative total cost (USD)		68,746,295	26,736,295		42,010,000		

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

Component	Output	Activities	GCF financing Amount USD (\$)
Capacity Building	Output 1.1 Strengthen the enabling environment for IUS in each of the three participating BMCs	Capacity building and technical assistance support to address regulatory barriers, strengthen policy framework, etc.	1,547,600
	Output 1.2 Enhance targeted outreach to	Capacity building and technical assistance support to raise customer awareness on the	2,015,000

	<u>customers to stimulate demand for renewable energy and energy efficiency and interest in the IUS model</u>	opportunities and benefits; promote the IUS model and associated services; provide customers with hands-on support to develop sub-projects, etc.	
	<u>Output 1.3 Enhance regional capacity, knowledge exchange and networks to promote IUS and accessible distributed renewable energy, energy efficiency, and DERs.</u>	Capacity building and technical assistance support to facilitate knowledge exchange among participating and regional (non-participating) countries; train TSPs; create community of practice, etc.	<u>789,750</u>
<u>Technology Development</u>	<u>Output 2.1 Accessible and inclusive financing is provided for sub-projects for different customer groups under IUS model</u>	Low-cost interest financing is provided to the end-beneficiaries through participating utilities to prepare and implement sub-projects in the participating countries.	<u>20,800,000</u>

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

Mitigation impact potential

D.1.1. The analysis on mitigation impact potential accounts for key differences in the national context in the partner countries, including i) the estimated carbon intensity of electricity generation based on the Harmonized Grid Emission Factor data set published by United Nations Framework Convention on Climate Change, as well as ii) the anticipated changes to the emissions trajectory based on the countries' unconditional targets for climate change mitigation. A detailed analysis is provided in Annex 2: Feasibility Study and Annex 3: Integrated Financial Model. Detailed methodology to assess the mitigation impact potential is provided in Annex 22: Assessment of GHG emission reductions and their monitoring and reporting.

D.1.2. The climate change mitigation impact for the sub-projects may arise from the grid electricity consumption avoided due to the electricity generation of solar PV and the energy savings of the more energy-efficient technologies. The IUS programme has the potential to avoid a total of 601,550 tCO₂e in GHG emissions over the productive lifespan of the priority/preapproved technologies that were used as a basis for preparing the impact potential estimates. The assumed lifespans are: 25 years for solar rooftop PV; 7 years for batteries; 15 years for efficient A/Cs; 5 years for efficient lighting; and 20 years for solar water heaters. The following table shows the total GHG emissions avoided under the IUS programme dis-aggregated by country and type of sub-project/technology. Detailed calculations are further elaborated in Annexes 3 and 22.

Table 10: GHG emissions avoided, dis-aggregated by country and sub-projects

Type of subproject	GHG emissions avoided over the sub-project's useful life (tCO ₂ e in 000')		
	Barbados	Belize	Jamaica
Mitigation impact			
Solar rooftop PV	20,687	11,756	53,140
Solar rooftop PV with batteries	4,398	1,606	7,786
Efficient A/C	144,488	16,415	219,654
Efficient lighting	13,798	4,323	19,054
Solar water heater	31,849	11,034	41,562
Total	215,219	45,134	341,197

Source: CPCS

Overall, the mitigation impact potential for commercial sub-projects is higher than that of residential sub-projects because of the larger size of sub-projects (e.g. larger system size of solar rooftop PV). The only exception is for solar water heaters where the size of commercial sub-projects is smaller than that of residential sub-projects based on water usage assumptions determined according to local contexts. The differences between countries are due to differences in the carbon intensity of grid electricity.

Adaptation impact potential:

D.1.3. The direct beneficiaries are expected to include the occupants of buildings in households, businesses and other utility customers who develop (and secure financing from their utility for) sub-projects that enable investments in efficient cooling systems. As a result of such investments, these utility customers will become more resilient to the increasing risk of heat stress that is outlined in more detail in Annex 2b: Baseline Assessment Report. The estimated number of direct beneficiaries is approximately 10,655, including approximately 5,328 women and 5,327 men. This is based on the estimated number of residential customers who are expected to implement sub-projects through which they would invest in efficient ACs (multiplied by the average household size in each country) and the number of commercial customers who are similarly expected to implement such sub-projects under the programme. A more detailed breakdown is included below.

Table 11: Estimated number of direct beneficiaries

Type of sub-project	Number of beneficiaries (individuals or businesses)		
	Barbados	Belize	Jamaica

Number of beneficiaries – Residential			
Efficient A/C	3,101	1,226	5,117
Number of beneficiaries – Commercial			
Efficient A/C	436	131	644

Source: CPCS

Indirect beneficiaries

Indirect beneficiaries are those individuals or entities who, although not directly participating in the programme, demonstrate a significant interest in the priority adaptation technologies and are likely to invest in or adopt these technologies independently or engage with related initiatives post-programme. These beneficiaries reflect the broader impact and awareness generated by the programme, contributing to market development and demand stimulation. The estimated number of indirect beneficiaries is approximately 30,000.

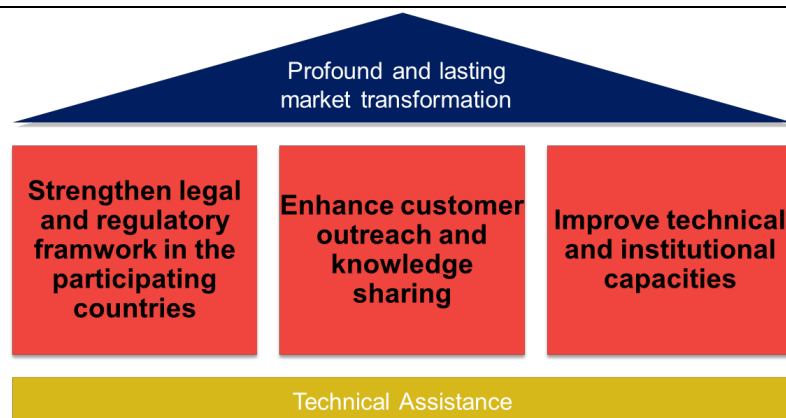
D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

D.2.1. The proposed programme aims to contribute to the process of **transforming the electricity systems and energy sectors in the three participating countries toward low-emissions and climate-resilient development pathways**. The IUS model serves as both an "implementation mechanism" for demand-side RE, EE, and DER projects and a **novel business model for utilities** within a rapidly changing landscape. In implementing this innovative model, the programme will support sector transformation by changing the way utilities and customers think about, engage in and shape the energy sector. In particular:

a. Utilities: The programme will assist each of the three participating electric utilities to offer a new and innovative DER-focused service as part of their business model. The utilities will fundamentally change (and expand) their role in the energy sector of their respective countries, shifting from primarily being a provider of energy to also becoming a provider/facilitator of energy services. In so doing, the utilities will become catalysts in the transition toward a more decentralized electricity grid. This is a fundamental (and necessary) step in the broader process of decarbonizing the electricity system and propose alternative solutions to the increasing demand associated from cooling systems in the participating countries – a process that the programme will equip the utilities to champion in the years to come.

b. Customers: The programme also aims to reshape attitudes about energy (and associated behaviours and decision-making) and energy resilience and increase adaptation to heat stress among utility customers in the participating countries. By working with customers to better understand their energy use/needs and the significant potential of distributed RE, EE and other DERs (i.e. to generate savings, contribute to mitigating climate change, increase resilience and improve local air quality), and provide them with a simple and low-risk way to invest, the programme will empower these individuals to play a more active role in their respective electricity systems. This shift toward more active and engaged households and businesses is fundamental to changing the broader patterns of electricity generation, transmission, distribution, and consumption such that they make the greatest possible contribution to decarbonizing and enhancing the resilience of the energy sector over the medium/long term. This requires that distributed RE, EE and other DER technologies/measures be regarded not simply as consumer goods but rather as critical elements of a country's energy infrastructure – a shift in views and behaviour that the programme will actively support.

D.2.2. While aiming to change the ways that utilities and customers view (including their own roles in) and support the energy transition, the programme also includes other measures to help ensure a **profound and lasting market transformation**. This includes: 1) strengthening legal and regulatory frameworks in the participating countries to better support IUS roll-out as well as enhance the broader enabling environment for DERs; 2) enhancing customer outreach and knowledge sharing to stimulate demand over the medium/long term; and 3) improving technical and institutional capacities of key energy sector stakeholders (including utilities, regulators and TSPs) to ensure IUS services can continue to be provided – with increasing efficiency – beyond programme closure.



Source: CPCS

More specifically, the programme will aim to address **legal, regulatory, and operational challenges** by supporting the preparation and amendment of legal and regulatory documents to facilitate IUS implementation, the development of necessary regulatory frameworks, the modernization of administrative processes, and the optimization of energy accounting procedures. It will also **prioritize capacity building** for regulatory authorities and national stakeholders, aiming to refine policies, incentive frameworks, and investments related to IUS. Furthermore, the programme will **boost customer awareness**, create customer-friendly tools such as energy savings calculators, and produce marketing materials. Critical efforts also include hosting knowledge sharing sessions, developing digital application platforms for IUS, ensuring adherence to environmental and social standards, and encouraging regional cooperation to exchange insights and best practices.

D.2.3. The programme has considerable potential to be scaled up within the participating BMCs and replicated across the Caribbean region in a way that would allow each country to prioritize the RE, EE and other DER technologies that will be most beneficial within their local contexts. To actively support these efforts, the programme will **facilitate regional learning and knowledge sharing among Caribbean governments, the CCREEE, and other regional organizations**. As countries adapt the model to their specific contexts, they can exchange insights and best practices, **accelerating the transition to sustainable energy** solutions across the region and **contributing to global efforts to combat climate change**. Furthermore, CDB plans to develop and submit a 'Phase II' IUS programme proposal over the coming years through which the Bank will actively support additional BMCs to implement and operate IUS at scale.

D.3. Sustainable development (max. 500 words, approximately 1 page)

D.3.1. The programme is expected to deliver significant sustainable development co-benefits. It will not only make important contributions to achieving Sustainable Development Goals (SDGs) **7 (affordable and clean energy)** and **13 (climate action)**, but also **SDGs 3 (health and well-being), 5 (gender equality), 8 (decent work and economic growth), 9 (industry, innovation & infrastructure), and 17 (partnerships for the goals)**. These co-benefits are described in more detail below.

D.3.1.1. Economic: The IUS model will enable utilities in the BMCs to diversify their revenue streams, engage in new business models, and adapt to a changing energy landscape. Sub-projects will generate savings for the utilities' customers, which can in turn be used to repay the investment costs to the utilities and cover other consumption and/or investments needs as required. As outlined in Annex 3: Integrated Financial Model, the estimated net financial savings (NPV) for the final beneficiaries/utility customers are expected to amount to over USD230 million over the full productive lifespan of the technologies to be supported through programme-financed sub-projects. These sub-projects will also generate revenue for (and potentially lead to employment creation among) TSPs in the participating countries. The broader process of decarbonizing the electricity systems and energy sectors (to which the programme will make an important contribution) also holds significant potential to generate income and employment over the medium/long term. Indeed, available evidence suggests that considerable (generally local) labour will be required to support widespread installation and maintenance of distributed RE, EE and other DERs – jobs that will continue to be in demand as the utilities expand IUS after programme closure. Economic co-benefits will also be accrued by the central governments in the participating BMCs, as existing energy imports are a significant drain on foreign exchange (FX) reserves – a challenge that the programme will help address by reducing the demand/need for imported fossil fuels.

D.3.1.2. Environmental: The transition to an electricity system and energy sector based on renewables can significantly reduce the potential for environmental damage associated with the importation, handling, and combustion of fossil fuels. Moreover, this shift can greatly enhance air quality in participating countries, given that burning diesel and other fuels generates substantial particulate pollution. In this context, it is also important to note that the nature of the investments to be made under the programme are not expected to result in negative ecological impacts (e.g. from ground-mounted renewable energy installations). Due to the small-scale nature of the envisaged investments, sub-projects are expected to have little/no detrimental effect on land use and biodiversity.

D.3.1.3. Social: Decreased particulate pollution, resulting from reduced reliance on diesel fuel for electricity generation, can improve health outcomes for residents of participating countries. These social co-benefits may, in turn, lead to additional economic advantages, as healthier populations are often more economically productive. In addition, the IUS programme presents a significant opportunity for groups that have had limited access to traditional financing opportunities, including entrepreneurs, youth, individuals facing gender discrimination, and low-income individuals. As current utility customers, the IUS programme will allow them to bypass these hurdles by leveraging their existing contracts with the utility. By utilizing on-bill financing, the IUS model will enable these otherwise marginalized groups to invest and obtain RE and EE technologies without the need for conventional credit-based financial products. This inclusive approach not only promotes widespread adoption of sustainable energy technologies but also fosters social equity and facilitates a more inclusive and 'just' sustainable energy transition. To further support these efforts at greater inclusivity, the TA component will offer free energy audits to eligible utility customers (e.g. low-income residential customers).

D.3.1.4. Gender: The programme will take a proactive approach to supporting gender-equitable development. This is guided by the programme-level Gender Action Plan (GAP) and informed by a comprehensive Gender Assessment. Specific actions to be taken by the programme include: (i) working with policymakers to mainstream gender considerations into energy sector policy development and planning, including by developing gender mainstreaming guidelines, conducting gender sensitization training for policymakers and regulators, and establishing a working group among the participating countries' gender bureaus to empower them to learn from one another as it relates to integrating gender more prominently into energy sector planning and development; (ii) supporting women and other vulnerable and marginalized groups to directly engage with/benefit from the Programme, including by administering free energy audits for low-income customers, and by preparing and delivering outreach and awareness-raising campaigns that are specifically tailored to the needs of marginalized groups; (iii) supporting the electric utilities/national Executing Entities to utilize the IUS model to better serve and empower women and other vulnerable and marginalized groups, including by conducting gender-sensitivity training for the utilities participating and TSPs, supporting the utilities to develop IUS products/service offerings that are specifically tailored to the needs of marginalized groups, and supporting the utilities to adopt/strengthen systems for capturing and analyzing beneficiary data disaggregated by sex, age, income, location and household type; and (iv) directly (and independently) monitoring the number of Programme beneficiaries who are female-headed households (FHH), persons with disabilities (PWD) and low-income households in accordance with the target set in the GAP. The emphasis on supporting gender-equitable development is also reflected in the logical framework for the programme, which includes gender-specific and gender-disaggregated indicators where relevant/appropriate.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

D.4.1. The BMCs are acutely vulnerable to climate risks due to their geographic exposure to increasing temperatures and extreme weather events, and their economic dependence on climate-sensitive sectors like tourism and agriculture. These risks are exacerbated by high reliance on fossil fuels in their electricity sectors, contributing to significant emissions and compromising climate resilience. The IUS programme directly addresses this vulnerability by promoting the adoption of renewable energy and energy efficiency technologies, thereby aligning with each country's commitment to reduce emissions through their NDCs. Moreover, by advancing the installation of DERs, such as solar rooftops, the project contributes to enhanced energy security, resilience to heat stress, and grid resilience in the face of increasing climate threats.

At an economic and social level, the three participating countries are grappling with significant fiscal constraints and high debt-to-GDP ratios. Although the countries are all classified as 'middle income', this categorization masks significant social and income inequality within the countries. Such income-focused classifications also fail to account for the countries' pronounced vulnerability to climate change and other hazards, all of which can

leave the public and private sectors less well-resourced to address future needs/challenges (including the sustainable energy transition) than other middle-income countries that are less vulnerable. Indeed, in all three participating countries, governments' restricted fiscal space limits their ability to offer the necessary concessionality to launch and implement the IUS model – including by stimulating initial customer engagement and addressing other non-financial barriers. Furthermore, local capital markets lack the depth and history needed to support the sustainable energy transition (including through the IUS model) without external assistance. To address these challenges, the programme will provide external financial assistance to address the range of financial and non-financial barriers that (as outlined in Section B.1 and B.2) are inhibiting the launch and implementation of IUS.

D.4.2. It is also important to note that within the participating countries, lower-income households, smaller (and more informal) businesses and marginalized groups have thus far mostly been unable to participate in (and benefit from) the sustainable energy transition. The programme has provisioned for over 300 free energy audits for such customer groups. These energy audits will ensure that the programme is able to provide a just energy transition opportunity for these customers. Furthermore, the countries are all classified as 'middle income' but are simultaneously grappling with significant social and economic inequality - dynamics that are described in more detail in Annex 8: Gender Assessment and Gender Action Plan (GAP). To the extent that households and businesses have thus far been able to invest in DERs, this has mostly been limited to those that are on the upper end of the socio-economic distribution. Unless concerted action is taken to address the specific barriers faced by lower-income and marginalized groups, this dynamic is unlikely to change. The IUS programme therefore responds to these challenges, in part by providing a more accessible and inclusive way for utility customers to access the financing needed to invest. The IUS model (and the way it will be deployed under the programme) is designed to overcome barriers to entry into the sustainable energy sector for marginalized groups, such as women, PWDs, the LGBTQIA+ community, and indigenous peoples. The Gender Action Plan (GAP) under this programme seeks to provide targeted interventions to improve governance, participation, and access to resources and information. This will enable these groups not only to benefit from the prioritized technologies but also to become key players and entrepreneurs in this space, fostering economic development and social empowerment. The programme will simultaneously provide training and capacity-building initiatives to: (i) ensure that energy sector stakeholders (including utilities) are more adept at identifying, mitigating, and managing environmental & social risks; and (ii) support the participating electric utilities to be inclusive and responsive to the needs of marginalized groups and to reinforce their policies for the successful rollout of the model.

D.5. Country ownership (max. 500 words, approximately 1 page)

Alignment with national priorities

D.5.1. As outlined in Section B.1, the proposed programme is closely aligned with the priorities identified in the NDCs submitted by Barbados, Belize, and Jamaica. All three countries assign considerable importance to reducing GHG emissions from the energy sector by scaling up investment in renewable energy and energy efficiency. The proposed programme will therefore make important contributions to achieving countries' NDC targets and priorities, as well as those stipulated in the policies, strategies and plans that underpin their NDCs. This includes, inter alia:

- a. **Barbados:** National Energy Policy (2019-30); National Climate Change Policy Framework.
- b. **Belize:** Horizon 2010-2030; National Energy Policy Framework; Sustainable Energy Strategy and Action Plan (2014-2033); National Climate Change Policy, Strategy and Action Plan.
- c. **Jamaica:** National Energy Policy (2009-2030); Climate Change Policy Framework; Vision 2030 Jamaica – National Development Plan.

D.5.2. National stakeholders have also been actively involved in efforts to get their countries 'ready' for IUS, including by working with GIZ and the CARICOM Secretariat to implement upstream (pre-investment) IUS institutional support activities over the preceding years. These same stakeholders – including the utilities, energy ministries, regulators, and National Designated Authorities (NDAs) – have been working closely with CDB and the programme preparation team to develop the programme proposal. This is indicative of their commitment to launch and implement IUS and contribute to programme implementation as outlined in Section B.4.

Capacity of executing entities to deliver

D.5.3. All three participating electric utilities (as National Executing Entities) are already experienced with DER initiatives. The National Executing Entities will build on/draw from this experience when executing their ‘projects’ (portfolios of sub-projects) using their blended GCF-CDB lines of credit, and will do so in a fair, transparent and equitable manner. With support from the TA component, the National Executing Entities will also: (a) facilitate all necessary related regulatory approvals; (b) change their billing systems as needed to enable on-bill financing; (c) adopt and implement a standardized and timebound application process; and (d) utilize and distribute programme-generated outreach materials, among other responsibilities. CDB and the programme preparation team have assessed the utilities’ policies and procedures to confirm that they have sufficient capacity to fulfil their Executing Entity roles (see Annex 2: Feasibility Study). These assessments also identified specific areas where additional capacity and support would be needed – all of which has informed the design of Component 1 of the programme.

The role of NDAs and other (regional) stakeholders

D.5.4. Similar to other ‘behind the meter’ transactions, the IUS programme involves a complex process with multiple actors. These include public entities such as government ministries and electricity regulators, as well as private entities like service providers and customers. However, the IUS programme facilitated by the CDB stands out due to its regional scope and the involvement of the GCF. This uniqueness necessitates the participation of various critical actors not typically found in standard ‘behind the meter’ transactions. For example, NDAs will play a pivotal role by ensuring the programme’s alignment with the local context, engaging stakeholders, and acting as liaisons between key parties, among other roles’. The NDAs have also been closely involved in the programme preparation process, often directly supporting data gathering and stakeholder consultations. All NDAs firmly support the final programme proposal, as evidenced by the No-Objection Letters that are included in Annex 1.

D.5.5. Moreover, the programme places a strong emphasis on engaging with regional stakeholders such as CCREEE and CARILEC, along with other national stakeholders including national energy agencies, business associations, and customers. Particular emphasis is placed on engaging underserved communities, as described in prior sections.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

D.6.1. The IUS programme is poised to significantly reduce greenhouse gas emissions in the three participating BMCs, with an estimated avoidance of **601,550 tCO₂e**. This results in an estimated total cost of about **USD 70 per tCO₂e avoided** for the overall programme (including GCF financing and CDB co-financing).⁴⁸ The cost-per-tonne for the GCF is lower, at approximately **USD 43 per tCO₂e avoided**⁴⁹ as outlined in Table 8 below. These are considered to be competitive cost-per-tonne rates in a SIDS context, where emissions reductions are often more costly than in larger countries.

Table 12: Mitigation impact estimates

Mitigation impact estimates		
Total GHG avoided	tCO ₂ e	601,550
Estimated cost per tCO ₂ e avoided	US\$ / tCO ₂ e	\$70
Estimated GCF cost per tCO ₂ e avoided	US\$ / tCO ₂ e	\$43

D.6.2. The programme will also deliver important adaptation benefits. As outlined above in Section D.1, approximately 10,655 people, including approximately 5,328 women and 5,327 men, are expected to directly benefit from enhanced resilience to the increasing risk of heat stress. The cost per direct beneficiary would therefore be about USD 3,938. Although this figure seems high when compared to adaptation-focused projects financed by GCF, it needs to be evaluated within the context of the broader programme. In particular, it is important to bear in mind that: (i) the programme is not strictly focused on climate change adaptation, and will also produce considerable mitigation benefits as outlined above; and (ii) the programme focuses not only on scaling up investment during its implementation period *but also* aims to catalyse (and equip national

⁴⁸ Estimated cost per tCO₂e avoided refers to the total programme cost (USD39.9 mil) divided by the total GHG emissions avoided.

⁴⁹ Estimated GCF cost per tCO₂e avoided refers to the total GCF cost (USD28.7 mil) divided by the total GHG emissions avoided.

stakeholders to lead) the sustainable energy transition over the medium/long term, during which the utilities will continue to deliver IUS services and thereby support many more customers to invest in technologies that enhance resilience and reduce GHG emissions. Moreover, the programme is expected to develop or strengthen awareness on technologies to increase resilience to heat stress for approximately 30,000 indirect beneficiaries

D.6.3. Another important metric is the level of co-financing mobilized under the programme. As outlined in Section C, the financial contribution from the GCF is expected to directly mobilize additional resources from the CDB. In particular, the GCF's USD 20.80 million contribution to the loan component of the programme is expected to be blended with approximately USD 16.2 million in financing from the CDB. Furthermore, the participating electric utilities/national Executing Entities are expected to mobilize an additional USD 25 million as in-kind contributions in the form of investments in smart meters/AMI. This constitutes a favourable co-financing ratio, particularly for this type of programme (wherein new and innovative business models are being established) in a SIDS context – all of which requires a higher overall level of concessionality than may be needed for other types of projects in other jurisdictions. In this context, it is also important to note that metrics such as the co-financing ratio do not fully capture the benefits of the IUS programme. As outlined above, the sub-projects to be financed during programme implementation are simply an initial round that will provide the utility with a strong track record while also providing a strong demonstration effect within the local markets. Considerable additional investments are expected to occur downstream from the programme, which is regarded as a 'bridge to bankability' that will equip the utilities to secure other (less concessional) sources of financing for follow-on rounds of IUS sub-projects after the programme implementation period has ended.

D.6.4. Other important performance and efficiency-related metrics are outlined below and further elaborated in Annex 3: Integrated Financial Model. These are the estimatesd (aggregate) portfolio-level results that may be achieved through the programme, which are based on reasonable assumptions about the number of different types of sub-projects to be financed by the electric utilities using their respective lines of credit.

Table 13: Estimated IUS programme performance

Metric	Notes/Explanation	Estimated IUS programme performance
Net financial benefit for customers (NPV)	This is the Net Present Value (NPV) of the total financial benefits that CDB estimates may be accrued by customers over the productive lifespan of the technologies in which they may invest through the Programme.	USD 263,024,251
Installed capacity of rooftop solar PV	This is the total amount of new generating capacity of the rooftop solar PV that CDB estimates may be installed in the three countries during Programme implementation.	5,622 kW
Amount of renewable energy generated	This is the total amount of electricity that CDB estimates may be generated from the newly installed solar PV (above) attributable to the Programme over the course of the productive lifespan of these technologies.	220,511 MWh
Electricity consumption avoided through EE measures	This is the total amount of electricity consumption that CDB estimates may be avoided over the productive lifespan of the efficient ACs that may be installed during Programme implementation.	1,166,259 MWh
Installed capacity of energy storage	This is the total storage capacity that CDB estimates may be installed in the three countries as a result of the support provided by the Programme.	1,083 kWh

D.6.5. Furthermore, as detailed in Section B.3 and further elaborated in Annex 2: Feasibility Study, the technologies that have been preapproved under the programme were prioritized because they fulfil the following important performance-related criteria: they provide direct mitigation and adaptation benefits; their capital cost is low enough to ensure financial feasibility; they are established technologies with a likelihood of local capacity; they are scalable and replicable; and they are relevant to the IUS programme, aligning with existing legal or regulatory frameworks.

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☒ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Currently, adoption of distributed clean energy and energy efficiency is hindered due to several barriers including high cost of financing, regulatory hurdles, limited capacity of the service providers, limited capacity of customers to fully understand technologies and their climate and cost benefits, among others. Stakeholders also lack	<u>Low</u>	Paradigm shift will involve a tangible increase in the availability and uptake of loans for customers, increase stakeholders' and service providers' capacity, standardize regulatory policy framework thereby developing a conducive market system that facilitates continued investment in the clean energy	The paradigm shift requires a change in the way the distributed clean energy market functions in the participating countries. There is a need to provide accessible, affordable, and inclusive financial products for eligible customers, develop regulatory policy framework, create customer awareness, strengthen service providers capacity, etc. To contribute to this, the programme will design capacity building tools and procedures and deliver awareness-raising training to the key stakeholders including customers, etc. The programme will also invite global experts to share their experience on behind the meter energy solutions, support development of regulatory policy documents, connect customers

	necessary tools, procedures, and capacities to integrate clean energy investment into their operations. Furthermore, access to finance is still difficult for many customers as local commercial lenders lack tools to assess customers creditworthiness of low-income and vulnerable demographics, among other socio-cultural reasons.		solutions beyond programme period.	with service providers, among others. Detailed outreach strategies will be developed to ensure customers can be made aware of both the technologies and the programme offerings. The programme will also conduct outreach sessions with local commercial banks to help them understand how they can facilitate programme scale-up through tailored financial products and services for the distributed energy sector. The programme efforts coupled with the strong demonstration effect that is expected as a result of strong sub-project performance under the programme (attributable in part to the direct support that will be provided for sub-project preparation), this is expected to increase interest in these technologies/practices both during and beyond the programme implementation period, and thus similarly stimulate demand.
Replicability	At present, distributed energy adoption in the three participating countries and the broader Caribbean region is still in nascent stages. Moreover, due to limited customer awareness, difficulty in accessing finance, high cost of finance combined with limited technical and regulatory capacity of the participating countries, there is significant scope of sub-project replicability as a direct result of programme intervention. The programme will enable robust sectoral development and establish a strong foundation for replication.	<u>Low</u>	If adequate customer awareness, strengthened regulatory policy environment, institutional, and service providers' capacity could be achieved as envisaged, the programme can help strengthen the overall market ecosystem and successfully deploy clean energy sub-projects in the participating counties. Furthermore, with the lessons learnt from this programme, the success story could then be replicated seamlessly in other Caribbean countries as they are facing similar climate change and energy	The programme will support the participating countries, particularly the regulator and the utility, to develop, market and deploy distributed renewable energy and energy efficiency solutions. At the same time, the programme also aims to establish a robust regulatory policy framework that governs behind the meter distributed energy sector for the long-term. Furthermore, the programme has also envisaged technical skill enhancement of the local TSPs, strengthen the existing institutional landscape through trainings/workshops on E&S, GESI, M&R, etc. that enables long-term sectoral development. The tools and procedures developed as part of the programme can be leveraged to deploy similar programmes and/or technologies beyond the programme period. Adequate local and regional stakeholder outreach has been planned to share lessons learnt, global and local best practices, potential hurdles and remedies during programme implementation which will demonstrate a successful model for scaling up fit-for-purpose financing for climate-related investments, while also helping to establish strong branding for such financial products. This in turn will provide a strong foundation for replication in multiple ways:

			challenge. The programme's approach and financial solutions can support more widespread adoption of climate change mitigation and adaptation technologies/ practices.	- Customers outside the programme scope – many customer groups such as industrial, large commercial and institutional customers are outside the ambit of the current programme. Although these customers do not necessarily face financial challenges to adopt or implement behind the meter energy solutions, however may face issues or administrative delays due to the lack of a robust regulatory policy framework for the sector. The programme addresses this barrier by developing a long-term market ecosystem through the planned TA support. This will increase the likelihood of large customers increasing their investments in distributed clean energy solutions due to regulatory policy certainty. - Local commercial banks currently outside the programme scope – as elaborated in the barrier assessment, accessing low-cost finance and availability of financial products to service the behind the meter distributed energy solutions is limited. Among other reasons, perceived technology risks, assessing customer's creditworthiness, etc. may be attributed for the current state. The programme has envisaged outreach sessions and material for the local commercial banks to help them understand the IUS model and its benefits, technology awareness, etc. This understanding of the model, technology, and risk will increase the likelihood that local banks will similarly integrate considerations into their lending operations. - CDB can easily replicate the programme in other Caribbean countries, drawing on the developed elements, including the established programme brand, as well as the programme approach, tools and capacity-building materials.
Sustainability	Currently, the regulatory policy framework, availability of affordable finance, streamlined application procedures, among	<u>Low</u>	Paradigm shift involves establishing long-term solutions such as digital tools, transparent and accountable	The programme aims to build a robust market ecosystem in the participating countries to ensure distributed clean energy capacity addition using a utility centric and led business model. This ensures that a streamlined capacity addition could be planned through a programme that not only promotes various

	others is limited in the participating countries. Furthermore, there is a need for institutional strengthening on cross-cutting areas such as E&S, GESI, among others.		processes, developing robust regulatory policy frameworks, strengthening institutional capacity, and ensuring adequate and accessible low-cost financial products for the distributed clean energy solutions.	clean energy technologies but also provides the customers with necessary capacity building and tools for decision making. The programme also envisages to lower TSPs customer acquisition costs, increase market competition, as well as quality of service and customers' experience. In addition, the programme also aims to ensure that end beneficiaries in partner countries will have positive experiences with their programme-supported sub-projects (i.e. achievement of the expected energy cost savings), in part because of the support provided for sub-project preparation. This in turn will contribute to ensuring that the programme-supported technologies/practices are sustainably utilized beyond programme closure, and that the interest in such technologies/practices among other local private sector stakeholders is similarly sustained.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁵⁰	
<u>All mitigation results areas</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Utilities, with the help of National Energy Consultant, will collect, aggregate and report sub-project information as per the programme	0	Average CO2 saving (tonnes/year): 24,062	Average CO2 saving (tonnes/year): 24,062	Quantity of GHG emissions associated with a unit (MWh) of electricity generated in the respective

⁵⁰ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective. Total lifespan period (here and below) – is defined as 25 years for solar PV (lifespan of technology), 20 years for solar water heaters, 15 years for efficient ACs, and about 5 years for efficient lighting.

		monitoring & reporting requirements (either digitally or in an excel file), drawing on the following (non-exhaustive) list of methods which will vary depending on the sub-project type: • Equipment/technology data sheets • Equipment/technology purchase/service contracts • Project technical assessment reports (in case of complex projects involving more than one technology) • Energy audit reports (were necessary) • On-site visits if necessary Information gathered through the above-listed methods will be independently verified through baseline, mid-term and endline surveys		CO2 saving (tonnes) – cumulative: 72,186 total, including: Barbados (25,826) Belize (5,416) Jamaica (40,944) Mid-term period (here and below) is defined as the end of the 3rd year after initiation of the programme Targets for GHG emission reduction are presented per year, as well as cumulative at the end of the period.	CO2 saving (tonnes – cumulative: 144,372 total, including: Barbados (51,653) Belize (10,832) Jamaica (81,887) CO2 saving (tonnes – cumulative during total lifespan of technologies): 601,550 Final period (here and below) is defined as the end of the 6th year (end of programme implementation – period during which all funds shall be allocated) Targets for GHG emission	IUS country and provided to customers via the grid. Ex-ante mitigation estimation: For the purpose of the IUS programme, the GEF for each country has been modelled across the years of the IUS implementation. The modelling was done to account for the possible increase share of renewable energy in the national grid of the respective country over the years. Hence, the value of the GEF per country and per year was used in computing the ex-ante. Ex-post mitigation estimation: For ex-post emissions estimation, current and updated GEF values will be employed. Source of data: • UNFCCC IFI values and IUS Financial Model (modelled values for ex-ante) • Utilities (for ex-post)
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					reduction are presented per year, as well as cumulative at the end of the period.	Latest UNFCCC IFI values published GEF values (for ex-post); currently Harmonized IFI Default Grid Factors 2021 v3.2 ⁵¹
<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Utilities, with the help of National Energy Consultant, will collect, aggregate and report sub-project information as per the programme monitoring & reporting requirements (either digitally or in an excel file), drawing on the following (non-exhaustive) list of methods which will vary depending on the sub-project type: • Equipment/technology data sheets • Equipment/technology purchase/service contracts • Project technical assessment reports (in case of complex projects involving more than one technology)	0	Average CO2 saving (tonnes/year): 3,975 CO2 saving (tonnes) – cumulative: 11,925 total, including: Barbados (3,010) Belize (1,603) Jamaica (7,311) Mid-term period (here and below) is defined as the end of the 3rd year after initiation of the programme	Average CO2 saving (tonnes/year): 3,975 CO2 saving (tonnes – cumulative): 23,849 total, including: Barbados (6,020) Belize (3,207) Jamaica (14,622) CO2 saving (tonnes – cumulative during total lifespan of technologies): 99,373	Quantity of GHG emissions associated with a unit (MWh) of electricity generated in the respective IUS country and provided to customers via the grid. Ex-ante mitigation estimation: For the purpose of the IUS programme, the GEF for each country has been modelled across the years of the IUS implementation. The modelling was done to account for the possible increase share of renewable energy in the national grid of the respective country over the years. Hence, the value of the GEF per country and per year was used in computing the ex-ante. Ex-post mitigation estimation: For ex-post

⁵¹ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

		• Energy audit reports (were necessary) • On-site visits if necessary Information gathered through the above-listed methods will be independently verified through baseline, mid-term and endline surveys		Targets for GHG emission reduction are presented per year, as well as cumulative at the end of the period.	Final period (here and below) is defined as the end of the 6th year (end of programme implementation – period during which all funds shall be allocated) Targets for GHG emission reduction are presented per year, as well as cumulative at the end of the period.	emissions estimation, current and updated GEF values will be employed. Source of data: • UNFCCC IFI values and IUS Financial Model (modelled values for ex-ante) • Utilities (for ex-post) • Latest UNFCCC IFI values published GEF values (for ex-post); currently Harmonized IFI Default Grid Factors 2021 v3.2⁵²
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.2: Installed energy storage capacity</u>	At the time of commissioning of the installed solar PV (with battery storage) by the utility, the installed capacity of the system will be recorded – this will be an ongoing process within the life of the IUS programme. Each customer will be issued a unique ID/Code against which the	0	Total estimated installed storage capacity: 0.54 MWh total, including: Barbados (0.068 MWh) Belize (0.09 MWh)	Total estimated installed storage capacity: 1 MWh total, including: Barbados (0.137 MWh) Belize (0.181 MWh)	The estimated energy storage capacity is based on the estimated numbers of subprojects or customers that can benefit from the financing of the IUS programme. These numbers are calculated using the estimated country-level financing allocation, project level financing allocation, estimated average unit

⁵² <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

		installed capacity will be recorded in an Excel spreadsheet.		Jamaica (0.382 MWh)	Jamaica (0.765 MWh)	costs, and assumptions on the size of subprojects.
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.3: Installed renewable energy capacity</u>	At the time of commissioning of the installed solar PV by the utility, the installed capacity of the PV system will be recorded – this will be an ongoing process within the life of the IUS programme. Each customer will be issued a unique ID/Code against which the installed capacity will be recorded in an Excel spreadsheet.	0	Total estimated installed capacity: 2.2 MW total, including: Barbados (0.5 MW) Belize (0.5 MW) Jamaica (1.2 MW)	Total estimated installed capacity: 5.6 MW, including: Barbados (1.3 MW) Belize (1.3 MW) Jamaica (3 MW)	The estimated capacity is based on the estimated numbers of subprojects or customers that can benefit from the financing of the IUS programme. These numbers are calculated using the estimated country-level financing allocation, project level financing allocation, estimated average unit costs, and assumptions on the size of subprojects.
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.4: Renewable energy generated</u>	Meter readings from the solar PV sub-projects, will be obtained at the time of normal meter reading as per the existing billing cycle and approach of the utilities for meter readings. The National Energy Consultant embedded in the utility of each country will be responsible for compiling the renewable energy data obtained by the utility into an Excel spreadsheet in a user-friendly manner. The	0	Total estimated renewable energy generated: 26,461 MWh total, including: Barbados (6,203 MWh) Belize (6,118 MWh) Jamaica (14,139 MWh)	Total estimated renewable energy generated: 52,922 MWh total, including: Barbados (12,406 MWh) Belize (12,237 MWh) Jamaica (28,278 MWh)	The estimated renewable energy generation is based on the estimated numbers of subprojects or customers that can benefit from the financing of the IUS programme. These numbers are calculated using the estimated country-level financing allocation, project level financing allocation, estimated average unit costs, and assumptions on the size of subprojects.

		consultant will transmit the data to the CDB annually or at any other frequency to be agreed upon.			Cumulative estimated renewable energy generated (over total lifespan of technologies): 220,511 MWh total, including: Barbados (51,969 MWh) Belize (50,990 MWh) Jamaica (117,826 MWh)	
<u>MRA3 Buildings, cities, industries and appliances</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Utilities, with the help of National Energy Consultant, will collect, aggregate and report sub-project information as per the programme monitoring & reporting requirements (either digitally or in an excel file), drawing on the following (non-exhaustive) list of methods which will vary depending on the sub-project type: Equipment/technology data sheets	0	Average CO2 saving (tonnes/year): 20,087 CO2 saving (tonnes) – cumulative: 60,261 total, including: Barbados (22,816) Belize (3,813) Jamaica (33,632)	Average CO2 saving (tonnes/year): 20,087 CO2 saving (tonnes – cumulative): 120,523 total, including: Barbados (45,632) Belize (7,625) Jamaica (67,265)	Quantity of GHG emissions associated with a unit (MWh) of electricity generated in the respective IUS country and provided to customers via the grid. Ex-ante mitigation estimation: For the purpose of the IUS programme, the GEF for each country has been modelled across the years of the IUS implementation. The modelling was done to account for the possible increase share of renewable energy in the

		• Equipment/technology purchase/service contracts • Project technical assessment reports (in case of complex projects involving more than one technology) • Energy audit reports (were necessary) • On-site visits if necessary Information gathered through the above-listed methods will be independently verified through baseline, mid-term and endline surveys		Mid-term period (here and below) is defined as the end of the 3rd year after initiation of the programme Targets for GHG emission reduction are presented per year, as well as cumulative at the end of the period.	CO2 saving (tonnes – cumulative during total lifespan of technologies): 502,178 Final period (here and below) is defined as the end of the 6th year (end of programme implementation – period during which all funds shall be allocated) Targets for GHG emission reduction are presented per year, as well as cumulative at the end of the period.	national grid of the respective country over the years. Hence, the value of the GEF per country and per year was used in computing the ex-ante. Ex-post mitigation estimation: For ex-post emissions estimation, current and updated GEF values will be employed. Source of data: • UNFCCC IFI values and IUS Financial Model (modelled values for ex-ante) • Utilities (for ex-post) Latest UNFCCC IFI values published GEF values (for ex-post); currently Harmonized IFI Default Grid Factors 2021 v3.2⁵³
<u>MRA3 Buildings, cities, industries and appliances</u>	<u>Supplementary 1.1: Annual energy savings</u>	Meter readings will be obtained as per the existing billing cycle and	0	Total estimated energy	Total estimated energy	The meter readings will be obtained by the utility in each country as per their

⁵³ <https://unfccc.int/climate-action/sectoral-engagement/ifi-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies>

		approach of the utilities based in meter readings. The difference of a customer's meter reading for a current billing cycle and that of the previous cycle represents the change in the quantity of units (kWh) consumed. The National Energy Consultant embedded in the utility of each country will be responsible for compiling the data obtained by the utility into an Excel spreadsheet in a user-friendly manner. The consultant will transmit the data to the CDB annually or at any other frequency to be agreed upon.		consumption avoided from energy efficiency services and appliances: 139,951 MWh total, including: Barbados (49,322 MWh) Belize (19,111 MWh) Jamaica (71,517 MWh)	consumption avoided from energy efficiency services and appliances: 279,902 MWh total, including: Barbados (98,644 MWh) Belize (38,223 MWh) Jamaica (143,034 MWh) Cumulative estimated energy consumption avoided from energy services and appliances (over total lifespan of services): 1,166,259 MWh total, including: Barbados (411,019 MWh) Belize (159,263 MWh)	respective billing cycle. To estimate the quantity of avoided emissions, the measured energy consumption will be compared with the baseline consumption prior to the implementation of the sub-project – this will be determined using three years historical data for the customer/premises, if available or by using historical data for similar premises. Annual energy savings estimates only includes the expected energy consumption avoided from energy efficiency initiatives under the IUS programme.
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					Jamaica (595,976 MWh)	
<u>ARA3 Infrastructure and built environment</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	The following sources of information and methods can be used (but not limited to) for collection of data to measure the progress against the targets: • Loan files and technical support documentation. To ensure timely and accurate reporting, required data obtained via above-mentioned should be stored in dedicated reporting tool.	0	Total estimated direct beneficiaries: 3,729 (of whom at least 50% women), including: Barbados (1,237) Belize (475) Jamaica (2,016) Total estimated indirect beneficiaries: 14,921 (of whom at least 50% women)	Total estimated direct beneficiaries: 10,655 (of whom at least 50% women), including: Barbados (3,537) Belize (1,357) Jamaica (5,761) Total estimated indirect beneficiaries: 29,752 (of whom at least 50% women)	Direct beneficiaries = total number of individuals in residential settings benefiting from AC sub-projects combined with the number of AC sub-projects for commercial customers. Average household sizes are: 2.85 in Barbados; 3.75 in Belize; and 3.18 in Jamaica. Indirect beneficiaries = Number of beneficiaries attending workshops, conferences, training sessions, accessing IUS helpdesk, availing low-cost energy audits, etc. Detailed breakdown included in Section E.5.
<u>ARA3 Infrastructure and built environment</u>	<u>Supplementary 2.6: Beneficiaries (female/male) living in buildings that have increased resilience against climate hazards</u>	The following sources of information and methods can be used (but not limited to) for collection of data to measure the progress against the targets:	0	Total estimated beneficiaries: 3,266, including • <u>Barbados</u> Total: 1,117; Males: 558;	Total estimated beneficiaries: 9,444, including • <u>Barbados</u> Total: 3,101; Males: 1,550;	Beneficiaries = total number of individuals in residential settings expected to benefit from efficient AC sub-projects split across females and males (i.e. number of residential beneficiary customers x average

		• Loan files and technical support documentation. To ensure timely and accurate reporting, required data obtained via above-mentioned should be stored in dedicated reporting tool.		Females: 559 • <u>Belize</u> Total: 302; Males: 151; Females: 151 • <u>Jamaica</u> Total: 1,847; Males: 923; Females: 924	Females: 1551, • <u>Belize</u> Total: 1,226; Males: 613; Females: 613 • <u>Jamaica</u> Total: 5,117; Males: 2,558; Females: 2,559	household size). Average household sizes are: 2.85 in Barbados; 3.75 in Belize; and 3.18 in Jamaica.
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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	The IUS model is a novel model to accelerate distributed clean energy adoption in the participating countries by anchoring the programme delivery at the utility. Consequently, the institutional and regulatory landscape in the participating countries needs to be updated to cater to the	<u>medium</u>	Key stakeholder's i.e., the ministry, regulatory authority, and the utility will have strengthened their capacity to implement the IUS model and by extension serve the distributed clean energy market in the participating countries. Key stakeholders will have also developed additional capacity to	The target scenario will be achieved by: • Addressing regulatory barriers impeding the implementation of the IUS model (Activity 1.1.1) • Addressing broader policy framework needs to strengthen	<u>Multi-countries</u>

	programme needs as well as facilitate long term market development. All the participating countries are making positive institutional and regulatory improvements, there are a handful of outstanding issues that needs to be addressed to achieve complete legal, regulatory and institutional preparedness for seamless execution of the IUS programme. For instance, enhancing regulators capacity to wholistically understand IUS transactions its impact on the power sector landscape, process IUS related regulatory applications, establishing regulatory jurisdiction, etc. Furthermore, there is also the need to update and/or develop regulatory framework such as distributed energy regulations, feed-in-tariffs, etc.		address cross-cutting areas of programme delivery i.e., gender and social inclusion, environmental and climate change, among others. The regulatory framework for distributed renewable energy and energy efficiency will be strengthen in the participating countries. This will directly impact the sector's growth as a robust long-term market ecosystem is developed.	prospects and incentives for investment in distributed renewable energy, energy efficiency, and DERs (Activity 1.1.2) Enhance Executing Entities' and other stakeholders' capacity to implement the IUS model (Activity 1.3.2) Thorough the above-mentioned activities, the IUS programme aims to address a key barrier i.e., streamline the legal and regulatory environment in the participating BMCs. Additionally, the TA activities planned under these activities will not only help facilitate a robust market ecosystem development to fast-track distributed clean energy	
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				deployment under the IUS programme but also ensure a facilitative institutional, legal and regulatory environment for distributed clean energy post the programme period.	
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Distributed renewable energy and energy efficiency uptake in the participating countries is increasing; however, lack of standardized processes and framework, inequitable access to low-cost finance, limited capacity of the stakeholders, limited customer awareness, and high customer acquisition cost for the service providers etc. are some challenges that have resulted in a relatively slow technology deployment.	<u>medium</u>	All the participating BMCs develop the necessary framework(s) to ensure a robust market development, potential customers become aware about benefits of available technologies, customer acquisition cost for the TSPs is reduced, and there is a steady increase in the rate of adoption for behind the meter clean energy solutions.	The target scenario will be achieved by: • Raising awareness of renewable energy and energy efficiency opportunities and potential savings among customers (Activity 1.2.1) • Promoting the IUS model and associated services (Activity 1.2.2) • Providing customers with hands-on support to identify and develop sub-projects (Activity 1.2.3.)	<u>Multi-countries</u>

				• Enhance Executing Entities' and other stakeholders' capacity to implement the IUS model (Activity 1.3.2) • Training for qualified TSPs (Activity 1.3.3) • Prepare, finance and implement sub-projects (Activity 2.1.1) Through the above-mentioned activities, the IUS programme aims to sensitize customers on the benefits of distributed clean energy solutions and increase demand. The programme also ensures adequate support to the customers to avail programme benefits, improve technical competence of the service providers.	
<u>Core indicator 7: Degree to which GCF</u>	Key institutions and stakeholders in the	<u>low</u>	Key institutions and stakeholders in the	The target scenario will be achieved by:	<u>Multi-countries</u>

<u>Investments contribute to market development/transformation at the sectoral, local, or national level</u>	participating countries are aware of the climate risks as well as understand the opportunities available for low-carbon technologies; however, there is a lack of inclusivity and understanding of cross-cutting areas such as gender and social inclusion, etc. Furthermore, there are several pending issues hampering the market development for distributed renewable energy and energy efficiency solutions in the participating countries, such as (a) lack of accountable and transparent processes and procedures, (b) unavailability of affordable and inclusive financing options, (c) limited technical capacity of the service		participating countries are more aware on improve business operations and deliver programme services to even the most vulnerable sections of the society through a robust market. The behind the meter energy distributed energy market in the country is developed through adoption of accountable and transparent approval and administrative processes, locally developed affordable financing options for long term sectoral sustainability, adequate customer outreach to ensure technology and programme benefits are well understood, capacity building training of service providers, and capacity building workshops for key stakeholders i.e., energy ministry, regulator, etc. on cross-cutting areas.	• Addressing regulatory barriers impeding the implementation of the IUS model (Activity 1.1.1) • Addressing broader policy framework needs to strengthen prospects and incentives for investment in distributed renewable energy, energy efficiency, and DERs (Activity 1.1.2) • Raising awareness of renewable energy and energy efficiency opportunities and potential savings among customers (Activity 1.2.1) • Promoting the IUS model and associated	
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	providers, (d) limited customers' awareness on cost and climate benefits of the proposed energy solutions,			services (Activity 1.2.2) • Providing customers with hands-on support to identify and develop sub-projects (Activity 1.2.3.) • Enhance Executing Entities' and other stakeholders' capacity to implement the IUS model (Activity 1.3.2) • Training for qualified TSPs (Activity 1.3.3) The IUS programme is designed to address both the market needs by creating a conducive business environment and provide low-cost finance to install technologies and services. The above-mentioned activities directly provide the much-needed market	
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				support through the TA component and address barriers and root causes that have hindered sector's growth in the country.	
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	The IUS model is a relatively newer model in the Caribbean market. Globally, similar, or slightly modified versions of the proposed model have been implemented in many jurisdictions. Therefore, the participating countries are relatively well versed with the renewable energy technologies, energy efficiency services and their climate benefits, there is still a need for knowledge sharing, experiential learning, understand and implement global best	<u>low</u>	Stakeholders from the participating countries will engages with international experts to understand issues, challenges, and remedies for smooth implementation of the programme. This experiential learning from experts of more developed markets will enhance stakeholders' knowledge around good practices, etc. Participating stakeholders will engage with regional stakeholders form non-participating Caribbean nations to share their experience, lessons learnt, implementation issues and remedial actions, etc. This will ensure long-term	The target scenario will be achieved by: - Facilitating knowledge exchange among targeted BMCs (Activity 1.3.1) - Supporting information exchange with non-participating Caribbean countries, including governments, regulators, utilities, etc. (Activity 1.3.4.) - Creating a Community of Practice within the CDB borrowing member countries. (Activity 1.3.5.)	<u>Multi-countries</u>

	practices, methodologies, and standards, for smooth implementation of the IUS programme.		regional programme sustainability. The knowledge products created under the IUS programme, such as regulatory frameworks, digital application tools, administrative processes, etc. could be leveraged by non-participating countries to develop their local distributed clean energy markets.	The IUS programme design ensures that the programme lessons and learnings are shared among regional Caribbean members. This will help other Caribbean nations to leverage the IUS programme, adopt good practices, and develop their own programme to accelerate clean energy transition.	
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E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the “**Project/programme co-benefit indicators**” section in table below. Add rows as needed.*

Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 1.1: Strengthened regulatory and institutional frameworks	<i>Number of legal and/or regulatory frameworks developed or updated, and notified</i>	Regular progress report(s) by CDB TA Execution Team.	0	3	6	At this stage it is difficult to envisage if there is a need for any regulatory approval or amendment for expedited implementation of the IUS programme in participating countries.

facilitating the deployment of distributed RE, EE and other DER technologies.		Progress report(s) by TA Implementation Firm Copy of the framework(s) developed, updated and notified				Participating countries have enabling regulatory framework(s) to a varying degree. For instance, Barbados and Jamaica have been publishing renewable energy FIT, on the other hand Belize is still finalizing them. Based on the feasibility assessment and stakeholder consultations, provision to develop the most relevant regulatory framework specific to the country's requirement has been provisioned that help strengthens enabling environment for the IUS.
	<i>Number of updated and streamlined administrative/ application procedures published</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copy of the procedures published	0	3	3	Currently, some participating countries have put in place an administrative process that is followed by interested customers to avail distributed energy services. These administrative processes need to be updated to include both renewable energies as well energy efficiency solutions as envisaged under the IUS programme.
	<i>Number of TSP technical qualification guidelines published</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA	0	3	3	One qualification guideline document will be prepared for each participating country during PY-1 and PY-2

		Implementation Firm				
		Copy of the technical guidelines published				
	<i>Number of staff members of the Ministry and the Regulator trained to address cross-cutting issues such as GESI, E&S, etc.</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copy of the training/workshop notes Copy of the attendance sheet	0	30	60	It is assumed that each participating country will nominate at least 10 staff members (5 each from Ministry and the Regulator) per training/ workshop.
Output 1.2: Enhanced capacity of customers to invest in distributed RE, EE and other DERs through the IUS Model.	<i>Number of customers attending the renewable energy and energy efficiency awareness sessions</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copy of the workshop notes	0	150 (Males: 75; Females: 75)	300 (Males 150; Females: 150)	It is assumed that about 50 customers attend each awareness session per country. Two session will be organized per country, and thus 6 sessions in total.

		Copy of the attendance sheet				
	<i>Number of low-cost energy audits completed.</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Energy audit report by the TSP	0	180	300	A total of 300 low-cost energy audits (pre-feasibility) are planned for the most vulnerable/low-income residential customers at subsidised rates, i.e., 100 customers per participating country. One customer eligibility criterion per country will also be developed during PY 1.
	<i>Number of customers helped through the strengthened utility helpdesk</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Number of customers helped by the helpdesk (either through phone call, email, online chat, etc.)	0	2,250	4,500	It is assumed that a total of 250 customers per country per PY, i.e., 750 customers per PY across the participating BMCs, reach out to the customer helpdesk seeking clarification on the IUS programme.
	<i>Number of downloads (knowledge products, audience</i>	Regular progress report(s) by CDB TA Execution Team.	0	2,250	4,500	It is assumed that a total of 250 customers per country per PY, i.e., 750 customers per PY across the participating BMCs, download some programme related

	<i>reached by the communication strategies, etc.)</i>	Progress report(s) by TA Implementation Firm				information from the programme's website, etc.
	<i>Number of IUS programme website visitors</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copy of the website visit log	0	9,000	18,000	It is assumed that a total of 1,000 customers per country per PY, i.e., 3,000 customers per PY across the participating BMCs visit the programme's website to understand more about the programme.
Output 1.3: Increased technical capacity of utilities, TSPs and other stakeholders to deploy and maintain distributed RE, EE and other DER technologies.	<i>Number of participants attending the knowledge sharing sessions among participating countries.</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Session reports and material used Copy of the participation or attendance list.	0	45	90	A total of two virtual knowledge exchange sessions are planned over the programme implementation period. Knowledge exchange material, primarily presentations and notes, will also be developed and shared among stakeholders prior to the session. Local Regulatory Consultant (part of the TA Implementation Firm) and National Energy Consultant will provide inputs on the lessons learnt by the regulator and the utility respectively during programme implementation. It is assumed that at least 5 stakeholders per country (15 total) will participate in each knowledge sharing session.

	<i>Number of capacity building strategies for business development that are being utilised.</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copy of the strategy	0	3	3	One business development strategy and action plan will be developed during PY 1 per utility.
	<i>Number of digital application platforms that are operational</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Weblink of the digital platform	0	3	3	One digital application submission and approval platform will be developed in PY 1 as per the approved administrative process.
	<i>Number of Executing Entity's staff trained on cross cutting areas</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Training report	0	30	30	A total of at least six capacity building trainings, i.e., 2 per participating country, are planned for the Executing Entity on cross-cutting topics like environmental and social management, gender and social inclusion. It is expected that each Executing Entity nominates at least 5 staff members per training session, i.e., a total of 10 staff members per BMC.

	<i>Number of TSP training</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Training report	0	60	60	At least three TSP-focussed trainings (1 per country) are planned in PY 1, with refresher trainings in PY2 and PY4. These trainings will cover topics such as IUS programme design and implementation; technical session on design, standards, etc.; legal and regulatory session on key insights of the local laws and framework; E&S risk and mitigation; GESI, M&R, etc. It is estimated that around 20 TSPs will attend a training session per country, i.e., a total of at least 60 TSPs across the three participating countries. These same TSPs are then expected to attend the refresher trainings in PY2 and PY4.
	<i>Number of participants attending the regional knowledge exchange sessions with non-participating countries</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Session report	0	80	160	A total of four virtual regional exchange sessions to be organized to promote programme benefits, learnings, and experiences. There are 20 member states and associate members of the CARICOM. It is assumed that at least 2 participants attend the regional session per CARICOM member, i.e., about 40 participants per session.
	<i>Number of regional in-person knowledge sharing summits/ side events</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA	0	1	1	CDB to explore the possibility of organizing one regional in-person knowledge sharing summit or side event on IUS programme at either CDB's or CARICOM's or GCF's energy event in the Caribbean.

		Implementation Firm				
		Session report and copy of the content/ material developed for the session				
Outcome 3 (intermediate): Increased access to financing for clean energy investments, allowing utility customers to adopt RE, EE and other DER technologies at scale.	<i>Amount (in USD) of financing approved for IUS sub-projects.</i>	Progress report(s) by TA Implementation Support Firm Copies of sub-project application approvals Electric utility reports, including loan files	0	Total: USD14.8 million, including: - Barbados (USD4.67 million) - Belize (USD3.87 million) - Jamaica (USD6.26 million)	Total: USD37 million, including: - Barbados (USD11.67 million) - Belize (USD9.67 million) - Jamaica (USD15.66 million)	These estimates assume full and timely draw-down of the three lines of credit, which is contingent on there being sufficient demand for the financing offered through the programme.
Output 2.1: Installation of distributed RE, EE and other DER technologies among a critical mass of utility customers, including households, businesses and public sector entities.	<i>Number of new units of eligible technologies installed via the implementation of IUS sub-projects financed/executed by utilities.</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copies of sub-project application approval	0	- Barbados: 3,866 - Belize: 2,848 - Jamaica: 5,431	- Barbados: 10,714 - Belize: 7,892 - Jamaica: 15,052	Estimated number of new units is based upon the unit costs and total expected CAPEX per type of technology in each country. The estimate includes all types of customers and all types of preapproved technologies.

	<i>Number of additional utility customers benefiting from smart meters/AMI investments during the Programme implementation period</i>	Electric utility reports	0	15,000	50,000	Estimate is based on the total number of existing meters/connections across all three countries to be replaced with a smart meter/AMI.
Project/programme co-benefit indicators						
Co-benefit 1: Socio-economic benefits	<i>Average annual cost reductions (nominal) per beneficiary due to energy savings</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Sub-project reporting	0	Residential customers: USD500 per year Commercial customers: USD4,000 per year	Residential customers: USD550 per year Commercial customers: USD4,400 per year	Average (nominal) annual cost reductions are expected to gradually increase over time in line with inflation.
Co-benefit 2: Socio-economic benefits	<i>Number of female beneficiaries of the IUS Programme</i>	Regular progress report(s) by CDB TA Execution Team. Progress report(s) by TA Implementation Firm Copy of sub-project	0	Total estimated beneficiaries: Residential - Barbados: 4,868 - Belize: 4,607 - Jamaica: 8,109	Total estimated beneficiaries: Residential - Barbados: 7,969 - Belize: 7,354 - Jamaica: 13,123	Estimated number of female beneficiaries in the residential customer segment availing IUS programme sub-project benefits either as the sub-project applicant or as a part of an IUS sub-project beneficiary household.

		application approval				
E.6. Project/programme activities and deliverables						
<i>All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.</i> <i>Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.</i>						
Activities	Description	Sub-activities	Deliverables			
Activity 1.1.1: Immediately address regulatory barriers impeding the implementation of the IUS model under Component 2	Immediately address regulatory barriers impeding the implementation of the IUS model under Component 2.	Sub-activity 1.1.1.1 Prepare a legal and regulatory opinion on existing electricity laws/frameworks for each utility/BMC (i.e. utility as financier). Sub-activity 1.1.1.2 Assist national stakeholders in each BMC to prepare and secure approval of legal and regulatory amendments if/as needed. Sub-activity 1.1.1.3 Assist national stakeholders in each BMC to strengthen at least 1 aspect of the regulatory framework (e.g. RE/FIT regulations or rules, grid codes, etc.). Sub-activity 1.1.1.4 Update/draft administrative, energy accounting and settlement process. Sub-activity 1.1.1.5 Support and capacity building of the	D. 1.1.1.1. Note on legal and regulatory opinion on existing electricity laws/frameworks from an IUS roll-out perspective - 3 Notes (1 for each BMC) D. 1.1.1.2 Legal and/or Regulatory amendments (expected - 3 amendments, one in each BMC) D. 1.1.1.3 RE/EE and/or FIT regulations or rules (3 regulations - 1 in each BMC) D. 1.1.1.4 Administrative/application process (1 for each BMC - the administrative process will include energy accounting and settlement process) D. 1.1.1.5 Written inputs/ feedback to the IUS related applications/filings. D.1.1.1.6 Regular progress reports submitted to the CDB. D. 1.1.1.7 Field mission/ stakeholder consultation report(s)			

		regulatory authorities to review and approve IUS related regulatory applications/filings. Sub-activity 1.1.1.6 Execute the TA, including procurement, contract management, supervision, quality assurance/enhancement, direct reporting and invoice/payment processing. Sub-activity 1.1.1.7 Stakeholder consultations, in-person meetings, etc.	
Activity 1.1.2: Address broader policy framework needs to strengthen prospects and incentives for investment in distributed RE, EE and other DERs	Address broader policy framework needs to strengthen prospects and incentives for investment in distributed RE, EE and other DERs.	Sub-activity 1.1.2.1 Assist national stakeholders (notably in the Ministry responsible for energy, but also the regulatory authority and other stakeholders if/as needed) to strengthen at least one aspect of the national policy and incentive framework to better support investment in priority technologies and (by extension) IUS programme implementation. Sub-activity 1.1.2.2 Execute the TA, including procurement, contract management, supervision, quality assurance/enhancement,	D. 1.1.2.1 Consultant report summarizing updates/changes in broader distributed/behind the meter or associated energy policy (both generic and gender compliant update) and/or design of additional incentive framework as per local needs to facilitate IUS implementation (1 for each BMC) D. 1.1.2.2 Regular progress reports submitted to the CDB. D. 1.1.2.3 TSP technical eligibility/qualification criteria: 3 sets of qualification criteria (1 per BMC)

		direct reporting and invoice/payment processing. Sub-activity 1.1.2.3 Develop TSPs technical qualification guidelines.	
Activity 1.2.1: Raise awareness of RE/EE opportunities and potential savings	Raise awareness of RE/EE opportunities and potential savings.	Sub-activity 1.2.1.1 Develop technology specific marketing material for programme sustainability Sub-activity 1.2.1.2 Develop IUS programme's marketing material for programme sustainability. Sub-activity 1.2.1.3 Develop a comprehensive and inclusive marketing, communications and outreach strategy for the Ministry focussed on the priority technologies. Sub-activity 1.2.1.4 Develop material/content for customer awareness/ outreach sessions on IUS programme. Sub-activity 1.2.1.5 Outreach sessions to create customer awareness of RE and EE opportunities and potential savings.	D. 1.2.1.1 Marketing material like technology specific infographics, pamphlets, social media posts, etc. to promote the IUS technologies (at least 2 infographics per technology/ service per BMC) D. 1.2.1.2 IUS programme marketing/promotional/outreach material like infographics, social media posts, multimedia material, etc. (at least 3 infographics/ multimedia material per BMC showcasing the programme benefits, high level programme design, etc.) D. 1.2.1.3 One (1) strategy per BMC - the strategy must also cater to vulnerable groups [such as blind, deaf or speak different local languages such as the indigenous populations] D. 1.2.1.4 Materials for outreach sessions including presentations, including awareness campaigns for adaptation to heat stress, FAQs, etc. (multiple material as per the session/BMC needs - broadly to cover the overall programme design and benefits - outreach material must be gender and socially inclusive) D. 1.2.1.5 Training/workshop reports.
Activity 1.2.2: Promote an inclusive and just IUS model and associated services	Promote the IUS model and associated services.	Sub-activity 1.2.2.1 Develop IUS programme's marketing material for programme sustainability beyond Phase	D. 1.2.2.1 Marketing material targeted towards identified stakeholders i.e., commercial banks, large commercial and industry customers, TSPs, etc. to showcase IUS model and its benefits (2 infographics/ documents/ multimedia/ other material per BMC as per their requirement)

		1 beneficiaries and stakeholders. Sub-activity 1.2.2.2 Develop a comprehensive and inclusive marketing, communications and outreach strategy for the EE focussed on promoting the IUS model	D. 1.2.2.2 One (1) strategy per BMC - the strategy must also cater to vulnerable groups [such as blind, deaf or speak different local languages such as the indigenous populations]
Activity 1.2.3: Provide customers with hands-on support to identify and develop distributed RE, EE and other DER sub-projects	Provide customers with hands-on support to identify and develop distributed RE, EE and other DER sub-projects.	Sub-activity 1.2.3.1 Located at the Executing Entity's office to support day to day activities of the IUS programme, support to the implementation consultant, support outreach events, data collection, reporting, and reviewing, QA/QC, etc. Sub-activity 1.2.3.2 Set-up or strengthen utility help desk/customer support to handle IUS queries Sub-activity 1.2.3.3 Develop beneficiary eligibility criteria to support energy audits for lower income residential customers by TSPs Sub-activity 1.2.3.4 Low-cost/free energy audits by TSPs for lower income residential customers Sub-activity 1.2.3.5 Develop basic energy savings calculator for self-screening by customers	D. 1.2.3.1 Monthly progress reports to the Executing Entity highlighting project progress, tasks undertaken, planned activities for the next month, etc. D. 1.2.3.2 Strategy to set-up/strengthen the utility help desk for the IUS programme (virtual and in-person) - one for each national Executing Entity D. 1.2.3.3 Three beneficiary eligibility criteria - one for each BMCs D.1.2.3.4 300 low-cost free energy audits (100 per BMC) D. 1.2.3.5 One (1) energy savings calculator per national Executing Entity (three total)

Activity 1.3.1: Facilitate knowledge exchange among the participating BMCs	Facilitate knowledge exchange among the targeted BMCs.	Sub-activity 1.3.1.1 Develop knowledge exchange material. Sub-activity 1.3.1.2 Organize virtual knowledge sharing sessions among key stakeholders of the three BMCs.	D. 1.3.1.1 Knowledge exchange materials, primarily presentations and notes, to be used for the knowledge sharing session (2 to 3 documents to be prepared per BMC). Knowledge sharing material to include inputs from the local regulatory consultant and national energy consultant on the lessons learnt by the regulator and the utility which could be shared with other BMCs. D. 1.3.1.2 Consultant's summary report for each knowledge-sharing session.
Activity 1.3.2: Enhance Executing Entities' and other stakeholders' capacities to implement the IUS model in a just transition context	Enhance Executing Entities' and other stakeholders' capacities to implement the IUS model.	Sub-activity 1.3.2.1 Develop utility-/country-specific programme operational manuals including the final financial and organizational modalities for the IUS model at each national Executing Entity. Sub-activity 1.3.2.2 Support each national Executing Entity to develop a business development strategy and action plan. Sub-activity 1.3.2.3 Assist each utility/National Executing Entity to establish a digital sub-project application submission and approval platform. Sub-activity 1.3.2.4 Support National Executing Entities with sub-project appraisal if/as needed (e.g. when Executing Entities seek to approve sub-projects that	D. 1.3.2.1 Three operational manuals (one for each national Executing Entity/BMC). D. 1.3.2.2 One business development strategy and action plan per national Executing Entity (3 in total). D. 1.3.2.3 One digital application submission and approval platform per BMC (3 in total). D. 1.3.2.4 Consultant reports summarizing the support provided to the sub-project appraisal. D. 1.3.2.5 Training reports

		include non-preapproved technologies). Sub-activity 1.3.2.5 Deliver capacity building trainings to National Executing Entities on ESMF and Gender, monitoring and reporting.	
Activity 1.3.3: Train qualified (technology) service providers, notably TSPs, to support consumers	Training for qualified (technology) service providers, notably TSPs, to support consumers.	Sub-activity 1.3.3.1 Develop training modules for TSPs covering the following topics: IUS programme design and implementation; technical session on design, standards, etc.; legal and regulatory session on key insights of the local laws and framework; E&S risk and mitigation; GESI, M&R, etc. - multiple modules to be developed for each BMC based on local needs Sub-activity 1.3.3.2 Deliver TSP trainings using the modules developed under sub-activity 1.3.3.1.	D. 1.3.3.1 TSP training modules D 1.3.3.2 Training reports
Activity 1.3.4: Support information exchange with other (non-partner) countries, incl. governments, regulators, TSPs and utilities	Support information exchange with other (non-partner) countries, incl. governments, regulators, TSPs and utilities.	Sub-activity 1.3.4.1 Develop programme specific information exchange materials. Sub-activity 1.3.4.2 Deliver (virtual) information exchange/ knowledge sharing sessions. Sub-activity 1.3.4.3 Organize an in-person	D. 1.3.4.1 Materials such as pamphlets, leaflets, presentations, etc. to be developed as per the programme needs and target non-partner stakeholder(s). D. 1.3.4.2 Consultant's summary report for each information exchange/knowledge sharing session. D. 1.3.4.3 Report summarizing the results of the knowledge sharing summit and how the programme-sponsored participants contributed.

		regional knowledge sharing summit/ side event in CARICOM's or CDB's or GCF's Energy Events.	
Activity 1.3.5: Create a Community of Practice (COP) in the BMCs, inviting governments, regulators, utilities, TSPs and NGOs to participate	Create a Community of Practice in the BMCs, inviting governments, regulators, utilities, TSPs and ESCOs to participate.	Sub-activity 1.3.5.1 Set-up the regional Community of Practice headed by the CDB. Sub-activity 1.3.5.2 Promote south-south learning / peer-to-peer learning.	D. 1.3.5.1 Report summarizing the structure/outline of the Community of Practice elaborating the guidelines, rules of engagements, etc. D.1.3.5.2 Report summarizing the activities and results (including how programme-supported participants contributed) for each event/trip.
Activity 2.1.1: Prepare, finance and implement sub-projects utilising distributed RE, EE and other DERs among households, businesses and public sector entities, while supporting complementary investments in smart meters/AMI	Preparation, financing, and implementation of sub-projects utilising distributed RE, EE and other DERs among households, businesses and public sector entities.	Sub-activity 2.1.1.1 Finance and oversee the implementation of IUS sub-projects through which RE, EE and/or other DER technologies will be installed at the premises of utility customers. Sub-activity 2.1.1.2 Install smart meters/AMI at the premises of electric utility customers.	D.2.1.1.1 Documentation confirming the completion of sub-projects. D.2.1.1.2 Documentation confirming the installation of smart meters/AMI.
E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)			
E.7.1. The implementation of the IUS programme will involve:			
- Monitoring and reporting on progress in relation to the programme's logical framework, as outlined section E.5; - Monitoring and reporting on financial progress (i.e. approvals/commitments, disbursements, reflows);			

- Monitoring and reporting on sub-project compliance with environmental and social risk management requirements, guided by the Environmental and Social Management Framework (ESMF) that is included in Annex 6;
- Monitoring and reporting on the implementation of the GAP that is included in Annex 8;
- The consolidation of the above information in regular progress reports; and
- Mid-term and final evaluations to assess implementation progress and challenges and propose corrective actions.

E.7.2. Additional information about the monitoring and reporting processes is included in Annex 21: Programme Operations Manual. Further information about the monitoring and reporting frequency and distribution of roles & responsibilities is also included in Annex 11: M&E Plan. Meanwhile, a detailed overview of the methodology for preparing the ex-ante estimates of emissions to be reduced/avoided, as well as the methods to be used to calculate actual emissions reductions during the implementation of the IUS programme, is included in Annex 22: Assessment of GHG Emission Reductions and their Monitoring and Reporting.

E.7.3. Monitoring

Under Component 1, monitoring will be led by the CDB TA Execution Team, which will be overall responsibility for procuring and contracting almost all consultants and other service providers that will deliver support under the TA component. The CDB TA Execution Team will ensure that the consultants/service providers complete their assignments in a timely and efficient manner and will gather the data and documentation needed to demonstrate progress (in line with the logical framework for the programme) during the implementation of the consultancy and non-consultancy contracts. This will include monitoring to ensure that the ESMF and GAP and are being fully and effectively implemented.

Under Component 2, the monitoring of sub-projects will be led by the electric utilities (as National Executing Entities). Initial data on sub-project implementation (i.e. installation of eligible technologies) will be sourced from the contracted TSPs, as will information about adherence to necessary environmental and social risk management standards. Specifically with regards to sub-project performance, the electric utilities (with support from their programme-financed National Energy Consultants) will utilize the customer's electricity meter to establish a customer-specific baseline for each sub-project that they have decided to finance. The utilities will then continue to use the customers' meters to monitor progress after the technologies have been installed and throughout the programme implementation period, in line with the methods outlined in Annex 22.

To supplement the meter-based monitoring, the CDB TA Execution Team will also contact a firm to assist with gathering additional data/information from sub-project beneficiaries as needed, including through telephone interviews, surveys and site visits. This will enable the utilities and CDB to cross-check the data from the meters at regular intervals throughout the programme implementation period, and thus validate the meter data and/or explain anomalies. The firm will gather data from a random sample of beneficiary customers each year *while also* following up with those whose meter data deviates from the changes/benefits that the utility would have projected at the time of sub-project approval.

E.7.4. Reporting

Reporting from electric utilities (National Executing Entities) to the CDB Accredited Entity Team. CDB will enter into a subsidiary agreement with each electric utility that clearly specifies the utility's reporting obligations. This will include both financial reporting (information on disbursements, reflows, etc.) and progress reporting (information on the sub-project pipeline, approvals, implementation status and progress vis-à-vis performance metrics), covering each utility's full portfolio of sub-projects financed using the blended line of credit provided under the IUS programme. The utilities will be required to prepare and submit these reports to the

CDB Accredited Entity Team at regular intervals to be specified in the subsidiary agreements. The utilities' reporting to CDB will also include information on compliance with environmental and social risk management requirements, as well as GESI requirements, for the portfolio of sub-projects. The utilities will be expected to gather/collect the data needed for these reports, including using the methods/processes outlined above.

Reporting from CDB to GCF. Using the data contained in the reporting provided by the National Executing Entities and the CDB TA Execution Team, as well as other sources of data/information as relevant, the CDB Accredited Entity Team will provide the GCF Secretariat with:

- Annual Performance Reports (APRs) that provide detailed information on the status of the programme throughout its implementation period;
- A mid-term evaluation (interim evaluation) report at the midpoint of the implementation period of the programme, which will be prepared by an independent external evaluator contracted by CDB; and
- A final evaluation report at the end of the implementation period of the programme, which is expected to be prepared by the same independent external evaluator as the mid-term evaluation.

E.7.5. Evaluation

In line with GCF requirements, CDB will contract an external evaluator to conduct independent evaluations halfway into the programme implementation period (mid-term review) and at the end of the programme (final evaluation). The CDB Accredited Entity Team will procure the evaluator using an open tender process and will supervise and quality-assure the work of the evaluator in close collaboration with the Bank's Office of Independent Evaluation (OIE). The mid-term and final evaluations will be conducted in line with, *inter alia*, the following principles: independence; competencies and capacities; impartiality; credibility; transparency; integrity and gender equality.

The evaluation methods for the mid-term and final evaluations are to be determined, but are tentatively expected to include: (i) key informant interviews with beneficiary customers and institutional beneficiaries (i.e. utilities, regulators, TSPs); and (ii) detailed review of relevant programme documentation, including monitoring data provided by the electric utilities and CDB TA Execution Team, additional monitoring data (from interviews, surveys and site visits) collected by the data gathering firm contracted by the CDB TA Execution Team, APRs and other project reports, based on an agreed list of evidence to be provided by CDB to the contracted evaluator.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1 Challenges in establishing efficient systems and procedures for IUS service delivery

Category	Probability	Impact
<u>Credit</u>	<u>Medium</u>	<u>High</u>

Description

F.1.1. There is a moderate likelihood that utilities may encounter challenges in putting in place systems and procedures to deliver the IUS programme. For instance, there is a risk that utilities may not complete the start-up phase during PY1 as envisaged.

Mitigation Measure(s)

F.1.2. To mitigate the challenge of establishing efficient systems and procedures for IUS service delivery, the TA component focuses on comprehensive support and capacity building for utilities. This includes addressing regulatory barriers by assisting national stakeholders in updating legal and regulatory frameworks and strengthening regulatory capacity. External consultants will provide direct assistance and expert advice on regulatory enhancements. A digital application portal will streamline processes, while training will cover best practices and effective procedural frameworks.

Selected Risk Factor 2 Capacity constraints among TSPs

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>

Description

F.1.3. There is a risk associated with the quantity and technical capabilities of technology service providers (TSPs) in the participating countries to fully address the high/increasing demand for RE and EE technologies.

Mitigation Measure(s)

F.1.4. While this risk is noteworthy, the feasibility assessment did not identify it as a major concern. Additionally, the TA component includes training for TSPs to help them improve their services/products and enhance the efficiency with which they deliver them. The TA component will also help establish a Community of Practice within the BMCs through which the TSPs can share knowledge and best practices with one another, as well as with government officials, regulators, and utilities.

Selected Risk Factor 3 Regulatory and Policy Barriers

Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>High</u>

Description

F.1.5. Inconsistent regulatory frameworks and policies across the Caribbean can impede the implementation of the IUS programme. Challenges include slow permit processes, lack of supportive policies for renewable energy, and

varying standards across countries. Utility regulators in several of the participating BMCs have also indicated that they would need additional training and capacity to effectively regulate IUS implementation in their respective countries.

Mitigation Measure(s)

F.1.6. The IUS programme is a comprehensive initiative that involves collaboration with local and regional authorities to create spaces that contribute to the harmonization of energy policies and regulations supporting the deployment of RE and EE. Additionally, the TA component will assist the participating countries to develop and implement supportive policies. Finally, the programme will provide support to: (i) address pending regulatory barriers in each participating country; (ii) strengthen the capacity of the utility regulators in the participating countries to understand and regulate IUS; (iii) directly assist the utilities to apply for regulatory approval in PY1 to enable them to implement IUS sub-projects starting in PY2; and (iv) establish a process for regular regulatory reviews will be established to promptly identify and address potential barriers, ensuring that regulatory frameworks remain aligned with the programme's goals and the evolving needs of the energy sector.

Selected Risk Factor 4 Risk aversion/reluctance to engage among potential beneficiaries

Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>High</u>

Description

F.1.7. Some potential beneficiaries may be reluctant to adopt/invest in new RE and EE technologies. This often stems from concerns over high initial costs, operational disruptions and cultural resistance to change, as well as regulatory uncertainties. Potential beneficiaries may also be aware of/interested in sustainable energy technologies but (initially) reluctant to invest in them through their electric utility given that the IUS model is new and thus not (yet) well established/known.

Mitigation Measure(s)

F.1.8. The IUS programme includes a robust strategy to increase stakeholder engagement and buy-in. This will involve targeted educational campaigns to raise awareness of the benefits and practical applications of RE and EE technologies, as well as enhance awareness of (and interest in) the utility-led IUS model. Additionally, workshops and seminars will be facilitated to allow stakeholders to interact directly with these technologies, addressing concerns and demonstrating efficacy and safety. These sessions will also provide an opportunity for interested customers to learn more about the IUS service offering and address any questions/concerns they may have. Regular feedback sessions will also be implemented to ensure that stakeholder concerns and suggestions are continually incorporated into programme implementation, thereby enhancing acceptance and minimizing resistance.

Selected Risk Factor 5 Risk related to compliance, including money laundering/terrorist financing (ML/TF)

Category	Probability	Impact
<u>ML/FT</u>	<u>Medium</u>	<u>Medium</u>

Description

F.1.9. The electric utilities will be establishing new business models through which they will interact with customers (as beneficiaries) and TSPs (as contractors) in new ways. There are numerous compliance issues to be addressed in this context, including KYC due diligence, AML/CFT and environmental & social risk management, among others.

Mitigation Measure(s)

F.1.10. CDB and the programme preparation team have conducted a detailed assessment of the electric utilities' policies, procedures and capacities with a view to determining/confirming their ability to serve as Executing Entities without posing unacceptable/unmanageable risks related to compliance issues. These assessments included reviews of the entities' governance and management structures, financial management and accounting systems, internal controls and policies/procedures related to prohibited practices and AML/CFT. The two entities for which detailed assessments were completed (BLPC and BEL) were all found to be satisfactory in this regard, as outlined in Annex 2: Feasibility Study. JPS will be subject to the same type of review during PY1.

In addition, CDB will enter into subsidiary agreements with the utilities (as National Executing Entities) that include provisions related to AML/CFT, as well as other compliance-related risks. This is guided by CDB's own policies and procedures in this area, which were reviewed during the GCF accreditation process and found to be satisfactory and consistent with GCF standards and requirements.

In this context, it is also important to note that Jamaica is the only country from the BMCs that is currently in the "Increased Monitoring" list (grey list) (FATF (2024). [Link](#))

Selected Risk Factor 6 Risks related to default/non-payment rates by participating utility customers.		
Category	Probability	Impact
<u>Credit</u>	<u>Low</u>	<u>Medium</u>
Description		
F.1.11. Based on the experiences of stakeholders in other jurisdictions that have implemented IUS-type models (i.e. utility-based financing for RE, EE and other DERs), CDB has estimated a very low default/non-payment rate among Programme end beneficiaries. This is partly because the use of the on-bill financing model ties repayment to the electric utilities/national Executing Entities to customers' regular utility bills, which are a priority for most customers given the potential consequences associated with non-payment (i.e. being disconnected). Nevertheless, there is a risk that non-payment rates in the three participating countries may be higher than expected, particularly in the event of an exogenous shock such as a broader economic downturn.		
Mitigation Measure(s)		
F.1.12. To mitigate the risk of higher-than-expected default/non-payment rates, the Programme has prioritized technologies that deliver direct and immediate financial benefits to the participating customers. This will ensure that savings from these investments help offset the repayment charges immediately. In addition, the use of on-bill financing will enable customers to see both the savings that they have accrued because of the investment and the additional charges associated with financing the investment on the same bill, which will help beneficiaries – including those with more limited financial literacy – to understand the real-time benefits of continuing to pay these charges and thus meet their repayment obligations. Furthermore, the Programme TA component includes activities to support enhanced customer awareness raising and outreach, through which customers will be able to better understand the costs and benefits of the investments and how they will produce significant economic benefits over the medium/long term as long as they meet their repayment obligations over the short/medium term. The user-friendly energy savings calculators to be supported through the TA component will similarly help reinforce these efforts to enhance understanding among customers and thereby mitigate the risks associated with higher-than-expected default/non-payment rates.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

G.1.1. An ESMF was developed to ensure that the programme adequately identifies, mitigates, manages and monitors environmental and social risks and potential impacts during programme implementation, and complies with the relevant national legislation and guidelines in the participating countries (Belize, Barbados and Jamaica). By extension, the ESMF will ensure compliance with the environmental and social standards of both the GCF and CDB.

G.1.2. The programme is classified as Category B. This is because the nature of the investments to be made through the sub-projects have the potential for limited adverse environmental and social impacts and/or risks, all of which can be readily identified, mitigated, and managed. Screening was first done at the conceptual phase of the programme by the CDB. A more in-depth environmental and social risk assessment and management analysis was done during the programme preparation stage, which was also informed by detailed consultations with relevant stakeholders in the three countries. The risks and associated risk mitigation/management measures outlined in the ESMF will serve as a guide for the National Executing Entities, which will (with support from the Implementation Support Consulting Firm) each prepare a utility-specific Environmental and Social Management Plan (ESMP) and corresponding waste disposal plan during the first year of programme implementation (PY1). The ESMPs will provide more detailed guidance to the national stakeholders – notably the electric utilities given their roles as national Executing Entities – on how to identify, assess, mitigate and manage E&S risks throughout the sub-project workflow/cycle. They will also provide/include:

- detailed guidance on the specific E&S risks associated with each type of eligible technology;
- a detailed checklist to be used by the electric utilities when reviewing sub-project applications to ensure such risks are clearly identified and sub-projects can be properly categorised, which will be an adapted version of the draft checklist that is appended to this ESMF;
- actionable guidance on which associated risk mitigation/management measures should be implemented in each country, how they should be implemented, and by whom;
- templates and associated guidance to support the development of the corresponding Environmental, Social, Health and Safety (ESHS) plan for any sub-project for which the screening identifies notable E&S risks and thus is categorised as a Cat B sub-project;
- templates and associated guidance for TSPs on how to adhere to the E&S requirements outlined in the ESHS plans, and how to report to the utilities to demonstrate compliance with such requirements;
- guidance for utilities on how to integrate the ESHS plan implementation and reporting requirements into their contracts with TSPs; and
- a utility-specific disposal plan that specifies the requirements associated with the collection, storage and final disposal of waste (e.g. old AC units and other technologies that are removed during installation, as well as programme-supported RE, EE and other DER equipment after their productive lifespan has ended).

G.1.3. The ESMF that is attached at Annex 6 also includes an initial overview of the sub-project workflow (with clearly defined roles and responsibilities for E&S management) to demonstrate when/how the identification and assessment of E&S risks, as well as the implementation and monitoring of associated risk mitigation and management measures, will be implemented.

G.1.4. In addition to the ESMF, an Indigenous Peoples Planning Framework (IPPF) has been developed to guide the development of an Indigenous Peoples Plan (IPP) in Belize and Jamaica – see Appendix C to Annex 6. The IPPs will be developed in tandem with the ESMPs for Belize and Jamaica, and in close consultation with indigenous groups in the country, and (once finalized and validated) will guide all relevant actors to ensure the execution of programme activities in Belize and Jamaica is done in an inclusive and culturally appropriate manner.

G.1.5. Although the E&S risks and potential impacts associated with the IUS programme are likely to be limited and manageable, it will be important to strengthen and supplement the capacity of the National Executing Entities and other relevant stakeholders (e.g. TSPs) to enable full and effective compliance with the ESMF (and later, ESMP) requirements. The TA component of the programme therefore includes key support measures, including training for electric utilities and other relevant stakeholders (e.g. TSPs) on E&S risk identification, mitigation, and management monitoring, and reporting during programme implementation. The TA component of the programme will also supplement the capacity of the electric utilities (in their role as National Executing Entities) to fulfil their

E&S-related responsibilities, notably by contracting: (i) a National Energy Consultant to work with/in each electric utility; and (ii) an Implementation Support Consulting Firm that will include an E&S Specialist.

G.1.6. To enhance accountability and thereby help ensure that E&S risks are properly mitigated and managed, the programme will provide easily accessible avenues for national stakeholders (including sub-project beneficiaries and non-beneficiaries who are affected by the programme and/or sub-projects) to submit complaints/grievances. This includes: (i) working with each national Executing Entity to establish their own channels through which grievances/complaints can be submitted; (ii) CDB's Projects Complaints Mechanism (PCM), which is further described below; and (iii) the GCF Independent Redress Mechanism, which is also elaborated below.

CDB Projects Complaints Mechanism

The CDB PCM is managed by the CDB's Office of Integrity, Compliance and Accountability (ICA). Through this PMC, complaints can be submitted through six channels: a webform; a hotline; email; telephone; mail; and office visits. Guided by the CDB Projects Complaints Mechanism Policy⁵⁴, such complaints are reviewed and handled independently by the CDB ICA in accordance with its detailed PCM Procedures.

Allegations may be made by a variety of sources including staff, consultants, contractors, and other people who may consider themselves participants in, witness to or victims of wrongdoing. Detailed information about the channels through which complaints can be filed anonymously are included in the ESMF and also outlined below for ease of reference:

- online through a portal: <https://secure.ethicspoint.com/domain/media/en/gui/55678/index.html>
- through a whistle blower hotline (770-409-5029), via telephone (246-539-1777),
- via email (ica@caribank.org), and
- via mail to:

Head, Office of Integrity Compliance and Accountability
Building A
Caribbean Development Bank
P.O. Box 408
Wilkey
St. Michael
Barbados, W. I. BB11000

Complaints related to the programme can also be filed to anticorruption@caribank.org, Projectcomplaints@caribank.org

GCF Independent Redress Mechanism

The GCF Independent Redress Mechanism can be used by project workers, communities and other affected stakeholders as needed to submit complaints/grievances directly to the GCF. More information about the GCF Independent Redress Mechanism is accessible at the following link:

<https://irm.greenclimate.fund/>

Specific complaints can be submitted to the GCF Independent Redress Mechanism via the following link:

<https://irm.greenclimate.fund/case-register/file-complaint>

G.1.7. The ESMF was developed in close consultation with relevant national stakeholders. The programme preparation team consulted with, *inter alia*, government officials, utility regulators, and the electric utilities via a questionnaire-based survey. The team also organized virtual face-to-face interviews with TSPs, whose insights were of critical importance given their central role in implementing and monitoring (and reporting on) risk mitigation and management measures. Consultations were organized between April and May 2023. Apart from responses obtained through questionnaire-based surveys, during interviews, notes were taken, and each interview lasted for about thirty-five minutes. To minimize the potential for biased responses, the programme preparation team avoided leading the interviewees during the interviews and reacted in a neutral manner by not signalling preferred answers.

⁵⁴ This policy remains in effect and is publicly accessible here: <https://www.caribank.org/about-us/policies-and-strategies/projects-complaints-mechanism-policy>.

G.1.8. Stakeholder engagement will be similarly important during programme implementation. The ESMF provides clear guidance to the Executing Entities in this regard. The above-described trainings for electric utilities and TSPs will also further strengthen their capacity to support inclusive stakeholder engagement in line with ESMF requirements. the programme will further supplement the electric utilities' capacities in this area by contracting (i) a National Energy Consultant to work with/in each electric utility; and (ii) an Implementation Support Consulting Firm that will include an E&S Specialist.

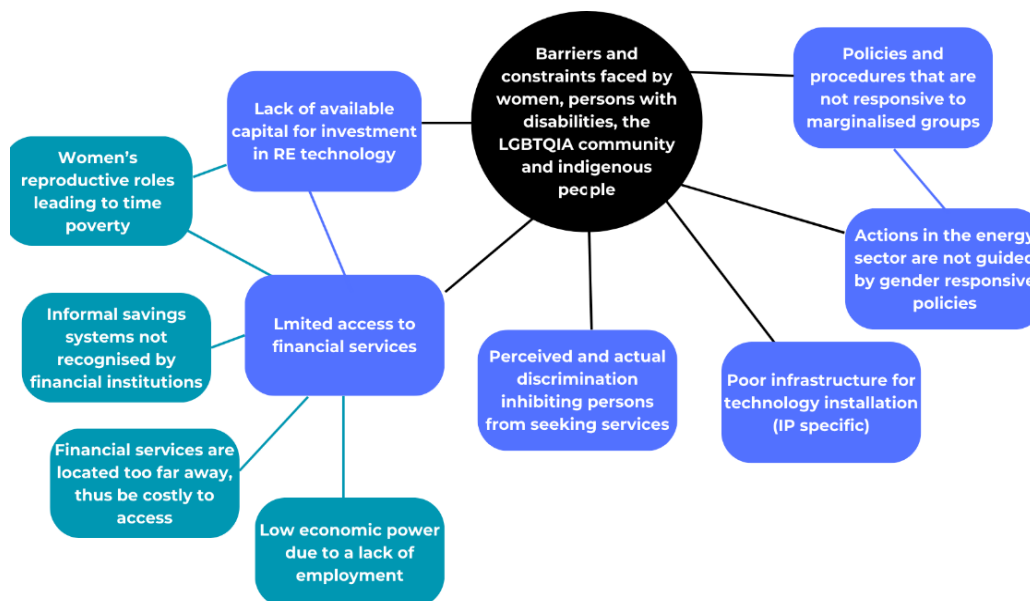
G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

G.2.1. The design of the programme has been informed by a detailed Gender Assessment, which focused on four key groups as relevant per country i.e. women, persons with disabilities (PWDs), the LGBTQIA+ community, and indigenous peoples (specific to Belize). The universal challenge experienced by all groups was the limited access to finance and financial services. Although the ways in which this challenge may manifest for each group often differ due to unique circumstances such as locality (more specific to indigenous people), discrimination, low economic power and lack of capital, the resulting effect is the same across the board - namely that financial and non-financial barriers are inhibiting the extent to which marginalised groups can participate in/benefit from the sustainable energy transition, and thus will not automatically benefit from the services and opportunities presented by this programme.

G.2.2. An additional barrier observed primarily in Belize and Jamaica is the weak enabling environment for the adoption of renewable energy technologies. These countries have seen less work and campaigning done around promoting the use and uptake of said technologies. This can be a deterrent as persons may be unaware of the benefits and may perceive these technologies to be more costly and burdensome to implement. Additionally, current attitudes on these technologies may not be the most conducive to the successful implementation of this programme. This information and empowerment gap is a critical one to be tackled through the TA component on several levels.

G.2.3. An additional and unique barrier facing indigenous people is that their traditional homes may not be compatible with some of the technologies on offer. PWDs and the LGBTQIA community also face access and affordability barriers including those shaped by bias and discrimination and that take the form of inaccessibility of services, though the reasons for these between the two groups.

Figure 22: Summary of barriers and constraints experienced by marginalised groups attempting to utilise or engage with RE technologies⁵⁵



G.2.4. Through the Gender Action Plan (GAP), the programme will address the following gender needs specific to this programme: **governance and participation** (through targeted action focused on policies, the utilities and other key actors), and **access, education and assets** (through targeted action focused on the marginalized beneficiaries). More specifically, the GAP – the contents of which are coherent with, and have been integrated into, the broader programme (notably the TA component) – will support the following types of actions:

(a) *Improving the enabling environment:* The programme will support electric utilities and energy ministries to better understand gender and social inclusion issues in the context of the energy transition and integrate this understanding when supporting them to strengthen relevant policies, planning and procedures. In addition, training will be provided for utilities, TSPs and other key energy sector stakeholders to (i) improve their understanding of the programme GAP, its requirements, etc.; and (ii) enhance their awareness of broader GESI and Sexual Exploitation, Abuse and Harassment (SEAH) issues in the energy sector. In so doing, these GAP measures will improve the extent to which local (sustainable) energy services are responsive to the needs of marginalized groups.

(b) *Enhancing awareness among marginalised groups:* The programme will support the development of marketing and outreach strategies to promote the prioritized technologies (to be led by the energy ministries) and the IUS programme (to be led by the electric utilities). These strategies will place a specific emphasis on reaching, and raising awareness among, marginalized groups. The materials to be developed through the programme TA component to support implementation of these strategies will similarly emphasize inclusivity, as outlined in Section B.3 (above). Furthermore, the programme partners (including energy ministries and utilities) will work with local NGOs to implement their marketing and outreach strategies, particularly to help disseminate awareness-raising materials among marginalized groups with which NGOs are better connected. In so doing, these GAP measures will ensure that marginalized groups are aware of both the prioritized technologies and how the programme can assist them to invest.

(c) *Promoting a more inclusive model:* Complementing the two above-described areas is the fact that the IUS model itself provides a more inclusive way to support sustainable energy investments among marginalized groups. By eliminating the need for beneficiaries to secure loan financing from a local commercial bank, and by allowing repayment on favourable terms through on-bill financing, the IUS model will directly address several of the financial barriers that otherwise often prevent marginalized groups from investing in sustainable energy technologies.

(d) *Direct sub-project support:* While the above-described areas are expected to make meaningful contributions to improving the inclusivity of the IUS programme, lower-income households and other marginalized groups may still

find it difficult to identify and develop sub-project ideas/proposals that would be eligible to be financed by their electric utility. To help address this, the TA component will: (i) provide free (simple) energy audits for up to 180 eligible (low-income/marginalized) utility customers in each of the three participating countries; (ii) assist each utility to establish and operate a user-friendly help desk and energy savings calculator to support customers with sub-project application troubleshooting (which is expected to be particularly beneficial for lower-income and more marginalized groups that may have more limited technical capacity and financial literacy); and (iii) provide the funding needed for each utility to contract a full-time National Energy Consultant who can support business development and customer outreach efforts, with a particular emphasis on reaching and supporting more marginalized groups.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Legal arrangements

G.3.1: CDB will sign a subsidiary agreement with each of the National Executing Entities that will govern the use of funds throughout the programme implementation period, as well as the terms and conditions for repayment of their respective lines of credit. The agreements will reflect CDB standards and requirements as well as any supplementary standards and requirements that may have to be added to account for the provisions in the Funded Activity Agreement to be signed between CDB and the GCF. The process of preparing such subsidiary agreements will be led by the CDB Legal Department in co-ordination with the CDB Accredited Entity Team.

Fiduciary standards

G.3.2: Each electric utility (as a National Executing Entity) will bear overall responsibility for executing its respective line of credit under this programme in line with the subsidiary agreement that it will sign with CDB. When doing so, the utilities will administer resources using their internal systems and procedures (e.g. for financial management, accounting, internal controls, etc.) while also accounting for supplementary programme-specific requirements. These are captured in the overall Programme Operations Manual and will be elaborated in more detail in the utility-specific Operations Manuals that will be developed during PY1. As outlined in Annex 2: Feasibility Study, the programme preparation team has already assessed two of the utilities' capacities to fulfil their respective Executing Entity roles and found them to be satisfactory. A similar assessment will be completed for the third utility during PY1, the satisfactory completion of which will be a condition precedent to first disbursement of the line of credit to that entity.

G.3.3: The overall financial management of the programme will be done in accordance with relevant CDB policies, procedures and standards. These policies, procedures and standards are consistent with international standards, as evidenced by CDB's accreditation to the Adaptation Fund and Green Climate Fund, as well as its track record of managing resources from bilateral partners (e.g. Government of Canada, Government of Italy, European Commission) and other Multilateral Development Banks (e.g. European Investment Bank, Inter-American Development Bank). The CDB OPPM will therefore guide all aspects of financial management under the programme, including disbursements to each electric utility/National Executing Entity under their respective lines of credit which are generally expected to use the CDB 'float' approach.

Procurement

G.3.4: Procurement under this programme will be managed in accordance with the standards and requirements outlined in the "Procurement Policy for Projects Financed by CDB" (2019)⁵⁶ as well as the "Procurement Procedures for Projects Financed by CDB" (2021)⁵⁷. The CDB Procurement Policy and Procedures are aligned with international standards for public procurement, and cover the procurement of Goods, Works, Non-Consulting Services and

⁵⁵ The three turquoise shapes emerging from "Limited access to financial services" are subcategories of this barrier experienced by marginalised groups. The one turquoise shape "Women's reproductive roles leading to time poverty" emerging from both "Lack of available capital for investment in RE technologies" and "Limited access to financial services" contributes to these barriers.

⁵⁶ The CDB Procurement Policy is publicly available here: <https://www.caribank.org/sites/default/files/publication-resources/Procurement%20Policy%20for%20Projects-final.pdf>

⁵⁷ The CDB Procurement Procedures are publicly available here: https://www.caribank.org/sites/default/files/publication-resources/Procurement%20Procedures%20for%20CDB%20financed%20projects_Jan%202021-final_0.pdf

Consulting Services. Note that most of the participating utilities are familiar with the CDB Procurement Policy and Procedures, although under this specific programme they are each only expected to manage one procurement process. Most of the consulting and non-consulting services under the TA component will be procured and managed directly by the CDB TA Execution Team, as outlined in Annex 10: Procurement Plan. Note that the act of purchasing specific technologies in the context of sub-projects will not constitute a procurement action under CDB's procurement framework, and thus (as outlined in Section 7.1 of the CDB Procurement Policy) will instead be guided by established commercial practices as well as the IUS procedures and TSP eligibility criteria.

Audit

G.3.5: Auditing under this programme will be managed in accordance with relevant CDB policies and standards, which adhere to International Accounting standards, and were reviewed by the GCF Secretariat and Accreditation Panel and found to be consistent with those of the GCF. To improve efficiency and economies of scale, the CDB TA Execution Team will directly contract independent external auditors to perform annual financial audits of the programme and prepare English-language audit reports. Quality assurance of such audits will be performed by the CDB Accredited Entity Team.

G.4. Disclosure of funding proposal

Note that in addition to the full feasibility study being provided for review by the GCF Secretariat, iTAP and Board, CDB will provide an updated version of the feasibility study in which some sensitive/confidential information (e.g. about the electric utilities' internal policies and procedures) is redacted. The latter can be published on the GCF website along with the funding proposal and all other annexes.

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁵⁸
- ☐

⁵⁸ Annex 22 is mandatory for mitigation and cross-cutting projects.

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*



GOVERNMENT OF BARBADOS

Senator The Hon. Chad J. Blackman

Minister – Economic Affairs and Investment
Ministry of Finance, Economic Affairs and Investment
3rd Floor East Warrens Office Complex,
Warrens, St. Michael, Barbados, West Indies

Our Ref.: I051/1 Vol. I

Tel.: (246) 535-1302/17

September 5, 2024

Ms. Mafalda Duarte
Executive Director
Green Climate Fund
175 Art-Center Daero
Yeonsu-gu, Incheon 22004
REPUBLIC OF KOREA

Dear Ms. Duarte,

**Funding Proposal for the Green Climate Fund by the
Caribbean Development Bank regarding Scaling up the
Deployment of Integrated Utility Services (IUS) to Support
Energy Sector Transformation in the Caribbean (Phase 1) Programme**

We refer to the programme titled **Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase 1) Programme** in Barbados, Belize, Guyana, Jamaica as included in the funding proposal submitted by Caribbean Development Bank to us on August 9, 2024.

The undersigned is the duly authorized representative of Senator the Hon. Chad Blackman, LLB, LLM, ACI~~Arb~~, the National Designated Authority of Barbados.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) the Government of Barbados has no-objection to the programme as included in the funding proposal;
- (b) the programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Barbados; and

- (c) in accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely,

A handwritten signature in black ink, reading "Chad Blackman". The signature is fluid and cursive, with the first name "Chad" and last name "Blackman" clearly distinguishable.

SENATOR THE HON. CHAD J. BLACKMAN, LLB, LLM, ACI Arb

Minister
and National Designated Authority



GOVERNMENT OF BELIZE

Ministry of Finance, Economic Development and
Investment

ECONOMIC DEVELOPMENT

P.O. Box 42

Ground Floor, Sir Edney Cain Building

Belmopan City

Belize, Central America

Tel: (501) 880-2526

(501) 880-2527

Email: econdev@med.gov.bz

Our Ref: IA/GCF/1 (35)

October 21, 2024

Ms. Mafalda Duarte

Executive Director

Green Climate Fund

Songdo Business District

175 Art center-daero

Yeonsu-gu, Incheon 22004

Republic of Korea

Dear Ms. Duarte:

Subject: Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase I)

We refer to the programme titled “Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase I),” submitted by the Caribbean Development Bank on July 8, 2024.

The undersigned is the duly authorized National Designated Authority of Belize.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, I hereby communicate our no-objection to the above-mentioned GCF proposal.

By communicating our no-objection, it is implied that:

- (a) The Government of Belize has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national

- priorities, strategies and plans of Belize;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Mr. Carlos Pol
Ag. Chief Executive Officer and National Designated Authority-GCF
Ministry of Economic Development

- C. Mr. Jose Urbina, Chief Executive Officer, Ministry of Public Utilities, Energy, Logistics & E-Governance

Mr. Dean Molina, Chairman, Public Utilities Commission

Mr. John Mencias, Chief Executive Officer, Belize Electricity Limited



ANY REPLY OR SUBSEQUENT
REFERENCE SHOULD BE ADDRESSED
TO THE PERMANENT SECRETARY AND
THE FOLLOWING REFERENCE NUMBER
QUOTED:

Tel Number: 876-926-1690-3
926-1590-9 FAX 754-0975

MINISTRY OF ECONOMIC GROWTH AND JOB CREATION
25 DOMINICA DRIVE
KINGSTON 5
JAMAICA

25 September 2024

Ms Mafalda Duarte
Executive Director
Green Climate Fund
Korea

Re: Funding proposal for the GCF by Caribbean Development Bank, regarding Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean

Dear Ms. Duarte,

We refer to the programme titled **Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean** as included in the funding proposal submitted by **Caribbean Development Bank** to us on 22 August 2024.

The undersigned is the duly authorized representative of Ministry of Economic Growth and Job Creation, the National Designated Authority of Jamaica.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Jamaica has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Jamaica;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Omar Alcock (Mr)
Senior Technical Officer (Mitigation)
Ministry of Economic Growth and Job Creation
Jamaica

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase 1) programme
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Caribbean Development Bank (CDB)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Barbados, Belize, and Jamaica
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 30, 2025
Language(s) of disclosure	English
Explanation on language	English is the official language of the target countries.
Link to disclosure	https://www.caribank.org/publication_types/documents/environmental-and-social-impact-assessments/scaling-deployment-integrated-utility-services-ius-support-energy-sector-transformation-caribbean
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B programme is contained in the "Environmental and Social Management Framework (ESMF)".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 30, 2025
Language(s) of disclosure	English
Explanation on language	English is the official language in all three countries.
Link to disclosure	https://www.caribank.org/publication_types/documents/environmental-and-social-impact-assessments/scaling-deployment-integrated-utility-services-ius-support-energy-sector-transformation-caribbean
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B programme is contained in the "Environmental and Social Management Framework (ESMF)".
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A

Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Indigenous Peoples Planning Framework (IPPF)
Date of disclosure on accredited entity's website	Friday, May 30, 2025
Language(s) of disclosure	English
Explanation on language	While some Indigenous Peoples in the participating countries (particularly in Belize) speak indigenous languages, English is also commonly spoken; it is the official language in all three countries.
Link to disclosure	https://www.caribank.org/publication_types/documents/environmental-and-social-impact-assessments/scaling-deployment-integrated-utility-services-ius-support-energy-sector-transformation-caribbean
Other link(s)	N/A
Remarks	An IPPF consistent with the requirements for a Category B programme is contained in the "Environmental and Social Management Framework (ESMF)".
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	From Friday, May 30, 2025
Place	Affected peoples may access the ESMF at the premises of their electric utilities, where they can request support to print a hard copy. In particular: <u>In Barbados:</u> Barbados Light and Power Company (BLPC) Highway 7 Bridgetown, Saint Michael Barbados <u>In Belize:</u> Belize Electricity Limited (BEL) 2 ½ Miles Philip Goldson Highway Belize City Belize <u>In Jamaica:</u> Jamaica Public Service Company (JPS) 6 Knutsford Boulevard Kingston 5 Jamaica Given that households and businesses must opt-in to participate in/benefit from this programme, and thus specific sites and beneficiaries within the countries are not yet known, there are no other specific place(s) or location(s) where the ESMF could be printed and made available that would be more convenient for potentially affected peoples.
Date of Board meeting in which the FP is intended to be considered	

Date of accredited entity's Board meeting	TBD
Date of GCF's Board meeting	Monday, June 30, 2025

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP275

Proposal name:	Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase 1) Programme
Accredited entity:	Caribbean Development Bank
Country/(ies):	Barbados, Belize and Jamaica
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. This funding proposal, submitted by the Caribbean Development Bank (CDB) as accredited entity (AE), seeks to establish and scale an Integrated Utility Services (IUS) Programme in three Caribbean countries – Barbados, Belize and Jamaica – with the goal of accelerating decarbonization and enhancing energy resilience. The proposal represents Phase I of a larger regional initiative and is focused on enabling regulated national utilities to evolve their business models to provide distributed renewable energy (RE), energy efficiency, and other demand-side energy services directly to customers.
2. The IUS model builds on the concept of utilities acting as energy service companies, bundling financing, installation and maintenance of clean energy solutions, such as solar photovoltaic (PV), battery storage, efficient lighting and appliances, into a service offer that can be repaid through on-bill financing. The utilities would serve as key delivery agents, supported by regulatory reforms, technical service providers and capacity-building activities. CDB serves as the central coordinating entity and lender, channelling GCF resources and co-financing from its own balance sheet.
3. The programme is structured around two interlinked components:
4. Component 1 (Technical Assistance) focuses on enabling conditions. It provides grant financing to support the creation of regulatory frameworks, institutional readiness, customer awareness, and technical capacity among utilities, regulators and technology service providers. CDB will establish a dedicated technical assistance team, supported by three national teams, to oversee implementation, stakeholder engagement, and knowledge transfer. These upstream, non-revenue-generating activities are essential to market formation and require full concessionality.
5. Component 2 (Loan Component) provides a concessional financing facility for the actual deployment of distributed RE, energy efficiency, and distributed energy resources (DERs) at the customer level. A USD 20.8 million loan from GCF will be blended on a pari passu basis with USD 16.2 million from CDB to establish credit lines to the participating national utilities. Utilities, acting as executing entities (EEs),¹ will on-lend these resources to eligible subprojects –

¹ Barbados Light and Power Limited is a regulated electric utility in Barbados. It is a limited liability company incorporated under the laws of Barbados and is a wholly (100 per cent) owned subsidiary of Emera Inc., a Canadian utilities conglomerate.

Belize Electricity Limited is a regulated electricity utility in Belize. It is a limited liability company incorporated under the laws of Belize, and is majority owned by the Belize public sector. The Government of Belize owns 32.58 per cent, while the Social Security Board owns 31.27 per cent. The remaining shares are owned by Fortis Cayman Inc. (33.3 per cent) and 1,500 small shareholders (2.85 per cent).

residential, commercial and public sector – covering capital expenditures for clean energy technologies. The subloans are expected to be offered at interest rates in the 4.5 to 6.0 per cent range, which is below prevailing commercial rates and made possible through the concessional GCF-CDB blend.

6. Each utility will retain the credit risk associated with the customer loans and is expected to mobilize an additional USD 25 million in in-kind contributions, largely in the form of advanced metering infrastructure. CDB has committed to entering subsidiary agreements with each utility to ensure proper fiduciary controls, eligibility criteria and mechanisms for monitoring and reporting. Regulatory engagement will be critical, as the design and terms of the on-lending will be finalized in collaboration with national regulators during the first programme year.

7. While the Caribbean region has significant technical potential for distributed RE and strong public support for energy affordability, uptake has been hampered by high upfront costs, credit access constraints, and underdeveloped regulatory environments. This programme attempts to address these barriers by empowering the utilities that are trusted and well-established actors, to become the primary facilitators of behind-the-meter clean energy deployment. By integrating concessional financing, technical assistance and regulatory reform, the programme is designed to create a replicable, utility-led delivery model for the clean energy transition in the context of small island developing States (SIDS).

8. The proposal also positions this initial phase as a pilot for a broader regional roll-out, with Phase II expected to extend the IUS model to other CDB borrowing member countries. Tools, data systems, regulatory approaches and institutional lessons developed under Phase I will be leveraged to reduce costs and streamline future implementation efforts. Knowledge-sharing and regional cooperation are embedded into the design, with the intention of promoting replication and scale.

9. While distributed RE, energy efficiency and other DERs often offer compelling savings and relatively short payback periods, uptake in the Caribbean remains constrained. Many households and businesses are risk-averse and face limited access to affordable finance. Similarly, utilities in the region have historically been conservative in adopting new business models. The IUS Programme seeks to address this by ensuring that financing is offered on favourable terms, with modest initial on-bill repayment rates that help demonstrate value to end users. A key element of the programme design is the structured transfer of concessionality from GCF to utilities, and from utilities to customers. Several mechanisms are included to safeguard this pass-through, including caps on the spreads that utilities can apply to the blended GCF-CDB credit lines. This is essential to making clean energy technologies more accessible than traditional financing channels, while maintaining market discipline.

10. In summary, the IUS Programme proposes an innovative model to unlock distributed clean energy investments through utility leadership. Its success will depend on several factors, including the utilities' operational readiness, the effectiveness of the regulatory and institutional reforms, and the extent to which concessionality is passed through to end users. While the approach is novel for the region, it rests on a sound rationale and responds directly to the structural barriers facing clean energy deployment in Caribbean SIDS.

1.2 Impact potential

Scale: Medium

11. The proposed IUS Programme seeks to deliver both mitigation and adaptation benefits through the utility-led deployment of distributed RE, energy efficiency and other demand-side

Jamaica Public Services Company Limited is a regulated electricity utility in Jamaica. It is a limited liability company incorporated under the laws of Jamaica, and is majority owned by a consortium comprising the Marubeni Corporation in Japan and the East-West Power Corporation in the Republic of Korea (combined 80 per cent). The Government of Jamaica owns 19 per cent of the shares while a small group of minority shareholders control the remaining 1 per cent.

energy services in Barbados, Belize and Jamaica. These three Caribbean SIDS face increasing climate risks such as intensified storms, prolonged droughts, sea level rise and rising average temperatures, which are already affecting the reliability and cost of power generation and distribution. The programme responds to these structural challenges by enabling households and businesses to access and finance distributed energy solutions that are otherwise difficult to adopt due to cost, credit constraints and regulatory barriers.

12. Mitigation Potential: The mitigation impact is based on avoided grid electricity consumption through the installation of rooftop solar PV devices, solar water heaters, efficient air conditioning, lighting and battery systems. The AE estimates that the programme could avoid approximately 601,550 tonnes of carbon dioxide equivalent (t CO₂ eq) over the lifetime of these technologies, using grid emission factors published by the United Nations Framework Convention on Climate Change Harmonized Grid Emission Factor data set and adjusted for each country's national context and nationally determined contribution trajectory.

13. This figure reflects differentiated assumptions about technology lifespans, ranging from 5 years for lighting to 25 years for rooftop solar installations, and incorporates projected technology uptake across both residential and commercial users. Importantly, the mitigation potential varies significantly by country due to differing grid carbon intensities and market baselines. Jamaica, which has a relatively carbon-intensive grid, accounts for over half of the projected greenhouse gas savings (341,197 t CO₂ eq), followed by Barbados (215,219 t CO₂ eq) and Belize (45,134 t CO₂ eq).

14. The independent Technical Advisory Panel (iTAP) finds the mitigation estimation methodology to be sound and in line with Independent Fiscal Institution best practices, as detailed in annexes 3 and 22. However, as with many distributed energy programmes, realized emissions reductions will depend heavily on the rate of actual subloan uptake, sustained use of the technologies, and ongoing maintenance over time. Reliance of the IUS Programme on utilities to manage origination, installation and monitoring provides a plausible delivery channel, but performance should be closely tracked in implementation.

15. Adaptation Potential: The IUS Programme also articulates a clear, though modest, adaptation impact. The primary adaptation impact is framed around increasing access to efficient cooling technologies- particularly air conditioning systems- which can help mitigate the health and productivity risks associated with intensifying heat stress across the region. Based on uptake projections, the AE estimates that approximately 10,655 individuals (5,328 women and 5,327 men) will benefit directly from these adaptation measures, primarily through improved thermal comfort and energy cost savings in homes and small businesses.

16. An additional 30,000 indirect beneficiaries are projected, based on broader awareness, demonstration effects and potential market spillovers. While the indirect beneficiary figure is not backed by rigorous market modelling, it is a reasonable proxy for the broader ecosystem impacts expected through regulatory reform, stakeholder engagement and replication.

17. The iTAP notes that the adaptation logic is credible but would benefit from a more detailed vulnerability analysis and a clearer articulation of the causal pathways linking cooling access to resilience outcomes, particularly for marginalized or climate-sensitive populations. Annex 2b provides some baseline data but monitoring and evaluation at the implementation stage will need to track adaptation benefits more rigorously, ideally with disaggregated reporting and engagement of health or resilience-related institutions.

18. Enabling Environment and Structural Impact: Beyond direct mitigation and adaptation impacts, the IUS Programme contributes to long-term market development by supporting policy reform and institutional strengthening under Component 1. This includes establishing a regulatory framework for behind-the-meter solutions, enhancing utility and technology provider capacity, and building tools and systems that can be leveraged for future DER deployment. The emphasis on local stakeholder outreach and regional knowledge-sharing is

appropriate and consistent with the goal of scaling the model across additional borrowing member countries in a potential Phase II.

19. The AE has also proposed several mechanisms to ensure that concessionality from GCF is passed through to end beneficiaries, including limits on utility spreads and regulatory oversight of retail interest rates. This is particularly important given the AE's own limitations in offering long-tenor, low-interest financing from its balance sheet. The concessionality is expected to bring end-user loan rates down to approximately 4.2 to 6.0 per cent, below prevailing commercial offerings in the region, thereby enabling access for customer segments that would otherwise be excluded.

20. Risks to Impact Delivery: There are several risks to impact realization. First, the financial profile of many potential end users, especially low-income households, raises the possibility of loan default despite the economic case for investment. While utilities are experienced in customer screening, repayment performance will need to be monitored carefully, particularly in the context of inflationary pressure and income volatility. The iTAP felt that the responses of the AE concerning the ability of the EEs to perform the role of credit underwriters were not entirely convincing. We note that low-income households- those who would benefit most from energy efficiency and solar solutions- may present greater credit or payment risks. Utilities are not structured like banks and generally do not have strong underwriting capabilities. This dynamic can lead to higher default rates and, ultimately, regulatory pushback.

21. To mitigate the risk of defaults at higher rates than expected, or non-payment, the programme places emphasis on technologies that offer clear, near-term financial benefits to customers. This design choice is efficient, as it increases the likelihood that energy cost savings will offset repayment obligations from the outset. The use of on-bill financing also provides a degree of transparency by presenting both the savings and the repayment charges on the same utility bill, which may be particularly useful for customers with limited financial literacy. That said, it remains to be seen whether this level of transparency alone will be sufficient to ensure sustained repayment, particularly in lower-income segments. The technical assistance component includes customer outreach and awareness-raising activities, including the development calculators of energy savings, intended to reinforce repayment behaviour by illustrating the medium-term financial advantages of participation. These are helpful features, though their actual effectiveness in reducing credit risk will depend on the quality and reach of implementation.

22. In addition, political changes could disrupt alignment between national designated authorities, regulators and utilities. Although upstream work by development partners (e.g. Deutsche Gesellschaft für Internationale Zusammenarbeit and the Caribbean Community) has helped build institutional readiness, the IUS Programme remains exposed to shifts in public sector commitment or regulatory support. This risk is partially mitigated by the programme design, which includes technical assistance to regulatory bodies and deep utility engagement.

23. The proposal presents a robust mitigation case, a plausible adaptation narrative, and a credible enabling environment for long-term structural transformation in energy service delivery. While actual outcomes will depend on effective execution, customer uptake and institutional continuity, the programme design provides a sound foundation for achieving meaningful and scalable climate impact in the Caribbean SIDS context.

24. The iTAP assesses the impact potential of the IUS Programme as medium.

1.3 Paradigm shift potential

Scale: Medium

25. The proposal offers a well-considered approach to addressing entrenched structural barriers in the Caribbean energy sector and exhibits several characteristics consistent with the GCF definition of paradigm shift. The IUS Programme is designed to catalyse systemic

transformation by altering the roles of utilities, consumers and regulators in how distributed energy services are delivered, financed and valued.

26. The proposed IUS Programme aims to advance structural transformation in the energy sectors of Barbados, Belize and Jamaica by embedding distributed RE, energy efficiency and other demand-side technologies within utility operations. The central innovation of the programme lies in positioning regulated utilities as facilitators of customer-level DER deployment, blending financing, installation and service provision into a single utility-led offering. This shift from the traditional supplier model towards a more service-oriented role is both operationally novel and aligned with the broader decentralization of energy systems.

27. The IUS Programme also intends to empower customers through simplified access to clean energy technologies and behavioural tools, enabling greater participation in the energy transition. By integrating on-bill financing and user-facing applications (e.g. energy savings calculators), the programme seeks to normalize investment in DERs among households and micro, small and medium-sized enterprises (MSMEs) that have historically faced high barriers to participation.

28. In parallel, the IUS Programme includes a robust enabling environment component aimed at updating the legal, regulatory and administrative frameworks governing behind-the-meter DERs. This includes support for permitting processes, energy accounting modernization, and the development of performance-based incentive structures. These reforms are critical for anchoring the model in long-term institutional practice and ensuring it can be sustained and replicated without ongoing external concessional support.

29. The iTAP notes that while the programme's absolute scale is relatively modest, both in terms of greenhouse gas abatement and financing volume, its design is intentionally structured to serve as a model for replication. A future Phase II is envisioned to extend the approach to other CDB borrowing member countries. Knowledge-sharing efforts, digital platforms, and regional institutional engagement (e.g. through the Caribbean Centre for Renewable Energy and Energy Efficiency) are embedded in the design to support regional uptake.

30. However, while the IUS Programme is well-conceived from a distributed energy market development perspective, its paradigm-shifting potential must be assessed considering broader sector planning trends in the region, particularly in Barbados. Many SIDS, including the target countries, have articulated highly ambitious national RE targets. Barbados, for instance, has set a goal of reaching 100 per cent RE generation by 2030, and currently has over 500 megawatts of new RE capacity in various stages of development. This figure is more than double the country's current installed capacity and over three times its peak demand of ~141 megawatts.

31. In this context, there is a legitimate concern that a programme focused on customer-level DERs, particularly rooftop PV installations and efficiency, may draw attention and capital away from utility-scale RE investments that, in theory, offer greater scale and grid-level impact. Moreover, there is a risk that the most financially viable customers – those with creditworthiness and high consumption – will exit the grid or reduce their net load via DERs, potentially undermining the financial case for centralized RE investments and straining utility revenue models.

32. When asked about this, the AE acknowledged that many of the utility-scale RE plans articulated by national governments are aspirational and may not materialize due to permitting, financing and political constraints. In this light, the IUS Programme is presented as a more grounded pathway to climate action with greater ease of implementation. While this may be pragmatically valid, it nonetheless raises questions about coherence between distributed and centralized planning efforts, and whether the programme might inadvertently exacerbate systemic tensions within the power sector, particularly around grid stability, cost recovery and equitable allocation of subsidies and incentives.

33. The IUS Programme does include design features intended to mitigate some of these risks, such as utility ownership of the DER deployment process and integration with national

regulatory bodies. However, these alone may not be sufficient to reconcile the divergence between bottom-up distributed energy approaches and top-down utility-scale targets.

34. The long-term success of the IUS Programme will depend on how effectively it is integrated into broader sector planning, including decisions around tariff structures, capacity payments, and the sequencing of new utility-scale capacity additions. For instance, the iTAP notes that the distributed approach advanced here can address a distinct set of policy and infrastructure objectives that bolster paradigm shift, listed below.

35. **Resilience:** Distributed systems provide redundancy in the face of climate-related disruptions and offer a more granular pathway to energy security, particularly in storm-prone regions.

36. **Inclusivity:** Many Caribbean households and MSMEs have historically lacked access to financing for clean energy investments. The IUS model is specifically designed to address this gap, offering more equitable participation in the energy transition.

37. **Land-use efficiency:** In land-constrained island contexts such as Barbados, where siting large-scale RE assets is increasingly difficult, rooftop deployment is not only desirable but necessary.

38. **Energy efficiency:** The IUS Programme prioritizes not just RE but also efficient cooling and lighting – technologies that are typically underserved by both public procurement and private financial intermediaries.

39. **Local ownership:** Unlike many utility-scale RE projects, which are typically led by foreign investors, the distributed model has the potential to foster greater domestic ownership of clean energy assets.

40. Taken together, these attributes reinforce the programme's contribution to broader systemic change in how energy is produced, delivered and consumed. However, the success of this transformation will depend, to a significant extent, on the capacity of the participating utilities to adapt to new roles and business models, the regulatory follow-through in each country, and the eventual cost-effectiveness of scaling the model.

41. For that reason, the iTAP assesses the paradigm shift potential of the IUS Programme as medium as it introduces a promising and replicable utility-led model for distributed energy deployment and has the potential to alter how utilities, customers and regulators approach clean energy investment. However, these gains must be weighed against unresolved tensions with national utility-scale RE ambitions, particularly in contexts like Barbados where DER proliferation may affect the bankability of planned large-scale projects. Absent clearer articulation of how the IUS model complements broader grid expansion strategies, its contribution to systemic transformation remains promising but not assured.

1.4 Sustainable development potential

Scale: Medium to high

42. The funding proposal outlines a range of expected sustainable development co-benefits, many of which are plausible and well-integrated into the IUS Programme core design. These span environmental, social, economic and gender dimensions, and align with multiple Sustainable Development Goals (SDGs), notably SDG 7 (affordable and clean energy) and SDG 13 (climate action), as well as secondary contributions to SDG 3: good health and well-being; SDG 5: gender equality; SDG 8: decent work and economic growth; SDG 9: industry, innovation and infrastructure; and SDG 17: partnerships for the goals. While some outcomes will depend heavily on effective implementation and uptake, the programme includes mechanisms that could support meaningful and measurable development co-benefits.

43. **Environmental co-benefits:** The IUS Programme supports a shift away from fossil fuel-based generation through the deployment of distributed RE and energy efficiency technologies. Unlike utility-scale RE installations, the proposed subprojects – rooftop PV, efficient cooling and lighting, and solar water heaters – are small-scale and embedded in existing structures,

minimizing risks of ecological disruption or land-use conflicts. This is particularly relevant in island contexts such as Barbados, where land-use constraints limit the feasibility of large-scale RE development.

44. In addition to emissions reduction, the transition to cleaner distributed generation is expected to improve local air quality, particularly in urban areas reliant on diesel-based generation. These environmental benefits are reasonably substantiated and consistent with regional energy and health policy objectives.

45. Economic co-benefits: The IUS Programme presents a credible pathway to delivering financial benefits to both utilities and consumers. On the customer side, the estimated net present value of savings across subproject lifetimes exceeds USD 230 million, according to the AE's integrated financial model. These savings could be reinvested by households and MSMEs or used to offset consumption-related costs. Utilities may also benefit from the diversification of revenue streams and the opportunity to develop service-based business lines that may prove more resilient in the face of structural changes in the electricity market.

46. At the broader macroeconomic level, the IUS Programme is expected to reduce fossil-fuel imports, which remain a significant pressure on foreign exchange reserves across the three participating countries. Local job creation is another potential benefit, particularly through the installation, maintenance and support services associated with DER deployment. While exact employment figures are not specified, the focus on distributed solutions may support a more geographically dispersed and locally embedded labour market than utility-scale alternatives.

47. Social co-benefits: The IUS Programme aims to improve household-level access to clean energy technologies through inclusive financing mechanisms, notably on-bill repayment. This feature is particularly relevant in the Caribbean, where access to commercial credit is limited and many potential users, especially low-income households and informal businesses, are effectively excluded from the clean energy transition. By anchoring repayment in existing utility relationships, the programme removes the need for traditional collateral or credit history, potentially widening participation.

48. Health-related co-benefits: Albeit indirectly, health co-benefits are also likely from reduced diesel usage which may lower particulate matter and other pollutants, with attendant benefits for respiratory health. The technical assistance component also includes targeted support for low-income households in the form of free energy audits, which could enhance equity and ensure greater inclusion in benefit distribution.

49. Gender equality and social inclusion: The proposal is supported by a detailed Gender Action Plan and gender assessment and includes gender-disaggregated indicators in its results framework. The IUS Programme commits to concrete actions across four areas: mainstreaming gender in policy and planning; tailoring outreach to women and other marginalized groups; developing inclusive IUS product offerings; and strengthening utility systems for sex-disaggregated beneficiary tracking.

50. These measures are designed to improve both participation and outcomes for women, youth, persons with disabilities, and other underserved populations. However, as with other co-benefits, their impact will depend on the depth and consistency of implementation. The plan to monitor uptake among female-headed households and vulnerable groups is appropriate, though this data should be regularly reported and independently verified.

51. The proposal integrates credible pathways to social inclusion, local economic benefits, and environmental health improvements, with appropriate attention to gender and equity considerations. The inclusion of customer-facing support tools and technical assistance for marginalized groups is a strength. While many co-benefits are contingent on implementation quality and uptake, the programme design creates a strong enabling framework for their realization.

52. The iTAP assesses the sustainable development potential of the IUS Programme as medium to high.

1.5 Needs of the recipient

Scale: High

53. Barbados, Belize and Jamaica, the participating countries, face well-documented climate and development challenges that align with the GCF prioritization of SIDS and climate-vulnerable populations. All three are highly exposed to climate risks such as extreme weather events, sea level rise, heat stress, and disruptions to key economic sectors like tourism and agriculture. At the same time, their energy sectors remain heavily reliant on imported fossil fuels, with limited local capacity to self-finance the transition to more resilient and sustainable systems.

54. Despite being classified as middle-income countries, this status belies substantial underlying vulnerabilities. High debt burdens and restricted fiscal space limit public investment capacity, while shallow domestic capital markets constrain private sector financing of distributed energy technologies. The IUS Programme directly addresses these structural limitations by providing concessional financing and technical assistance that the national governments and utilities would not otherwise be able to mobilize at scale. This is particularly relevant considering the countries' stated commitments under their nationally determined contributions, which include ambitious emissions reduction and energy access goals but often lack credible, funded implementation pathways.

55. Importantly, the IUS Programme targets persistent equity gaps within these countries. While DER adoption has accelerated among higher-income customers, lower-income households and informal MSMEs, which are often those most exposed to climate and economic shocks, have been largely excluded from participation in the energy transition. The programme's inclusive on-bill financing model, combined with free energy audits for lower-income customers, offers a pragmatic response to these access barriers. By leveraging the existing utility-customer relationship, the model bypasses traditional credit-based lending channels that have historically disadvantaged marginalized populations.

56. The IUS Programme also integrates a social inclusion and gender framework. Through the Gender Action Plan and associated training and outreach, its aim is to improve governance, participation and responsiveness among energy sector institutions – including utilities – towards groups that have been traditionally underserved, such as women, persons with disabilities, LGBTQIA+ communities, and Indigenous Peoples. While the implementation of these measures will require close monitoring, the aim to position these groups as active participants and not just passive recipients is well-aligned with the GCF emphasis on a just and inclusive transition.

57. The iTAP notes that the geographic targeting is limited to three countries in this first phase, and its reach in absolute terms remains modest relative to national populations. However, the selection of countries is appropriate given both their acute vulnerability and the potential for institutional readiness and scale-up through a planned Phase II. The programme design reflects an awareness of the constraints faced by the governments and utilities in launching new delivery models without external support, particularly in a high-risk, resource-limited context.

58. The programme responds directly to structural vulnerabilities in climate exposure, fiscal capacity and financial access, particularly for low-income and marginalized groups. While limited in geographic scope and initial coverage, the model is appropriately targeted and well-aligned with GCF priorities for SIDS and inclusive development. The success of the programme will depend in part on its ability to meaningfully reach underserved populations and support institutional capacity-building that persists beyond the life of the programme.

59. The iTAP assesses the needs of the recipient as high.

1.6 Country ownership

Scale: High

60. The IUS Programme demonstrates strong alignment with the national climate and energy priorities of the three participating countries – Barbados, Belize and Jamaica – all of which have issued no-objection letters. The objectives and design of the programme are consistent with the countries’ nationally determined contributions, as well as sectoral strategies and policy frameworks, including the National Energy Policy (2019–2030) of Barbados, the Sustainable Energy Strategy and Action Plan (2014–2033) of Belize and the Vision 2030 Development Plan of Jamaica. The focus on scaling up distributed RE, energy efficiency and DERs corresponds with each country’s stated intention to reduce emissions and enhance energy resilience.

61. The proposal reflects broad-based engagement with national stakeholders. Regulators, energy ministries, national designated authorities, and utilities were closely involved in programme preparation. Previous upstream work, supported by partners such as Deutsche Gesellschaft für Internationale Zusammenarbeit and the Caribbean Community Secretariat, has also contributed to country-level readiness for programme implementation. These engagements signal meaningful institutional support and suggest that the programme has been shaped by local priorities and context. The regional scope is further strengthened by engagement with organizations such as the Caribbean Centre for Renewable Energy and Energy Efficiency and the Caribbean Electric Utility Services Corporation, which are well positioned to support knowledge exchange and promote replication.

62. The iTAP also notes positively that all three utilities proposed as EEs have prior experience with DER deployment and have undergone a capacity assessment to confirm their suitability. Under the programme, the utilities will implement billing reforms, oversee subproject execution, and support customer engagement, with targeted technical assistance to address identified institutional capacity gaps.

63. That said, the iTAP has some reservations regarding the level and nature of the national co-financing contribution. The proposal identifies an in-kind contribution of USD 25 million from the utilities, in the form of investments in advanced metering infrastructure (AMI). While AMI can support grid modernization and enable more granular DER integration over time, it is not directly linked to the financed subprojects nor to the programme’s estimated climate impact. As such, there is a legitimate question as to whether this investment constitutes true co-financing or rather reflects a continuation of business-as-usual capital expenditure. The IUS Programme documents assert that the AMI deployment is complementary to the IUS objectives and forms part of the broader enabling ecosystem. However, from a concessionality standpoint, the iTAP typically looks for more direct financial commitment, ideally with measurable risk-sharing or matching funds, when substantial concessional financing is requested.

64. While this does not undermine the overall alignment with country priorities, it does point to an opportunity for deeper national financial participation in future phases of the programme. In particular, there may be scope for governments or utilities to take a more active financial role as the IUS model matures, and implementation risks decrease.

65. The iTAP assesses country ownership as high, reflecting strong policy alignment, inclusive stakeholder engagement and the active involvement of national institutions in programme design and execution.

1.7 Efficiency and effectiveness

Scale: Medium to high

66. The IUS Programme presents a reasonably strong case in terms of cost-efficiency, alignment of financial instruments, and structuring of concessionality. This is particularly notable given the geographic and structural constraints associated with SIDS, where distributed climate mitigation and adaptation solutions often entail higher marginal costs than in continental or higher-density settings.

67. Cost-effectiveness metrics: The programme is expected to achieve mitigation benefits of approximately 601,550 t CO₂ eq avoided, resulting in an estimated cost of USD 70 per tonne for

the overall programme and USD 43 per tonne for the GCF contribution. These figures are broadly consistent with GCF precedents for distributed energy interventions in SIDS contexts and are deemed competitive when compared to mitigation programmes with similar technical profiles.

68. On the adaptation side, the estimated cost per direct beneficiary is approximately USD 3,938, based on the 10,655 direct beneficiaries targeted through improved resilience to heat stress. While this is relatively high by GCF adaptation standards, it must be understood within the broader context of the dual objectives of the programme. The costs are associated not only with direct technology uptake but also with the establishment of institutional models, digital tools and customer-facing systems expected to deliver benefits beyond the initial implementation window.

69. Importantly, the programme also estimates over USD 260 million in financial savings for participating customers over the productive lifespan of the financed technologies. These net benefits are expected to accrue through avoided electricity costs from distributed solar energy and savings from energy-efficient appliances and systems. This represents a meaningful value proposition for both customers and the utilities, although, as CDB acknowledges, actual results will depend on technology performance, repayment behaviour and subproject uptake.

70. Concessionality structure and co-financing: The financial structure involves a USD 20.8 million GCF concessional loan blended with USD 16.2 million from CDB on a *pari passu* basis. Participating utilities will on-lend these funds to end users through structured credit lines, with interest rates anticipated to fall in the range of 4.5 to 6.0 per cent, depending on local regulatory determinations and risk-sharing discussions during the first year of the programme. These terms compare favourably to prevailing market rates in the region, which typically begin at 6.5 per cent or higher for comparable unsecured loans.

71. In-kind contributions of USD 25 million from the utilities in the form of AMI are also included in the co-financing total. While CDB maintains that AMI is a critical enabler of the IUS model and aligned with the programme's theory of change, the iTAP notes that this investment may have been pursued independently of the IUS Programme and appears to reflect business-as-usual utility modernization. As such, its inclusion as co-financing, while defensible, may overstate the degree of additionality and direct financial commitment from national stakeholders.

72. The IUS Programme is nonetheless likely to catalyse further investment downstream. It is framed explicitly as a "bridge to bankability", designed to establish proof of concept and performance track record for utility-led DER delivery, thereby enabling the utilities to secure additional finance with less concessionality in future cycles. This ambition is credible, particularly given the integrated capacity-building support and the regional focus on replication, but it remains contingent on execution quality and early success.

73. Risk allocation and incentive alignment: CDB will act as both lender and programme manager, receiving lending returns while benefiting from substantial GCF-funded technical assistance, which comprises over 13.5 per cent of the total GCF contribution. While this structure may appear to raise moral hazard risks, whereby CDB captures upside returns while GCF bears a disproportionate share of de-risking, the AE has provided a clear rationale. It argues that the technical assistance is essential to establish a novel delivery model in the Caribbean, where such utility-led energy service offerings have not previously been implemented. Furthermore, grant-financed activities include compliance with GCF safeguards, social inclusion outreach and programmatic knowledge dissemination, all of which extend beyond direct support to the utilities.

74. That said, the iTAP would have preferred to see a more differentiated risk structure, such as a subordinated tranche, wherein CDB exposure would be more explicitly aligned with performance outcomes. The current *pari passu* structure may be acceptable, but the rationale rests heavily on the need for both financiers to maintain equivalent tenors and exit points. While CDB emphasizes that its own lending is not commercially priced and is adjusted to reflect

its borrowing costs, the absence of a more risk-sensitive capital stack reduces the strength of the incentive alignment framework.

75. To mitigate the risk of utilities over-capturing the benefits of concessional financing, CDB has indicated that spreads charged to end users will be capped in the subsidiary agreements signed with each utility. Regulatory authorities will also play a role in rate-setting during the programme's first year, with CDB monitoring subloan terms through regular reporting. These are positive design features, but enforcement over time will require sustained regulatory engagement and transparent monitoring of on-lending terms and customer-level impact.

76. Customer uptake and delivery risk: A key determinant of programme success will be the ability to generate sufficient customer uptake. In other IUS contexts, customers have rejected offers when savings from installed technologies did not clearly outweigh the costs of loan repayments, or when participation processes were too complex. The programme attempts to address these challenges by providing:

- (a) On-bill financing mechanisms that combine costs and benefits in a single, transparent billing statement;
- (b) User-facing tools, including energy savings calculators and online portals, to demystify technology performance and financial implications;
- (c) Technical assistance for low-income customers, including free energy audits and help desk support; and
- (d) Prequalification processes and performance standards for Technology Service Providers (TSPs) to ensure service quality.

77. While these are thoughtful features, the multi-stakeholder structure and regulatory dependencies introduce a degree of complexity that may affect implementation. Customer satisfaction and referral effects will be critical to achieving scale, particularly given the relatively modest direct beneficiary target (10,655 people) in the initial phase. The programme includes provisions to delay or suspend further technical assistance disbursements if uptake stalls – an important accountability mechanism that improves delivery discipline.

78. Coordination with other initiatives: The iTAP notes that the IUS Programme does not appear to be meaningfully coordinated with other major climate finance initiatives in the region, including the GCF-supported Barbados Blue Green Bank. While CDB maintains that these initiatives are complementary and focus on more traditional, bank-led financing models, it would have been helpful to include more detail on how pipeline sharing will be managed to avoid duplication or market distortion, especially in relatively small economies like Barbados.

79. Accredited entity fees: The AE fee of 6.7 per cent is within the range of comparable programmes, though slightly higher than the iTAP typically expects for an intervention with a large technical assistance component. CDB justifies the fee through a bottom-up costing model and points to higher personnel and travel costs in the Caribbean. While these factors are valid, the iTAP notes that the programme's relatively high cost of coordination should be monitored and assessed alongside actual disbursement and performance metrics over time.

80. The design reflects a strong understanding of regional delivery constraints and incorporates several safeguards to promote cost-effectiveness, manage concessionality and enforce performance-based disbursement. However, questions remain around the depth of national co-financing, the absence of differentiated risk-sharing between lenders, and the need for clearer coordination with parallel financing mechanisms. The programme is well-structured to catalyse downstream investments and shift market norms, but sustained execution and oversight will be critical to realizing its full potential.

81. The one area we would have preferred to see more ambition involves the feasibility of introducing a more risk-sensitive capital structure in the funding proposal, such as a subordinated tranche for CDB financing. This would enhance incentive alignment by ensuring that CDB retains greater exposure to performance outcomes. If the *pari passu* structure is

retained, we would encourage the Secretariat should seek a more robust justification, including an assessment of whether alternative structuring options could better achieve alignment while preserving the IUS Programme financial viability and implementation timeline.

82. The iTAP assesses the efficiency and effectiveness potential of the IUS Programme as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

83. The funding proposal makes a solid attempt to address a significant climate finance need in the Caribbean. The iTAP finds that the IUS Programme presents credible climate rationale, addresses structural financing gaps and demonstrates potential for long-term paradigm shift.

84. The iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP275)

Proposal name:	Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase 1) Programme
Accredited entity:	Caribbean Development Bank (CDB)
Country/(ies):	Barbados, Belize, Jamaica
Project/Programme size:	Medium

Impact potential
The iTAP assessment is noted with thanks. CDB appreciates the favourable review of the impact potential of this programme.
Paradigm shift potential
The iTAP assessment is noted with thanks. CDB appreciates the favourable review of the paradigm shift potential of this programme.
Sustainable development potential
The iTAP assessment is noted with thanks. CDB appreciates the favourable review of the sustainable development potential of this programme.
Needs of the recipient
The iTAP assessment is noted with thanks. CDB appreciates the favourable review with regards to the needs of the recipients.
Country ownership
The iTAP assessment is noted with thanks. CDB appreciates the favourable review with regards to country ownership.
Efficiency and effectiveness
The iTAP assessment is noted with thanks. CDB appreciates the favourable review of the efficiency and effectiveness of the programme.
Overall remarks from the independent Technical Advisory Panel:
CDB appreciates the thorough and constructive review, and thanks the iTAP for recommending that the proposal be approved by the GCF Board without conditions.

Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase I) Programme

Annex 8: Gender Assessment and Gender Action Plan (GAP)

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Acronyms / Abbreviations

A/C	Air Conditioning
AE	Accredited Entity
BEL	Belize Electricity Limited
BIT	Board of Industrial Training
BLP	Barbados Light and Power
BMC	Borrowing Member Country
BREA	Barbados Renewable Energy Association
BZD	Belize Dollars
C-COOL	Caribbean Cooling Initiative
CARICOM	Caribbean Community
CCREEE	Caribbean Centre for Renewable Energy and Energy Efficiency
CDB	Caribbean Development Bank
CDR	Culturally Defined Responsibilities
CEDAW	Convention for the Elimination of All forms of Discrimination Against Women
CEI	Caribbean Energy Initiative
CHPA	Central Housing and Planning Authority
CoC	Code of Conduct
DFC	Development Finance Corporation
EE	Energy Efficiency
ELPA	Electric Light and Power Act
EMISDE	Energy Matrix Diversification and Institutional Strengthening of the Department of Energy
EU	European Union
FHH	Female Headed Household
FIT	Feed In Tariff
FPIC	Free, Prior and Informed Consent
GAP	Gender Action Plan
GBV	Gender-Based Violence
GCCCSAP	Gender and Climate Change Strategy and Action Plan
GCF	Green Climate Fund
GDP	Gross Domestic Product
GEPOS	Gender Equality Policy and Operational Strategy
GESI	Gender Equality and Social Inclusion
GHG	Greenhouse Gas
GRM	Grievance Redress Mechanism
GSDS	Green State Development Strategy
HOPE	Home Ownership Providing Energy
HFS	Hydrofluorocarbons
HPS	High-Pressure Sodium
HVAC	Heating, Ventilation, and Air Conditioning
HWI	Hot Water Immersion
ICT	Information and Communications Technology
IDB	Inter-American Development Bank
ILO	International Labour Organisation

IP	Indigenous People
ITVET	Institute for Technical and Vocational Education and Training
IUS	Integrated Utility Service
J-CCCP	Japan-Caribbean Climate Change Partnership
JMD	Jamaican Dollars
JNCS	Jamaica's National Cooling Strategy
JPS	Jamaica Public Service
LCDS	Low Carbon Development Strategy
LCP	Local Content Policy
LED	Light Emitting Diode
LFS	Labour Force Survey
LGBTQIA	Lesbian, Gay, Bisexual, Transgender, Queer, Intersex, Asexual
MICS	Multi-Indicator Cluster Survey
MOHA	Ministry of Home Affairs
N/A	Not Applicable
NDA	National Designated Authority
NDC	Nationally Determined Contribution
NGO	Non-Governmental Organisation
NHT	National Housing Trust
NOAU	National Ozone Action Unit
NPGE	National Policy on Gender Equality
NPV	Net Present Value
OCC	Office of Climate Change
POWA	Productive Organization for Women in Action
PUC	Public Utilities Commission
PV	Photovoltaic
PWD	Person With Disability
R2RP	Roofs to Reefs Program
RAC	Refrigeration and Air Conditioning
RE	Renewable Energy
RGA	Regional Gender Analysis
SBB	Small Business Bureau
SDG	Sustainable Development Goal
SIDS	Small Island Developing State
STEM	Science, Technology, Engineering, and Math
SWH	Solar Water Heater
TNES	Transitioning to National Energy Security
TOR	Terms of Reference
UN	United Nations
UNICEF	United Nations Children's Fund
USD	United States Dollars
VAWG	Violence Against Women and Girls
WEEP	Women Empowerment in Energy Programme
WIRE	Women in Renewable Energy

1 Introduction

In accordance with GCF gender policies and guidance, the gender assessment is done as a critical element of the project proposal to outline how the project while attempting to meet its goal of supporting the target countries' transition to renewable energy states, will benefit vulnerable groups and how in particular will vulnerable groups be able to participate in this project. The gender assessment will elucidate on factors that will impair vulnerable groups' ability to benefit fully and identify areas of entry to address these issues while advancing gender equality and supporting a Just Transition.

The Caribbean Development Bank (CDB) is in the process of implementing the “Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean” (Phase I) Programme in the Caribbean, which is a Green Climate Fund (GCF) funded activity. This Phase of the project will have three Borrowing Member Countries (BMCs) participating in the programme and they are as follows: Barbados, Belize, and Jamaica. These three BMCs have pledged to reduce greenhouse gas (GHG) emissions through the adoption of renewable energy (RE) to facilitate energy transition and expanding the scale of energy efficiency (EE) measures.

The GCF is committed to maximising the impact of its funding for adaptation and mitigation while ensuring environmental, social, economic and development co-benefits are advanced and a gender-sensitive approach is taken¹. With its updated 2020 gender policy, the GCF is committed to promote and integrate gender equality in its investment criteria. The Gender Policy has three main objectives which are as follows:

1. Utilise a comprehensive gender approach applied by the institution and its network of partners which include accredited entities (AEs), national designated authorities (NDAs) and focal points to support climate change interventions and innovations.
2. Promote climate investments that advance gender equality and minimise social, gender and climate related risks through climate change mitigation and adaptation measures.
3. Contribute to shrinking the gender gap caused by climate change-exacerbated social, economic and environmental vulnerabilities through mainstreaming gender equality in climate investments.

The gender policy's application to the IUS Project would entail:

- Gender inequality is addressed/reduced and through enhanced stakeholder engagement can generate sustainable livelihoods opportunities, promote health and wellbeing, and increase resilience to climate-induced shocks and risks for men and women.
- An enabling environment among GCF stakeholders built on existing policy commitments, social and environmental standards and redress mechanisms is supported and sustained in order to achieve gender equality.
- The comprehensive, systematic, and participatory collection of sex-disaggregated data and the execution of qualitative and quantitative gender analysis is supported to effectively monitor, report and evaluate project impact.

¹ “Gender Policy 2020-2023” retrieved from <https://www.greenclimate.fund/document/gender-policy>

The Assessment builds on the previous work in the baseline assessment and seeks to provide a deeper understanding of how the proposed technologies (solar water heaters, rooftop solar photovoltaic, efficient cooling and HVAC, and LED lighting) will impact populations of the target countries as well as identify barriers to uptake.

The Assessment is expected to inform:

- the gender mainstreaming approach for the GCF proposal;
- capacity-building activities in the overall proposal;
- gender-sensitive monitoring and evaluation tools that may be needed; and
- Institutional policy, programming and operational actions/steps that may be needed in the short, medium and or long-term to support the IUS model implementation in an equal, inclusive and just manner.

1.1 Limitations

This assessment utilised a mixed method approach i.e. a desk-based research and stakeholder engagement. However, this process was still hindered by a few factors including:

- **Outdated data** – the region has limited human resources funding and infrastructure available to execute frequent data collection and analysis activities which results in the use of data at least 5 years old in this assessment. Despite the recent efforts of Caribbean Centre for Renewable Energy and Energy Efficiency (CCREEE) to narrow this gap, some of the essential questions for the gender assessment still are not generally covered or readily available.
- **Data availability** – access to data is a weak point in the Caribbean as many governments and agencies possess updated data but have issues with processing, analysing, and publishing said data for public access, also due to limited human resources and infrastructure. Due to demand also for consultations and engagement, there was limited time to follow up in detail with the Statistics Departments of each country to inform this draft assessment.
- **Limited access to critical documents** – throughout various engagements with government stakeholders, we became aware of policies in development that could affect and possibly inform this project but despite our best efforts, we were unable to secure access to these documents (drafts included). This includes the draft Energy Policy for Jamaica as well as the new access policy for Belize, amongst others.
- **Inaccessibility of key stakeholders** – some of the Executing Entities (the utilities) did not participate in the stakeholder engagement process and in some cases cited other priorities or the need to see a project document prior to their engagement on questions relating to gender and social issues. The latter suggests a lack of understanding of the requirements and the process flow for GCF proposals. Their engagement, as the proposed executing agencies, is critical to obtaining information that would inform aspects of the gender action plan that follows the Gender Assessment including technical assistance and capacity-building actions as possible elements of more than one Component of the proposed project.

As it stands therefore, the Gender Assessment reflects a few key limits and limitations which are also of value to the Assessment itself in terms of indirectly assessing knowledge, practice and behaviours on gender, gender mainstreaming, women's empowerment and social inclusion in the sector as well as for this specific project.

1.2 Challenges

This assessment was meant to be informed by desk-based research and stakeholder engagement which would give insight into the enabling environment in each country by

identifying the current circumstances the IUS would be operating in while also gaining knowledge on key stakeholders' capacities and practices on gender equality and social inclusion mainstreaming in energy sector activities and transformations. Executing the stakeholder engagement aspect of the process within the given tight timeframe was greatly hindered by various factors. Engaging with key stakeholders was inhibited by time conflicts, limited personnel availability and the clearance of engagements of key government stakeholders. Due to the difficulty in scheduling meetings and the limited time available to continue consultations and complete the assessment, stakeholder engagement was closed off on June 9th to facilitate the preparation of this Draft Report.

2 Methodology

Key chapter takeaway

A diverse range of stakeholders were met, yielding responses from 50/50 split of male and female voices. The insights and information gained from these engagements specifically information centered around engagement and processes observed and performed by key stakeholders. Stakeholder engagement also captures the lived experiences of the stakeholder and raised awareness on ongoing work that was not available to the public.

This assessment used a two-step approach. The first step entailed the desk-based research which involved gathering existing analysis on energy, renewable energy, energy and gender, and renewable energy transformations to obtain information that could shed light on how this project may impact persons, identify any possible lessons learned in previous works and analysing the existing enabling environment for renewable energy. The research applied a renewable energy lens to the current socio-economic conditions within each country to understand how varying factors affect renewable energy usage, uptake and appetite for renewable energy technologies.

A gender analysis rating tool was used to assess relevant policies based on how gender-responsive they are (see Figure 2-1). This diagnosis involved reviewing policies and identifying the elements that speak to gender, women, and other marginalised groups and the commitments/intent made by those policies on issues of (i) engagement, (ii) equity and justice, (iii) inclusion, (iv) protection/mitigation of risks/impacts. This is an important perspective in addressing issues of a just transition and in meeting GCF requirements for mitigation investments including those with adaptation co-benefits.

Figure 2-1: Policy Gender Rating Tool

Gender Rating	Description
Gender Neutral (no mention)	The policy makes no mention or does not reference gender or women.
Gender Aware	Gender specific norms, roles and issues are acknowledged within the policy.
Gender Sensitive	Measures are outlined within the policy to ensure gender inequalities are not exacerbated as a by-product of the policy's actions.
Gender Responsive	The policy acknowledges and identifies differences and inequalities between men and women and sets out actions which seeks to rectify these issues.
Gender Transformative	The policy seeks to resolve fundamental structural inequalities and challenges social and cultural norms.

The second step builds on the first step and used those findings to shape the stakeholder engagement to follow. Key stakeholders were identified, and a stakeholder engagement plan was developed (see Annex A). The findings were used to develop PowerPoint presentations to be used with stakeholders that help guide the conversation by providing context and key questions for stakeholders to answer ([see presentations here](#)). These questions were tailored for engagement with non-governmental organisations (NGOs) as well as government stakeholders. Similar presentations were prepared for engagement with utilities. Engagement with diverse stakeholders sought to gain an understanding of their engagement on gender and

energy and with energy activities, their capacities and experience with gender and gender mainstreaming particularly within a renewable energy project and their perspectives on issues that need consideration in the proposed IUS model approach. In addition to this, engagement with government stakeholders helped to identify existing, past or possible approaches and policies used when responding to the needs of the general public and vulnerable groups.

Executed in three phases, we consulted with twenty-eight organisations in total (13 NGOs and 09 governmental organisations) across the three countries (see Annex B for the list of stakeholders engaged). Phase 1 entailed engaging NGOs who worked with vulnerable groups and/or in renewable energy. In Phase 2, outreach was made to government stakeholders including energy ministries/agencies, public utilities commissions and gender bureaux. Phase 3 was intended to target the utilities (the Executing Entities of the project) but was ultimately postponed due to a lack of response. Twenty-eight (44) persons, in total, were consulted – 14 men and 14 women (see Figure 2-2).

Figure 2-2: Stakeholder Breakdown By Organisation and Gender

Stakeholders	Female	Male
Barbados		
NGOs	3	2
Government	3	3
Belize		
NGOs	2	4
Government	0	2
Jamaica		
NGOs	4	1
Government	2	2
Total	14	14

Stakeholder Engagement Focus

From the desk-based research, issues of finance and access to finance were identified as barriers to this project, and the engagement sought to explore additional factors that might limit a person's access to finance that may not have been covered in the desk-based review/research? (see Figure 2-3). Obtaining information would help to inform possible financial strategies that the project may need to adopt to ensure access to those who need financial assistance.

Figure 2-3: Financial Barriers identified in research



Financial Barriers

Difficulties in accessing loans from financial institutions based on requirements in place for approval

- Apart from strict requirements that may be difficult for some persons to meet, are there other existing barriers to accessing loans from financial institutions (e.g. client discrimination)?
- What are some accessible financial measures that are utilised by low-income persons?

Hesitancy based on upfront, ongoing and future maintenance costs

Disinterest in investing DER technologies based on recent purchases on related technologies

Key to ensuring the project meets its GCF gender requirements was to identify the existing capacities of those who will be charged in delivering and supporting this project and their experience with mainstreaming gender throughout their projects as well as identifying potential partners to support gaps. This insight will be used to determine whether there will be a need for gender sensitisation and capacity-building in the gender action plan to empower key project actors. Figure 2-4 and Figure 2-5 provide the questions used to obtain this information from stakeholders.

Figure 2-4: Gender Mainstreaming Questions for NGOs



Gender Mainstreaming in the IUS Project

Gender mainstreaming will not be possible without first performing a baseline gender capacity assessment on all related actors (ministries, utilities, regulatory bodies).

Gender mainstreaming is the process of integrating gender equality principles into policies, plans and programmes. This entails accounting for the unique aspects that can be attributed to the diverse, social groups people can belong to in planning and development which ensures resources are allocated to meet the needs of all individuals.

Does your organisation have any experience with mainstreaming gender in its past work?

Does your organisation have the capacity to execute gender mainstreaming activities?

Has your organisation ever been engaged by government ministries/agencies or development partners to support in the development or implementation of energy/renewable energy projects?

Does your organisation possess the capacity to provide gender sensitisation training or gender-related technical inputs?

Figure 2-5: Gender Mainstreaming Questions for Government Stakeholders



Gender Mainstreaming in the IUS Project

Gender mainstreaming will not be possible without first performing a baseline gender capacity assessment on all related actors (ministries, utilities, regulatory bodies).

Gender mainstreaming is the process of integrating gender equality principles into policies, plans and programmes. This entails accounting for the unique aspects of men and women in planning and development which ensures resources are allocated to meet the needs of men and women.

Does your organisation have any experience with mainstreaming gender in its past work?

Does your organisation have the capacity to execute gender mainstreaming activities?

Does your organisation possess a gender policy or is it in the process of developing a gender policy?

Has your organisation ever engaged local women or gender groups, organisations or experts in the development or implementation of projects particularly energy/renewable energy projects?

Does your organisation utilize a grievance redress mechanism for projects?

One out of the three countries have large indigenous populations, and these groups tend to live very differently from regular citizens in their country. It, therefore, became important to gain an understanding of the context within which they live as they face unique barriers to their way of life and location. Figure 2-6 shows the questions aimed at gaining insight into these areas of interest.

Figure 2-6: Understanding the Indigenous Context and their needs



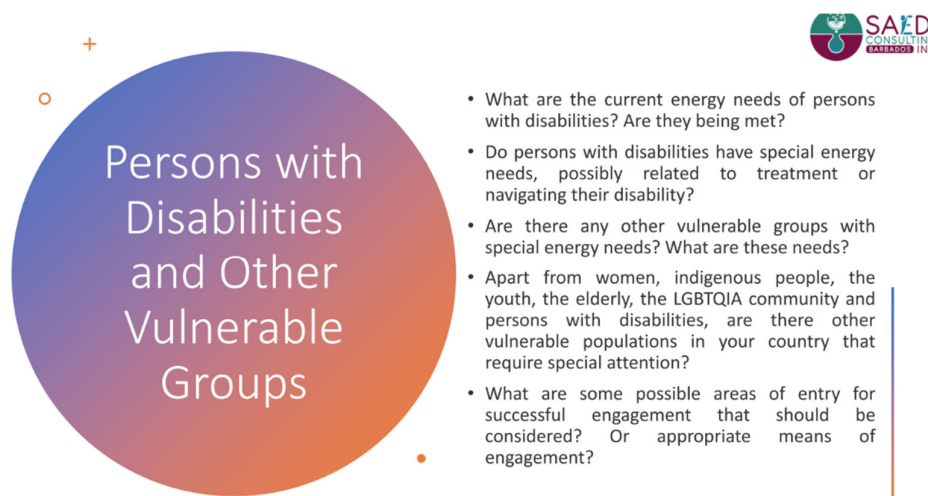
Indigenous People and Communities

- Noting that most indigenous peoples communities tend to be located in more rural parts of their countries and tend to have higher poverty rates, do indigenous people face any discrimination in your country and any specific access risks or challenges?
 - What are the energy needs of indigenous people? Are they being met?
 - Are there any systems (formal or informal) in place that limit or affect their livelihoods and economic ability in your country?
 - Are there any social/cultural factors related to indigenous people that need to be considered within a project of this type? Issues or specific dynamics in communities?
 - Are there good practices for engaging with indigenous people on RE/EE that we can build on?

Persons with disabilities are another special group of interest to be considered within the setting of this project as they as traditionally ignored and not consulted in the typical project development process, often leaving them unable to benefit from a project due to the lack of consideration made on their behalf. Acknowledging this and unique circumstances of PWDs, it was critical to

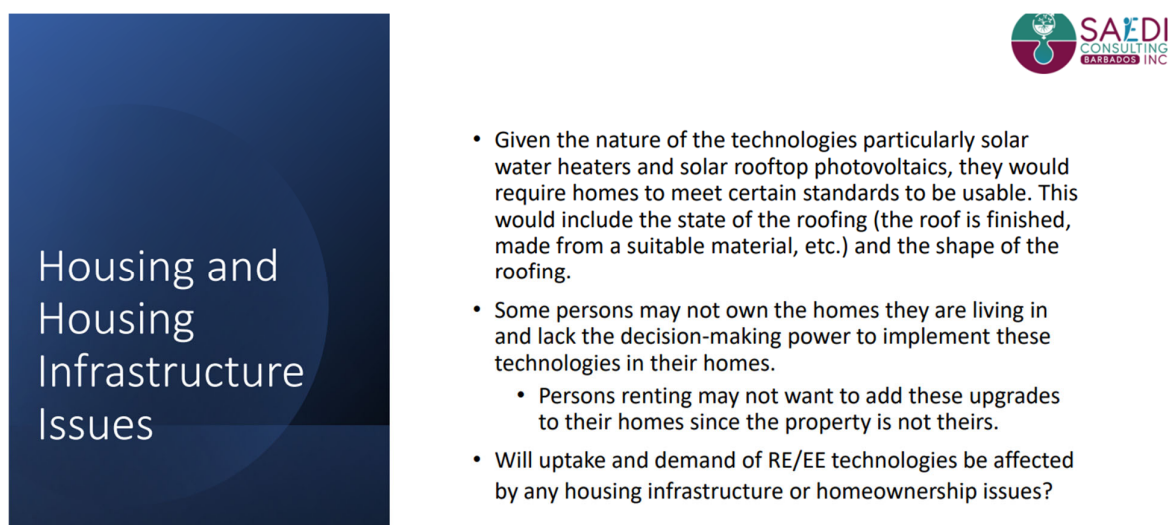
obtain information that could be used to shape project activities and processes to ensure that PWDs are not left behind by this project (see Figure 2-7).

Figure 2-7: Understanding the needs of PWDs



Housing structure and homeownership issues also emerge from the desk-based research and given the nature of the project wherein installations and modifications may be needed for homes, it was important to understand factors that might prevent the smooth execution of that process (see Figure 2-8).

Figure 2-8: Issues related to Housing



3

Country Level Analysis

Key chapter takeaway

Although, the three countries have their own individual cultures, economies and other aspects of society, the renewable energy sector in each country and their populations' interactions with the sector are affected by similar factors and barriers. The adoption of the designated renewable energy technologies will reap numerous benefits for persons, though barriers that reduce their ability to implement do exist.

This section explores the gendered usage and implications of the technologies of focus in this project across the three countries and identifies existing and potential barriers that would affect the uptake and adoption by these technologies by women and other marginalised communities. The assessment will explore the current state of usage of LED lighting, solar rooftop photovoltaic (PV), efficient air conditioning (A/C) and refrigeration differentiated by gender and any specific gender-differentiated needs. Noting from the project's TOR that more equitable access to these technologies is a goal of this project, the assessment considers/explores factors that may impact this outcome.

In so doing, it considers the following questions:

- Are the usage and uptake of these technologies differentiated between men and women? If there is a difference, is it related to gender roles?
- In what ways do they affect the daily activities of women, and can they bring increased quality of life?
- Will the increased distribution and adoption of these technologies support a just transition for all?
- What institutional, social and economic barriers may influence the rate of uptake of these technologies by women and other marginalised communities? Will these factors affect the benefits women and other marginalised communities will be able to receive from this project?

Differences based on sex and age are considered as well as other intersectional attributes such as income, geography, (dis)ability and ethnicity where relevant. IP issues are expected to be considered there in the context of a vulnerable group in the general population rather than the specific needs for indigenous and hinterland communities as it may apply in Jamaica and or Belize.

This section also reviews the enabling environment for gender mainstreaming in the energy sector and existing efforts to apply a gender lens to renewable energy and energy efficiency transformations. The review of existing legal, institutional, policy, strategy and planning frameworks that are summarised individually and collectively exposes any major gender gaps, barriers and constraints. Additionally, relevant international commitments and regional policies are mentioned. Furthermore, comparisons between countries, key issues as well as common and differentiated needs are summarised including any evidence or likelihood of a gender inequality of risk that target beneficiaries may be exposed to.

3.1 Barbados

3.1.1 Policy Environment

Barbados is a Caribbean leader in the energy sector. Its leadership has seen progress in the sustainable transformation of the energy matrix in its ambitious goals of being 100% renewable by 2030. This goal has resulted in the acquisition of 49 electric buses and the installation of rooftop solar photovoltaic systems across various government buildings. In achieving its goal to be 100% renewable, the Government of Barbados has enacted legislation and policies to improve efficiency with LED Lighting and harness the sun's energy ultimately reducing the need for fossil fuel-generated electricity.

One programme being implemented is the Roof to Reefs project as noted in the country's Nationally Determined Contributions (NDC). The project accounts for the use of IUS Model technologies such as rooftop solar photovoltaic on homes and hopes to improve the social development of vulnerable communities through resilience-building, post-disaster cost deduction and improving the standard of living and quality of life². Though this programme promotes equity and addresses resilience concerns for vulnerable groups such as women and girls, several other policies do not enhance equity and a just transition and transformation of the energy sector under the IUS Model. This is a result of policies being gender neutral, or simply gender aware, with no mention or reference to gender or women or the acknowledgement of gender-specific norms (See Figure 3-1).

Take, for example, the Barbados National Energy Policy which establishes the roadmap for the renewable energy transformation in the country. This policy has no specific mentions of gender equality, or how the different uses of energy by women would be considered and or have influenced within the roadmap. Figure 3-1 highlights relevant legislation and policies to the landscape of Energy and Gender within Barbados, and the gender content and rating of each policy.

Figure 3-1: Energy and Gender Policy Environment for Barbados

Policy	Summary	Gender Content	Gender Rating
National Policy on Gender (Draft)	The draft seeks to provide the framework to improve the status of men and women, as well as to indicators for measurement in areas such as gender-based violence, women in decision-making, employment and youth affairs. ³	No public draft is available.	N/A
National Determined Contribution (NDC), 2021⁴	It provides an update on the country's progress and outlines actions for meeting its commitments to the Paris Agreement	It recognises gender plays a role in vulnerability and commits to promoting gender equality.	Responsive

² "2021 Barbados NDC Update" retrieved from <https://unfccc.int/sites/default/files/NDC/2022-06/2021%20Barbados%20NDC%20update%20-%202021%20July%202021.pdf>

³ "Gender Policy To Be Introduced" retrieved from <https://gisbarbados.gov.bb/blog/gender-policy-to-be-introduced/>

⁴ Under the NDC 2021, the programme Roofs to Reefs is noted. The Programme seeks to build resilience in the most vulnerable communities and environmental areas through restoration, and energy and water efficient technologies, highlighting the importance of the social and gender impacts, looking at elements such as the quality of life and standard of living of those vulnerable persons.

Policy	Summary	Gender Content	Gender Rating
Electric and Power Act 2013	This act stands as one of the governing legislation pieces, which governs the supply and use of electricity, promoting the use of renewable energy. This legislation also works hand in hand with the Electric (Fees) Regulations. Amended in 2019, this amendment seeks to improve the process of householder's installation of renewable energy systems. ⁵	No mentions of gender, women, or marginalised communities.	Neutral
Sustainable Energy Framework for Barbados (2009)	This framework under the facilitation of the Inter-American Development Bank explores various viability opportunities in Barbados's journey to improve energy security whilst keeping core the protection of its environment. It provides very clear analysis on renewable energy sources, noting their existence and availability in the lens of the island for investments.	No mentions of gender, women or marginalised communities.	Neutral
Electric (Fees) Regulations	This legislation governs the fees charged to consumers under the Barbados Light and Power and the application and licensing fees for licenses.	No mentions of gender, women, or marginalised communities.	Neutral
Barbados National Energy Policy 2019- 2030	This Policy seeks to showcase the direction of the 100% renewable energy goal by the Barbados Government by 2030. It uses dated information from 2015 as many of its baseline measurements, however, does account for the vulnerabilities the nation faces as a	No mentions of gender, women, or marginalised communities.	Neutral

⁵ Refer to The Householder's Right to Renewable Energy Policy

Policy	Summary	Gender Content	Gender Rating
	Small Island Developing State, mainly its place within the International Political Economy and Climate Change.		
Control of Inefficient Lighting Act 2021	This policy outlines the phased approach to the elimination of inefficient electrical lamps, in the country's approach to be energy efficient, granting the Barbados National Standards Institution governance in the supervision of the outlined within the Act.	No mentions of implications of gender, women, or marginalized communities.	Neutral
The Householder's Right to Renewable Energy Policy	This policy reduces the barriers to entry for residents allowing them to install up to 10 KW of domestic renewable energy systems without a license.	No public publication is available. ⁶	N/A
Employment Rights Act	This policy affords employees rights such as fair dismissal, equal opportunity, and days off – sick days.	In response to pay there is no mention that a female and male employee should be paid equally.	Aware

Of the 9 policies assessed, 6 are gender-blind, 1 gender aware and 1, the Barbados NDC, gender responsive. Those most directly relevant to the country's energy sector were all found to be gender blind. The NDC, subject to international requirements for gender mainstreaming, includes greater provisions for gender than the other policies reviewed.

As noted in the 2021 NDC for Barbados, there is a commitment by the government towards "equality for all, especially as it relates to gender", as recognised in Barbados' Constitution. The NDC notes this commitment through emphasis on facilitating and supporting female entrepreneurship, business development to address unemployment and economic recovery, post-pandemic.⁷ In terms of a more direct focus on the energy sector in Barbados, the NDC references gender within the context of the Roofs-to-Reefs Program (R2RP), with a view to the positive social and gendered impacts from rooftop solar PV installation programmes. Aiming to provide support to low- and medium-income households to become more resilient to climate change related events, the programme seeks to protect both private and community roofs from the impacts of extreme weather and promotes renewable energy use, including solar, as well as rainwater harvesting, and nature-based solutions aimed at reducing runoff into coastal environments.⁸

⁶ This action of this policy was made possible through the 2019-2020 Electric and Power Act (Amendment)

⁷ "Barbados 2021 Update of the First Nationally Determined Contribution" retrieved from <https://unfccc.int/sites/default/files/NDC/2022-06/2021%20Barbados%20NDC%20update%20-%202021%20July%202021.pdf>

⁸ "Roofs to Reefs" National Resilience Plan. Barbados: 2021" retrieved from https://www.climate-laws.org/document/roofs-to-reefs-national-resilience-plan_fb59

Beyond overarching national and international policies and agreements, a recently implemented policy, at the Ministry level through the Ministry of Energy and Business, is the 'Right to Renewable Energy Policy'. This policy gives householders, more generally those intending to use domestic renewable energy systems, the right to own and install renewable energy systems (up to a total of 10kW) without requiring a formal application and license under the Electric Light and Power Act (ELPA). The purpose of the policy is primarily timesaving, with a simplified and assured process for individuals wishing to use this service. The act makes no notable mention of female-headed households or provisions for women using the service, however, the direct link toward time saving and improved renewable energy systems can indirectly positively benefit women in terms of household tasks, time allocation and potential cost savings depending on the return from investment in adopting renewable energy grid technologies.

3.1.2 Usage

Patterns and Trends

Culturally, women carry out three times (3x) as much unpaid care and domestic work as men, including household chores and cooking; both unpaid and invisible⁹. These culturally defined responsibilities are closely tied to energy and energy access. Energy access saves time and allows for some of these tasks to be carried out simultaneously and with planning and preparation while still ensuring safety. Moreover, through energy access, technologies employed by many women reduce the physical burden that may be otherwise required for these tasks. When electricity is not available either due to a blackout or to affordability challenges these daily required activities are more likely to become a burden to women.¹⁰

Men, on the other hand, carry out protective as well as economic and decision-making roles within the household. These are linked to gender stereotypes and norms which dictate that a man should: (i) not do unpaid domestic work, (ii) have the final say in household decisions, (iii) control household assets, and (iv) protect and exercise guardianship of women in the household.¹¹ In the Barbadian context, often times these classical gendered roles are mixed with those of women (cooking and cleaning). Such traditional norms, provide only a shared level of control over the acquisition of energy resources and energy access, as equal acquisition of assets is noted with the country scoring 100 in the WBL Index 2023.¹²

LED Lighting

Culturally defined responsibilities (CDRs) such as caretaking, may lead to an increased use of LED Lighting in matrifocal households. LED lighting enables women to perform these CDRs alongside their economic responsibilities, such as employment), often taking up these CDRs during times of inactive work (before or after work, during periods of no sunlight e.g. early morning or late evening)¹³ or even during remote or hybrid work arrangements.

As seen in Figure 3-2, the increase in consumer price inflation exceeds the amount of household disposable income real growth in developing countries like Barbados.

⁹ (UN Women as cited in Energia, 2021)

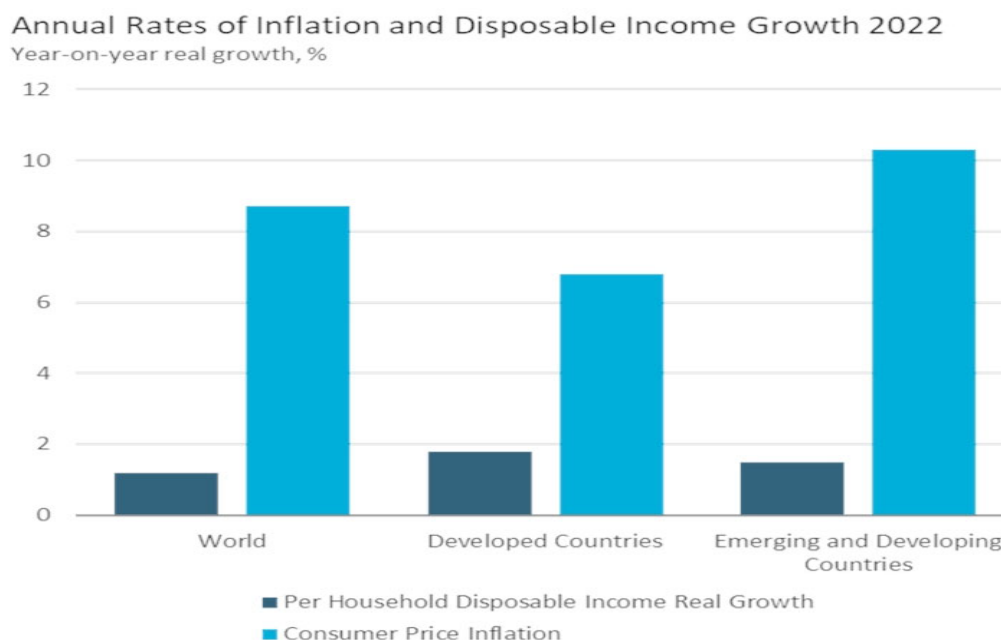
¹⁰ "How clean energy can empower women in rural communities" retrieved from <https://www.theigc.org/blogs/gender-equality/how-clean-energy-can-empower-women-rural-communities>

¹¹ "Man Enough? Measuring Masculine Norms to Promote Women's Empowerment" retrieved from <https://euneighbourseast.eu/wp-content/uploads/2021/07/oecd-masculinity-1.pdf>

¹² "Women, Business and the Law" retrieved from <https://openknowledge.worldbank.org/server/api/core/bitstreams/b60c615b-09e7-46e4-84c1-bd5f4ab88903/content>

¹³ As noted within Stakeholder Engagements

Figure 3-2: Showing Annual Rates of Inflation and Disposable Income Growth in 2022



Source: Euromonitor International Macro Model; national statistics

Note: Data are forecast. Inflation data were updated on 15 August 2022

This translates into an increase in the need for employment as a means of (additional) income to meet basic needs (food, clothing, shelter, essential services, access to drinking water, sanitation, education, health care facilities and public transport).¹⁴ The increased demand for employment can lead to the increase in non-traditional working hours (outside of 8 AM – 5 PM) which would, in turn, increase the demand for LED Lighting for safety and security as women (and others) journey to and from work or school, whether using public transportation, or private vehicles.

As women generally feel less safe in a public environment than most cis-heterosexual men (especially when it is dark outside), women often adjust their schedule, considering ways of transport and daily routes with safety in mind.¹⁵ There is a need for better lighting on Barbadian roads, on which 24, 250 existing high-pressure sodium (HPS) streetlights have been replaced with LED streetlights in a partnership with the Government of Barbados and the Barbados Light and Power Limited.¹⁶ Additionally, the Government's acquisition of electric buses, can not only improve access to public transportation at non-traditional times of work but provide a comfortable, monitored and Wi-Fi-accessible place for women and members of the disabled community to traverse during these non-traditional working hours.¹⁷

Men share some similarities in LED Lighting usage when compared to women¹⁸. The 2010 Census notes that there are 41,335 male-headed households, with 48, 036 children dependents. Similar safety concerns are a factor in the provision of LED Lighting, whilst journeying after daylight hours. Road safety is a possible area of need for the usage of LED Lighting by men, with the island reporting an upward trend in road fatalities (before the COVID-19 pandemic) of

¹⁴ "Basic Needs" retrieved from https://link.springer.com/referenceworkentry/10.1007/978-94-007-0753-5_150#:~:text=Typically%2C%20the%20list%20includes%20basic, facilities%2C%20and%20to%20public%20transportation.

¹⁵ "Public Lighting" retrieved from <https://thisisgendered.org/entry/public-lighting/>

¹⁶ "The LED Streetlight Project" retrieved from <https://www.blpc.com.bb/index.php/company/our-projects/led-streetlight-project>

¹⁷ "Electric Vehicles are Accelerating Barbados' Energy Transition" retrieved from <https://ifc.shorthandstories.com/electric-vehicles-are-accelerating-barbados-energy-transition/index.html#group-section-The-Shift-to-E-Buses-filDeSTOJQ>

¹⁸ Women's safety remain higher than men due to gender norms

which men are often the victims. However, the use of LED Lightning is far more expanded than just the use of road fatalities.

Solar Water Heating

The island is commonly applauded for its use of solar water heating in households and businesses, with over 50,000 solar water heaters (SWH).¹⁹ SWH technology combines the use of energy, and water in a highly water-scarce country to provide heated water used within households for bathing, cleaning, cooking, and washing. Outside of the benefits hot water has for CDRs such as cooking, cleaning and caretaking such as easy removal of stubborn stains such as dirt and grease and aid in the provision of safe and nutritious meals for the household, the use of SWH improves the sanitary hygiene of women.²⁰

The ageing population (60+) in Barbados amounts to 18% of the population, according to the 2010 Census. This group within the population remains vulnerable to illnesses such as diabetes²¹ and rheumatoid arthritis.²² Consequently, warm water baths generate added health benefits for the island's seniors giving them the added ability to combat low-grade non-chronic inflammation, reduce blood sugar, better circulation and soothing of the muscles and joints.²³ better circulation and soothing of the muscles and joints. Particularly, as it relates to blood sugar, Hoekstra et al noted within their study that, observations of a lowering in fasting glycosylated hemoglobin and blood glucose concentrations following hot water immersion (HWI) in individuals with type 2 diabetes support the notion of improved insulin sensitivity following HWI (33), the mechanisms that underlie this beneficial effect are currently unclear. better circulation and soothing of the muscles and joints.²⁴ Particularly, as it relates to blood sugar, Hoekstra et al noted within their study that, observations of a lowering in fasting glycosylated hemoglobin and blood glucose concentrations following HWI in individuals with type 2 diabetes support the notion of improved insulin sensitivity following HWI (33), the mechanisms that underlie this beneficial effect are currently unclear.²⁵

These benefits can allow the elderly who have fixed incomes to continue to support positive health practices at a reduced cost (as the island benefits from yearly sunshine). These savings, in turn, allow for further investments in their diet (particularly, healthy food consumption) which can be costly to these individuals as the region experiences the highest cost of a healthy diet when compared to other world regions.²⁶

Rooftop Solar Photovoltaic

Rooftop solar photovoltaics remain niche across genders, because of the economic barriers for a rooftop solar photovoltaic system (See Economic Barrier Section) as well as physical barriers due to rooftop types. With no specific data on how rooftop solar photovoltaic may be used differently across vulnerable groups within the country, it is generally presented as a cost-saving benefit to local households across the island.

¹⁹ "Solar Water Heaters" retrieved from <https://smartenergybarbados.com/smart-products/solar-water-heaters/>

²⁰ "Menstrual Hygiene" retrieved from <https://www.cdc.gov/hygiene/personal-hygiene/menstrual.html>

²¹ "The Barbados Health of the Nation Survey: Core Findings" retrieved from https://extranet.who.int/ncdccs/Data/BRB_C7_HotN_report_FINAL2.pdf

²² "The Aging Caribbean: 20 Years of the Madrid Plan of Action" Retrieved from https://www.cepal.org/sites/default/files/events/files/mipaa20_subregional_review_rev2_0.pdf

²³ "The Importance of Warm Water for Elderly Bathing" retrieved from <https://showerbay.com/importance-warm-water-elderly-bathing/>

²⁴ "The Importance of Warm Water for Elderly Bathing" retrieved from <https://showerbay.com/importance-warm-water-elderly-bathing/>

²⁵ "Acute and chronic effects of hot water immersion on inflammation and metabolism in sedentary, overweight adults" retrieved from <https://journals.physiology.org/doi/epdf/10.1152/japplphysiol.00407.2018>

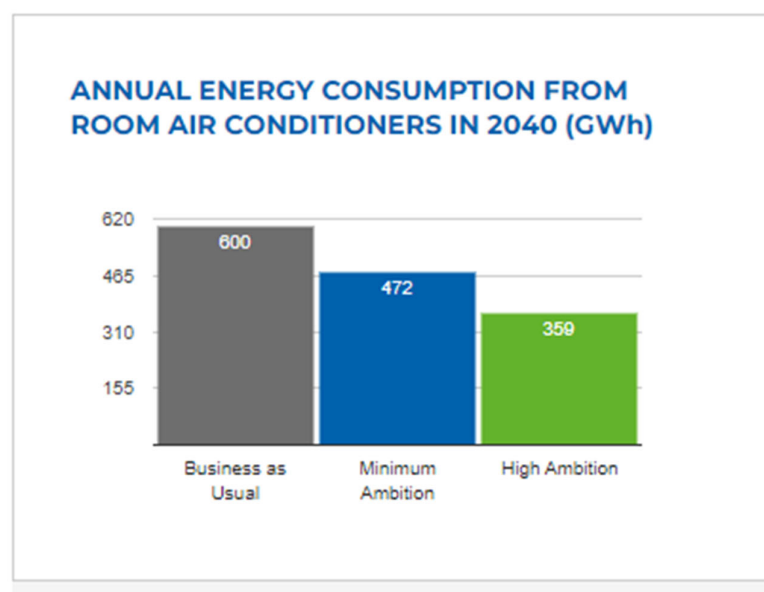
²⁶ "Regional Overview of Food Security in Latin America and the Caribbean" retrieved from <https://www.fao.org/3/cc3859en/cc3859en.pdf>

The installation of a rooftop solar PV system provides collective savings to Barbadian users, over USD 2.5 million a year on their electricity bills (equivalent to 5,000 barrels of oil, or over 2,000 tonnes of CO2 emissions).²⁷ As women reinvest 90% of their income in their families and communities, this extra income can aid in the reduction of poverty locally²⁸ and expand the scope of investment in other forms and more complex RE, EE and DER technologies.

Efficient Cooling and Refrigeration

The island has two weather seasons due to its geographic location. These include the dry season from December to May and the wet season from June to November. Increasing rates of climate change have caused intensified storms and extreme heat during dry periods²⁹, but the use of cooling remains mostly equal between men, women, and the elderly population. During these high dry periods, temperatures can be high temperatures with the Barbados Met Office predicting temperatures of 34 to 35 degrees Celsius in September 2023.³⁰ This extreme heat causes an increased need for hydration and cooling.

Figure 3-3: Estimate Annual Energy Consumption of Room Air Conditioners in 2040. Source: United for Efficiency³¹



Under three scenarios, the UNEP United for Efficiency programme estimates that room air conditioners will consume 600 GWh with business as usual, 472 GWh with minimum ambition and 359 GWh with high ambition.³² Business as usual considers that the energy efficiency of the technology improves at 1% per year, further increasing market share of variable speed models is assumed to improve overall efficiency at an additional 2% per year until new sales are entirely from that product type (see Figure 3-3).³³

²⁷ "Consumer Guide Solar PV" retrieved from <https://brea.bb/wp-content/uploads/2016/06/BREA-Solar-guide.pdf>

²⁸ "The case of a gender perspective on energy access" retrieved from <https://www.energja.org/case-gender-perspective-energy-access/>

²⁹ "Country Profile – Barbados" retrieved from <https://rcc.cimh.edu.bb/files/2018/06/Country-Profile-Barbados.pdf>

³⁰ "Barbados Meteorological Services Weather Update – April 2023" Retrieved from <https://gisbarbados.gov.bb/blog/barbados-meteorological-services-weather-update-april-2023/#:~:text=Of%20high%20interest%20is%20the,%CB%9AC%20observed%20in%202022.>

³¹ "Country Assessments" retrieved from <https://united4efficiency.org/country-assessments/barbados/>

³² United for Efficiency. 2017. "Barbados" retrieved from <https://united4efficiency.org/country-assessments/barbados/>

³³ UNEP United for Efficiency Programme. 2017. "Barbados" retrieved from <https://united4efficiency.org/country-assessments/barbados/>

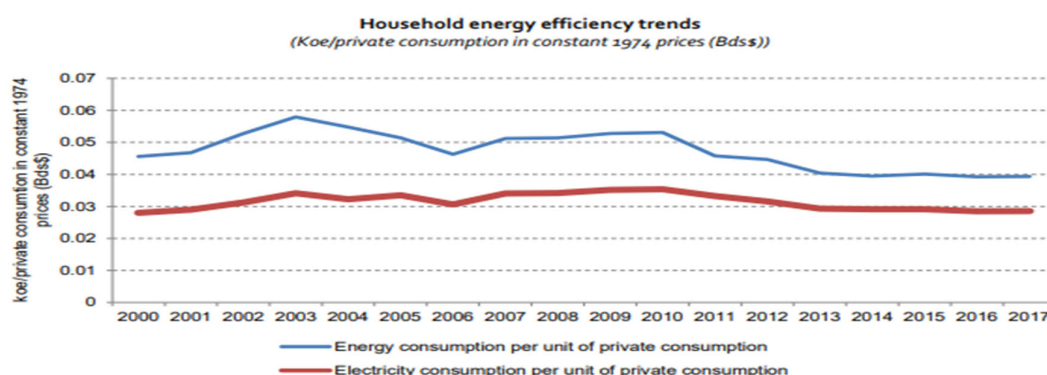
The usage of technologies such as efficient air conditioning and refrigeration can aid in women's cooking responsibilities and within their natural ageing journey including menopause. The use of an efficient refrigerator, for example, allows women to better balance their CDRs and job responsibilities by facilitating meal preparation in advance and reducing the time burden of these activities. At home, efficient air conditioning and cooling also stand to benefit 28 to 63% of middle-aged women who are affected by sleep disturbances such as hot flashes and night sweats.³⁴ This technology allows women to get a good night's sleep at home, which is key to physical and mental health. Within the workforce, air conditioning can aid in the management of job tasks allowing women to better cope with menopausal symptoms at work and provide these women with fulfilment, self-esteem, identity, and social needs.³⁵

With benefits from using efficient air conditioning and cooling during menopause (which traditionally occurs between the ages of 45 to 55), the use of these technologies creates an important demand cohort of over 20,000 thousand women within the age range of 45 to 59 years old, based on the 2010 Census, that may not currently be considered in terms of energy use and demand.

3.1.3 Demand

The demand for energy in Barbados over the last 10 years is characterised by a declining trend, with an average demand for primary and secondary energy estimated at 11,297 BOE per day.³⁶ This trend has been partially attributed to the improvement of household energy efficiency through the usage of IUS Model technologies like SWH and rooftop solar PV as seen in Figure 3-4.³⁷

Figure 3-4: Household Energy Efficiency Trends



At the macro level, demand for IUS Model technologies is influenced by the Government of Barbados' efforts to reduce the import fuel bill to improve the country's energy security and stability in achieving two of the following goals: (1) To convert 75% of the island's energy supply to renewable sources by 2030 and (2) reduced consumption of electricity by 22% by 2029³⁸. Though there is no data measuring the degree to which barriers noted in the Barrier Section of

³⁴ "Women's Health Concern Fact Sheet – Menopause and Insomnia" retrieved from <https://www.womens-health-concern.org/wp-content/uploads/2022/12/17-WHC-FACTSHEET-Menopause-and-insomnia-NOV2022-B.pdf>

³⁵ "Menopause and work. What does one have to do with the other?" retrieved from <https://menopauseintheworkplace.co.uk/articles/menopause-and-work-its-important/>

³⁶ "Barbados National Energy Policy 2019 – 2030" retrieved from <https://biblioteca.olade.org/opac-tmpl/Documentos/cg00883.pdf>

³⁷ "National energy efficiency monitoring report for Barbados" retrieved from https://repositorio.cepal.org/bitstream/handle/11362/45053/S1900970_en.pdf?sequence=1&isAllowed=y

³⁸ "National energy efficiency monitoring report for Barbados" retrieved from https://repositorio.cepal.org/bitstream/handle/11362/45053/S1900970_en.pdf?sequence=1&isAllowed=y

this report have influenced demand, it can be noted that barriers noted in the section below have impacted macro demand to some degree.

Notably, the island is also prone to natural hazards such as hurricanes. Using a mix of RE, EE and DER technologies can offer resilience, particularly through self-efficiency. For example, the Reefs to Roof Programme (noted in the Policy Environment Section) where the installation of rooftop solar PV systems can provide electricity independent of the grid, which provides households with electricity in the event of power outages caused by hurricanes or any other natural disaster³⁹. With these capabilities, households can maintain a source of energy that may be required for the care of members of the disabled community and enable key functions such as lighting and communications to (reliable sources of) media in a post-disaster scenario.

3.1.4 Barriers

This small island developing state (SIDS) has been taking steps in reducing its dependence on fossil fuel-generated electricity since the uptake of solar water heating in 1974. In its formal declaration to be 100% renewable, the Government of Barbados, and its local (private sector e.g., Barbados Light, and Power Limited), bilateral and multilateral partners (e.g., Inter-American Development Bank) have been actively reducing some barriers in accessing the renewable energy and reducing the country's reliance on fossil fuel generated electricity.

These barriers range from gaps in legislation addressing inefficient air conditioning to social and cultural influences on women's education and experience within the renewable energy sector and should be addressed to promote an equitable and just transition in this goal.

Institutional

The island has reduced institutional barriers in the uptake of the target technologies of the IUS project, for the most part. Legislation and policies have provided some reduced barriers for solar water heating LED Lighting, and solar rooftop photovoltaic, as the Government of Barbados journeys towards its goals through the NDCs and the Barbados National Energy Policy.

Though tax incentives and the streamlining of services have been used as tools to reduce barriers, other barriers exist. Tax incentives through the Fiscal Incentive Act of 1974, caused a 30% increase in the cost of electric water heaters 30% and a 20% decline in the cost of solar water heating.⁴⁰ However, this tax rebate failed to address the consumer awareness (consumers were not aware of the SWH technology, as it was new to the region) at the time.

The streaming line of services is addressed with the Electric and Power Act (Amendment) 2019 and the Household Right to Renewable Energy⁴¹ with an online application process for householders. Though still a form of licensing, the online application process - which requires roof dimensions, the type and size of the renewable energy system and PV Diagrams⁴² - improves the rate of approval for a landowner to have their 10 KW system within the Barbados Grid. This process, however, brings about a CDRs-related barrier linked to gendered stereotypes (See Social Barrier Section). Specifically in reference to the skillset required for the technical creation of blueprints being predominantly held by men as seen in the Barbados Labour Survey 2022 where there are 10.9 thousand men within the construction, mining and quarrying industry in comparison to 0.4 thousand women, enabling men to benefit more from this barrier than women.

³⁹ "More Than Just Cost Savings: How Rooftop Solar Energy Can Improve the Resilience of Local Energy Systems" Retrieved from <https://energy5.com/more-than-just-cost-savings-how-rooftop-solar-energy-can-improve-the-resilience-of-local-energy-systems>

⁴⁰ "Seizing the sunshine: Barbados' thriving solar water heater industry" retrieved from https://cdkn.org/sites/default/files/files/Barbados-InsideStory_WEB_1.pdf

⁴¹ This legislation and policy work hand in hand.

⁴² "Householders' Rights to RE" retrieved from <https://energy.gov.bb/licensing/householders-right-to-re/>

Another barrier at the institutional level is the absence of legislation requiring landlords to implement IUS Model technologies such as solar water heating, and rooftop solar on their properties. Currently, as noted from stakeholder engaged as part of this assessment, renters who may be willing to install and use rooftop solar PV, solar water heating and efficient air conditioning can be blocked by the landlord from installation and usage. As of 2010, the Barbados Census reported 16,316 renters, with an increasing need for rental units⁴³ seeing the National Housing Corporation receiving over three thousand applications for rental accommodations during a seven year period.⁴⁴ Such legislation would require conversations between representatives of the community, landlord, IUS model technology experts and government to determine best practices accounting for any unforeseen cost, such as water damage and fire.

In addition, the absence of legislation governing the use of inefficient air conditioning and refrigeration sector is also a barrier, as noted by the Barbados Renewable Energy Association (BREA). Additionally, a lack of enforcement of the existing legislation and regulation also create a barrier. These barriers currently enable the use of air conditioning systems which are consuming heavy amounts of electricity (as noted in the Usage Section). This gap in legislation and lack of enforcement could hinder the adaptation of the IUS Model and reduce mitigation efforts on fossil fuels with inefficient systems using more energy than an efficient system.

Notably, the Government of Barbados has been making strides to implement projects that improve access and investment incentives within the island. Two of these actions and projects include the HOPE Project and the acquisition of electric buses. Figure 3-5 shows partnerships and projects implemented with or by the Government of Barbados which can aid in the transition to and within the IUS Model.

The introduction of Electric Mobility into Public Transportation Project is one such example. These buses attract riders with air-conditioning, wi-fi, and wheelchair lifts, all of which the old diesel buses lack. There are now 49 electric buses, with about a dozen on the way, and the government has requested pricing for even more.⁴⁵ This project aims at reducing the fossil fuel reliance on the transportation sector (which is responsible for 33% of energy consumption⁴⁶) whilst providing more accessibility in the island's transportation system for members of the disabled community.

Figure 3-5: Government Projects Increasing Public Access and Investment Incentives within the IUS Model in Barbados

Programme	Programme Summary	Associated Partner	Targeted Technologies
Public Sector Smart Energy Programme	The programme is designed to promote and implement the use of viable Renewable Energy and Energy Efficient measures in the public sector ⁴⁷ . The programme also financed 2 electric buses.	IDB and EU	LED Streetlights and Solar Photovoltaic Systems
Energy SMART Fund Programme I and II	Programme provides financial and technical support to renewable energy and energy efficiency projects in Barbados.	Government of Barbados and IDB	LED Lightning and Solar Photovoltaic Systems

⁴³ Confirmed within Stakeholder Enagagements

⁴⁴ "Special Audit on the National Housing Corporation High Rise Apartment at the Grotto and Valeris" retrieved from <https://www.babadosparliament.com/uploads/document/f5e2f92770000faa74584e72ca8742b2.pdf>

⁴⁵ "Electric Vehicles are Accelerating Barbados' Energy Transition" retrieved from <https://ifc.shorthandstories.com/electric-vehicles-are-accelerating-barbados-energy-transition/index.html#group-section-The-Shift-to-E-Buses-filDeSTOJQ>

⁴⁶ "Barbados National Energy Policy" Retrieved from <https://biblioteca.olade.org/opac-tmpl/Documentos/cq00883.pdf>

⁴⁷ "Public Sector Smart Energy Programme (PSSEP)" retrieved from <https://energy.gov.bb/our-projects/public-sector-smart-energy-programme-pssep/>

DREAM Project	This project seeks to reduce GHG emissions from fossil fuel-based power generation by demonstrating the exploitation of renewable energy sources for electricity generation in Barbados.	Government of Barbados and Global Environmental Facility	Solar Photovoltaic Systems
Smart Energy Building – Demo	The project seeks to deploy Smart Energy Buildings on islands, reducing the power consumption and peak local demand and the concomitant reduction in CO2 emissions.	Government of Barbados and the Republic of Korea	EMS
The Accelerated Investment Premium Programme	The programme provides a 25% premium for investments made in solar (and wind) technology systems above 1 MW and up to 3 MW, and a 20 % premium for investments made in solar (and wind) technology systems above 3MW and up to 5MW.	Government of Barbados	Solar Technology Systems

The Home Ownership Providing Energy Project (HOPE) Project seeks to protect households with a gross net income of around USD 2,500 (BBD 5048) per month^{48,49}, however, the most vulnerable and those who are more vulnerable to poverty and those in poverty unfortunately are unable to benefit from such a project as the average person makes just 10% of the amount required to be a part of the HOPE Project. Concern for young people, in this context, arises from their lack of adaptive capacity – economically and financially – due to low-income status or underemployed status.

More generalised vulnerability is further exacerbated by post-disaster shocks such as rapid increases in house insurance premiums and increased house costs from damages and loss of homes. An example is the passing of Hurricane Elsa in 2021, with reports indicating 1,333 roofs were damaged, 326 other houses damaged otherwise, and 145 houses collapsed.⁵⁰

Economic

Due to its geo-political position i.e., Latin America and the Caribbean, Barbados like other Caribbean SIDS has been identified as being at risk to money laundering and the use of funding for and from terrorism and gangs. Consequently, the number of protocols and requirements for access to finance and to international funding flow create an even higher barrier to both men and women in applying for and establishing a local bank account. This is also conditioned by the fact that most banks in the region are of Canadian origin with some of these pre-conditions built in. These layers of review and risk factors act as a potential barrier to accessing financing for the installation and usage of all technologies under the IUS model, as an account will most likely be required at a recognised financial institution.

This problem of creating a bank account further intensifies when there is no law mandating equal remuneration for work of equal value.⁵¹ With no law, employers can discriminate based on gender within the workforce, which may lead to women being paid less. This is noted in Figure

⁴⁸ "HOPE Programme" retrieved from <https://www.hopebarbados.com/>

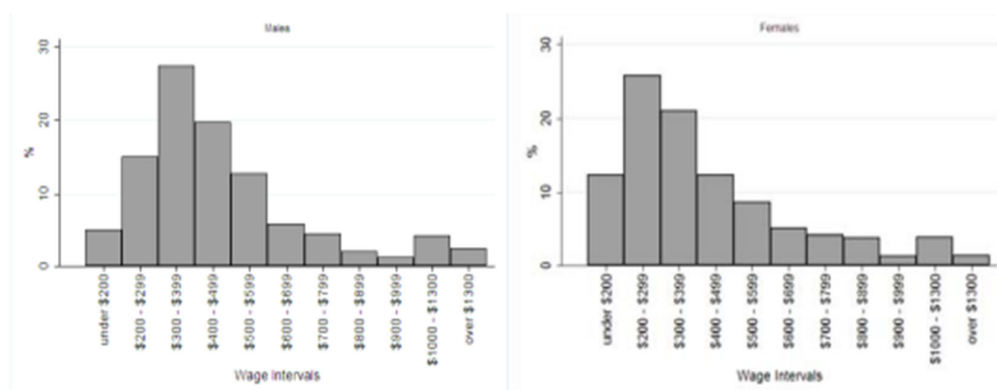
⁴⁹ If it is a double-income household, the net income cannot exceed USD 5,000 per month.

⁵⁰ "DEM Reports on Damage from Hurricane Elsa" retrieved from <https://gisbarbados.gov.bb/blog/dem-reports-on-damage-from-hurricane-elsa/>

⁵¹ "Women, Business and the Law: Barbados" retrieved from <https://wbl.worldbank.org/content/dam/documents/wbl/2022/snapshots/Barbados.pdf>

3-6 where there are more women being paid within low-income ranges in comparison to their male counter parts.

Figure 3-6: Distribution of Weekly Earnings Intervals by Gender in Barbados. Source⁵²



This provides difficulty in having the required income level required to be considered for a loan from these financial institutions as was confirmed within stakeholder engagements.

Although LED Lightning is generally affordable, with the bulb costing approximately USD 4.42 to USD 17.70 at one of the main hardware stores on the island and streetlighting remaining a public good,⁵³ the use of solar water heating, rooftop solar photovoltaic and air conditioning is more of a middle-class phenomenon. Persons within this economic and social class usually more easily take up and sustain the use these technologies (SWH and rooftop solar PV) within their homes. A 3.78 Litres solar water heating system costs USD 2,000 (BDS 4,000),⁵⁴ a rooftop solar photovoltaic system⁵⁵ starts at around USD 5,000 (BDS 10,000)⁵⁶ and a 12,000 BTU air conditioning unit costing at about USD 546 (BDS 1,092).⁵⁷

This high cost of entry is more likely to serve as a barrier to uptake for low-income women and women-headed households – limiting their capacity in accessing these technologies – as well as vulnerable households within the low-income category of USD 100 to USD 250 a week. These are the realities for the 21% of females who are experiencing poverty and the 24, 800 women whose income is USD 100 (BDS 200) to USD 250 (BDS 500) per week or less.⁵⁸

Though financing through commercial banks is an option, the interest rates and the biases towards stable income sources when granting loans act as a determinant to financing outside of credit unions. Credit unions have lower interest rates than commercial banks and its member-based organisational structure allows members to have a greater say in its operations.

It is known that commercial banks tend to favour stability, and secure employment for loans, excluding, or having less favourability to the 3,800 female entrepreneurs on the island.⁵⁹ These

⁵² "Gender Earnings Gaps in the Caribbean: Evidence from Barbados and Jamaica" retrieved from <https://publications.iadb.org/publications/english/viewer/Gender-Earnings-Gaps-in-the-Caribbean-Evidence-from-Barbados-and-Jamaica.pdf>

⁵³ "LED Bulbs" retrieved from <https://www.cartersonline.bb/electrical/bulbs/led-bulbs.html>

⁵⁴ "Get the facts right about Solar Heating Systems" retrieved from <https://sunpowr.com/the-facts/>

⁵⁵ These systems start at 2.5 kW and go up to 10kW for residential purposes.

⁵⁶ "Photovoltaic in Barbados – The Rise" retrieved from <https://terrared.com/photovoltaic-in-barbados-on-the-rise/#:~:text=Currently%20the%20installation%20costs%20for,the%20system%20and%20equipment%20installed.>

⁵⁷ "Air Conditioning" retrieved from <https://www.shopcourts.com/barbados/products/home-appliances/air-conditioning.html>

⁵⁸ "2022 Annual Statistical Bulletin - Continuous Household Labour Survey" retrieved from <https://stats.gov.bb/subjects/social-demographic-statistics/labour-market-statistics/>

⁵⁹ "2022 Annual Statistical Bulletin - Continuous Household Labour Survey" retrieved from <https://stats.gov.bb/subjects/social-demographic-statistics/labour-market-statistics/>

barriers limit women and other vulnerable communities such as youth and low-income earners from accessing these technologies. As a result, low-income earners tend to focus on covering core needs⁶⁰ in comparison to enhancing lives through solar water heating or rooftop solar photovoltaics, perceiving the high cost and difficulty in accessing finance as determinants in the uptake and usage of these IUS Model technologies, particularly women as female income is significantly lower when compared to a man as seen in Figure 3-8).

However, there are some options for women and other vulnerable groups to reduce these barriers to better promote a just transition and sustainable adaption to the technologies within the IUS model including collaborations through community projects or cooperatives (See Figure 3-7).

Figure 3-7: Alternative and Programmes for Accessing Technologies under the IUS Model

Alternative or Programme	Programme Summary	Associated Partner	Targeted Technologies
Credit Union Loans ⁶¹	These loans under various names such as Green Loans, and Green Energy Loans provide more favourable rates than commercial banks when accessing financing for these projects.	City of Bridgetown, Credit Union Limited Barbados Public Workers Cooperative Credit Union Barbados Worker's Union	Rooftop Solar Photovoltaic, Efficient Cooling and Refrigeration, LED Lightning, Solar Water Heating ⁶²
Hire Purchase or Long-Term Financing	This allows customers to make payment medium to short-term arrangements for the payments for these items	Solar Dynamics Ltd, Massy Stores Barbados, Unicomer Barbados Limited (Courts)	Solar Water Heating, LED Lighting, Efficient Cooling and Refrigeration
Home Ownership Providing Energy (HOPE) Inc.	This provides land to potential owners with land at no cost to the potential owner. The house construction is paid for by the landowner with HOPE arranging a 100% mortgage loan.	Government of Barbados and Private Sector	Rooftop Solar Photovoltaic Systems
Community Projects	This enables a group of up to 15 persons to work together on a renewable energy project and obtain a special rate tariff of 10% premium.	Government of Barbados	Solar Photovoltaic Systems
Energy Cooperative	Provides reduced risks through grouped investment. Membership barriers remain relatively low seeing an application fee of USD	Barbados Sustainable Energy Cooperative Society Ltd.	Solar Photovoltaic Systems

⁶⁰ "How Do Low-Income Families Spend Their Money?" Retrieved from <https://econofact.org/how-do-low-income-families-spend-their-money>

⁶¹ Credit Unions are membership-based institutions seeing loans and other financial services being tailored to its membership. In the 2023 Budget, Prime Minister Mottley describes these institutions as the embodiment of poor people money.

⁶² Though not under some green loan coverage, IUS Model technologies such as Solar Water Heating and Efficient Cooling and Refrigeration can be granted alone through the credit unions regular loaning system.

Alternative or Programme	Programme Summary	Associated Partner	Targeted Technologies
	50 and a minimum share requirement of USD 500.		
Grouped Investment through associations such as BREA	Provides a group investment into renewable energies, reducing some insecurities to investment.	Barbados Renewable Energy Association	Solar Photovoltaic Systems

These alternatives help to reduce the cost of investing and the purchase of these technologies, however, all these technologies within the IUS Model require the correct roof structure as was noted by the BREA. Even with these alternatives the investment and being a part of the economic side of the adaptation of technologies for the IUS Model remain biased towards a specific level of income earner.

Heavy economic barriers affect a just transition and have been reinforced by key stakeholders' consultations with the Bureau of Gender Affairs – Barbados, the Business and Women's Professional's Club, the Ashley Lashley Foundation, and the Ministry of Energy and Business – Energy Division. Today, a large portion of Barbados' energy transition remains in the hands of those who are not low-income earners.

Social

The 2010 Census highlights that 48,860 out of 79,300 or 61.61% of women had children, with 73.5% of children found in households experiencing poverty, where females are the breadwinners. Moreover, many of these women make up most low-income workers across many sectors (see Figure 3-8) with 21% of women facing poverty.⁶³

⁶³ "COVID-19 Macroeconomic and Human Impact Assessment Data" retrieved from <https://caribbean.unwomen.org/sites/default/files/Field%20Office%20Caribbean/Attachments/Publications/2020/Human%20and%20Economic%20Impact%20Assessment%20-%20Barbados.pdf>

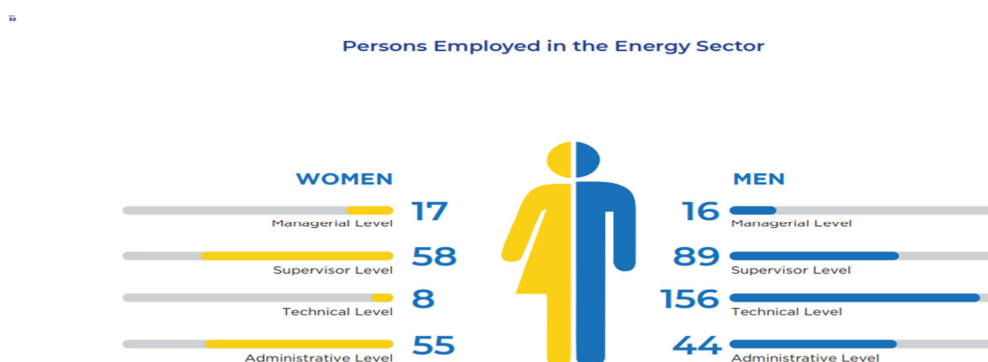
Figure 3-8: Vulnerable and Poor Incomes Segregated by Industry and Sex

	Type	\$700 <Income<\$900 (vulnerable)			Income<\$700 (poor)		
		Total	Men	Women	Total	Men	Women
Accommodation/Food Services	Tourism	4.96	0.77	4.19	4.97	2.22	2.75
Administration	High-flex	0			4.33	3.08	1.25
Agriculture/Fisheries	Essential	3.40	0.95	2.45	12.42	8.14	4.28
Construction/Mining/Quarrying	Low-flex	2.78	2.78	0	2.25	2.25	0
Education	Low-flex	0			3.69	1.78	1.91
Electricity/Energy/Water	Essential	0			0		
Financial Services	High-flex	1.00	0	1.00	0		
Health and Social Work	Essential	1.82	1.82	0.00	3.2	0	3.2
Information and Communication	Essential	0	0	0	0		
Manufacturing	Low-flex	0.95	0	0.95	8.67	1.43	7.24
Professional/Technical Activities	High-flex	3.04	3.04	0	1.99	1.99	0
Transport	Essential	0.76	0.00	0.76	8.75	8.75	0
Wholesale/Retail Trade	Low-flex	3.93	1.09	2.84	7.63	1.73	5.9
Other	Low-flex	3.63	2.45	1.18	12.31	3.36	8.95

Source: IADB's Barbados Survey of Living Conditions 2016/2017. These categories are discrete, but closely related to the IADB's definitions of poverty and vulnerability.

Education allows people to emerge from poverty, reduces inequalities and creates a climate of peace and tolerance in each society. Education remains subsidised by the Government of Barbados up to the tertiary level providing equal access to all, however, there are some areas, or fields of study that may require some cost inputs from the student. This subsidised cost enables vulnerable communities to have a chance at accessing education up to tertiary level. However, with limited data on poverty, the 21% of women facing poverty in Figure 3-8, will experience some level of opportunity cost should they opt to pursue any form of tertiary level education. A woman facing poverty may have to reduce payments on a bill or defer a payment.⁶⁴

Figure 3-9: Persons Employed in the Energy Sector.⁶⁵



Another factor of consideration is the volatility of the industry, women are mostly employed in. Take for example, the accommodation and food services industry which employ the second highest number of women after wholesale and retail totalling 9.100 thousand women, exposed

⁶⁴ However, it must be noted that they are scholarships available at tertiary levels to offset or reduce this cost.

⁶⁵ Barbados Energy Report Card for 2021" retrieved from https://cekh.ccreec.org/wp-content/uploads/2023/01/ERC_Barbados_final_002.pdf

to the external shocks of the global economy, for example, the COVID-19 pandemic.⁶⁶ This volatility increases financial stress for women, particularly those in matrifocal households to provide for their dependents in down seasons of the industry.

Gendered stereotypes are a barrier to the uptake of these technologies as women continue to be underrepresented within the technical capacities of these technologies⁶⁷ (See Figure 3-9) and enrolment into courses and programmes for these technologies are under-enrolled by young women and women (as noted within Stakeholder Engagements).

Gender stereotypes affect the enrolment of women in technical areas related to these technologies seeing 1,500 men within the electric, gas, steam, water, and air conditioning supply in comparison to 1,000 women of the 2022 Labour Force Survey (LFS). Moreover, they limit women's employment opportunities, access to resources, and agency in decision-making processes.⁶⁸ These stereotypes act as a major barrier for women being able to benefit from the employment opportunities that arise with the increased use and demand for these technologies. This reduces the ability for a just transition within this sector, as men continue to disproportionately be able to reap the rewards of employment and knowledge of these technologies.

Women in male-headed households can find themselves dependent on men for the maintenance and upkeep of these technologies. This could lead to an increase of gender-based violence cases due to the power dynamics within these households as the male head of household takes on more financial stress by purchasing these relatively high upfront costs for RE technologies wherein they are the sole income generator.⁶⁹ Such dynamic is highlighted amongst several men who thought that women's lack of economic power made them vulnerable to abuse.⁷⁰

The use and demand for technologies under the IUS model, i.e., solar water heating, rooftop solar photovoltaic, efficient cooling, and refrigeration, remain differentiated by gender due to gender norms such as caretaking and cooking.⁷¹ Though there is a notable need for the technologies and the movement of the Government of Barbados to reduce barriers to these technologies, there are some gaps in legislation, high economic barriers and gender stereotypical barriers affecting the adaptation and the just transition of these technologies.

Some of these gaps illustrate the following key issues within the adaption and just transition for the IUS Model:

- Indirect gender inequality within loan accessibility for women, persons with disabilities and other vulnerable groups with a low income.
- The absence of gender-responsive and transformative policies and capacity strengthening in gender mainstreaming across government and other key stakeholders.
- Strong gender stereotyping affects women's participation at technical levels within the energy sector.
- Suitable roofs for the installation of these technologies, i.e., solar water heating and rooftop solar photovoltaic.

⁶⁶ "Labour Force Survey Annual 2022." Retrieved from <https://stats.gov.bb/wp-content/uploads/2023/03/LFS-Bulletin-Annual-2022.pdf>

⁶⁷ Barbados Energy Report Card for 2021" retrieved from https://cekh.ccreee.org/wp-content/uploads/2023/01/ERC_Barbados_final_002.pdf

⁶⁸ "More than Words: Dismantling the Gender Myths that Hinder Women's Leadership" retrieved from <https://womendeliver.medium.com/more-than-words-dismantling-the-gender-myths-that-hinder-womens-leadership-8a12051d787d>

⁶⁹ "Associations of financial stressors and physical intimate partner violence perpetration" retrieved from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4771826/>

⁷⁰ Joseph and Jones. 2022. "Understanding Violence Against Women in the Caribbean Through an Exploration of Men's Perspectives". Retrieved from <https://journals.sagepub.com/doi/10.1177/10778012221104845>

⁷¹ In same examination, with LED Lightning gender norms play a role in the use of the technology differentiated by gender norms and responsibilities.

- The unclear relationship between a renter and the landowner, with the lack of legislation. The rights of both landlord and tenant is needed to be addressed within legislation.

To ensure a just and fair transition in adapting these technologies, there is a need for the strengthening of centralised gender mainstreaming institutions in Barbados, improving the organisational and institutional gender mainstreaming capacities. Educational and economic empowerment is also needed for women to take up these capacity-building and decision-making level opportunities. This empowerment can address the culturally defined roles and responsibilities such as caretaking, better-improving participation, and engagement of women within the sector, which may act as a barrier to participation.

3.1.5 Institutional Gender Mainstreaming Capacity

Institutionally gender mainstreaming capacity remains limited. The main organisation/body with established gender mainstreaming capacity the Bureau of Gender Affairs in Barbados, has been faced with limited capacity and has taken efforts to address this gap by the hiring of consultant(s) and through non-governmental organisation (NGO) partnerships. Presently, the two ways of boosting capacity at the Bureau has intersected with a member of an NGO being hired as a consultant for the Bureau.

Similarly, most stakeholders also highlighted the use of consultant(s) in strengthening their gender capacity due to their current limited or no existing capacity. This becomes an area of concern, particularly in achieving an equitable and just transition of the IUS Model as gender mainstreaming in energy projects supports business models that aid women and other vulnerable groups to go beyond their traditional roles and work; as job-creators, employees, entrepreneurs, and leaders, in the energy sector.⁷² Such support at a macro level enables equal participation and equal distribution of benefits of women and vulnerable groups would see the reduction in poverty especially the feminisation of poverty from unequal access.

⁷² "The blackbox of gender mainstreaming in the energy sector" Retrieved from <https://www.energia.org/the-blackbox-of-gender-mainstreaming-in-the-energy-sector/>

Figure 3-10: Stakeholder's Gender Mainstreaming Capacity

Stakeholder	Does your organisation have any experience with mainstreaming gender in its past work?	Does your organisation have the capacity to execute gender mainstreaming activities or provide gender sensitisation training/gender-related technical inputs?	Does your organisation possess a gender policy or is it in the process of developing a gender policy?	Has your organisation ever been engaged by government ministries/agencies or development partners to support in the development or implementation of energy/renewable energy projects?	Does your organisation utilize a grievance redress mechanism for projects?
The Ashley Lashley Foundation	Yes	No ⁷³	No	Yes	No
Business and Professional Women's Club (Barbados)	Yes	Yes	Yes	Yes	Yes
Barbados Council for the Disabled	Yes	Yes	No	No	No
Barbados Renewable Energy Association	No	No	No	Yes	No
Barbados Light and Power Limited ⁷⁴	N/A	N/A	N/A	N/A	N/A
Bureau of Gender Affairs	Yes	Yes	Yes	Yes	Yes
Ministry of Energy and Business – Energy Division	Yes	Yes	Yes	Yes	Yes ⁷⁵

⁷³ Organisation noted while internally, they do not have the capacity, the organisation does utilise the support of consultants.

⁷⁴ Consultant was unable to meet with the BLP and as such will not be able to provide answers for the gender mainstreaming questions for this organisation (N/A will be applied in response to the questions).

⁷⁵ External Capacity (Consultancy)

3.2 Belize

Belize, a small developing country, is highly vulnerable to the effects of climate change. Adapting to and mitigating the impacts of climate change will require large expansive financial investments at the governmental level and individual level. In 2022, Belize achieved a gross domestic product (GDP) of 6.7 billion USD, noting recovery from the financial impacts of the COVID-19 pandemic but despite this recovery, the country is experiencing an inflation rate of 6.3% which is almost double the inflation rate of 2021 (3.2%).⁷⁶ This has implications for the capacity and ability of the population to implement climate change adaptation measures as the cost of food and other essentials will rise. This means also that persons will have tough decisions to make as they balance having their needs met while still investing in mitigation measures. Female-headed households, particularly those in rural communities, will be hit harder by the rising cost of living due to various factors such as women earning 10% less than men⁷⁷, these women bearing the brunt of childrearing on their own and being employed in the informal sector.

In 2021, the Belize Statistical Office released the findings of a poverty study it undertook in 2018 and it found that 52% of the population is living in poverty which is an 11% (41% poverty rate) increase from the last study done in 2009.⁷⁸ The study notes that 9% of the population is critically poor and 11% of the population is vulnerable to falling into poverty. The Maya and Garifuna ethnic indigenous groups saw the highest rates of poverty at 78% and 52% respectively and these groups primarily reside in the Stann Creek and Toledo districts (noting that these two districts have the lowest electricity connection rates as seen in Figure 3-11).^{79,80} Toledo possessed the highest poverty rate at 82% while the remaining districts rate ranged from 45-57%. The findings of the study suggest there is a link between the level of education attained and poverty status as households with the head of the household having no formal education saw a rate of 64% while households with the head of household possessing tertiary or post-secondary education having a rate of 12%.

In 2020, 97.1% of the population enjoyed access to electricity⁸¹ which indicates significant work being done to provide all Belizean with reliable access to electricity because as seen in the UNICEF 2015 Multi-Indicator Cluster Survey (MICS), only 91.8% of the population had access to electricity with 96.9 of the urban population and 87.5% of the rural population having electricity access.⁸² As seen in Figure 3-11, of the 6 six districts Toledo district had the lowest rate of 70.8% in 2015.⁸³

Figure 3-11: Connection Rate by District. Sources Belize MICS⁸⁴

District	% of Population with Access to Electricity
Corozal	93.5
Orange Walk	93.2
Belize City	97.6
Cayo	91.3
Stann Creek	90.5
Toledo	70.8

⁷⁶ "Belize Staff Concluding Statement of the 2023 Article IV Mission" retrieved from <https://www.imf.org/en/News/Articles/2023/02/17/belize-staff-concluding-statement-of-the-2023-article-iv-mission>

⁷⁷ "Average Salary in Belize 2023" retrieved from <https://worldsalaries.com/average-salary-in-belize/>

⁷⁸ "Poverty Study 2018/2019" retrieved from <https://sib.org.bz/press-release-2021-06-30/>

⁷⁹ "History Of The Garifuna People" retrieved from <https://belize.com/history-of-the-garifuna-people/>

⁸⁰ "Belize Maya Culture & History" retrieved from <https://www.cahalpech.com/belize-maya-culture-history/>

⁸¹ "Access to electricity (% of population) – Belize" retrieved from <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=BZ>

⁸² "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from <https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

⁸³ Data from 2015 is used as more recent data with this level of breakdown is available. However, this data provides insight into which areas in Belize that requires more resources to enhance residents' access to electricity.

⁸⁴ "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from <https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

Noting Toledo as the area with the lowest connection, the district has been the target of many projects aimed at increasing access. Renewable energy is playing an important role in tackling this problem as renewable energy technologies are being utilised as the means of providing energy to rural communities while playing a part in the country's efforts to address and adapt to climate change. For example, in 2020, a solar plant was installed in Indian Creek Village in Toledo District which will also benefit Medina Bank and Golden Stream villages as well. This work was a part of the emPOWERing Rural Electrification Project funded by the government of the United Arab Emirates.⁸⁵

3.2.1 Policy Environment

Energy policies in general, in Belize, did not possess strong content that spoke to empowering women in the sector or implementing measures to address their needs; however, mention is made of health risks faced by women with the usage of fuelwood. At the same time, women's low participation within the sector is acknowledged (see Figure 3-12). The national gender policy seeks to expand opportunities for all and could serve as a good starting point when initiating any revisions or policy developments that would need to be specific to the energy sector.

Figure 3-12: Energy and Gender Policy Environment for Belize

Policy	Summary	Gender Content	Gender Rating
NDC, 2021	It provides an update on the country's progress and outlines actions for meeting its commitments for the Paris Agreement.	It recognises gender as a factor in vulnerability and sets out to assess and address issues that result from gender inequality.	Responsive
Revised National Gender Policy 2013	The policy identifies and examines the inequities experienced by both men and women and suggests strategies to correct gender disparities as well as gives direction for the co-ordination and implementation of the policy	Policy speaks to promoting and facilitating the equitable access and distribution of resources, services and opportunities for men and women.	Transformative
National Energy Policy Framework	This framework sets out a recommended roadmap that directs Belize to achieving energy efficiency, sustainability and transformation to renewable energy.	It noted the health risks posed to women and children using burning wood fuel at home.	Aware
Sustainable Energy Strategy	This strategy highlights Belize's unrealised sustainable energy potential and sets forth recommendations to address barriers preventing the full utilisation of this potential.	No mentions of gender, women or rural communities.	Neutral
Nation Energy Policy 2011	This policy outlines key areas the nation needs to develop and support to attain a secure and sustainable energy supply.	Policy notes the lack of education and promotion on energy issues affecting the public. It raises awareness on the lack of skilled and trained women in the energy sector. It also notes that gender-balanced training is	Aware

⁸⁵ "Ministers Break Ground for Rural Electrification Project in Toledo District" retrieved from <https://www.pressoffice.gov.bz/ministers-break-ground-for-rural-electrification-project-in-toledo-district/>

Policy	Summary	Gender Content	Gender Rating
		constrained by cultural and traditional influences. It recognizes that men and women have different energy needs given based on their different socio-cultural and traditional roles. It notes that women are underrepresented at all levels of energy generation, transmission and distribution. Gender imbalance exists at various levels of planning and decision-making. It notes the burden on women especially rural women related to searching for energy (specifically wood fuel). Fuel wood is the main source of energy for most households in rural areas. Noting the need for access to sustainable, secure fuelwood supplies and renewable energy technologies as energy greatly influences the socio-economic states of rural communities.	
The National Gender Based Violence Plan of Action, 2010	The plan provides a framework to guide action and work around GBV with goals, objectives, and activities to address the issue of gender-based violence in the country.	The plan outlines 4 key goals and associated objectives for responding to GBV and supporting survivors of GBV. The Plan addresses provisions for legal support to survivors and legal action toward perpetrators. It also addresses the need for greater public awareness of the root causes of GBV and greater awareness and understanding by men of the root causes of GBV.	Responsive
Belize National Child Labour Policy and Strategy 2022 to 2025	The policy and strategy lays out a plan to remove children from harmful and hazardous work, increase social protection and economic resilience for families, and increase school attendance.	The strategy utilizes gender disaggregated data and denotes the need to build awareness of legal access and support while tackling gender and social norms which place boys at much higher risk for child labour participation.	Transformative

Seven policies were analysed, those most directly relevant to the energy sector and including the National Gender Policy (all are summarised in Figure 3-12). Of the 7, 2 are transformative, 1 responsive and transformative, 1 responsive, 2 aware and one blind. The existence of the Gender Policy and the NDC in combination provide a solid foundation or defining gender and social dimensions for a just transition in the context of the IUS model implementation in Belize.

The updated NDC⁸⁶ of 2021 for Belize, was developed using a gender and vulnerable group analysis which provided recommendations for improving the gender sensitivity of both medium term and long-term strategies that exist within the commitments and activities outlined within the NDC. NDC targets and actions in the document were developed with the participation of the public, including consultations with local communities and indigenous persons, in a gender-responsive manner. Actions and targets in the updated NDC were submitted after undergoing a gender and vulnerable group scoring analysis. This aided in providing recommendations for improved gender sensitivity in NDC implementation and the longer-term low-emissions strategy. While gender responsiveness is imbedded within the strategy for the updated NDC, there exists only one direct reference to women in the document which falls under the targets outlined for the agricultural sector. This mention directs specific action toward increasing public awareness that targets women, youth and communities on reducing GHG emissions through alternate production methods.

In terms of consideration for indigenous persons within the actions of the NDC beyond consultations, specific references to indigenous groups include promoting the monitoring and stewardship of indigenous lands, as well as indigenous coastal lands, to be sustainably managed toward the goal of producing net carbon sink areas in over 10,000 hectares of indigenous land. The document also notes the development of area-specific strategies for adaptation that pay attention to indigenous communities more generally while also promoting and enhancing policies and frameworks that consider climate (and water) sensitive development and use of lands. In doing so, the NDC notes the need to enhance and encourage current practices in land stewardship that exist in indigenous communities. The document also notes the need to be inclusive of gender, indigenous communities, PWDs, and youth in a way that is balanced and inclusive in terms of climate financing. These efforts are supported by climate finance strategy developed and implemented for the period 2021 – 2025.

Belize's Revised National Gender Policy⁸⁷, updated in 2013, is focused on building upon and continuing the country's efforts to achieve and sustain gender equality and equity. It includes objectives for supporting efforts toward 'human-centered development', with equal participation of men and women, with equal co-benefits being distributed. The policy also includes the objective to encourage and facilitate the equal access to, and control over productive resources, services, and opportunities between men and women. The revised national gender policy is based on the vision, whereby there exists:

"A society in which all men and women, boys and girls are able to achieve their full potential through the enjoyment of their human rights; live together in mutual respect, dignity and harmony; and are equal partners as they participate in services and resources for realizing and sustaining their economic, social, political, and cultural development for equal enjoyment of all."

⁸⁶ "Belize Updated Nationally Determined Contribution" retrieved from <https://unfccc.int/sites/default/files/NDC/2022-06/Belize%20Updated%20NDC.pdf>

⁸⁷ "Revised National Gender Policy" retrieved from https://oig.cepal.org/sites/default/files/belize_2013national_gender_policy.pdf

Two additional policies were reviewed that give mention to women and vulnerable persons, specifically children within the country. These were the National Gender-Based Violence Plan of Action 2010-2013⁸⁸ and the Belize National Child Labour Policy and Strategy 2022-2015.⁸⁹

Both policies were shown to be gender transformative, while the GBV Plan of Action provides several specific gender-responsive actions in its framework. The policy outlines four key goals and associated objectives for responding to GBV and supporting survivors of GBV. The Plan addresses provisions for legal support to survivors and legal action toward perpetrators. It also addresses the need for greater public awareness of the root causes of GBV and greater awareness and understanding by men of the root causes of GBV. The National Child Labour Policy and Strategy (2022-2015) utilises significant gender disaggregated data in its development of actions and denotes the need to build awareness of legal access and support while tackling both gender and social norms that place boys at much higher risk for child labour participation. The policy and the plan provide a framework in which the protection and safety of these two vulnerable groups can be safeguarded in the project and avoided possible exploitation and harm that may arise intentionally or unintentionally.

A new energy policy is currently in development for the country as the previous energy as noted in the table above has limited mentions of gender and there exists opportunities to improve on that area for the new policy (access to the draft has not been received yet)⁹⁰. Additionally, New Service Access policy⁹¹ is a new policy being launched by Belize Electricity Limited (BEL) which aims to reduce the cost of electricity connections by removing the additional connection setup fees when connecting homes in areas with existing connections. This helps support low-income persons to gain access to electricity. In cases of areas with no existing connections, the policy has provisions for persons to pay a deposit instead of the entire connection setup cost upfront.

3.2.2 Usage

Rooftop Solar Photovoltaic

Within the Maya indigenous communities in the Toledo district (found in the southernmost part of the country) exist technical capacity to deploy solar energy technologies. Three Maya women were trained in solar energy technology at the Barefoot University in India.⁹² From their experience, the benefits of women being involved in the field could be seen. With this capacity gained and material support, they were able to provide the community of Santa Teresa solar power over a two-year timespan. Additionally, through a project called Indigenous Female Solar Engineers Scaling up Solar Energy to Machakilha and Graham Creek, a collaboration between Plenty Belize and the Ministry of Rural Transformation, these women will be helping to provide these communities with solar powered electricity while also equipping these communities with the knowledge to maintain the technologies on their own.

This ongoing project has a strong emphasis on ensuring all residents of the community are empowered through skills-building and other capacity building workshops made available to all. Another emphasis of this project is the promotion of and support women in non-traditional roles with women who were trained at Barefoot University serving as models for other women and girls of the nation. The solar electrification provided by this project generate several direct household benefits for all members. These benefits address children's need for education through the provision of light and power for technologies that aid in education. For the adults in

⁸⁸ "The National Gender-Based Violence Plan of Action 2010-2013" retrieved from <http://www.humandevlopment.gov.bz/wp-content/uploads/2013/07/GBV-PlanOfActionGBV-FINAL-revised-August-2010.pdf>

⁸⁹ "Belize National Child Labour Policy and Strategy (2022-2025)" retrieved from https://www.ruraltransformation.gov.bz/docs/50/NationalChildLabourPolicy-BELIZE-HighResolutionWithoutCropMarks_sNO45qE.pdf

⁹⁰ Information on this policy was shared during stakeholder engagement held on May 19th.

⁹¹ Information on this policy was shared during stakeholder engagement with Mr. Austin Castro from BEL on June 2nd, however access to the policy has not yet been obtained.

⁹² "Indigenous Engineers Provide Solar Solution for South" retrieved from <https://www.7newsbelize.com/sstory.php?nid=57240>

the household, access to electricity provide them with new opportunities for their livelihoods or enhancements to their livelihoods which in turn, enhances their ability to make decisions to increase their quality of life and implement climate change adaptation measures.⁹³ Alongside the direct household benefits, the project provides these communities with communal benefits such as 1) small shops with the ability to run a refrigerator/freezer ran by one or two systems and these refrigerators serve to enhance food security and food safety through the ability to store foods and extend food's shelf life, 2) a medium size power system installed at the primary school which would enhance the school's ability to provide quality education to these children, and 3) a solar water pump to provide potable water which would improve the quality of health in these areas by providing clean water and reduce the time spent by primarily women in fetching water for household use.

Access to electricity provides numerous benefits to those living in rural communities. It provides cleaner energy to use for lighting and other tasks and reduces exposure to smoke and other pollutants released from the burning of coal, firewood and kerosene that are the typical biomass or fossil-fuel-based energy sources typically used in rural communities. This allows them to save money as the need to purchase fuel is reduced and the savings can be used to increase the quality of their lives in other ways. With reliable access to electricity, persons in very remote areas are now able to utilise mobile/satellite phones and this provides many benefits to the community as it increases their security to an extent as calls for help can be sent out faster than persons having to journey out of the community, and this also applies to requesting medical aid. Additionally, having enhanced communication increases their access to information which would ideally expand their capacity for participation and to exercise rights in the context of energy as well as climate change mitigation. Similarly, the Garifuna communities in the Stann Creek District also utilize the combustion of fuels to meet their energy needs as well and face similar issues.

The abovementioned issues and concerns primarily affect those living in Toledo and Stann Creek districts. Those living in Cayo, Orange Walk, Corozal and Belize City districts which constitute the more urban areas of the country, enjoy better connections to electricity, thus reducing their need to rely on the burning of fuels to provide energy to carry out domestic duties or to support their livelihoods. In turn, they are less likely to be exposed to harmful pollutants released from the combustion of these fuels. A notable exception to this would be for the island of Caye Caulker with 1,700 persons residing on the island that would have utilised a diesel generator to supply the island with power. The island imports 730, 000 gallons of diesel annually to provide electricity for the entire island, however, in 2019, CDB approved a loan to fund the connection of the island to the national grid via a submarine cable.⁹⁴ This result in the island utilising renewable energy for 50% of its energy and reduces electricity costs from USD 0.26 per kWh to USD 0.11 per kWh. The submarine cable adds the benefit of being more reliable and resilient to storm surges and hurricane winds.

LED Lighting

In 2019, the energy unit of the Ministry of Public Service, Energy and Public Utilities launched an initiative in collaboration with electrical students at the Institute for Technical and Vocational Education and Training (ITVET) to install LED lights in public buildings.⁹⁵ The initiative saw the installation of 185 tubes across four locations in Stann Creek with the aim of promoting EE in public building and energy conservation measures to the public while also developing the

⁹³ "Indigenous Female Solar Engineers Scaling up Solar Energy to Machakilha and Graham Creek Villages- Belize's Most Rural and Remote Communities" retrieved from <https://sqp.undp.org/spacial-itemid-projects-landing-page/spacial-itemid-project-search-results/spacial-itemid-project-detailpage.html?view=projectdetail&id=29729>

⁹⁴ "CDB to connect island of Caye Caulker to Belize's national electricity grid, making gains in environment, energy security" retrieved from <https://www.caribank.org/newsroom/news-and-events/cdb-connect-island-caye-caulker-belizes-national-electricity-grid-making-gains-environment-energy>

⁹⁵ "Energy Conservation Measures by Installing LED lights in Public Buildings" retrieved from <https://www.publicservice.gov.bz/index.php/medias/news-and-events/item/130-energy-conservation-measures-by-installing-led-lights-in-public-buildings>

technical knowledge and skills of the ITVET students. The initiative noted that over a 5-year period, the cost of using fluorescent lighting would have amounted to USD 27,311.90 USD (BZD 55,302.91) while the use of LED lighting for the same period would only cost USD 13,655.95 (BZD 27,651.46). The switching of traditional (fluorescent) bulbs to LED bulbs can result in a reduction in electricity consumption which in turn, means households can save additional money on monthly basis as their electricity bill would be reduced. This reduction in monthly expenses can provide single income especially female-headed homes with some relief as their financial burdens are lessened which could possibly translate to better mental health as persons would have less issues to worry about. At the same time, these savings though small, could be utilised to improve other aspects of the household.

BEL carried out a renewable energy project nationwide, aimed at replacing 31,000 streetlights with LED lights in 2016.⁹⁶ This project benefited areas across the country that did not have well lit streets, noting that some lights were installed in strategic areas frequented by people on their way to work or home to help deter criminal activities. Residents of San Mateo (located on the island of Ambergris Caye) expressed satisfaction over the activity as recent crime in the area had residents fearful as many streets were not properly lighted. Lighting can be used as a passive measure to reduce crime as persons would feel less inclined to commit criminal acts when their identity is more exposed.⁹⁷ Additionally, women would feel a bit more empowered to participate in society, particularly in sectors that may have them working late at night as the light provides a sense of safety.

Solar Water Heating

The Development Finance Corporation (DFC) has partnered with the Belize Biltmore Plaza to promote the adoption of renewable energy in businesses. In 2019, they provided the plaza with resources to setup a solar water heating demonstration project.⁹⁸ The demonstration project was aimed to raise awareness of financial instruments that can be utilised to invest in SWH as well as other RE technologies and demonstrate the benefits of switching to renewable energy. This demonstration was targeted at hotels as they use a large amount of hot water for laundry and guest usage. Caribbean Regional Energy Development Programme of the CARICOM Secretariat hosted two one-day seminars in Belize aimed at introducing hoteliers to the benefits of using SWH in their businesses as well as promote CARICOM region SWH manufacturers and explore business opportunities.⁹⁹

While a lot of the promotion and awareness efforts for SWH has been targeted at the tourism sector, SWH can also be beneficial to the domestic household. The use of a SWH can lead to the reduction of electricity consumption as less electricity is used to heat water needed for showering and other domestic tasks. Through this reduction, persons could possibly save up to USD 597.58 (BZD 1204) per year depending on the usage.¹⁰⁰ This could lead to increased financial ability through the savings made from using a SWH. Additionally, it could lessen the domestic work performed by women by speeding up the process of heating water when needed.

Efficient Cooling and Refrigeration

The country experiences a moist tropical climate that slightly differs from other Caribbean Countries. The wet season occurs from May to October and its dry season occurs during November to April. The country sees a mean annual temperature range from 23-27°C, noting

⁹⁶ "San Mateo Sub-Division gets LED street lights" retrieved from <https://www.sanpedrosun.com/community-and-society/2016/12/02/san-mateo-sub-division-gets-led-street-lights/>

⁹⁷ "Can Enhanced LED Lighting Contribute to Reduce Violent Crime" retrieved from <https://newarkcollaborative.org/blog/can-street-lighting-reduce-crime>

⁹⁸ "The DFC & The Belize Biltmore Plaza Partner On A Solar Water heating Demonstration Project" retrieved from <https://www.ambergristoday.com/news/2019/02/11/dfc-belize-biltmore-plaza-solar-water-heating-demonstration-project>

⁹⁹ "Press Releases Caricom To Facilitate One-Day Seminars On Solar Energy" retrieved from <https://caricom.org/caricom-to-facilitate-one-day-seminars-on-solar-energy/>

¹⁰⁰ "Frequently Asked Questions" retrieved from <https://www.swhbelize.com/faq>

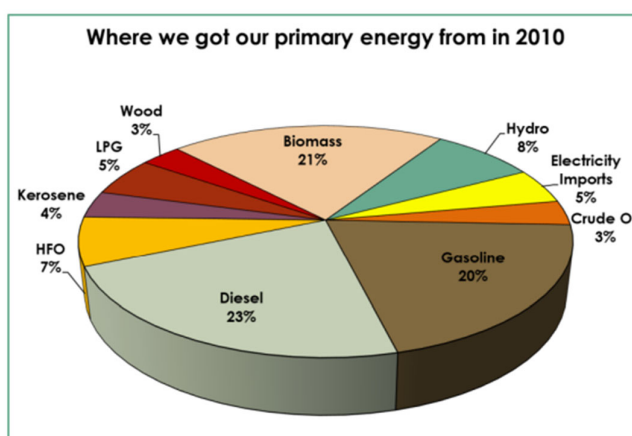
that mean annual temperatures have increased by 0.45°C at an average rate of 0.10°C per decade since 1960.¹⁰¹ As the effect of climate changes begin to hit harder and affect more persons, there is an increased need for cooling and thus efficient cooling for the cost saving aspects. Relief from rising temperatures is necessary as looking at forecast data from Belize's meteorological office shows temperatures from 31.9-34.5°C across the country with maximum temperatures reaching 38.8°C.

No data could be found on the differential usage of cooling between men, women, children and the elderly. It is important to understand that efficient cooling will be important in coming days to combat heat strokes as they may become common and rising temperatures may exacerbate discomfort women experience during menopause. Heat stress can be disruptive to productivity and quality of life. Caribbean Climate Outlook Forum warns that Belize among other Caribbean countries will be expected to deal with increased heat stress in the third quarter of 2023.¹⁰²

3.2.3 Demand

The demand for renewable energy and renewable energy technologies has been rising steadily over the years which has been supported by the government as it works towards achieving its renewable energy targets. Looking at the Belize's energy sources in 2010 43% came from fossil fuels (see Figure 3-13), the government has been working to develop policies¹⁰³ and strategies to guide the country towards more renewable energy sources and increase the adoption of EE to reduce energy consumption.¹⁰⁴

Figure 3-13: Belize's Energy Sources in 2010



Noting from the country's energy targets presented in 2021 (see Figure 3-14), the country is aiming to hit 70% of energy coming from renewable sources by 2030 and by 2021, the country sat at 43.3%.

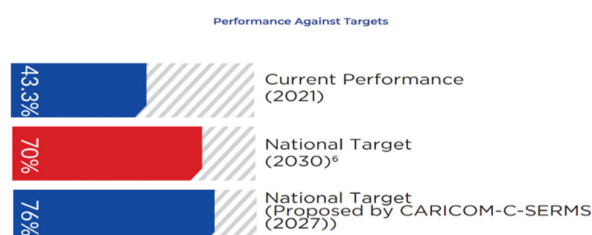
¹⁰¹ "Climatology" retrieved from <https://climateknowledgeportal.worldbank.org/country/belize/climate-data-historical>

¹⁰² Belize among regions forewarned of intensified heat stress in the upcoming three months" retrieved from <https://www.breakingbelizenews.com/2023/06/25/belize-among-regions-forewarned-of-intensified-heat-stress-in-the-upcoming-three-months/>

¹⁰³ In August 2023, the Ministry of Public Utilities, Energy, Logistics, and E-governance has concluded consultations on the Belize National Energy Policy 2023-2040 and is in the process of finalising before presenting it to Cabinet.

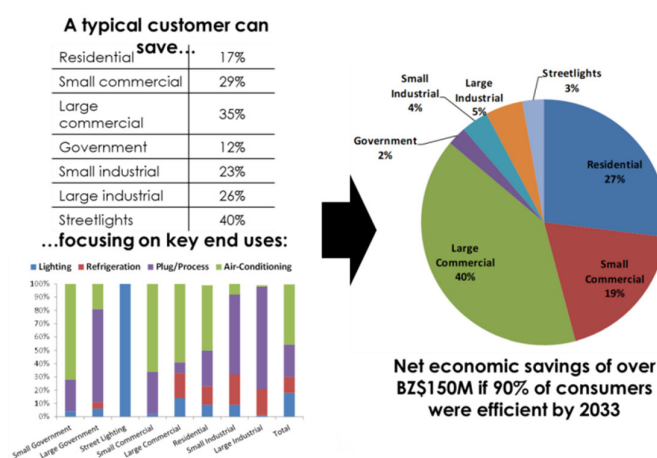
¹⁰⁴ "Sustainable Energy Roadmap 2021-2040" retrieved from <https://energy.gov.bz/wp-content/uploads/2022/02/ENERGY-PLAN-ROADMAP-1.pdf>

Figure 3-14: Belize's Renewable Energy Targets¹⁰⁵



Outlined in the Sustainable Energy Roadmap 2021-2040 were benefits Belize could reap through the adoption of EE (see Figure 3-15). The Ministry of Public Utilities, Energy, Logistics & E-Governance through the Energy Unit has been collaborating with international partners and local stakeholders to foster RE, EE and DER awareness, as well as develop a market for these technologies in the country. Outlined in the roadmap is the projected savings the country could gain by 2033 by having at least 90% of consumers utilizing EE technologies with the project savings being USD 74.5 million (BZD 150 million) as seen in Figure 14.

Figure 3-15: Projected Cost Benefit of EE Adoption in Belize



3.2.4 Barriers

3.2.3.1 Financial Considerations

Rising costs of living is not the only factor that inhibits the financial ability of Belizean people, particularly women to afford a range of energy sources and technologies. Employment greatly impacts the finances available to persons whether they are employed in seasonal sectors, working in underpaid jobs or working in the informal sector. Many persons operate in the informal sector, particularly in rural areas as they may lack qualifications (education) or fail to meet requirements (possessing a valid form of national identification) or simply unable to afford the costs of travel. Many places of employment require persons to possess a valid form of national identification, however, it must be noted this could be a barrier of entry to low-income persons as the national identification card costs USD 2.48 (BZD 5¹⁰⁶) to issue for the first time and USD 12.91 (BZD 26) to replace¹⁰⁷ while the driver's license costs USD 44.68 (BZD 90) to issue and

¹⁰⁵ "Energy Report Card 2021 – Belize" retrieved from https://www.ccreee.org/wp-content/uploads/2023/01/ERC_Belize_final_002.pdf

¹⁰⁶ The conversion rate used is 1 BZD equals 0.49 USD as of June 2023.

¹⁰⁷ "Card Information" retrieved from <https://www.socialsecurity.org.bz/card-information/>

USD 29.79 (BZD 60) to renew yearly.¹⁰⁸ These costs may be untenable for some as they balance costs for themselves and unemployed dependents.

The UNICEF 2015 Multi-Indicator Cluster Survey for Belize found only 37% of women and 40% of men have a secondary level education.¹⁰⁹ It also notes that two in five women (2 in 5) have only obtained a primary school level of education and a similar rate can be seen in men. These factors alone or working in conjunction together leave a noted proportion of persons (35.3% or 64, 268 persons out of the 182, 084 employed persons¹¹⁰) with only informal work as a means of generating income.

Working in the informal sector may provide persons with opportunities and certain freedoms particularly rural women who may not have the ability to access more formal employment due to ineligibility or conflicting responsibilities (such as family care). Despite these benefits, informal sector work can lack the security and integrity afforded to formal employment. This can have far-reaching consequences for persons as the lack of an employment letter or self-employed forms could pose as a barrier to accessing loans and other financial tools from financial institutions.

Having access to additional finance will be important to purchase renewable energy technologies. The average cost of a solar water heater in Belize ranges from USD 992.25 (BZD 1998.71) to USD 1984.50¹¹¹ (BZD 3,997.42) which is 7-14.1% of the average yearly income (USD 14,112¹¹²/BZD 28,426.13) in Belize. Noting this level of expense, access to financial aid will be a necessity for low-income female-headed households and other vulnerable groups to access these technologies, without which they would otherwise be left behind in the transition to green energy.

Similar barriers are experienced by indigenous communities, especially those in rural parts of the country. These communities, particularly the Maya, are located in areas that are quite difficult to access, often without fully developed roads and telecommunication infrastructure and/or coverage. These conditions pose a significant hindrance to the economic empowerment of indigenous people. The barriers posed by limited transportation options and long travel times severely limit economic opportunities, be it seeking employment or selling goods as examples of how livelihoods could be impacted. This leaves formal and informal agriculture as a source of livelihood for many and depending on the way agricultural activity is managed, many may be ineligible to apply for loans from financial institutions. Obtaining loans could entail needing valid identification, financial records, business plans, marketing plans¹¹³ and some rural indigenous farmers, especially female farmers, may not have the capacity to develop these critical documents to meet the scope of the institutions.

Persons with disabilities (PWDs) face their own unique barriers and plight when it comes to access to finance and the cost of living. Some PWDs live in squalor¹¹⁴ as they subsist on whatever aid they receive from relatives or organisations or government subvention. Many PWDs are unable to work due to (i) a lack of accommodations in the workplace to meet the needs of their disability and (ii) their limited access to education because of their disability thus the lack of necessary qualifications. Some PWDs live with relatives on whom they are entirely dependent financially, thereby creating an additional set of possible challenges to caregivers.

¹⁰⁸ "Types of License" retrieved from

https://belizecitycouncil.org/services.php?service_id=10#:~:text=Here%20are%20the%20License%20costs,%2460.00%20BZ%3A%20for%20annual%20renewal

¹⁰⁹ "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from <https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

¹¹⁰ "Labour Force Survey Summary Findings, October 2022" retrieved from <https://sib.org.bz/statistics/labour-force/>

¹¹¹ "Belize Technology Needs Assessment" retrieved from <https://c2e2.unepccc.org/wp-content/uploads/sites/2/2020/06/solar-water-heater.pdf>

¹¹² "Belize Salary" retrieved from <https://www.averagesalarysurvey.com/belize>

¹¹³ "Agriculture Loans" retrieved from <https://www.dfcbelize.org/loans/agriculture-loans/>

¹¹⁴ "Living With a Disability in Dire Conditions" retrieved from <https://edition.channel5belize.com/archives/251267>

Given this financial dependence, PWDs have limited economic power. In addition to the lack of finances due to unemployment, the cost of caring and treating for disabilities can be quite expensive - this could range from the price of obtaining mobility aids to requiring frequent medical treatments and/or the price of medicine to treat chronic conditions. PWDs' ability to implement renewable energy technologies is thus lowered due to these financial demands as they form additional financial hurdles that PWDs must deal with.

3.2.3.2 Housing

The state and quality of housing has been an issue identified in Belize even within urban areas of the country in relation to the implementation (installation) of RE/DER technologies identified for the IUS project particularly for solar water heaters, and rooftop solar. It must be noted that the project entails rooftop solar (although there is a possibility of this technology being implemented in other ways) and efficient cooling through air conditioning, there exists a requirement for homes to meet certain standards in order for these technologies to be effective. Figure 3-16 shows the characteristics of housing across the country derived from the MICS survey. From the MICS survey it can be seen that a low but still significant number of households would not be able to support some of these technologies efficiently and would require adjustments or support to bring their houses up to standard. Only 95.7% of urban households and 88% of rural households have finished floors, roofing and walls. 7.9% of urban households and 12% of rural households have rudimentary roofing which is of concern due to these types of roofs being unable to support the solar water heaters or solar photovoltaic and may even suffer damages from them.

From the survey, 67% of households are headed by men with the remaining 33% headed by women.¹¹⁵ The Mestizo/Spanish/Latino ethnic group accounts for the largest ethnic group for household heads at 49.1%, followed by Creole at 25.5% while the indigenous ethnic groups of Maya and Garifuna account for 9.1% and 5.7% respectively.¹¹⁶

The state of housing quality is especially of concern for PWDs in situations where they are living on their own and typically lack the finances to secure proper housing or to maintain and upkeep their homes.¹¹⁷ Another group of major concern, because of housing quality and housing infrastructure, is indigenous communities living in rural areas, especially those who live in traditional housing. Traditional indigenous housing structures are not typically suited to RE, EE and DERs and require an adapted approach (either installation or the technology itself). Rooftop solar, in particular, may compromise the structural integrity of the roofing over time without adjustments.

Figure 3-16: Housing Characteristics of Urban and Rural Homes in Belize¹¹⁸

Housing Condition	Urban Households	Rural Households
Access to electricity	96.9%	87%
Household with finished floors, roofing and walls	95.7%	88%
Rudimentary roofing	7.9%	12%
Possess air conditioning	13.2%	7.1%

¹¹⁵ Information on the distribution of these households across the six districts is not available nor is data on breakdown of these households by ethnicity and sex.

¹¹⁶ "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from <https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

¹¹⁷ "Living With a Disability in Dire Conditions" retrieved from <https://edition.channel5belize.com/archives/251267>

¹¹⁸ "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from <https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

An additional housing-related issue to be considered is the ownership of the home. 68.9% of all households are owned by a household member with 59.3% of urban households and 77.2% of rural households being owned by a household member.¹¹⁹ For persons in urban areas, this suggests more than 35% of persons are renting or living in a house owned by another relative outside of the home. For persons living in homes they do not own, there are likely limitations on what modifications they are allowed to make to the house without the consent of the property owner based on the rental agreement/lease they signed. The Landlord and Tenant Act does not have provisions¹²⁰ that would prevent renters acting on their own volition, this means the extent of modifications a renter is allowed would be dependent on the stipulations made in the legal contract between the landlord and tenant. Implementing rooftop solar photovoltaic, solar water heaters and air conditioning in rented homes may require having to navigate the power dynamics between renter and property owner and the legal requirements in place as they may require infrastructural work to be installed unlike LED lighting which is a simple installation.

With regard to leadership and decision-making within homes, it is notable that 67% of households are male-headed while only 33% are female-headed. Given that men and women have different energy needs and uses due to division of labour¹²¹, it is important to consider whoever is making the decision on the sources of energy to be used by the household as there can be implications for the household's energy costs and consumption patterns. Women in male-headed households may spend more time at home performing domestic work and will benefit more from the use of solar water heaters in assisting with their household tasks than the male head would, whom may have less use for hot water. Likewise, women in male-headed household would spend more time at home and would benefit more from the cooling provided by air conditioning, gaining relief from the rising temperatures than the male head who may spend a third or more of the day out of the home for work. Given these possibilities, the male head may deem these technologies as low priority as they do not directly see the immediate benefits.

On the opposite spectrum, female-headed households may have the same needs of the solar water heater and other technologies as women in male-headed households, however they must contend with conflicting responsibilities. In a male-headed household, there is a possibility for multiple income sources in cases such as married couple, common law couple or other types of relations. For female-headed households¹²², these are typically single income homes with a woman caring for children or other dependents. In this situation with more limited resources, women must weigh their needs and choose according to their priorities wherein acquiring solar water heater or the other technologies may be designated as a low priority action in favour of other pressing needs.

Livelihoods

A critical element to consider when assessing the barriers and the capacities of persons to implement the target technologies is understanding the source of their income generation. Livelihood plays a major role in a person's ability to build capacity and empower themselves and would have implications for the climate change adaptations, in this case the implementation of RE/EE/DER technologies they can afford and are capable of implementing. While livelihoods could be impacted by a person's ability to access education and opportunities, it influences the other barriers to accessing these technologies such as housing and finance. Income from livelihoods would then determine what technologies a person can afford, what loans they are

¹¹⁹ "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from

<https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

¹²⁰ "Landlord and Tenant Act" retrieved from https://www.belizejudiciary.org/download/LAWS-of-Belize-rev2011/Laws-of-Belize-Update-2011/VOLUME%209B_2/Cap%20189%20%20Landlord%20and%20Tenant%20Act.pdf

¹²¹ "Energy" retrieved from <https://eige.europa.eu/gender-mainstreaming/policy-areas/energy#:~:text=From%20a%20gender%20and%20human,heating%20and%20indoor%20air%20quality.>

¹²² Due to gender norms present and exhibited in the Caribbean, when speaking of female-headed households this usually entails a lone woman leading the household with no adult males (excluding her children) present in the household.

eligible and what other related measures they are able to adopt for the successful and effective implementation of the target technologies.

One of the biggest factors that affects a person's livelihood and their options for livelihood is their ability to access education and assets.¹²³ Having access to necessary resources and education, empowers one to capitalise on opportunities and enhances their ability to participate in climate change adaptation measures in this instance. From the highest education level attained by heads of households, it could be seen that almost half (49.2%) heads of households have only primary school level of education as their highest form education, 26.7% with a secondary school level, 16.9% with a tertiary school level and 7.2% with no formal education.¹²⁴ It is critical to acknowledge that affects the kinds of livelihoods persons engage in and it is reflected in the occupations held by persons as seen in the Belize Labour Force Survey. From the 2022 survey, services and sales workers had the largest share of employed persons at 58.1 and generally female-dominated (39.7% females to 18.4% males) while elementary occupations followed with 35.5, still male-dominated (21.5% males to 14% female).¹²⁵ A large percentage of services and sales jobs are associated with the tourism sector (with it accounting for 28% of employed persons) which, in turn, has significant implications for income generation for those tied to the sector. The seasonal nature of the tourism sector is of concern due persons' sources of income decreasing during the off-peak season which receives less visitors to the country (the high season for tourism in Belize is from December to April which coincides with its dry season¹²⁶) as this will affect the level of loan they might qualify for, the financial decisions they need to make during and before the off-season and other issues that may result from the lowered economic ability during this time.

Traditional gender norms are at play in the share of men and women in the tourism sector. Despite it being one of the more important sectors in the country, women still dominate this sector as jobs related to catering to and providing services to persons which are typically perceived as activities for women. This perception of some jobs being deemed for women, and some being deemed for men continues to perpetuate the labour market segmentation and impacts career/livelihood choices for men and women. This is seen above with the share of women employed in sales and services jobs accounting for double the number of men in this field. This implies that these traditional gender norms still hold root in society with the way persons choose careers. The domination of women in this field has one major consequence which is on a yearly basis; many women see reduced income between May to November when the country receives less visitors and business slows down. The gender division of labour can also be seen in the share of women and men in clerical support worker roles (13.5% for women and 5.6% for men). Meanwhile, in more technical/physical fields, men dominate more as seen in the following breakdown: plant and machine operators 9.1% men to 0.6% women, craft and related trades works 19.3% men to 3.8% women and skilled agricultural and forestry workers 7.8% men to 0.8 women%¹²⁷. Noted in 2020 Energy Report Card, women only accounted for 30.3% of the energy sector workforce which is comprised of the energy unit, BEL and the public utilities commission (PUC) (see Figure 3-17), further exhibiting the traditional division of labour seen in the country.¹²⁸

¹²³ "Education and Livelihood" retrieved by <https://govdocs.in/education-and-livelihood/>

¹²⁴ "Belize Multiple Indicator Cluster Survey, 2015-2016, Final Report" retrieved from <https://www.unicef.org/belize/media/936/file/Multiple%20indicator%20cluster%20survey.pdf>

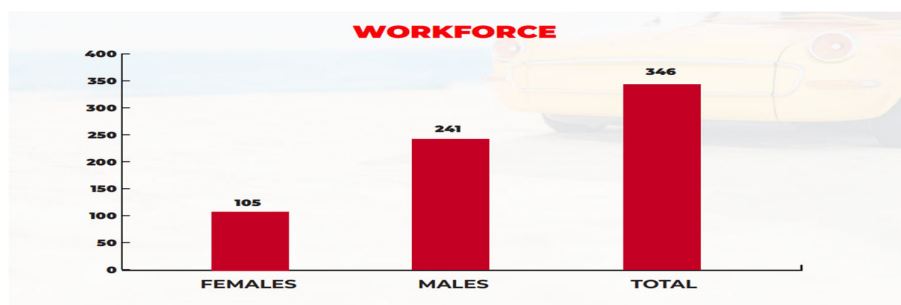
¹²⁵ "Labour Force Survey 2022" retrieved from <https://sib.org.bz/statistics/labour-force/>

¹²⁶ "Best Time to Visit Belize" retrieved from <https://www.belizeadventure.ca/travel-tips/when-is-the-best-time-to-visit-belize/#:~:text=the%20low%20season.-,High%20Tourism%20Season,for%20amazing%20weather%20in%20Belize.>

¹²⁷ "Labour Force Survey 2022" retrieved from <https://sib.org.bz/statistics/labour-force/>

¹²⁸ "2020 Energy Report Card – Belize" retrieved from <https://www.ccreee.org/wp-content/uploads/2022/01/CCREEE-ERC-BELIZE-2020.pdf>

Figure 3-17: Energy Sector Workforce¹²⁹. Source¹³⁰



Men enjoy more stable sources of income which would allow them to amass finances and be able to buy into the services more easily being offered via the IUS project and model. Issues of affordability would be less of a concern for many as they should be able to outright afford payments or be able to acquire loans. Investments into RE, EE and DER would pose less of a risk and liability to men. This is the case for the more immediate reality but also likely in the medium term as well. For women, challenges may exist due to the nature of their work and other responsibilities they bear particularly as single heads of households. For women outside of formal employment, unemployed and informally employed, family care and duties were identified as the main cause of those not actively seeking work in the 2022 LFS. The traditional division of labour would see men benefiting more from the IUS project in terms of employment opportunities as men are represented more in the associated engineering and energy fields. The three indigenous women who were trained in solar technology do represent an opportunity for more women to get involved in the sector.

3.2.5 Institutional Gender Mainstreaming Capacity

A limited gender mainstreaming capacity assessment was performed with stakeholders as part of the stakeholder engagement process conducted to support the gender assessment of Belize. These series of questions were designated to obtain some insight into the capacity of stakeholders to perform gender mainstreaming as well as delve into what institutional frameworks within their organisations exist for gender. Meeting with country's main body charged with gender would have been pivotal to understand its capacity and ongoing work that may be able to inform or feed into this project.

From the government institutions, it can be seen in Figure 3-18, there is no gender mainstreaming capacity¹³¹ and limited experience with gender mainstreaming capacities, highlighting a need for gender sensitisation and capacity- building. It is important to note that NGOs particularly Productive Organization for Women in Action (POWA) has capacity to support gender mainstreaming activities and capacity- building, given their history with supporting renewable energy initiatives from another Belizean NGO, Plenty Belize.

Noting from the Ministry of Energy that while projects they have carried out in the past had substantial participation from women, the ministry has never analysed these projects from a gender perspective. A competency-based approach is used in hiring and procurement.

Important to flag is the limited gender mainstreaming capacity of the Executing Entity (BEL) as they will be charged with executing and handling various aspects of the project. Their limited

¹²⁹ The workforce consists of the staff of BEL, PUC and the energy unit and does not provide a breakdown of the roles, men and women occupy.

¹³⁰ "2020 Energy Report Card – Belize" retrieved from <https://www.ccreee.org/wp-content/uploads/2022/01/CCREEE-ERC-BELIZE-2020.pdf>

¹³¹ Noting that the consultants were unable to meet with the Women and Family Support Department.

capacity will need to be a priority to address and resolve as early as possible in the project's lifespan.

Figure 3-18: Belize Stakeholders' Gender Capacity

Stakeholder	Does your organisation have any experience with mainstreaming gender in its past work?	Does your organisation have the capacity to execute gender mainstreaming activities or provides gender sensitisation training/gender-related technical inputs?	Does your organisation possess a gender policy or is it in the process of developing a gender policy?	Has your organisation ever been engaged by government ministries/agencies or development partners to support in the development or implementation of energy/renewable energy projects?	Does your organisation utilize a grievance redress mechanism for projects?
POWA	YES	YES	YES	YES	NO
PETAL	YES	YES	NO	YES	NO
Plenty Belize	YES ¹³²	NO	NO	YES	NO
Belize Assembly for Persons with Diverse Abilities	YES	YES	NO	YES	NO
Ministry of Energy	YES	NO	NO	YES	YES
PUC¹³³	N/A	N/A	N/A	N/A	N/A
Women and Family Support Department	N/A	N/A	N/A	N/A	N/A
BEL	NO	NO	NO	YES	YES

¹³² Plenty Belize noted it collaborates with POWA for gender mainstreaming needs in past work.

¹³³ Consultant was unable to meet with the PUC and the Women and Family Support Department and as such will not be able to provide answers for the gender mainstreaming questions for this organisation (N/A will be applied in response to the questions).

Executing Entity Gender Capacity Analysis

One of the key stakeholders for this project will be the utility, BEL (the Executing Entity). They will be charged with executing this project which would entail managing the funds to be allocated to support the distribution of the renewable energy technologies to customers and interacting with customers in general. Noting this will entail BEL interacting with vulnerable groups such as rural, low-income women, indigenous people and PWDs, it was important to understand how BEL operates, how they engage with customers, how complaints are handled and the capacity of the organisation and staff. Given BEL's importance to the project, it was key to engage with them to gain insight into these areas.

BEL shows flexibility in meeting and accommodating their customers through policies that allow customers to conduct all their business via WhatsApp communication. For those customers who do not utilise WhatsApp, BEL will go to their location in order to conduct business with said customers.

BEL frequently participates in trade shows which they use to educate customers on policies and procedures in order to boost customers' ability to engage the utility. It is noted that BEL has no specific engagement with PWDs or persons from LGBTQIA+ community but the utility does work with indigenous groups to increase access to electricity. BEL is aware of the Free, Prior, and Informed Consent (FPIC) and operations with indigenous people are guided by FPIC to a degree.

BEL has a grievance redress mechanism in place for employees, however for customers, grievance is handled by the regulator (PUC).

BEL notes that it is not in possession of a gender policy.

3.3 Jamaica

3.3.1 Policy environment

Jamaica's renewable energy capacity stood at 17.1% in 2020 with solar and wind contributing to most of the country's current renewable energy usage, which means a lot of work needs to be done in order to meet the new goal set out.¹³⁴

Jamaica's Integrated Resource Plan 2018 laid out the country's goal to reach 30% renewable energy usage by 2030.¹³⁵ However, the country has a new goal it is aiming to achieve following after Prime Minister Andrew Holness revealed intentions for a new standard of 50% renewable energy by 2030 instead.¹³⁶ The transition towards more energy efficient methods has the capacity to greatly minimise the time and labour constraints associated with gender inequality while also enhancing community health, income generating opportunities, and capacity-building efforts.¹³⁷ However, Jamaica faces issues of costing, an enabling policy environment and barriers to entry, particularly for women.

Figure 3-19 provides a review of current policies. The policies listed in the Jamaica context include no substantive provisions for advancing the area of gender equality for the country and can be deemed as negligible in terms of support for any necessary gender mainstreaming actions. However, the National Policy on Gender Equality (NPGE), the NDC and the Vision 2030

¹³⁴ "Jamaica Energy Report Card" retrieved from <https://www.ccreee.org/wp-content/uploads/2022/01/CCREEE-ERC-JAMAICA-2020.pdf>

¹³⁵ "2018 Jamaica Integrated Resource Plan" retrieved from <https://www.mset.gov.jm/wp-content/uploads/2020/03/2018-Jamaica-Integrated-Resource-Feb-21-2020.pdf>

¹³⁶ "Jamaica's Prime Minister Goes Solar [Press Release]" retrieved from <https://solarheadofstate.org/press-releases/2018/jamaicasolar>

¹³⁷ "Industrial Development Report 2011" retrieved from <https://www.unido.org/resources-publications-flagship-publications-industrial-development-report-series/industrial-development-report-2011>

all provide a good framework for gender mainstreaming action and could serve as an entry point for revisions to relevant energy policies.

Figure 3-19: Energy and Policy Environment for Jamaica

Policy	Summary	Gender Content	Gender Rating
NDC, 2020	It provides an update on the country's progress and outlines actions for meeting its commitments for the Paris Agreement.	It notes that will be guided by the gender equality strategy	Sensitive
Jamaica National Energy Policy 2009-2030	This policy sets forth steps to creating an enabling environment for achieving the goal of "a secure and sustainable energy supply for our country" that is laid out in the National Development Plan.	No mentions of gender, women or marginalised communities.	Neutral
National Energy Conservation and Efficiency Sub-Policy of 2010	This policy is intended to support the goals of the National Energy Policy by calling for the improvement of energy efficiency in key energy consuming sectors of power generation and manufacturing extraction and production processes.	No mentions of gender, women or marginalised communities.	Neutral
National Renewable Energy Sub-Policy 2009-2030	This policy is intended to support the goals of the National Energy Policy by enabling the development of a diversified renewable energy sector.	No mentions of gender, women or marginalised communities.	Neutral
Vision 2030 Jamaica	This plan sets forth the goals and pathways for Jamaica to achieve an internationally competitive economy, secure and cohesive society, healthy natural environment and promoting human capital development as well as greater opportunities for social and economic mobility and prosperity.	The plan recognizes inequalities experienced by women and persons with disabilities and seeks to rectify these issues by creating an environment that ensures future policies captures these issues while also outlining plans to address some of the underlying societal problems.	Transformative
National Policy for Gender Equality of 2021	This policy outlines Jamaica's commitment and actions to developing and fostering gender equality for the nation.	The policy addresses the gender inequalities prevalent in society but also provides actions to resolve these inequalities while guiding the nation to gender equality. There are no direct mentions to energy within the	Transformative

Policy	Summary	Gender Content	Gender Rating
		policy, but it does speak to enhancing women's access to resources.	
Jamaica's Long-Term Emissions Reduction and Climate Resilient Strategy and Operationalisation Plan¹³⁸	This strategy works in tandem with the goals of Vision 2030 Jamaica to attain sustainable, low carbon development and outlines targets and actions per sector achieve this goal.	The document recognizes that vulnerability varies across different groups and recognizes how vulnerability impacts climate resilience and outlines principles to be incorporated into future actions to ensure future activities are gender responsive	Transformative

Seven policies were identified in Jamaica: 3 neutral, 1 sensitive and 3 transformative. The latest, the LTS, combines several elements and provides a critical long-term vision for where energy fits within a broader climate change response and sustainable development pathway. While the energy policies are relatively gender-neutral (or gender blind), other broader policies tended to be more gender transformative.

The NPGE is quite detailed and one of the few across the region and among the three countries that is detailed across several sectors, including energy. The fact that two of the above-mentioned policy frameworks are transformative also lays a critical foundation for just transition elements of the proposed GCF proposal depending on what is agreed for Jamaica compared to other countries. It provides a framework on the country's commitments toward gender equality measures, providing an enabling environment to achieve fairness and socio-economic justice for both women and men, in line with the goals and objectives as defined by Jamaica's Vision 2030 plan. Within the policy, an overarching goal defines the need to transform existing gender ideologies and create transformation within the public sector. The policy's objectives support reducing gender-based discrimination and improving mainstreaming of gender within all cultural, social, political, and economic structures and systems. It also encourages the promotion of capacity-building and improved organisational behaviour throughout the development, implementation, and monitoring of gender responsive activities. Despite the extent of cross-cutting information across sectors: poverty, education, labour, health, trade, agriculture, and social protection systems, and culture, the author noted no specific references to renewable energy and women's involvement or provision of services, noted within the policy.

Jamaica's update of their NDC was published in 2020. In preparation of the document, consultation was conducted with local communities and indigenous populations in a gender-responsive way. The document notes the country's progress towards its measures for reaching SDG 5, gender equality. It also notes its commitment to adhering to policies and programmes aimed at improving environments for inclusivity and fairness as outlined in the National Policy for Gender Equality, 2021.

Jamaica's National Development Plan¹³⁹, Vision 2030, developed in 2010 in intended to provide a roadmap for making Jamaica a developed country by 2030. The plan consists of several goals and associated outcomes. Within the four main goals are measures for social protection, health,

¹³⁸ A draft of this document was received by the consultant who participated in a validation meeting for the document.

¹³⁹ "Vision 2030" retrieved from <https://www.vision2030.gov.jm/>

and education; a secure and cohesive society; a prosperous economy; and maintaining a thriving natural resource environment. The plan recognises inequalities experienced by women and PWDs and seeks to rectify these issues by creating an environment that ensures future policies captures these issues while also outlining plans to address some of the underlying societal problems. Within the energy sector, the plan promotes improved energy conservation within households, but does not make specific reference to gender or women. The same exists for promoting a technology-enabled society, while noting a desire to expand knowledge and development, creativity and development of products and services, does not specifically reference gender or improving the needs and livelihoods of women.

Jamaica has recently, as of July 2022, developed and launched a gender and climate change strategy and action plan (GCCSAP), guided by the results of a national gender and climate change assessment.¹⁴⁰ The assessment, in part, conducted an analysis of gender and climate change linkages, providing recommendations for action in the country going forward. Financed by the GCF, the GCCSAP includes provisions for institutional capacity for mainstreaming gender in climate finance programming.¹⁴¹

3.3.2 Usage

In general, women tend to face a disproportionate burden of labour compared with men, through their responsibilities taking care of children and other household tasks, in addition to any labour performed outside the home for earned income¹⁴². In terms of women's abilities to access financial assets and resources, infrastructure services can play a role in lessening the added burdens placed on women through their dual roles as caretakers and paid employees in the labour force.¹⁴³

Ensuring proper services through sustainable energy provisions can provide both cost and time-savings to consumers, especially women who already may be operating in a deficit in each of these areas. These economic and time savings may allow women to accumulate capital which can have additional benefits in terms of resource allocation toward better health, added income and education for women in most contexts.

LED Lighting

In recent years, the Jamaican Public Service (JPS) has been working on energy conversion projects and initiatives in Jamaica, including those for light-emitting diode (LED) lighting and high-pressure sodium (HSP) lamps. Currently, LED lighting and public buildings targeted for energy efficiency retrofits consume only 7% of Jamaica's total generated electricity.¹⁴⁴ In 2018, the JPS, converted 63% of the island's streetlights over to LEDs. In real terms, this covered a total of 67,000 – out of the 105,000 streetlights on the nation's roads. In 2022, the company was also reported to have converted all existing high-pressure sodium (HPS) lamps with LEDs within Jamaica. These are more energy efficient, have a longer service life and added intelligent-technology capabilities. JPS is continuing work on its streetlight general maintenance and repair programs in the interim.

Difficulties with the HPS lighting, including vandalism, have been an issue. These problems are confirmed with statistics showing that an average of 620 repairs per month were conducted by

¹⁴⁰ "Jamaica launches gender and climate change strategy and action plan" retrieved from <https://www.iisd.org/articles/iisd-news/jamaica-launches-gender-and-climate-change-strategy-and-action-plan>

¹⁴¹ "Jamaica launches gender and climate change strategy and action plan" retrieved from <https://www.caribbeannationalweekly.com/news/caribbean-news/jamaica-launches-gender-and-climate-change-strategy-and-action-plan/>

¹⁴² "A Quantum Leap for Gender Equality" retrieved from https://www.ilo.org/wcmsp5/groups/public/---dgreports/---dcomm/---publ/documents/publication/wcms_674831.pdf

¹⁴³ "Balancing The Burden? Desk Review of Women's Time Poverty and Infrastructure in Asia and The Pacific" retrieved from <https://www.adb.org/sites/default/files/publication/177465/sdcc-balancing-burden.pdf>

¹⁴⁴ "Energy Management and Efficiency in Jamaica" retrieved from <https://blogs.iadb.org/energia/en/energy-management-and-efficiency-in-jamaica/>

JPS, due to high rates of maintenance repairs due to crime.¹⁴⁵ Efforts by JPS on street lighting is a part of Jamaica's Energy Policy, to facilitate the introduction of energy-saving devices e.g., LED, solar panels, and solar street lighting under Goal 1.¹⁴⁶ Applications from persons for new streetlights can be made to the Ministry of Local Government, as Jamaica Public Service (JPS) must be authorized by the Parish Council to install new street lighting infrastructure¹⁴⁷ and outside of Jamaica's National Energy Policy¹⁴⁸, there are no noted implementing frameworks that support the energy transition to LEDs.

LED Lighting for Support for Women and Vulnerable Groups

Provision of improved lighting services can support various vulnerable groups, in the urban and rural context. Poor lighting services can keep women and girls out of public spaces, exacerbating their senses of vulnerability and insecurity, and placing restrictions of their ability to safely pursue independent living. Ample street lighting has been seen to reduce the prevalence of crime in areas with sufficient lighting.¹⁴⁹ Increased Street lighting can offer important economic advantages in addition to safety benefits. Better lighting services can enable longer business hours in some cases where work hours transition from daylight hours, this is especially the case for women who operate in the informal economy in roles such as street vendors.

By creating safe gathering spaces, lighting services have also been shown to support the reduction of social isolation, boost community cohesion, and decrease the number of people, especially women who stay indoors during the night.¹⁵⁰ For other vulnerable populations, improved lighting services has also shown to assist with children's education in terms of how well they perform and function by supporting their visual, cognitive, and behavioural abilities through focus, alertness, and mood management.¹⁵¹ It also makes it easier for those with visual disabilities to see better and allows a person with hearing impairments to make lipreading easier.

Problems and challenges, however, do exist in transitioning energy systems and improving lighting capacity. In 2021, the theft situation rapidly worsened in Jamaica, hampering government efforts to illuminate crime-ridden areas. The power company disclosed in 2019 that the thefts were harming its investments in public lighting.¹⁵² Within the first 5 months of the year, 525 streetlights have been stolen. The US Department of State reports that the municipalities of Montego Bay and Portmore have some of Jamaica's worst rates of violent crime. These locations are where streetlight theft was most prevalent. Theft also occurs in the Northern towns of Moneague and Claremont in St. Ann, Jamaica's largest parish.¹⁵³

In such environments, the lack of proper public lighting due to theft can further exacerbate feelings of insecurity and vulnerability, especially for women who may need to navigate these areas during nighttime hours. Moreover, women are often primary caregivers and responsible

¹⁴⁵ "JPS Smart LED Street lighting Programme to Improve Lighting Across Jamaica" retrieved from <https://jis.gov.jm/jps-led-street-light-programme-to-be-completed-next-year/>

¹⁴⁶ "Jamaica's National Energy Policy 2009-2030" retrieved from https://www.mset.gov.jm/wp-content/uploads/2019/07/Draft-Renewable-Energy-Policy_0.pdf

¹⁴⁷ "JPS hits 105,000-mark in LED streetlights" retrieved from <https://jamaica-gleaner.com/article/news/20220616/jps-hits-105000-mark-led-streetlights>

¹⁴⁸ "Jamaica's National Energy Policy 2009-2030" retrieved from https://www.mset.gov.jm/wp-content/uploads/2019/07/Draft-Renewable-Energy-Policy_0.pdf

¹⁴⁹ "Reducing Crime Through Environmental Design: Evidence from a Randomized Experiment of Street Lighting in New York City" retrieved from https://urbanlabs.uchicago.edu/attachments/e95d751f7d91d0bcbfeb209ddf6adcb4296868c12/store/cca92342e666b1ffb1c15be63b484e9b9687b57249dce44ad55ea92b1ec0/lights_04242016.pdf

¹⁵⁰ "Transformative power of innovative street lighting in Vanuatu" retrieved from <https://www.undp.org/pacific/blog/transformative-power-innovative-street-lighting-vanuatu>

¹⁵¹ "Impact Of Lighting on Children's Learning Environment: A Literature Review" retrieved from <https://www.witpress.com/Secure/elibrary/papers/SC21/SC21031FU1.pdf>

¹⁵² "JPS' Smart LED Streetlight Improvement Efforts Being Impacted by Theft" retrieved from <http://radiojamaicanewsonline.com/local/thieves-make-off-with-jps-street-lights-in-silversun-estate>

¹⁵³ "Jamaica Struggling with Constant Theft of Street Lights" retrieved from <https://insightcrime.org/news/jamaica-struggling-with-constant-theft-of-street-lights/>

for the safety and well-being of their families. Insufficient lighting in their communities can hinder their ability to protect themselves and their children, further compromising their sense of security.

As most of the country has already been converted to LED lighting, additional challenges can also occur during the repair and replacement process of stolen LEDs. The JPS may be reluctant to constantly invest in replacement, especially in high-crime areas as mentioned before. It may not be deemed feasible to fix streetlights, leaving the areas that need lighting the most, without. The safety of JPS employees is also a concern as installing and fixing streetlights in high crime areas is another potential constraint as employees become hesitant to enter areas for fear of persons stealing the lights from their possession and for their own personal safety.

Solar Water Heating

In 2009, it was estimated that more than 5,000 solar water heaters (SWH) were installed in Jamaica. As of 2020, Jamaica's use of renewables accounted for approximately 11 per cent of its energy: 6.5% wind, 3.5% hydropower, and 1% solar power.¹⁵⁴

The SWH market in Jamaica has the potential to be expanded due to the high levels of insolation and urbanism. The country receives an average daily peak of 4.6 hours of sunlight for solar panel absorption.¹⁵⁵ The combination of good climatic conditions and Jamaica's readiness and urgency to diversify its energy sector will create favorable demand for SWH. However, there are specific barriers to SWH adoption which include limited access to capital, incomplete information and data, lack of policies governing the injection and transitions of renewable energy resources into the power grid and economic regulatory distortion.¹⁵⁶ Each of these factors pose challenges to promoting gender equality and women's empowerment efforts due to the limited access to capital available for women, a lack of available sex disaggregated data to inform decision-making and the limitations on gender mainstreaming in policy across the energy sector more generally.

SWH systems in the Caribbean typically have short payback periods and a high net present value (NPV) which means over time, not only will systems pay for themselves but also generate profits or in this case immense savings.¹⁵⁷ They can also be extended depending on incentives, local electricity rates, and system sizing.¹⁵⁸ Prices vary based on sizes offered, materials used and other factors, but they generally begin at USD1,000 not including installation.¹⁵⁹ This cost can deter home and business owners from investing. These issues of costing can pose a significant barrier for women-headed households who, in general, de facto face income constraints. It is estimated that 47% of children live in single-parent homes with their biological mothers.¹⁶⁰ To address some of the issues around this cost prohibitive barrier to access, financial institutions have begun offering segmented renewable energy financing. Although the direct purpose of environmentally conscious financial initiatives are typically intended to support energy efficiencies more generally, conservation and alternatives, they can be utilised for the installation of SWH. The Development Bank of Jamaica, Exim Bank, National Housing Trust and

¹⁵⁴ "Jamaica Energy Snapshot" retrieved from <https://www.energy.gov/eere/articles/jamaica-island-energy-snapshot-2020>

¹⁵⁵ "Jamaica Solar Power Information & Peak Sun Hours" retrieved from <https://www.turbinegenerator.org/solar/iowa/jamaica/#:~:text=For%20Jamaica%20the%20number%20of,for%20a%202%2Daxis%20panel.>

¹⁵⁶ "Barriers of solar energy uptake and the potential for mitigation solutions in Barbados" retrieved from <https://www.sciencedirect.com/science/article/pii/S1364032118303095>

¹⁵⁷ "Break-even Cost for Residential Solar Water Heating in the United States: Key Drivers and Sensitivities" retrieved from <https://c2e2.unepccc.org/wp-content/uploads/sites/2/2020/06/solar-water-heater.pdf>

¹⁵⁸ "Solar Water Heating in the Tourism Industry" retrieved from <https://solar-dynamicslimited.com/wp-content/uploads/2011/11/UNEP-SWH-Extracts-SDL.pdf>

¹⁵⁹ These cost can range from USD 2 to 4 per Watt with most homes having up to 5kW systems.

¹⁶⁰ "Shifting the narrative of the single parent" retrieved from <https://www.jamaicaobserver.com/all-woman/shifting-the-narrative-of-the-single-parent/>

the National People's Cooperative Bank of Jamaica Limited all offer financial services specifically mentioning SWH.¹⁶¹

Heated water may be considered a luxury for some, but it is a part of many persons' everyday life. While the water is mainly used by everyone in the household for showering, women tend to use warm water more for themselves as well as for their children and household chores. As the current levels of water heating currently account for up to 30 per cent of a domestic light bill in Jamaica¹⁶², more sustainable methods of water heating should be considered.

SWH can provide financial benefits along with environmental sustainability. It lowers the carbon footprint and is expected to reduce electricity bills since the fuel is free sunlight.¹⁶³ Due to low operating and maintenance costs, solar water heaters over the long-term cover the huge initial fixed cost.¹⁶⁴ When all costs for purchase, installation, maintenance and operation are taken into account after the payback period, the SWH systems are expected to provide hot water for free with bare maintenance costs as sunlight is free.¹⁶⁵ Thereafter, electricity bills can be reduced by 30%, making it a cost-effective option for low-income homes including single-mother homes, persons with disabilities and retired persons.¹⁶⁶

Rooftop Solar Photovoltaic

The transition to a greener and more sustainable future can be facilitated by creating a sector for clean energy that is gender inclusive and gives women in rural communities more influence as both consumers and producers of clean energy. Targeted policy initiatives concentrating on improving loan access, skill development, education, and broader systemic gender issues like property and equal access laws, land rights, and political engagement are necessary to support women's success in the sector.¹⁶⁷

Women may play a crucial role in the transition to renewable energy by influencing patterns of energy use. They directly gain from using clean energy as consumers. With access to clean energy technologies, women's exposure to indoor air pollution can be eliminated.¹⁶⁸ Additionally, the ability to use time-saving appliances (like lighting, cookstoves, fans, and mobile phone chargers) are made possible due to increased connection to electricity and this in turns leads to a reduction in women's time poverty.¹⁶⁹ This involves running clean energy businesses that use energy directly (such as water pumps for irrigation and grain mills), as well as marketing these products and services (for example solar panels, clean cookstoves, solar lamps, etc.).¹⁷⁰

Clean energy such as rooftop solar can reduce the load on the electricity grid, increasing the rate of access to many. Jamaica's urban community enjoys generally full access to electric services, whilst those within the rural community have experienced increased access to electricity with 96.01% coverage in 2016 up from 94% in 2015. Many positive ripple effects on

¹⁶¹ "Financing institutions offering renewable energy financing" retrieved from <https://buildbetterja.com/2020/10/12/financial-institutions-offering-renewable-energy-financing-brochure/>

¹⁶² "Jamaica's National Energy Policy 2009-2030" retrieved from https://www.mset.gov.jm/wp-content/uploads/2019/07/National-Energy-Policy_0.pdf

¹⁶³ "The Importance of Using Solar Water Heater as an Alternative Eco-Friendly Technology in Global Market: Some Lessons of Experiences for Bangladesh Economy" retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2928814

¹⁶⁴ "Estimating the Cost and Energy Efficiency of a Solar Water Heater" retrieved <https://www.energy.gov/energysaver/estimating-cost-and-energy-efficiency-solar-water-heater>

¹⁶⁵ "Estimating the Cost and Energy Efficiency of a Solar Water Heater" retrieved <https://www.energy.gov/energysaver/estimating-cost-and-energy-efficiency-solar-water-heater>

¹⁶⁶ "How much of your Electricity Bill is for Hot Water" retrieved from <https://carbontrack.com.au/blog/hot-water-bill/>

¹⁶⁷ "How clean energy can empower women in rural communities" retrieved from <https://www.theiqc.org/blogs/gender-equality/how-clean-energy-can-empower-women-rural-communities>

¹⁶⁸ "How Does Clean Energy Consumption Affect Women's Health: New Insights from China" retrieved from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9266163/>

¹⁶⁹ "Powering Households and Empowering Women: The Gendered Effects of Electrification in sub-Saharan Africa" retrieved from <https://ipia.princeton.edu/news/powering-households-and-empowering-women-gendered-effects-electrification-sub-saharan-africa>

¹⁷⁰ "Applying for Street Lights" retrieved from <https://localauthorities.gov.jm/services/%E2%80%8BApplying-street-lights>

women's lives and socioeconomic standing emerge from having access to reliable energy. This includes lowering the health hazards associated with some of the present energy practices, cutting back on the time and effort required for domestic activities, improving local job prospects, raising the standard of living, and boosting productivity. There is a compelling argument for empowering women by integrating and employing them as business owners, workers, innovators, designers, and even decision-makers at the highest levels of the energy system's value chain.¹⁷¹ This would provide women with the opportunity to realise their entrepreneurial potential and act as change agents on the front lines.¹⁷²

Low-income consumers, generally comprised of female-headed households within Jamaica, can use solar energy technology as one of several strategies to lessen their energy costs, but up to now, they have not adopted solar at the same rate as other income groups. Since they pay a larger percentage of their income toward utilities, low-income consumers are disproportionately more vulnerable to the effects of shifting electricity prices.¹⁷³ The 2017 Jamaica Survey of Living Conditions provided valuable data indicating that poverty disproportionately affects female-headed households. The survey revealed that 14.8% of such households experienced poverty. Furthermore, a significant majority of the poorest female-headed households, approximately 54.8%, were single-parent households.¹⁷⁴ These findings highlight the vulnerability of these households and individuals to various socio-economic challenges, including labour market fluctuations, energy insecurity, and affordability issues.¹⁷⁵ Moreover, they face additional difficulties when confronted with energy rate increases or temporary hikes. These circumstances further accentuate the economic pressures faced by female-headed households and underscore the need for targeted support and measures to address their specific energy-related concerns.

Efficient Cooling and Refrigeration

Energy-related behaviours and cooling energy consumption are influenced by gender, noting variations in consumption among different genders.¹⁷⁶

A general gender comparison study conducted by researchers in China noted that, particularly for women, using the air conditioner frequently is a regular practice that affects cooling energy use. However, thermal preferences have more noticeable influence on cooling energy consumption for males than females.¹⁷⁷ These differences exhibit themselves in common habits such as daily average hours of air conditioning use, the use of electric fans in place of air conditioners, and the occupancy ratio in room and these habits have an impact on cooling energy consumption for both men and women. Women are more likely to provide unpaid care for young children, elderly individuals, and those with disabilities, who spend more time indoors due to physical and mental conditions and would therefore require more cooling in poorly ventilated homes.¹⁷⁸ With this situation, they need to use more energy for cooling. In comparison, men are shown to be more likely to labour outdoors or in hot locations, making them more sensitive to

¹⁷¹ "What Is Happening With Gender Equality and How Is Women's Corporate Situation Today?" retrieved from <https://youmatter.world/en/equality-men-women-work-performances/>

¹⁷² "Solar power for women empowerment" retrieved from <https://www.undp.org/lebanon/stories/solar-power-women-empowerment>

¹⁷³ "How High Are Household Energy Burdens" retrieved from https://www.michigan.gov/-/media/Project/Websites/mpsc/workgroups/eaac/aceee_energy_burden_report_2020.pdf?rev=24565d21283d4b70993a6050c80513d4

¹⁷⁴ "Jamaica Survey of Living Conditions" retrieved from <https://www.pioj.gov.jm/product/jamaica-survey-of-living-conditions-jslc-2017-executive-summary/>

¹⁷⁵ "Jamaica Survey of Living Conditions" retrieved from <https://www.pioj.gov.jm/product/jamaica-survey-of-living-conditions-jslc-2017-executive-summary/>

¹⁷⁶ "Typical energy-related behaviours and gender differences for cooling energy consumption" retrieved from <https://www.sciencedirect.com/science/article/pii/S095965261932712X>

¹⁷⁷ "Typical energy-related behaviours and gender differences for cooling energy consumption" retrieved from <https://www.sciencedirect.com/science/article/pii/S095965261932712X>

¹⁷⁸ "Gender differences in caregiving among family – caregivers of people with mental illnesses" retrieved from <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4804270/>

the effects of climate change from an outdoor labour perspective which can also have similar implication towards health and livelihood management.¹⁷⁹

A refrigeration and cooling case study conducted in 2020 in Kenya found that the use of cooling technology was able to boost productivity and give women more options to earn money and more freedom in general. While this study was not specifically directed to the Jamaican context, refrigerators can provide women more power by eliminating tedious tasks and providing added time savings. The capacity to preserve food can reduce the time needed to shop and prepare meals and can save food costs by reducing the need to get rid of food that has spoiled.

As another method to provide cooling for labour and in the home, fans are a relatively low-cost appliance, which can help increase productivity and help those at most risk adapt to climate change.¹⁸⁰ A study conducted to look at the insights of electricity poverty in urban Jamaica found that electric fans were the most common electrical appliance used (at a rate of 94%); followed by television (90.8 per cent); and refrigerators (89.2%) in the communities of Kingston and St. Andrew, St James, Clarendon, and St Catherine.¹⁸¹ Electrical appliances that consume a large amount of electricity such as air conditioners and electric stoves were in use for only 7.8% and 2.5% of the households, respectively.

At present, there are currently no approved policies in place to implement and spur the use of efficient cooling methods and refrigeration in Jamaica. However, as of 2021, as part of the regional Caribbean Cooling Initiative (C-COOL), Jamaica's National Cooling Strategy (JNCS) is currently entering its final stage toward approval, and its subsequent implementation. The strategy's recommendations were refined and were undertaken by various inter-agency working groups before submission to the Cabinet. Once approved, would be the second such strategy of its kind adopted in the region.¹⁸²

3.3.3 Demand

In 2021, Jamaica's demand for energy saw the consumption of 35 terawatts-hours (TWh).¹⁸³ The demand for energy was projected to decline by 6 million barrels of oil equivalent in 2020.¹⁸⁴ This downward trend is the consequence of the implementation of an effective efficiency improvement and conservation program as seen in Figure 3-20 and Figure 3-21.¹⁸⁵

¹⁷⁹ "Case study: conducting a gender impact assessment on urban cooling community education" retrieved from (Efficiency for Access, 2020) <https://www.genderequalitycommission.vic.gov.au/case-study-conducting-gender-impact-assessment-urban-cooling-community-education>

¹⁸⁰ Case study: conducting a gender impact assessment on urban cooling community education" retrieved from (Efficiency for Access, 2020) <https://www.genderequalitycommission.vic.gov.au/case-study-conducting-gender-impact-assessment-urban-cooling-community-education>

¹⁸¹ "Creating Policies that Foreground Appliances in Achieving Gender Equality and Energy Access for All" retrieved from <https://www.sacreer.org/sites/default/files/documents/files/Creating%20Policies%20that%20Foreground%20Appliances%20in%20Achieving%20Gender%20Equality%20and%20Energy%20Access%20for%20All%20by%20Marena%20Ileri%20-%20Efficiency%20for%20Access.pdf>

¹⁸² "Jamaica's National Cooling Strategy Prepared For Cabinet Submission" retrieved from <https://united4efficiency.org/jamaicas-national-cooling-strategy-prepared-for-cabinet-submission/>

¹⁸³ "Energy". Retrieved from <https://ourworldindata.org/energy/country/jamaica>

¹⁸⁴ "Jamaica's National Energy Policy 2009 – 2030". Retrieved from https://www.mset.gov.jm/wp-content/uploads/2019/07/National-Energy-Policy_0.pdf

¹⁸⁵ Ibid.

Figure 3-20: Jamaica's Projected Energy Demand (2005-2021)

Jamaica Energy Demand Projections - 2030							
	GDP Growth	Energy Demand Growth (% pa)			Energy Demand (Million boe)		
Year	%	S 1	S 2	S 3	S 1	S 2	S 3
2005	1.0	6.5	6.5	6.5	28.0	28.0	28.0
2006	2.7	3.9	3.9	3.9	29.1	29.1	29.1
2007	1.4	0.7	0.7	0.7	29.9	29.9	29.9
2008	-0.6	-7.0	-7.0	-7.0	27.8	27.8	27.8
2009	-2.3	-5.0	-5.0	-5.0	22.0	22.0	22.0
2010	0.1	0.2	0.2	0.1	22.0	22.0	22.0
2011	3.0	5.9	5.1	3.9	23.4	23.2	22.9
2012	3.0	5.9	5.1	3.9	24.7	24.3	23.7
2013	3.0	5.9	5.1	3.9	26.2	25.6	24.7
2014	3.0	5.9	5.1	3.9	27.8	26.9	25.6
2015	5.5	10.9	9.4	7.2	30.8	29.4	27.5
2016	5.0	9.9	8.5	6.5	33.8	31.9	29.3
2017	5.0	9.9	8.5	6.5	37.2	34.6	31.2
2018	5.0	9.9	8.5	6.5	40.8	37.6	33.2
2019	5.0	9.9	8.5	6.5	44.9	40.7	35.3
2020	5.0	9.9	8.5	6.5	49.3	44.2	37.6
2021	5.0	9.9	8.5	6.5	54.2	48.0	40.1

Figure 3-21: Jamaica's Projected Energy Demand (2021-2030)

Jamaica Energy Demand Projections - 2030							
	GDP Growth	Energy Demand Growth (% pa)			Energy Demand (Million boe)		
Year	%	S 1	S 2	S 3	S 1	S 2	S 3
2022	5.0	9.9	8.5	6.5	59.5	52.0	42.7
2023	5.0	9.9	8.5	6.5	65.4	56.5	45.5
2024	5.0	9.9	8.5	6.5	71.9	61.3	48.4
2025	5.0	9.9	8.5	6.5	79.0	66.5	51.6
2026	5.0	9.9	8.5	6.5	86.8	72.1	54.9
2027	5.0	9.9	8.5	6.5	95.4	78.3	58.5
2028	5.0	9.9	8.5	6.5	104.8	84.9	62.3
2029	5.0	9.9	8.5	6.5	115.2	92.1	66.3
2030	5.0	9.9	8.5	6.5	126	100.0	70.7
Growth Scenarios:		S 1 - Business as Usual (BAU)					
		S 2 - Efficiency Improvement Measures and fiscal Regimes					
		S 3 - Efficiency improvement plus fuel diversification					

Like Barbados, Jamaica has also noted the benefit of reduction in oil generated energy to increase energy from IUS technologies to reduce its oil import bill. The 2009 Strategy projected

US 555 million in the energy import bill in 2020, with an additional US 1.7 billion with the implementation of a national diversification program.¹⁸⁶

2019 saw the increase in RE usage particularly from bioenergy (445 [Terajoules] TJ) with decreases in the industry, transport and household sectors and a significant increase in other sectors as seen in Figure 3-22. However, the country's capacity in 2021 has seen a different use of RE with Wind and Solar Energy capacities of 39% and 36% respectively, signally a shift in the demand of RE under the IUS models.

Figure 3-22: Jamaica RE Consumption by sector in 2014 and 2020¹⁸⁷

Consumption by sector	2014	2019
Industry (TJ)	2 018	1 607
Transport (TJ)	1 260	1 581
Households (TJ)	3 907	3 170
Other (TJ)	910	2 182

3.3.4 Barriers

Some of the specific challenges women face in terms of access to affordable energy, and more specifically with their roles as typically the primary care providers in the home, include having access to energy for maintaining domestic work, access to water and energy options that are both safe and affordable, and help ease the burden women already face by taking on more hours of labour, through their domestic duties and those outside the home. Various barriers also affect uptake and use of technologies.

Institutional

A weak enabling policy environment for SWH also poses a barrier to implementation. While Jamaica has multiple policies on renewable energy, there seems to be only one with stipulations specifically for SWH. Jamaica's National Energy Policy 2009-2030 provides incentives for the installation of SWH and solar lights in the public and private sectors and communities are included under Goal 1: Jamaicans use energy wisely and aggressively pursue opportunities for conservation and efficiency.¹⁸⁸ The incentives, include concessional loans and import duties on EE and RE, and the exemption of lithium-ion batteries from the General Consumption Tax.

Lack of public awareness also limits the usage of SWH systems. There are currently no specific outreach campaigns for SWH in Jamaica. There have been efforts to increase awareness of the merits of renewable energy more generally, however, in order for women to have access to this renewable source of energy, financial incentives for system installations, as well as low-interest rates loan programs, should be implemented.¹⁸⁹

Another barrier that arises in the administrative process involved when applying for the installation of new lights or the replacement of lights. The process includes many steps and can take a long time to implement, posing an obstacle to persons applying.¹⁹⁰ This process makes it more difficult to have consistent street lighting. The lack of street lighting increases the risk of domestic violence and other forms of gender-based violence to occur during times where

¹⁸⁶ Ibid.

¹⁸⁷ "Energy Profile – Jamaica" retrieved from https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Central%20America%20and%20the%20Caribbean/Jamaica_Central%20America%20and%20the%20Caribbean_RE_SP.pdf

¹⁸⁸ "Jamaica's National Energy Policy 2009-2030" retrieved from https://www.mset.gov.jm/wp-content/uploads/2019/07/National-Energy-Policy_0.pdf

¹⁸⁹ "The Role of Women in Sustainable Energy Development" retrieved from <https://www.nrel.gov/docs/fy00osti/26889.pdf>

¹⁹⁰ "Applying for Street Lights" retrieved from <https://localauthorities.gov.jm/services/%E2%80%8BApplying-street-lights>

sunlight is absent. With a high rate of violence within the country, the lack of this infrastructure increases the risk and probability of these events occurring and further burden the already burdened police and security services in the country.

Social

Without increasing the provision of support through the addition of reliable and sustainable services and infrastructure, it is more difficult for women in low-income households to improve their positions and livelihoods, their ability to earn decent livelihoods, and to improve their socio-economic situation and outlook. Without target actions in these areas, the imbalance women face in their economic status from abilities to engage in income-generating activities and benefit from time savings, may consequently lead to further negative impacts across all areas mentioned including health, housing, and education for both women and the families they provide for.

As primary consumers and users of domestic energy, for CDRs, e.g. washing, household cleaning and personal care, cooking etc., women need to be involved in energy efficiency and renewable energy projects from their direct and indirect impacts on levels and usage of household energy consumption. Women may also inform their children's future energy consumption and conservation practices through their own examples and sustainable behaviours in the home. They may also engage children in discussions around energy usage, incorporate energy-conscious practices into daily life, or utilise educational resources and involve children in energy-saving initiatives.¹⁹¹ These means of engaging with their environment impacts the way future generations will learn to engage in efficient and sustainable energy use and practices in a household.

There are other barriers faced in implementing rooftop solar, in particular. Due to the lower discretionary income in low- to medium-income communities, hefty upfront expenses for rooftop solar can be a significant deterrent. Even when upfront costs are decreased, low-income persons do not always benefit fairly from such offers. Single-family homeowners also frequently encounter housing-related obstacles to implementing rooftop PV, such as unsuitable rooftops, and shoddy or outdated electrical wiring. The best roofs for solar have a solid foundation, no leaks, and do not require much maintenance when first installed.¹⁹² Families with lower incomes are more likely to reside in older homes than those with higher incomes, necessitating more expensive home renovations to accommodate on-site solar arrays.¹⁹³

Renting also can present barriers to accessing solar. In 2008, almost 63% of Jamaicans lived in their own homes. Regarding other common household tenure types, the report says 19.1% of Jamaicans occupied rented or leased property.¹⁹⁴ A survey found that investors consider affordability issues to be just as relevant as renters' unwillingness to pay higher rents for buildings with solar PV as investment roadblocks. The focus of apartment or townhouse owners is one of the structural barriers.¹⁹⁵ Moreover, adding to this barrier is the unknown element of who bears responsibility for applicable, and any damages that installation or usage may cause to the property.

3.3.5 Institutional Gender Mainstreaming Capacity

From Figure 3-23, it can be seen the NGO organisations possess gender mainstreaming capacity that can be used to address the noticeable lack of gender mainstreaming capacity held

¹⁹¹ "The Role of Women in Sustainable Energy Development" retrieved from <https://www.nrel.gov/docs/fy00osti/26889.pdf>

¹⁹² "What are the Roof Requirements for Solar Panels" retrieved from <https://solarworksenergy.com/what-are-the-roof-requirements-for-solar-panels/>

¹⁹³ "Affordable and Accessible Solar for All: Barriers, Solutions, and On-Site Adoption Potential" retrieved from <https://www.nrel.gov/docs/fy21osti/80532.pdf>

¹⁹⁴ "Survey Shows Increase in Home Ownership" retrieved from <https://jis.gov.jm/survey-shows-increase-in-home-ownership-2/>

¹⁹⁵ "Solar for renters: Investigating investor perspectives of barriers and policies" retrieved from <https://www.sciencedirect.com/science/article/pii/S0301421523000022>

by the government agencies. These NGOs present an opportunity for government stakeholders to engage and bolster their own limited gender capacities while fostering relationships between the state and representatives of vulnerable groups.

Lack of information for the Executing Entity (JPS) is of concern ahead of the project development stage as understanding their capabilities will be key for designing and developing a gender action plan for this project and well as offering inputs into the TA component.

Figure 3-23: Jamaica Stakeholders' Gender Capacity

Stakeholder	Does your organisation have any experience with mainstreaming gender in its past work?	Does your organisation have the capacity to execute gender mainstreaming activities or provides gender sensitisation training/gender-related technical inputs?	Does your organisation possess a gender policy or is it in the process of developing a gender policy?	Has your organisation ever been engaged by government ministries/agencies or development partners to support in the development or implementation of energy/renewable energy projects?	Does your organisation utilize a grievance redress mechanism for projects?
J-FLAG	YES	YES	YES	NO	NO
TransWave Jamaica	YES	YES	YES	NO	NO
Her Flow Foundation	YES	YES	YES	NO	NO
Women's Media Watch (WMW)	YES	YES	YES	NO	NO
Jamaica Association for the Deaf	YES	YES	NO	NO	NO
Ministry of Science, Energy and Technology	YES	NO	NO	YES	YES
Office of Utilities Regulations	NO	NO	NO	YES	YES
Jamaica Public Service ¹⁹⁶	N/A	N/A	N/A	N/A	N/A

¹⁹⁶ Consultant was unable to meet with the JPS and as such will not be able to provide answers for the gender mainstreaming questions for this organisation (N/A will be applied in response to the questions).

4

Multi-Country Level Analysis

Key chapter takeaway

Though progress has been made across the region on gender equality, women's empowerment and social inclusion generally and in energy access and affordability, progress and sustainable progress is uneven. While the policy environment shows encouraging signs, still more needs to be done on financial and social barriers as well as instruments designed to improve an even promotion of a range of technologies.

4.1 Comparative Policy Environment Analysis

Looking across the three countries, based on the analysis executed, it can be determined that Barbados is presently lagging behind the other countries in terms of gender-responsive policies. From all the Barbadian policies reviewed, only the Barbadian NDC was found to be gender-responsive which may be due to how recently it was developed and the guidelines around NDC development and submission. Belize, and Jamaica (following closely behind) are making greater strides in incorporating gender mainstreaming into their gender policies to ensure the vulnerable are sufficiently covered within their future plans and activities.

It is important to note that Barbados's energy policy released in 2019 failed to mention gender, women or vulnerable groups at all putting the policy on par with Belize's and Jamaica's much older energy policies 2011 and 2009, respectively. This has implications for Barbados as the policy is meant to drive country's energy direction until 2030.

The gender policies reviewed for Belize and Jamaica were all transformative, indicating that there exists a good framework for gender-responsive action to occur within these countries. Two out of the 4 NDCs were gender-responsive which would be due to the acknowledgement of the differentiated impacts of climate change felt by different groups but as well as the need to outline actions that would mitigate and address these issues to ensure the needs of the populations are met.

A major focus of the desk-based research was delving into the policy environment for renewable energy and gender for each country to understand the enabling environment that exists within each. With this focus and the nature of the project in mind, the policy analysis sought to look at the countries' NDCs (outlines the countries' actions to combatting climate change which has implications for energy use and social impacts on persons), energy policies, gender policies (including policies related to the protection or support of vulnerable groups), utilities Acts and other development policies that would guide the countries directions.

Twenty-six policies were reviewed overall and while 12 were gender-responsive or transformative, in general, the policy environment was found to be gender-aware at best (See Figure 4-1). Specifically looking at the energy policies of the countries which would have major influence on the execution of this project, only the Guyanese draft policy was found to be gender-responsive.

Figure 4-1: Target Countries Comparative Policy Environment¹⁹⁷

Policy		Barbados	Belize	Jamaica
NDC		Responsive	Responsive	Sensitive
National Energy Policy		Neutral	Aware	Neutral
National Gender Policy		N/A	Transformative	Transformative
Utility Act		Neutral	N/A	N/A
Sustainable Energy/Energy Conservation Policy/Framework		Neutral	Neutral	Neutral
Employment Rights/Child Labour		Aware	Transformative	N/A
GBV Plan/Policy		N/A	Responsive	N/A
Barbados Tourism Master Plan 2014 – 2023		Neutral	N/A	N/A
Barbados' Control of Inefficient Lighting Act 2021		Neutral	N/A	N/A
Jamaica's National Renewable Energy Policy 2009-2030		N/A	N/A	Neutral
Jamaica's Vision 2030		N/A	N/A	Transformative
Jamaica's Long-Term Emissions Reduction and Climate Resilient Strategy and Operationalisation Plan		N/A	N/A	Transformative

Two policies from CARICOM were also assessed as they pertained to both gender and energy within the region i.e., the draft Gender Equality Strategy for the CARICOM region and the CARICOM energy policy (see Figure 4-2) and how they could inform multi country projects such as the IUS project.

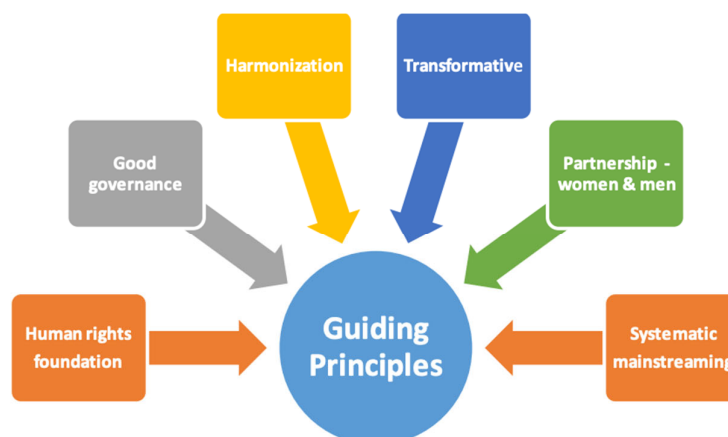
¹⁹⁷ N/A used in the comparative means that specific policy was not reviewed for the country, or the country does not possess a similar policy. For country specific policies, the countries were mentioned before the policy.

Figure 4-2: Policy Environment for CARICOM

Policy	Summary	Gender Content	Gender Rating
Draft Gender Equality Strategy, 2022	A framework of action for achieving gender equality and women's empowerment within the greater context of the Sustainable Development Goals (SDGs) and provides coordination to support Member States compliance with international commitments, including CEDAW.	The document recognises the systemic nature of gender inequality that is present in political, social, and economic institutions and substantiated through social/cultural norms, values and behaviours.	Transformative
Energy Policy, 2013	The Energy Policy for CARICOM is intended to guide the transformation of energy sectors in the Caribbean Member States, ensuring secure and sustainable energy supplies that are affordable and accessible to the region's population.	While the policy refers to the need for access to energy of poor and vulnerable groups, there is no specific reference or mention of women or gender within the policy.	Neutral

Consultations were started in 2022 for a draft gender equality strategy for the CARICOM region titled 'Stepping it up: a strategy to achieve gender equality in the Caribbean Community'.¹⁹⁸ The document comprises a series of strategic actions to achieve gender equality and women's empowerment in the context of the SDGs. The strategy reflects the work of the CARICOM Gender Equality Policy which aims to strengthen CARICOM secretariat capacity to promote gender equality and women's empowerment in partnership with member states and international and regional partners toward improving social and cultural prosperity while improving efforts to mainstream gender. The CARICOM Gender Equality Strategy is driven by six guiding principles and commitments as shown in Figure 4-3.

Figure 4-3: CARICOM Gender Equality Strategy Guiding Principles and Commitments



This work centred on a human rights approach, which recognises every person's right to freedom from discrimination and violence while operating in a just and equal society. It also upholds respect for diversity and social inclusion, including those most vulnerable: PWDs, elderly, youth, and people living with illness. The strategy also prioritises governance which further recognises the rights of women and men in terms of equal access and participation in decision-making in order to create effective governance that recognises the unique needs, barriers, and opportunities for both women and men. The harmonisation principle references the strategy's

¹⁹⁸ "Stepping it up: A Strategy to achieve gender equality in the Caribbean Community" retrieved from https://www.cepal.org/sites/default/files/events/files/stepping_it_up_draft_cges_synthesised.pdf.

position within guiding international frameworks including CEDAW and the SDGs, specifically SDG 5. These build on 2003 efforts to develop a 'Plan of Action to 2005: Framework for Mainstreaming Gender in Key CARICOM Programmes. This Plan was developed to provide more support for mainstreaming gender in CARICOM's work, including research and policy implementation and design.

The strategy firmly recognises the systemic nature of inequality, as based within social, political and economic institutions while sustained through the norms, values and behaviours present in society. It also notes the goal toward reaching substantive equality through equal access and equal benefits which are reached through sector coordination in order to influence/inform/support? substantive change. The strategy also calls for greater partnership between men and women as a shared responsibility toward gender equality through transforming traditional gender roles, in addition to attitudes and behaviours. Finally, the strategy calls for a systemic and cross-cutting approach to achieve the desired outcomes, including allocation of resources, both human capital and financial resources to encourage capacity toward mainstreaming gender in the Caribbean community.

Based on these guiding principles, the strategy outlines six key areas for action: equality and social inclusion, freedom from violence, economic empowerment, good governance, access to health services, and inclusive quality education – at least three of which are directly relevant to a just energy transition and the others enabling a clean and sustainable economy. While there is no specific reference within the policy to access to affordable and renewable energy with reference to women or other vulnerable groups, within the context of economic empowerment, the policy notes the need for equitable access to productive resources - where renewable energy and energy resources could potentially fit.

The other overarching policy reviewed from a regional perspective is the CARICOM Energy Policy.¹⁹⁹ The Energy Policy was approved in 2013 at the forty-first special meeting on energy, in Trinidad and Tobago. The purpose of the policy is to support the “fundamental transformation of the energy sectors of the Member States of the Community through the provision of secure and sustainable supplies of energy in a manner which minimises energy waste in all sectors”.

The Policy also supports ensuring that all CARICOM citizens can access 'modern', clean and reliable energy resources at prices that are both stable and affordable while ensuring the growth and facilitation of regional industries, competitive in the global market with the intention toward creating sustainable development for the Caribbean region. Within the policy's 15 objectives for expanding the energy sector, it recognises the need for having improved access to affordable energy for the region's poor and vulnerable populations. It also notes, as important for this project, the need to create strategies for maintaining sufficient energy supplies in the event of disasters and improving R and D in terms of clean and renewable energy sources. The policy, however, in terms of access to affordable and reliable energy, makes no specific reference to gender or women.

Two policies from CDB were also assessed as they pertained to gender, energy, youth and indigenous peoples within the region i.e. Energy Sector Policy and Strategy 2022²⁰⁰ and Gender Equality Policy and Operational Strategy.²⁰¹ These policies were analysed as they outline CDB's operating principles and future plans for working and supporting these areas as well as guiding how projects funded/investments supported by the CDB should operate.

From the analysis done on these policies, it can be determined that CDB's own enabling environment is geared towards fully supporting the implementation of this project (see Figure

¹⁹⁹ “Energy Policy – Summary” retrieved from https://caricom.org/documents/9360-energy_policy_nutshell.pdf

²⁰⁰ “Energy Sector Policy and Strategy 2022” retrieved from <https://www.caribank.org/energy-sector-policy-and-strategy-2022>

²⁰¹ “Gender Equality Policy and Operational Strategy” retrieved from [https://www.caribank.org/sites/default/files/publication-resources/GEPSOS-2019-FINAL%20\(1\)%20\(1\).pdf](https://www.caribank.org/sites/default/files/publication-resources/GEPSOS-2019-FINAL%20(1)%20(1).pdf)

4-4). It provides an example or even a framework that can be employed within the three target countries to facilitate the implementation of this project. The gender equality principles found within the gender equality policy should be adopted and applied to the project's operations and development as it would allow for gender equality to be effectively incorporated within the project.

Figure 4-4: Policy Environment for CDB

Policy	Summary	Gender Content	Gender Rating
Energy Sector Policy and Strategy 2022	The policy outlines the need for the transformation of the sector within BMCs and the shift towards more sustainable energy states while addressing issues of climate change and other factors that will impact the sector.	The document acknowledges gender imbalances found within the sector and outlines principles and actions aimed at addressing these issues and while further supporting gender equality within the sector	Transformative
Gender Equality Policy and Operational Strategy	The policy seeks to reinforce CDB's commitment to advancing gender equality among its BMCs and outlines principles and actions to support this advancement	The document outlines various aspects of society that have gender implications or are impacted by gender inequalities and sets out principles and actions aimed to address these issues and guide actions.	Transformative

4.2 Cross-cutting Stakeholder Findings

Despite being from different countries and operating in different contexts, stakeholders shared similar insights and concerns with regard to the issues discussed during direct engagement via key informant interviews and group interviews as well as focus group discussions. Below are some key findings that emerged and are applicable across all countries.

4.2.1 Finance

- A major hurdle to accessing loans particularly for women is time. Banking transactions typically require a person to allocate a fair amount of time and effort to engage these services and women especially those from female-headed households are usually constrained for time due to their domestic duties and responsibilities.
- Location plays a role in hindering access to financial tools, particularly for many who reside in rural locations (especially those living in indigenous communities). The nearest financial institution is usually very far away from where they reside, and the distance acts as a deterrent. This particular barrier can also involve time - the time it takes to get to and from the bank/financial institution and the cost to get there and back.
- Banking processes can be tedious and the requirements to engage in their services such as proof of address, form of national identification, lack of collateral often eliminate or discourage persons from seeking these options.
- Persons working in the informal sector and low-income persons utilise informal saving mechanisms such as syndicate, partner²⁰², boxhand, etc. to pool financial resources that allow persons to conduct large transactions when they are unable to meet requirements imposed by financial institutions²⁰³. In some cases, these pooled resources may be used to meet downpayments requested by financial institutions.

²⁰² "Jamaica's ROSCA: The Partner System" retrieved from <https://www.gdrc.org/icm/partner-sys.html>

²⁰³ "Building Economic Solidarity: Caribbean ROSCAs in Jamaica, Guyana, and Haiti" retrieved from https://link.springer.com/chapter/10.1057/978-1-137-60047-9_5

4.2.2 Gender Mainstreaming Capacity

- Efforts are being made by energy ministries to enhance their gender mainstreaming capacities. The Ministry of Energy and Business of Barbados noted that work on a gender equality strategy and action plan for the electricity sector is being advanced that could have bearing on this project (however, access to this document has not been realised).
- Many NGOs, particularly the disability-focused NGOs noted that they are rarely consulted for any project development or implementation, and it could be seen in the results of the project as their vulnerable groups do not benefit fully from these works.
- Gender mainstreaming capacity is low among government stakeholders, noting many do not possess gender policies and rarely engage with the gender bureaux of their countries. However, it must be stated that with increased engagement with international funds which require gender to be involved in any work they fund, capacity is slowly being developed and more awareness on gender is being raised.

4.2.3 Employment in the Energy Sector

- The energy sector is traditionally not one in which women participated significantly in the past but in more recent years, this has been changing with women involved across several areas. Numbers remain relatively low as men still dominate this field. Feedback received via stakeholder interviews (both KIIs and FGDs), suggest that a few technical roles in the Energy Ministry are now occupied by women.

4.2.4 Engagement with Vulnerable Groups

- NGOs', in their responses, indicate that they are rarely consulted by project actors on energy projects.
- Many government stakeholders noted that there are no explicit policies that guide stakeholder engagement to ensure its gender-responsiveness or inclusiveness.

4.2.5 Indigenous Persons (Belize)

- Many of the indigenous people are found in far/remote or rural areas of the countries and exhibit the highest rates of poverty in the country and share similar vulnerabilities and challenges with access.
- Respect must be paid to governing systems employed within the different communities and knowledge of these systems is critical to successfully navigating engagements with local indigenous populations.
- Recognising and acknowledging that a "one size fits all" approach will not work when engaging indigenous communities as they hold varying traditions and practices.

Exploring the technologies with the stakeholders, many shared similar perspectives and information on uses, benefits and demand for the target technologies (see Figure 4-5). The inputs gained from stakeholders were fairly aligned with what was discovered through desk-based research but also provided further country context based on lived experiences.

Figure 4-5: Key Concerns raised for the target technologies

Technology	Use	Benefits	Demand
LED lighting	Street lighting and household lighting	Street safety and safety in public spaces Small savings on monthly electricity bills.	Awareness needed on the cost benefits of switching to LED bulbs as the less informed may opt to continue using fluorescent bulbs due to the lower upfront costs.

Technology	Use	Benefits	Demand
Solar PV	Rooftop/household electricity access	Expanding access to electricity for persons in remote areas.	Outside of remote areas, demand is low as persons in more urban or rural areas with adequate connection may not see the need for this technology and may be dissuaded due to upfront costs.
SWH	Household cleaning and hygiene and sanitary purposes (personal)	Health benefits for elderly persons related to physical and mobility. Women's/girls' hygiene and sanitary care	Thriving market in Barbados but other countries, the demand is not as high and require some education on the benefits and how the initial costs will eventually return over a period. Likewise for SPV above.
Efficient Cooling and Refrigeration	Overall house and room cooling reducing ambient air temperatures keeping food products safe and sanitary	Combat rising temperatures provide relief to persons at home or work	ACs are expensive while there is a demand for them due to rising temperatures, the cost can be prohibitive to many.

From the stakeholder feedback (reflected also in the Table), there is uniformity in the uses and benefits of the technologies across the three countries. While there is demand for each technology, Barbados sees the most demand for SWH due to the extensive framework supporting the technology in the country with this framework being supported by the provision of incentives such as income tax rebates. Noted across the board by stakeholders is that the cost of the technologies being the main prohibitive factor to persons purchasing these items.

4.3 Barriers Across the Region

From the analysis of the three countries, a pattern of barriers has emerged. Despite differences in location, culture, economies, etc. the people particularly vulnerable groups all face similar barriers when it comes to accessing renewable energy technologies. While in Barbados, we see the most support for the adoption of renewable energy technologies by the population, vulnerable groups still encounter comparative issues faced by persons in countries with less of a supporting framework.

There is relatively poor access to and availability of sex and age-disaggregated data across the three countries and the Caribbean region in general, particularly on energy use, demand, needs and uptake. This results in challenges in differentiating energy usage among men, women, gender-diverse persons, PWDs, indigenous persons, children and the elderly. Current available energy data usually is household rather than individual specific and does not engender intersectional analysis. Also available to some extent is data/information on implications for the design and implementation of energy services, for education campaigns, and about financial measures. This means that existing and future actions utilising current data are less effective as the approaches would tend to be more generic. A generic approach is not desirable for many reasons including the likelihood that investments will not address the practical or the strategic gender needs that do exist amongst various groups and that it would make it difficult to also differentiate impacts across groups of beneficiaries.

Economic barriers are heavily present in all three countries as strict requirements for loans imposed by banks is the norm across the region and women are challenged to enhance their access and control of such resources because of reduced economic ability due to traditional divisions of labour or burdens imposed by unpaid care work. In Belize, indigenous people exhibit the highest rates of poverty while also being located in the most rural areas of their respective countries and with the greatest collective need for modern and accessible energy solutions. Not only do they have to navigate the loan requirements, but they must also contend with the challenges that time and distance imposes on them.

Socially, we can see in the three countries that work in the energy sector is skewed more towards men than women. Even the small percentage of women working in the sector mostly take on roles traditionally designated to women such as clerical and administrative work with few women taking on technical roles in the field. This gendered division of employment is heavily influenced by patriarchal norms²⁰⁴ and patriarchal norms' influence is not only restricted to formal work. The gendered division of labour can be seen in the allocation of domestic work which is skewed disproportionately to women and is a common occurrence in the region. This imbalance of domestic labour impairs women's economic ability through the limited finances they are able to attain due to the resulting time-poverty that comes from them performing the majority of domestic, primarily unpaid care work.

²⁰⁴ "Gender at Work in the Caribbean" retrieved from https://www.ilo.org/wcmsp5/groups/public/---americas/---ro-lima/---sro-port-of-spain/documents/publication/wcms_651944.pdf

5

Key Conclusions

Key chapter takeaway

While the demand and need for an IUS project is clearly present in the target countries and there exist enabling environments of varying degrees to support the project. At present, given the capacities of key stakeholders and state of policies, the ability of this project to successfully support and promote a Just Transition is highly questionable and will need several high level as well as practical interventions to ensure it is able to fulfill this goal. Key opportunities to enable the participation of low-income and single-headed households could be tapped into as well, leaving fewer persons behind in the energy transition in the short and medium-term. Gender and social considerations are relevant in all countries.

Based on the engagement with stakeholders and the research, there is both appetite and increasing need for renewable energy technologies, particularly affordable, easily maintained and those that generate multiple co-benefits. From available data and limited consumer analysis, demand is primarily rooted in potential cost and income savings and the additional potential for income-generation that may arise. As a result, the climate change adaptations benefits derived from these technologies are often viewed as additional benefits and does not serve the main motivation for driving the interest/desire of households and businesses for technologies. For government actors, a combination of these factors appears relevant, including the combined mitigation-adaptation co-benefit.

Despite this demand, many persons across the three focus countries are restricted from investing in these technologies primarily due to financial barriers. Uptake of these technologies for low-income persons is often restrained by limited financial resources or inability to access additional finance from financial institutions due to various factors such as limited collateral, time conflicts, and competing priorities. This project's ability to market and distribute these technologies in a way that minimises strain on low-income persons will be key. Doing so in ways that respond to social, cultural and livelihoods needs is crucial to meet multiple SDGs.

Consideration must be given to involving vulnerable groups in the further development and implementation of this project. The goal of this project supporting a Just Transition cannot be accomplished without involving those typically left behind and who are generally underrepresented in the development and expansion of the sector and in reaping the benefits from significant public sector and public finance investment. Moreover, the project should be actively gender-responsive paying close attention to intersectional gender needs as well as intersecting energy demand.

For Belize with large rural populations and significant indigenous sub-populations, customer outreach is crucial. Coupled with the financial issues mentioned, these communities also deal with the issues related to their location meaning their isolation and distance from the rest of the country and from technology hubs. Likewise, the needs of PWDs require greater attention and a specialised response.

Noting the lack of gender mainstreaming capacity amongst the government stakeholders who will be involved in the project and critically the lack of engagement with most of the executing entities, there are clear gaps that will need to be bridged. Efforts to boost gender awareness and sensitivity in ways that shape implementation, sustainability and scaling-up will be critical in the pre-implementation period of the project, as well as the first year of implementation.

6

Recommendations

Key chapter takeaway

Ensuring the project can fulfil its goal of supporting a just transition will require the execution of some key general and stakeholder-related capacity building activities, aimed at enhancing and addressing existing capabilities and gaps that were found in this gender assessment. Critical gender actions have been identified in 6 of 7 gender needs and some actions will be catalytic while others are more long-term in also setting a key good practice for other multi-country energy programmes. Utilities, less familiar with the requirements of international public finance, will experience a somewhat sharp learning curve but one that will boost the scope for multiple and expanding public-private sector partnership.

Emerging from the gender assessment, some of the recommendations to be explored in this section will be based on a 7-point gender needs framework. The recommendations touch on 6 of the 7: leadership and decision-making, gender-based rights and participation, governance, gender-based education and assets, access to and control over resources, and livelihoods. Recommendations are divided into two categories; 1) general recommendations that speak to actions that will boost programme actors' capacities and 2) stakeholder engagement recommendations that seek to guide programme actors' interactions with vulnerable groups and overall communication with beneficiaries and enhance the level of ownership and participation at the local, community, group, household, and individual level.

6.1.1 General recommendations

These recommendations seek to empower programme actors and address gender mainstreaming gaps identified in the assessment. The general recommendations are as follows:

1. Partnership guidelines aimed to serve as a framework to be used by programme actors (government stakeholders) to guide how they interact and form working relationships with NGOs.
2. Partnership building exercises aimed at building strong working relationships between government agencies and NGOs as well as between government agencies to eliminate potential siloing of activities.
3. Fostering collaboration between the three countries to enhance knowledge and experience sharing.
4. Execution of an in-depth gender capacity assessment of key government stakeholders and utilities involved in the programme. The development of gender sensitisation and gender capacity-building training will be based on the results of the gender capacity assessment.
5. More in-depth analysis of the financial tools being offered by the banks, credit unions, etc. to ensure the financial tools being offered by the utilities as part of this programme are competitive.

6. A review of the programme's logic framework and the design of gender targets of the prog including sex disaggregated indicators.
7. Development of training programmes that could help persons to seek employment in the energy sector while supporting the distribution and installation of technologies.
8. The IUS programme should provide support designed to address gender inequalities such as - gender sensitive repayment terms (no ID required; longer periods; lower collateral requirements; etc); special and appropriate technology offers for women in MSME business; adjustable rates for female headed households with children, targeted capacity building outreach to women-led businesses and other vulnerable groups to support their ability to develop sub-projects.
9. Consideration will need to be given to removing or mitigating inequity imbalances in terms of access to technologies including exploring more supportive pay back terms of loans, and or special process on AC/SWH for women customers.
10. Assess lessons learned from existing and past loan and grant programmes that target or engage vulnerable groups so that these can inform the GCF programme design and implementation and also shape financing options and how these can be accessed differently by such groups.

6.1.2 Stakeholder engagement recommendations

Given the outward-facing nature of this programme and the many potential actors that could be involved throughout the its activities, there is a need to create a framework for gender-responsive engagement with the beneficiaries. The stakeholder engagement recommendations are as follows:

1. Development of a code of conduct for all programme actors to follow when engaging the public and particularly direct beneficiaries.
2. Development of GBV checklist to reduce the potential of GBV occurring within the project due to the interactions between programme actors and the public and between different publics as well as at the household level.
3. Development a communication strategy that aims to provide the public with timely updates on the programme and ensures that communications including updates and technology awareness is crafted to ensure the information is easily understood and accessible to those who may be blind, deaf or speak different local languages such as the indigenous populations.
4. Development of a Grievance Redress Mechanism (GRM) for the programme that is readily accessible and easy to utilise by the public including multiple methods to access and use.
5. Development of a gender-responsive stakeholder engagement plan to guide all programme engagement including inclusive and gender-sensitive communications.
6. The hosting of townhall meetings or alternative approaches to engage specific vulnerable groups to ensure their issues and concerns are considered within the programme.

Figure 6-1 aligns the recommended actions to gender needs and identified barriers mapping out in summary key elements to be further expanded on and programmed through the Gender Action Plan (GAP). It responds to three main barriers: institutional, social and economic/financial.

Figure 6-1: Summary of Recommendations and associated gender needs

Barriers identified	Relevant Gender Needs	Proposed Gender Actions
Institutional	Gender-based rights and participation	Partnership guidelines aimed to serve as a framework to be used by programme actors (government stakeholders) to guide how they interact and form working relationships with NGOs.
Institutional	Gender-based rights and participation	Partnership building exercises aimed at building strong working relationships between government agencies and NGOs as well as between government agencies to eliminate potential siloing of activities.
Institutional	Gender-based education and assets	Fostering collaboration between the three countries to enhance knowledge and experience sharing.
Institutional	Gender-based education and assets	Execution of an in-depth gender capacity assessment of key government stakeholders involved in the programme.
Institutional	Gender-based education and assets	Development of gender sensitisation and gender capacity building training based off the results of the gender capacity assessment.
Economic /Financial	Leadership and decision-making	Analysis of the financial tools being offered by the banks, credit unions, etc. to ensure the financial tools being offered by the utilities as part of this programme are competitive
Institutional	Governance	A review of the programme's logic framework and the design of gender targets including sex disaggregated indicators.
Social	Livelihoods	Development of training programmes that could employ persons to support in the distribution and installation of technologies.
Institutional	Governance	Development of a code of conduct for all programme actors to follow when engaging the public.
Institutional	Governance	Development of GBV checklist to reduce the potential of GBV occurring within the programme due to the interactions between programme actors and the public.
Institutional	Gender-based education and assets	Development a communication strategy that aims to provide the public with timely updates on the programme and ensures that communications including updates and technology awareness is crafted to ensure the information is easily understood and accessible to those who may be blind, deaf or speak different local languages such as the indigenous populations.
Social	Governance	Development of a grievance redress mechanism for the programme that is readily accessible and easy to utilise by the public.
Social	Governance	Development of a gender-responsive stakeholder engagement plan to guide all programme actors' engagement with the public.
Social	Gender-based rights and participation	The hosting of townhall meetings or alternative approaches to engage specific vulnerable groups to ensure their issues and concerns are considered within the programme.

7

Gender Action Plan

Key chapter takeaway

Ensuring the programme can full its goal of supporting a just transition will require the execution of some key general and stakeholder-related capacity building activities, aimed at enhancing and addressing existing capabilities and gaps that were found in the gender assessment. Critical gender actions have been identified in 6 of 7 gender needs and some actions will be catalytic while others are more long-term in also setting a key good practice for other multi-country energy programmes. Utilities, less familiar with the requirements of international public finance, will experience a somewhat sharp learning curve but one that will boost the scope for multiple and expanding public-private sector partnership.

In the preparation of the proposal for the “Scaling up the Deployment of Integrated Utility Services (IUS) to Support Energy Sector Transformation in the Caribbean (Phase I)” Programme, it is essential to ensure the programme meets both the Green Climate Fund’s (GCF’s) and Caribbean Development Bank’s (CDB’s) requirements for gender equality. As a result, a gender assessment was done and from this assessment, emerged areas of concern and weakness that need to be addressed to ensure gender is adequately mainstreamed in this programme so it can support a Just Transition.

Gender mainstreaming is the process by which gender equality can be achieved through the integration of a gender perspective into key processes, policies and frameworks that usually adopt a gender-neutral stance, and by creating the necessary provisions and relevant supporting documents to facilitate the removal of gender inequalities. Within the context of the IUS programme, this entails ensuring the services and financing models employed by the programme are gender-responsive and provide benefits to marginalised groups as it pertains to the requirements of the GCF and CDB.

For the GCF, its updated 2020 Gender Policy (GP) seeks to ensure the institution and its network of partners adopt a comprehensive gender approach to support climate change interventions and innovations, advance gender equality and minimise social, gender and climate related risks through the promotion of climate investments and support the closing of the gender gap through mainstreaming gender equality in climate investments.

Applying the GCF GP to the IUS Programme would involve:

- Addressing and reducing gender inequality resulting in sustainable livelihoods opportunities, improved health and wellbeing, and increased resilience to climate-induced shocks and risks for men and women.
- Creation of an enabling environment among programme stakeholders founded on existing policy commitments wherein social and environmental standards and grievance redress mechanisms are supported and sustained in order to achieve gender equality.

- Support the comprehensive and systematic collection of sex-disaggregated data and execute qualitative and quantitative gender analysis to monitor, report and evaluate project impact.

GCF's 2018 Indigenous Peoples' Policy seeks to guide the institution's engagement with IP in climate change policies and action while acknowledging IP as a distinct stakeholder group that needs to be involved. The policy is guided by the following principles:

- Develop and implement free, prior and informed consent.
- Respect and enhance the rights of IP to their lands, territories and resources.
- Recognize key international human rights and principles.
- Respect the right of IP under voluntary isolation.
- Respect and recognize traditional knowledge and livelihood systems.
- Enhance the capacity for IP issues within GCF.
- Facilitate access to GCF resources for IP.
- Respecting the system of self-governance.

In the context of the IUS Programme and its GAP, the IP policy would apply to the following areas:

- Programme actors properly seek free, prior and informed consent when engaging with IP throughout the programme's lifespan, and
- Ensure that adverse impacts are avoided or minimized and compensate for residual impacts.

The CDB's Gender Equality Policy and Operational Strategy (GEPOS: 2019) outlines CDB's commitment to advancing gender equality through recognising the equal rights, responsibilities and opportunities of women, men, girls and boys which include the equal power to shape their own lives and contribute to society. The policy seeks to achieve gender equality in BMCs based on the following action pillars:

- Equitable access to infrastructure services for all,
- Economic empowerment of all,
- Education and training for all,
- Eliminating gender-based violence for all, and
- Resilience for all.

The guiding principles of the GEPOS that are applicable in the IUS programme include:

- Advocacy, dialogue and partnership,
- Intersectionality, participatory and inclusive perspectives and approaches,
- Institutional strengthening in BMCs,
- Enhancing leadership for and increasing compliance with gender equality, and
- Results monitoring.

7.1 IUS Programme Approach to Gender Action Planning

Noting that the programme will be implemented in three countries with differing contexts to be considered, the action plan will outline actions that are generally applicable to all three countries and where relevant, actions that are tailored to a specific country will be indicated.

7.1.1 Implementing Partners

The activities outlined in the GAP will be headed and supported by various partners. For clarity, the stakeholders who will be involved in the implementation of the GAP are outlined below. In the GAP, the specific stakeholders will be allocated to each activity, also specifying whether they are the lead or support on the activity.

Stakeholders with responsibilities in the GAP:

- Regional Executing Entity (CDB Technical Assistance/TA Execution Team)
- National Executing Entities (the electric utilities in the BMCs),
- Implementing Partners (government agencies such as ministries of energy, utilities regulators, gender bureaus),
- The Accredited Entity (CDB Accredited Entity Team),
- Consultants (hired for specific activities and roles), and
- Other partners (includes NGOs that represent marginalised groups).

7.1.2 Policy Environments

The gender assessment (drafted and submitted) which informs and accompanies this GAP highlights that the policy environment for energy is relatively gender-neutral for most countries. Accordingly, a universal approach or broad-brush strategy based on model policy and/ or legislation or even approaches to policy reform are not suitable and unlikely to be successful. Any such effort needs to be grounded in capacity-building and behaviour change to be viable, sustainable and scalable into other projects in the future. Each country has its own unique societal circumstances and existing systems that position them differently in how, where and when reforms are suitable, viable and likely to be impactful during the life of the programme and this is an important pre-requisite in defining starting points for policy reform.

There is also fertile ground upon which to identify and achieve win-win opportunities via the policy placeholders set in place in existing policies and frameworks including NDCs which outline operational strategies and guidance for gender mainstreaming in order to achieve tangible gender impact and a gender-just transition through mitigation investments in RE/EE.

7.2 Gender Action Plan

Given this context, the activities in this GAP are divided into (i) general activities that are foundational, required for all countries and critical to creating the right conditions for gender-responsive action and (ii) activities critical for delivering specific gender-responsive actions per outcome/output based on the programme's Theory of Change. The second set of activities are catalysers that build on the foundational elements to dual effect – resulting in a general paradigm shift in utility-led energy transition and investments that is more people-centred and deliver adaptation co-benefits and expands the scope and innovativeness of DERs in terms of the design, scope and application that are tailored also to the common as well as differentiated needs of households, public sector service delivery actors as well as businesses.

Table 1 outlines the GAP in detail including indicators and targets, timelines, responsibilities and costs in keeping with GCF requirements for the GAP. It aligns with several key findings and recommendations outlined in the Gender Assessment.

Table 1: IUS Programme GAP

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
Impact Statement: increased access to clean, affordable energy and access to energy and cost saving technologies for women, low-income persons, persons with disabilities, indigenous people and the LGBTQIA community that meets their needs and aids in building their climate resilience while improving the quality of their lives and livelihoods.				
General Recommendations				
Develop a Code of Conduct (CoC) for engagement with the public (i.e. utility customers). ²⁰⁵	<i>Indicators:</i> Number of CoCs produced. % of contractors and TSPs that sign the CoC. # of GRM Reports regularly produced and actioned. <i>Target:</i> 3 CoCs prepared within 6 months of programme start. 100% of contractors and TSPs sign the CoC. 1 GRM report per quarter.²⁰⁶	Inception /Year 1 first quarter	Overall: CDB Technical Assistance (TA) Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	12,000.00
Develop a GBV Checklist for programme actors' engagement with the public. ²⁰⁷	<i>Indicators:</i> Number of GBV Checklist developed and operational	Inception /Year 1 first quarter	Overall: CDB TA Execution Team Direct:	4,000.00

²⁰⁵ Given the different national and operational context in each country, a code of conduct will need to be developed for each country for energy projects to mitigate and prevent sexual and other forms of exploitation, abuse and harassment, particularly in any engagements involving marginalised groups, vulnerable groups and low-income households where power dynamics will be at play. Interesting work out there exists on sextortion (images.transparencycdn.org/images/2020_Report_BreakingSilenceAroundSextortion_English.pdf) and there have been some discussions in the Caribbean on this as well.

²⁰⁶ This report would note that no incidences occurred for that quarter. However, this number will change if there are cases occurring.

²⁰⁷ One checklist can be used across all three countries but there will be specific provisions for certain groups depending on unique groups such as indigenous people in Belize, and Jamaica.

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
	# of GRM Reports produced and actioned # of GBV Incidence Reports <i>Targets:</i> 1 GBV Checklist developed and operational within 6 months of programme start 1 GRM report per quarter 0 GBV incident reports 0 Grievances reported		Gender Consultant from the Implementation Support Consulting Firm	
Develop and implement an IP operational engagement framework. ²⁰⁸	<i>Indicator:</i> Number of Indigenous Peoples Plans (IPPs) developed. <i>Target:</i> Two (2) IPPs developed – one each in Belize and Jamaica.	Year 1	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	50,000.00
Output 1.1: Strengthened regulatory and institutional frameworks facilitating the deployment of distributed RE, EE and other DER technologies.				
Develop gender-mainstreaming guidelines to support policy reform in line with Activity 1.1.1 and 1.1.2. ²⁰⁹	<i>Indicators:</i> Number of country-focused gender mainstreaming guidelines Number of policies revised to include provisions that advance GESI goals # of SOPs that have been updated with GESI elements	Year 1-6	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	12,000.00

²⁰⁸ Two out of the three countries have significant IP population, there is still a significant amount of IPs living in semi-urban and rural areas that may interact with the programme. This would link into the marginalized group's engagement activity.

²⁰⁹ This activity will be expected to include stakeholder consultation and validation.

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
	<i>Targets:</i> 2 country focused mainstreaming guidelines by Year 1, 3 by Year 2 3 revised policies showing GESI content by Year 6 2 SOPs by Year 6			
Establish a working relationship among BMCs' gender bureaus for oversight of GAP implementation to underpin the broader policy-related support provided under Activity 1.1.1 and 1.1.2.	<i>Indicator:</i> Number of meetings held with gender bureaus <i>Target:</i> 3 meetings held per year, starting in PY2	Year 1-6	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	N/A
Include a review of TSPs' internal policies and practices related to gender, GBV and SEAH when preparing the TSP technical qualification guidelines under Activity 1.1.2.	<i>Indicator:</i> Number of TSP technical qualification guidelines that cover TSPs' policies and practices related to gender, GBV and SEAH. <i>Target:</i> Three (3) TSP technical qualification guidelines.	Year 1-2	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	3,000.00
Output 1.2: Enhanced capacity of customers to invest in distributed RE, EE and other DERs through the IUS model.				

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
In collaboration with relevant NGOs, develop and deliver outreach/marketing strategies and multi-media awareness raising campaigns that are tailored to the needs of marginalized groups as part of the broader outreach and marketing campaigns organized under Activities 1.2.1 and 1.2.2. ²¹⁰	<i>Indicators:</i> Number of outreach/marketing strategies developed that are tailored to the needs of marginalized groups.²¹¹ Number of awareness-raising materials that are designed to target marginalised groups. <i>Targets:</i> 2 marketing/outreach strategies developed per country (6 total) that are responsive to the needs of marginalized groups by Year 2. 2 awareness-raising methods/materials per country (6 total) that are responsive to the needs of marginalized groups by Year 3.	Year 1-3	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm Implementing partners: NGOs National Executing Entities Energy Ministries	25,000.00 ²¹²
Conduct free energy audits for low-income customers under Activity 1.2.3.	<i>Indicator:</i> Number of low-income, vulnerable utility customers receiving a free energy audit <i>Target:</i>	Year 1-6	Overall: CDB TA Execution Team	600,000 ²¹³

²¹⁰ Engagement with representative marginalized groups NGOs for the purpose of tailoring awareness content to meet the needs of their groups.

²¹¹ Cost for this activity would also be inclusive of necessary language translations for developed material.

²¹² This cost is associated with the time spent by the Gender Consultant to work with and support the EEs, NGOs and other stakeholders to design and deliver the campaigns, as well as the costs of preparing the GESI-related materials.

²¹³ This cost is based on an estimated USD 2000 per audit for 300 audits (100 per country).

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
	300 audits (100 per country) done by Year 6			
Conduct a Knowledge, Attitudes, Perception and Beliefs assessment with customers when administering free energy audits under Activity 1.2.3.	<i>Indicator:</i> % of low income/FHH customers who receive free energy audits who express interest and confidence in RE/EE technologies <i>Target:</i> At least 20% of low income/FHH customers who receive free energy audits are interested and confident in the technologies by the end of Year 4	Year 4	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	Costs are included in the budget for the free energy audits (see above). ²¹⁴
Output 1.3: Increased technical capacity of utilities, TSPs and other stakeholders to deploy and maintain distributed RE, EE and other DER technologies.				
Host a (virtual) regional utilities conference during which lessons learnt are exchanged regarding gender-responsive engagement as part of the virtual knowledge sharing sessions organized under Activity 1.3.1. ²¹⁵	<i>Indicator:</i> Number of conferences held with utilities to share new customer engagement practices. <i>Target:</i> 1 regional conference organized with at least 3 utilities in attendance.	Year 2	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm Implementing partners: BEL Other electric utilities	10,000.00

²¹⁴ Focus will be on a broad survey across all countries that allows disaggregation of customers by sex, age, income group and income source.

²¹⁵ This conference would focus on sharing the good practices currently employed by BEL when engaging with their customers.

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
Develop and deliver training modules on gender, social inclusion, and gender-sensitive engagement (including related to GBV and SEAH ²¹⁶) for the participating electric utilities as part of the broader trainings organized under Activity 1.3.2.	<i>Indicator:</i> Number of utilities whose staff have been trained.²¹⁷ <i>Target:</i> Staff from all 3 utilities have received training in PY1 and a refresher training in PY2.	Year 1-2 ²¹⁸	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	50,000.00 ²¹⁹
Develop and deliver training modules on gender, social inclusion, and gender-sensitive engagement (including related to GBV and SEAH ²²⁰) for TSPs as part of the broader TSP trainings organized under Activity 1.3.3.	<i>Indicator:</i> Number of trainings delivered. <i>Target:</i> Nine (9) trainings delivered in total: - 1 training per country in PY1. - 1 follow-up training per country in PY2. - 1 further follow-up training per country in PY4.	Year 1-4.	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	30,000.00 ²²¹
Support the design and development of gender-	<i>Indicator:</i>	Year 1	Overall: CDB TA Execution Team	10,000.00

²¹⁶ Note that this training would pick up elements from the Code of Conduct and GBV checklist (developed in earlier activities) and provide stakeholders with the necessary training to utilize those items.

²¹⁷ Staff should include all levels of staff (including management/board of directors), this is to ensure that while technical staff may be the ones applying knowledge in their work, gender mainstreaming is supported at the higher, ensuring that the enabling environment for gender-responsive engagement is supported at all levels.

²¹⁸ Two cycles of training with the second cycle being a refresher as well as to capture new persons that would have joined later on in the project's lifespan (potentially involving an online course).

²¹⁹ This cost relates to the support to be provided by the Gender Consultant in the development and delivery of the training modules under Activity 1.3.2, as well as the proportion of the total training costs that are estimated to relate to the GESI modules.

²²⁰ Note that this training would pick up elements from the Code of Conduct and GBV checklist (developed in earlier activities) and provide stakeholders with the necessary training to utilize those items.

²²¹ This cost relates to the support to be provided by the Gender Consultant in the development and delivery of the training modules under Activity 1.3.3, as well as the proportion of the total training costs that are estimated to relate to the GESI modules.

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
responsive and inclusive IUS products when developing the IUS business development strategy for/with each national Executing Entity under Activity 1.3.2. ²²²	Number of countries with at least 1 IUS product that is tailored to marginalized groups. <i>Target:</i> At least 2 countries with IUS products tailored to marginalized groups by Year 3.		Direct: Gender Consultant from the Implementation Support Consulting Firm Implementing partner: National Executing Entities	
Support electric utilities to prepare standard forms of contract (under Activity 1.3.2) that include provisions that oblige TSPs to implement measures to mitigate the risk of SEAH and GBV during installation of RE, EE and other DER technologies, and require them to report on the implementation of such measures ²²³ .	<i>Indicator:</i> Number of standard forms of contract that include provisions to mitigate risks related to SEAH and GBV. <i>Target:</i> At least one (1) standard form of contract per country – thus three (3) in total.	Year 1-2	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm Implementing partner: National Executing Entities	10,000.00
Develop and implement guidelines for capturing and analysing beneficiary data disaggregated by sex, age, income, location, and household type when developing the IUS	<i>Indicator:</i> Number of guidelines developed. Number of utilities using/applying the guidelines to	Year 1-4	Overall: CDB TA Execution Team Direct: Gender Consultant from the Implementation Support Consulting Firm	10,000.00

²²² This could entail incentives, procedures, etc tailored to the marginalised groups which are meant to 1) garner interest and confidence in investing into RE/EE technologies and 2) support their ability to invest while considering their economic abilities, constraints and unique needs/circumstances.

²²³ For example: daytime installations; 2-person installation teams; communicating/coordinating planned installation times well in advance to ensure beneficiary customers can have multiple people at the premises during installation.

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
business development strategy for/with each national Executing Entity under Activity 1.3.2.	collect disaggregated data on customers. <i>Target:</i> 3 guidelines developed by Year 2. 3 utilities are utilizing the guidelines and collecting disaggregated data on customers by Year 4.			
Output 2.1: Installation of distributed RE, EE and other DER technologies, as well as complementary smart meters/AMI, among a critical mass of utility customers, including households, businesses and public sector entities.				
Receive, process, approve and finance sub-projects from FHH, PWD and low-income households under Activity 2.1.1.	<i>Indicator:</i> Number of FHH, PWD, low-income households receiving financial support to purchase RE/EE technologies. <i>Target:</i> At least 40 FHH, PWD and/or low-income households in each country (120 total) are benefiting from programme-financed sub-projects by Year 6.	Year 1-6	National Executing Entities	Based on the target for the number of sub-project approvals as well as reasonable assumptions about the cost/value of different types of sub-projects, it is tentatively estimated that this activity could result in total investments made by FHH, PWD and/or low-income households of around USD650,000. Note that this is a rough ex-ante estimate that is subject to change during programme

Activities	Indicators and Targets	Timeline	Responsibilities	Costs (USD)
				implementation. The number of sub-project approvals and associated values will be monitored throughout programme implementation.
TOTAL				1,476,000

Noting the GCF and CDB's gender requirements stated earlier, it is critical that much of the gender mainstreaming work needed for this project (totalling **USD 1,5488,000**), occurs as early as possible during the project lifespan (Table 1). This is needed to ensure that the necessary capacity- building, measures, and systems are in place to support this project's ability to meet these gender requirements while also meeting its own objectives of facilitating access to RE/EE technologies and supporting a Just Transition in the target countries. Understanding this, the Inception Phase and Year 1 of the project will be when most activities occur. The Inception phase activities accrue a cost of **USD 16,000** with activities centred on building capacity and developing systems to support gender-responsive stakeholder engagement which will be essential in developing later actions.

Appendix A Stakeholder Engagement Plan

Background/Introduction to Stakeholder Engagement Process

Stakeholder engagement is aimed to address gaps emerging from desk-based research. Noting the limited work done in the region on gender and energy (particularly gender and renewable energy) and the limited online availability of what has already been done, the main goal is to address gaps such as qualitative data on gender mainstreaming capacity and experience particularly for renewable energy projects, and barriers and constraints to accessing stable, reliable and affordable energy.

This approach builds on the Baseline Assessment and addresses issues considered in the outlines for the Gender Assessment and the Gender Action Plan previously submitted to the Caribbean Development Bank (CDB) for consideration and comment. In addition to this, we will be seeking insight on barriers to and best practices for women's participation and economic empowerment in a just energy transition as well as the requirements for a socially inclusive approach, noting the different marginalised communities and the intersection of different vulnerabilities. This is in keeping with the policies of the Green Climate Fund (GCF) policy and guidelines as well as the CDB's environment and social review procedures as well as the Terms of Reference (TOR) for this assignment.

Gendered and Socially Inclusive Stakeholder Needs being considered

While the project will focus on technology types and supply, a gender and social inclusion analysis also need to consider demands, needs and interests even in a limited fashion and what will or may drive these including uptake of proposed interventions. Accordingly, areas to be explored during these more direct consultations include:

- Barriers and Constraints to accessing stable, reliable, affordable energy and lessons learnt from the Covid19 pandemic.
- Gender Mainstreaming Knowledge, Exposure, Experience and Capacity particularly related to renewable energy projects (particularly for implementing actors)
- Gaps, Barriers and Constraints and Best practices for women's participation and economic empowerment in a just, energy transition
- Requirements for an inclusive approach that considers the specific needs of persons with disabilities, the elderly, youth, LGBTQIA+ persons, indigenous people and other marginalised groups as well as intersecting vulnerabilities²²⁴.

Given the focus of the project, the baseline findings on gender equality and initial gaps as well as the economic context of the three countries – Barbados, Belize, and Jamaica, the following marginalized groups need to be considered in our engagements in the context of the four technologies – rooftop solar, solar water-heating, LED lighting and cooling:

- Women,
- Young people,
- School-aged children for whom alternative and consistent access to energy was a significant issue during the COVID-19 pandemic,
- The elderly for whom cooling can be a critical asset given the NCDs prevalent in the region,
- Persons with Disability,

²²⁴ This refers to persons who meet all or some of these groups at the same time.

- LGBTQIA+ persons or representative groups, and
- Indigenous People.

These inputs, from them directly where possible and from others in terms of how they currently engage with these needs, are critical to the quality of analysis as well as the recommendations that may emerge from the assignment to inform the design of interventions, the gender mainstreaming focus of the project as well as the Gender Action Plan (GAP).

Stakeholder Engagement Type

Stakeholder engagement will take three forms, key informant interviews (KII), group interviews (GI) and focus group discussions (FGD). All forms will be guided by structured questions or presentations and will primarily occur online except in cases where it would be better suited to conduct engagement in the country.

KIIs are planned primarily for government agencies and utilities. These interviews are intended to provide the representatives from these organisations with a safe space to speak about the topics from their organisation's perspective without any outside distractions.

Group interviews are planned for some governmental organisations and non-governmental organisations (NGOs) that work in a closely related field and would benefit from being present with each other during the interview or may feel comfortable in a limited group setting (e.g. LGBTQIA+ organisations).

Focused Group Discussions are planned for NGO stakeholders with the idea that discussions with these organisations working in a similar/the same field would create an engaging, safe space that facilitates discussions and possibly elicit different perspectives when presented with a talking point that they may relate to but have a different experience with.

A summary of the number of planned engagements per country and the type of engagement is provided in Figure A-1.

Figure A-1: Summary of proposed engagements and type of engagement

Country	Total Number of Engagements	KII	GI	FGD
Barbados	8	3	5	0
Belize	7	5	1	1
Jamaica	8	4	2	2

Proposed Consultation Schedule

Stakeholder engagements are planned from **April 3rd to April 21st**. The plan takes note of the public holidays of Good Friday and Easter Monday occurring on **April 7th and 10th** respectively for all countries. Each country will be allotted specific days to hold consultations, this will allow for following up on information with relevant organisations within the same period as necessary.

To facilitate scheduling [Calendly](#) to facilitate appointment setting and to limit back and forth regarding availability. This also allows more control over the stakeholder in identify and selecting a time based on availability in the schedule. This will be designated by the country days identified below.

Country Days Designation

Barbados – 3rd -6th

Belize – 11th – 13th

Jamaica – 19th to 21st

A draft of the email to be sent out to stakeholders follows.

Draft Email to Stakeholders

Dear Sir/Madam:

The Caribbean Development Bank (CDB) in collaboration with the governments and utilities of three Borrowing Member Countries (BMCs) is preparing a funding proposal to the Green Climate Fund (GCF) for the “Scaling Up The Deployment of Integrated Utility Services (IUS) To Support Energy Sector Transformation In The Caribbean” programme. This programme arises out of the needs and requests for support from BMCs in transitioning their energy sectors to more sustainable, renewable energy sources. This first iteration of the programme focuses on Barbados, Belize, and Jamaica. The programme will support these countries to (1) reduce greenhouse gas emissions (GHG) and (2) enhance resilience through the increased uptake of distributed renewable energy (RE), energy efficiency (EE) and distributed energy resources (DER) among customers of the countries’ electric utilities. The programme also aims to enable widespread investment in distributed RE, EE and other DERs while promoting equitable access to such technologies and enhancing electric utilities to be able to continue applying the IUS model after the end of the programme.

CDB has contracted CPCS (in collaboration with FOKABS Inc and SAEDI Consulting Barbados Inc) to support the development of the full funding proposal that meets the requirements of the GCF. SAEDI Consulting’s role is to bring gender and social inclusion expertise to the process. This involves conducting a gender assessment and developing a gender mainstreaming action plan in compliance with GCF guidelines.

To accurately reflect the target countries’ gender equality and socio-economic context and provide recommendations that deliver meaningful, inclusive results for the benefit of the countries, particularly for those most vulnerable, SAEDI Consulting will hold stakeholder consultations between **April 10th to April 21st**. Stakeholders targeted include key governmental and community-level actors in the energy transformation and social development space.

The following days are allocated to the three countries for stakeholder engagement (excludes weekends):

Barbados – 3rd -6th

Belize – 11th – 13th

Jamaica – 19th to 21st

The stakeholder consultations will explore the following:

- Barriers and Constraints to accessing stable, reliable, affordable energy and lessons learnt from the Covid19 pandemic.
- Gender Mainstreaming Knowledge, Exposure, Experience and Capacity particularly related to renewable energy projects (particularly for implementing actors)
- Gaps, Barriers and Constraints and Best practices for women's participation and economic empowerment in a just, energy transition
- Requirements for an inclusive approach that considers the specific needs of persons with disabilities, the elderly, youth, LGBTQIA+ persons, indigenous people and other marginalised groups as well as intersecting vulnerabilities²²⁵.

Your voice counts in this process. We look forward to your earliest response and any information the organisation or the identified representatives may be able to provide that is relevant to these issues.

We encourage confirmations by **April 2nd** as we begin planning for these engagements. To facilitate this process, we encourage you to use our Calendly link to book a suitable time within your country's allocated days. If you require any assistance or have any queries, please contact Kwasi Pierre at juniorassociate1@saediconsulting.com. Thank you in advance for your kind consideration.

Yours Sincerely,

SAEDI Consulting Barbados Inc

Stakeholders by Country

A provisional list of stakeholders per country has been defined and is provided below by country. A total of 30 engagements are planned with an average of 7.5 per country. They build on an expand engagements carried out in October 2022 and are more specific now with a focus on technologies and a just transition.

²²⁵ This refers to persons who meet all or some of these groups at the same time.

Barbados

Figure A-2: Barbados Stakeholders

Stakeholder	Organisation	Contact	Stakeholder Engagement Type	Grouping
Charmaine Gill-Evans	Barbados Light and Power	Charmaine.gill-evans@blpc.com.bb	KII	N/A
Stephen Worme	Barbados Renewable Energy Association	info@brea.bb	GI	N/A
TBC	Gender Bureau	genderaffairs@barbados.gov.bb	GI	Group 1
TBC	Business & Professional Women's Club of Barbados	bpwbarbadosonline@gmail.com	GI	Group 2
Ashley Lashley	The Ashley Lashley Foundation	executivedirector@theashleylashleyfoundation.org	GI	Group 2
	Barbados Council for the Disabled	Tel: 1 246 629-0574 bcd@caribsurf.com	GI	Group 3
John Hollingsworth	National Disabilities Unit	disabilities.unit@barbados.gov.bb	GI	Group 3

Belize

Figure A-3: Belize Stakeholders

Stakeholder	Organisation	Contact	Stakeholder Engagement Type	Grouping
Ryan Cobb	Ministry of Public Utilities, Energy and Logistics	energy@energy.gov.bz	KII	N/A
Dean Molina	Public Utilities Commission	chairman@puc.bz	KII	N/A
Sean Fuller	Belize Electricity	sean.fuller@bel.com.bz	KII	N/A
Juan Chub	Plenty Belize	solarbelize@gmail.com	KII	N/A
TBC	The National Women's Commission	adminassistant.nwc@humandev.gov.bz	GI	Group 1
Ava Pennill	Women and Family Support Department, Ministry of Human Development	senior.secretary@humandev.gov.bz	GI	Group 1
Michelle Irving	POWA (Productive Organisation for Women in Action)	powa2grow@gmail.com	FGD	Group 2
TBC	Belize Audubon Society	info@belizeaudubon.org	FGD	Group 2
Charrice Talbert	PETAL	petalbze@gmail.com	FGD	Group 2

Jamaica

Figure A-4: Jamaica Stakeholders

Stakeholder	Organisation	Contact	Stakeholder Engagement Type	Grouping
Cheryl Lewis	Office of Utilities Regulations	clewis@our.org.jm	KII	N/A
Dionne Nugenet	Jamaica Public Services Limited	dnugent@jpsco.com	KII	N/A
TBC	Consumer Affairs Commission	info@cac.gov.jm	KII	N/A
TBC	Bureau of Gender Affairs	srobinson@mcges.gov.jm	KII	N/A
TBC	WE- CHANGE	wechangeja@gmail.com	FGD	Group 1
TBC	Women's Resources & Outreach Centre	wroccommunications@gmail.com	FGD	Group 1
Tamisha Lee	Jamaica Network of Rural Women Producers (JNRWP)	jnrwpnetwork@gmail.com jnrwppresident@gmail.com	FGD	Group 1
TBC	Woman Incorporated	w.i.crisiscentre@cwjamaica.com	FGD	Group 1
TBC	I'm Glad I'm a Girl Foundation (IGIG)	mshgladgirl@gmail.com	FGD	Group 2
TBC	Sistren Theatre Collective	sistrentheatre@gmail.com	FGD	Group 2
TBC	Jamaica Youth Advocacy Network (JYAN)	jyan@youthadvocateja.org	FGD	Group 2
Hellen Atkins	Women's Media Watch (WMW)	wmwjam@hotmail.com	FGD	Group 2
TBC	Her Flow Foundation	embraceherflow@gmail.com	FGD	Group 2
Rena Green	TransWave Jamaica	transwave876@gmail.com renae@transwaveja.org Nas@transwaveja.org	GI	Group 3

Stakeholder	Organisation	Contact	Stakeholder Engagement Type	Grouping
<i>Glenroy Murray</i>	Jamaica Federation for Lesbian And Gays (J-FLAG)	admin@equalityjamaica.org	GI	Group 3
Kimberly Sherlock Marriott-Blake	Jamaica Association for the Deaf	ksherlock@jamdeaf.org.jm admin@jamdeaf.org.jm	GI	Group 4
Gloria Goffe	Combined Disabilities Association	glo.goffe@gmail.com	GI	Group 4

Appendix B Stakeholders Consulted

Figure B-1 to Figure B-3 present the list of stakeholders engaged for each country with Table 29 additionally presenting stakeholders of that group that were not engaged. These stakeholders that were not engaged were due to time constraints and conflicting priorities that made meeting difficult. Stakeholders from the stakeholder country that were engaged on the same day were done in group interviews. 28 organisations were met during the period of April to June.

Figure B-1: Phase 1 Stakeholder Engagement (NGOs)

Stakeholder	Meeting Date (2023)
Barbados	
Business & Professional Women's Club of Barbados	May 13 th
The Ashley Lashley Foundation	May 13 th
Barbados Council for the Disabled	May 5 th
Barbados Renewable Energy Association	May 10 th
Belize	
POWA (Productive Organisation for Women in Action)	April 13 th
PETAL	April 13 th
Plenty Belize	April 14 th
Belize Assembly for Persons with Diverse Abilities	May 16 th
Jamaica	
Jamaica Federation for Lesbian And Gays (J-FLAG)	April 13 th
TransWave Jamaica	April 13 th
Her Flow Foundation	April 13 th
Women's Media Watch (WMW)	April 13 th
Jamaica Association for the Deaf	April 24 th

Figure B-2: Phase 2 Stakeholder Engagement (Government Agencies and Ministries)

Stakeholder	Meeting Date (2023)
Barbados	
Bureaus of Gender Affairs	June 1 st
Ministry of Energy and Business	May 19 th
Fair Trading Commission	May 19 th
Belize	
Ministry of Energy	May 19 th
Jamaica	
Ministry of Science, Energy and Technology	May 8 th
Office of Utilities Regulations	May 8 th

Figure B-3: Phase 3 Stakeholder Engagement (Utilities)

Stakeholder	Meeting Date (2023)
Barbados	
Barbados Power and Light Company	Unable to meet
Belize	
Belize Electricity Limited	June 2 nd
Jamaica	
Jamaica Public Service	Unable to meet

Appendix C Unconsulted Stakeholders

Figure C-1 presents a list of stakeholders that the consultant attempted to engage as part of the stakeholder engagement process to support the gender assessment. Failure to obtain dates and times to meet or a lack of response from stakeholders were the reasons for the inability to meet with these stakeholders. This list primarily consists of NGOs and a few government organisations from Belize (and excludes the utilities, see Figure C-1).

Figure C-1: List of Stakeholders that were not met during stakeholder engagement

Organisation	Country
Public Utilities Commission	Belize
The National Women's Commission	Belize
National Garifuna Council	Belize
Woman Incorporated	Jamaica
Women's Resources & Outreach Centre	Jamaica
WE-CHANGE	Jamaica
Jamaica Network of Rural Women Producers	Jamaica
I'm Glad I'm a Girl Foundation	Jamaica
Sistren Theatre Collective	Jamaica
Jamaica Youth Advocacy Network	Jamaica
Combined Disabilities Association	Jamaica